

ClearPath OS 2200

Exec System Software Operations Reference Manual

ClearPath OS 2200 Release 19.0

July 2020

7831 0281-034

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Section 1

Operator Environment

This section gives you an overview of the operator environment by introducing the hardware, concepts, and tools you need to operate the system effectively. It describes the responsibilities that are typically handled by the system operator, provides an overview of the system hardware, and introduces the system software. Detailed information about the hardware and software is available in other manuals. Refer to the manuals listed in Appendix B.

1.1. About This Manual

This manual describes operator responsibilities and procedures for OS 2200 systems. It provides basic concepts and objectives of system operations and gives specific operating procedures.

Documentation Updates

This document contains all the information that was available at the time of publication. Changes identified after release of this document are included in problem list entry (PLE) 19251489. To obtain a copy of the PLE, contact your Unisys service representative or access the current PLE from the Unisys Product Support website:

<http://www.support.unisys.com/all/ple/19251489>

Note: *If you are not logged into the Product Support site, you will be asked to do so.*

Audience

System operators are the primary audience for this manual. System administrators and programmers might also need to read and refer to this manual.

Anyone using this manual must be familiar with mainframe computer operation and terminology, and have a basic understanding of Executive Control Language (ECL) and File Utility Routines/Program File Utility Routines (FURPUR).

How to Use This Manual

This manual is designed to be used primarily as a reference. Sections 1 through 9 include the basic concepts and objectives of system operation and specific operating procedures. Section 10 lists the unsolicited keyins according to function, and Section 11 provides detailed information about each keyin.

The *System Console Messages Reference Manual* (7833 3275) must be used in conjunction with this manual. It lists and explains console messages, provides system error codes, and describes other types of errors.

Notation Conventions

This manual uses these conventions for keyins and messages:

- Uppercase letters
Indicates the exact name of the operator command, also known as a *keyin*, for example: UP
- Uppercase letters with underscores
Indicates a system generation parameter, for example: SENTRY_CONTROL
- Lowercase italicized letters
Indicates a variable name, for example: *cmod*
- Leading zero on addresses
Indicates an octal address, for example: 077
- Small braces and bar
Indicates that the parameter can have one of two values, for example: {UPIDN}
- Brackets
Indicates optional parameter, for example: [,options]
- Brackets and bar
Indicates optional parameter; you can select one of the indicated values for example [STDISHARED]
- Subscript numbers on variable names
Denotes that there can be several values for the same parameter, for example: *devnam₁*
- Ellipsis (...)
Indicates that more values for the same parameter can be added, for example: *devnam₁,devnam₂,...,devnam_n*

Variable Names

Following is a list of the most common variable names used throughout this manual.

Symbolic Name	Meaning
<i>Cmod</i>	channel module
<i>Cu</i>	control unit
<i>Devnam</i>	device name
<i>dir-id</i>	directory identifier (for example, SHARED# or STD#)
<i>F-cycle</i>	file cycle
<i>host-id</i>	host identifier (for example, A, B, C, and so forth)
<i>iop</i>	input/output processor
<i>ip</i>	instruction processor
<i>mmod</i>	memory module
<i>pack-id</i>	pack identifier
<i>run-id</i>	run identifier
<i>site-id</i>	site identifier
<i>sname</i>	symbiont name
<i>user-id</i>	user identifier

This is an example of a keyin using a variable name:

```
DN ip
```

1.2. Operator Responsibilities

The operator responsibilities described here relate to a Unisys OS 2200 system. Although this manual contains specific information to adequately perform some of the responsibilities, you have to refer to other Unisys publications for detailed information.

The operator responsibilities for a Unisys system include the following:

- Initialize the system software and install and initialize products and application software

System start-up and initialization is the process that brings a system to the ready state. It includes powering up system components and loading software. Section 3 provides general information about system start-up and references manuals that contain specific information for some systems. See Section 8 for system security file initialization procedures.

- Respond to system messages

System messages are sent to the system console to inform the operator of system activity or to request that the operator perform some action, such as loading a specific tape. Section 2 includes information about the system console, system messages, and other aspects of console operation.

- Enter jobs into the system and check their status

This activity is done as part of monitoring system operations and in response to requests from system users. Section 4 includes an overview of the keyins used for controlling job input and checking their status.

- Monitor system status and throughput

General system status information is continuously displayed on the system console. Keyins let the operator display more detailed information, make entries in the system log concerning system activity, and adjust the job mix for efficient operation. The status of system components, such as printers (and their print queues) and disks (and storage allocation), is displayed in response to keyins. Section 4 includes information about the keyins used for status and throughput operations.

- Mount and dismount tapes

Tapes are mounted and dismounted in response to system messages. Section 5 describes how to respond to tape messages and includes information on Tape Automatic Volume Recognition (TAVR), a feature that lets the operator mount (premount) tapes before they are requested.

- Prep disks

Disks can be either fixed or removable. They can be reserved for specific assignments, and removable disks can be removed from one system and put on another. Information about disk packs and keyins used with disks is included in Section 5.

- Collect and sort printed output

If your system is configured to label user printouts, see Section 8.

- Back up system software

Backing up system software on a regular basis minimizes damage that can be done by system crashes, failed recoveries, and other system problems. It lets you recover more quickly and with greater confidence. See the *File Administration System (FAS) Operations Guide* (7830 7972) for more information on doing system software backups.

- Dump memory

System dumps can be performed to troubleshoot system errors. They can be started either manually or automatically. Section 6 describes the procedures for performing and processing system dumps.

- Recover the system if it fails

Recovery of the system generally refers to software recovery after a system failure. Integrated Recovery and the Integrated Recovery utility, as well as audit trails, are some of the features that streamline system recovery. Section 3 describes general recovery procedures and Section 7 provides information on audit trails and Step Control options, which are used in Integrated Recovery.

In addition to the tasks just listed, you need to be familiar with the following:

- Objectives, procedures, and standards at your site
- Efficient operation of hardware components
- Input, output, and processing requirements of the operating system
- Recognizing inefficient system operation and determining corrective action
- System operations in a multihost operations environment and how it differs from a one-host environment, if your site has a multihost environment.

1.3. System Hardware Components

Operating a Unisys computer system requires you to interact with the system hardware in a variety of ways: system start-up, system shutdown, loading tapes, preparing disks, and so on. This subsection provides a general overview of system hardware, identifying the components and their function. Specific information about powering on the components, identifying the location of switches and indicators, and so on is covered in the hardware manuals shipped with the component.

1.3.1. Central Electronics Complex (CEC)

The two major components of a central electronic complex (CEC) are the processing complex and the I/O complex, described below. The system can optionally contain an Extended Transaction Processor (XPC). Refer to the *XPC-L System Operations Guide* (3850 7422) for more information.

Processing Complex

The processing complex contains Processing Complex Cabinets (PCC), Processor Cooling Units (PCU), and Modular Power Centers (MPC) or Power Peripheral Distribution units (PPDU).

Each processing complex cabinet contains

- Instruction processors: The instruction processor executes program and system software instructions.
- Storage controller: The storage controller interfaces the instruction processors and the I/O complex with main storage. Storage controllers allow multiple PCCs and I/O complex cabinets to be configured as balanced multiprocessor systems.

- Main Storage Units (MSU): Each storage controller has a direct interface with two independent MSUs.
- Network Interface Module (NIM): The NIM connects the PCC with the system control facility for system operations.

The processing complex also contains the processor cooling unit and the modular power center. The PCU provides liquid cooling for the PCC, circulating water cooled by an external source. One PCU is required for each PCC.

The MPC or PPDU provides an isolation transformer, circuit breakers, and voltage-matching facilities for the central electronics complex.

I/O Complex

The I/O complex includes I/O complex cabinets (ICC) and disk subsystem adapter cabinets (DSAC). These units provide the system interface and the I/O channels to support peripheral equipment. The channels provide the final hardware link to the disk, tape, printer, and communications peripherals. They also manage and control transfers to and from the peripherals.

Each I/O complex cabinet contains

- I/O remote adapter: The I/O remote adapter provides the main storage interface to the ICC.
- I/O processor modules: I/O processor modules control the initiation, transfer, and termination of data sequences to specific channels in the ICC.
- Block multiplexer channel interfaces: BMCs provide the final hardware link to the disk, tape, printers, and communications peripherals. They manage and control transfers to and from the peripherals.
- An interface for each ICC to support one optional DSAC.

Each DSAC contains word channel modules used for the connection and control of peripheral devices requiring word channel interfaces.

See your *System Capabilities Overview* for more information.

1.3.2. Extended Processing Complex (XPC-L)

Extended Processing Complex (XPC-L) is an optional hardware device that connects to all hosts in an Extended Transaction Capacity (XTC) system. In a one-host system, file and record locking is handled through software locking by TIP File Control for TIP files or by UDS Control for UDS/TIP files. In an XTC system, XPC-L provides hardware locking outside the hosts to allow access to shared databases regardless of the database type. In addition, the XPC-L provides interhost communications.

1.3.3. Server Sentinel

The Server Sentinel software provides system control processing. Server Sentinel software executes on the embedded Service Processors.

Through the CSE, which provides access to hardware, the Server Sentinel software also communicates with the OS 2200 operating system and the CEC hardware.

1.3.4. Peripheral Subsystems

The peripheral subsystems provide data storage and hard-copy output for user programs. Data storage is provided by tape subsystems and disk subsystems (mass storage). Printer subsystems provide hard-copy output.

Mass Storage (Disk Subsystems)

Disk subsystems consist of the control units and their attached disk devices. The components of the disk subsystem are available in many configurations to meet a wide variety of needs.

For specific information about the disk subsystem attached to your system, refer to the manuals provided with it.

Tape Subsystems

Tape subsystems consist of the control units and their attached tape devices. Like disk subsystems, they are available in many configurations to meet a wide variety of needs. There are different types of tape devices and control units to handle reel and cartridge tapes of different densities.

Section 5 contains information about the cartridge tape library (CTL), Tape Automatic Volume Recognition (TAVR), and the tape subsystem. Refer to documentation provided with the tape subsystem components for more information.

1.3.5. Communications Subsystems

OS 2200 systems can include PCIOP, which is driven by the CPCComm software product. Ethernet network interface cards can be attached to the PCIOP.

1.4. System Software

System software includes the operating system of which the Exec is a part, and the various utilities, tools, and diagnostics that keep the system running so that user programs run properly. This manual documents the Exec and its features as they relate to system operation.

The Exec controls the system operating environment. It processes runs, controls files, manages system resources, and performs input/output operations for users. Console software lets the operator interact with the Exec, through operator keyins, to monitor and control system operation.

The following subsections describe some of the Exec features you need to understand as an operator. Your system cannot have all these features because some of them are optional.

1.4.1. Operating System and the Exec

The OS 2200 operating system supports batch, interactive, and transaction processing.

The Exec provides the basic support for all central complex hardware and channel-connected peripherals. It also isolates user programs from direct interface with the hardware. At its lowest level, the Exec makes users aware of the hardware and lets them control it. Unisys device control software, such as Enterprise Output Manager, the printer control program, and the online diagnostic software generally use this level of interface. In higher level operations, the Exec takes complete control. Interfaces for tapes and disks, for example, do not require the user program to know the physical device characteristics at all.

The Exec also controls how system resources are allocated. The primary resources are memory space, instruction processor time, mass storage space, tape drive and printer allocation, and network channels. Once resources are allocated, the Exec makes sure that programs do not interfere with one another or simultaneously modify the same data.

Common security and limit controls are also handled by the Exec. Security prevents unauthorized access to programs or data. Limit controls (the Quota system) make certain that no single user exceeds the resource usage limits imposed by the system administrator.

The operator keyins described in Section 11 let you control and modify the operational parameters of the operating system and request information from it.

1.4.2. Communications Software

A suite of software products enables communications between programs executing on the ClearPath system and software executing in other computers, work stations, and devices. Communications Platform (CPCComm) and SILAS provide the network protocol handling and control of the network connections.

COMAPI provides a Sockets-like interface to CPCComm and the network. CITA provides a TCP/IP path for TIP programs. The Message Control Bank (MCB) is the message handling and message recovery component of the OS 2200 Integrated Recovery system.

Figure 1-1 shows the relationships among CPCComm, SILAS, and the other communications software components.

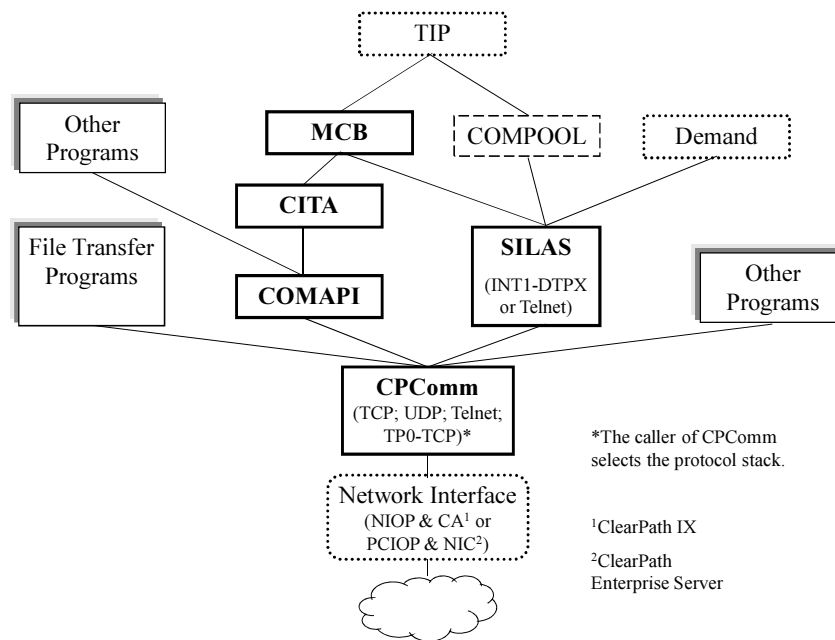


Figure 1-1. Communications Products with CPMComm

Refer to the following documents for operational instructions for the communications software products:

- *Communications Platform Configuration and Operations Guide* (7844 8438)
- *System Interface for Legacy Application Systems (SILAS) Configuration and Operations Guide* (7851 5475)
- *Communications Application Program Interface (COMAPI) User's Guide* (7851 5749)
- *Communications Interface for Transaction Applications Configuration and Operations Guide* (7862 6470)
- *Message Control Bank (MCB) Configuration and Operations Guide* (7833 1550)

1.4.3. Transaction Processing (TIP)

Transaction Processing (referred to as TIP) is an extension of the Exec that provides a high-performance system in which a user causes the execution of a predefined transaction program by entering an input message from a remote terminal. The program typically accesses the application program database and returns a response to the user.

The Exec supports the creation and maintenance of the database and program files for these applications programs. Step Control and Audit Control are important mechanisms for TIP that allow each program step to be completed and logged so that system failures do not jeopardize transactions.

For more information about TIP, refer to the *Transaction Processing Administration and Operations Reference Manual* (7830 7881).

1.4.4. Multi-Host File Sharing (MHFS)

Multi-host File Sharing (MHFS) is an Exec feature that allows file sharing between multiple OS 2200 systems. Each system in the MHFS environment is called a host. MHFS associates the files with a standard or shared master file directory and manages communications for file sharing within a host and across multiple hosts.

Keyins are available that enable you to change the MHFS configuration by bringing hosts or shared devices up or down. See Section 9 for operations procedures within a multihost environment. See the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469) for more information on MHFS.

1.4.5. Extended Transaction Capacity (XTC)

Extended Transaction Capacity (XTC) is an optional group of features that expands transaction processing capacity into multiple host systems. XTC provides for shared, synchronized, logical record access for multiple hosts between transactions and databases on shared mass storage. The application groups that simultaneously access the shared mass storage databases are called concurrent application groups. The XPC-L is a required central complex hardware device connected to all hosts in an XTC configuration that synchronizes locking for concurrent access to shared mass storage databases.

See Section 9 for operations procedures within a multihost environment. See the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919) for more information on XTC.

1.4.6. Integrated Recovery

Integrated Recovery synchronizes database and message recovery after a failure. Recovery features work in combination to expedite the recovery process, minimize recovery difficulties, regain system stability, and ensure data integrity. The major Exec components of Integrated Recovery are Exec Step Control, Exec Audit Control, and TIP File Control.

Integrated Recovery encompasses the features necessary to provide the database recovery needs of the Exec, TIP, and UDS, as well as the message recovery needs of MCB.

Integrated Recovery is available only for user sessions or programs associated with an application group. When you reboot the system after a failure or after bringing the system down, the Exec initiates most of the necessary recovery actions for each application group. The Exec either determines the appropriate action or prompts you to define the appropriate action. When necessary, the Exec also prompts you to start the Integrated Recovery Utility (formerly IRU) short recovery to bring up an application group.

When the recovery action initiated by the Exec cannot recover the system to its previous state, Integrated Recovery Utility provides other methods to recover the system, such as medium or long recovery.

Refer to the *Integrated Recovery Conceptual Overview* (7830 8186) for information about Integrated Recovery concepts and procedures.

Step Control

Step Control is an Exec Integrated Recovery component that synchronizes database and message recovery activities for user programs associated with a particular application group. It also provides central control information for the application group's Integrated Recovery software components so appropriate recovery actions can take place.

A step is a recoverable or nonrecoverable unit of work. It represents a portion of processing associated with a user program. A recoverable step is a unit of recoverable processing (for example, a sequence of user program commands that update a database). If a failure occurs before a recoverable step point is reached, the database updates associated with the step are discarded and the program rolls back (reverts) to the point just after the program's last recoverable step.

Section 7 provides information on the Step Control initialization and recovery options.

Audit Control and Audit Trail

Audit Control is an Exec component that initializes, creates, maintains, and controls access to an application group's audit trail. In addition, it manages audit trail recoveries and audit trail keyins. See Section 7 for Audit Control operations procedures.

An audit trail is a continuous and chronological sequence of audit records stored in mass storage or a tape file. In general, the audit record represents a recoverable event. Audit trails are associated with complexes within the Exec and manage all aspects of data storage on behalf of their associated complexes.

The complexes that use audit trails are:

- System log
- Integrated Recovery Step Control
- Integrated Recovery Transaction Performance Monitor (TPM)
- Capacity on Demand (COD)

Audit trails can be configured as simplex or duplex (one data file or two data files). Individual audit trails can also be configured to use tape or mass storage.

Integrated Recovery Utility

The Integrated Recovery Utility (IRU) plays a key role in an Integrated Recovery system. System software components such as the Exec, TIP File Control, UDS, and MCB have built-in procedures to handle most failures. If these recovery features fail, however, you can use Integrated Recovery Utility to perform alternate recovery procedures.

IRU is a software tool that combines and uses information available from other Integrated Recovery components to recover files and databases to ensure their integrity. Often, the work IRU does depends on the success or failure of the other Integrated Recovery system components. IRU might be considered a contingency-action product in recovery situations. You use it when automatic safeguard or recovery actions of the other components fail.

IRU supports two environments: single-host and Extended Transaction Capacity (XTC). IRU also offers three methods of recovery: short, medium, and long. Refer to the *Integrated Recovery Utility Operations Guide* (7830 8194) and the *Integrated Recovery Conceptual Overview* (7830 8186) for complete information.

1.4.7. Security

The security officer is responsible for security administration and planning. If you have specific questions about security operations, ask your site's security officer.

OS 2200 security is organized into following levels, with site configuration options within each level:

- **Fundamental Security**
Fundamental Security, a standard feature of OS 2200, offers basic protection for the ClearPath enterprise computing environment. Fundamental Security establishes security controls at the system level and provides limited additional security based on an individual user's identity.
- **Security Level 1** provides additional protections beyond Fundamental Security. It includes file ownership, clearance levels, access control records (ACRs), assignment of extra processing rights to individual users, TIP file security, enhanced auditing capabilities, and other security features.
- **Additional security levels (2 and 3)** can provide an even more rigidly protected environment for sites with more stringent security requirements.

For a comprehensive overview of OS 2200 security, see the *ClearPath Enterprise Servers OS 2200 Security Overview* (8205 7282).

As a system operator, you need to know about the level that is installed at your site and you need to understand the impact it has on your responsibilities, primarily in booting the system and labeling tapes and mass storage files. You must also be certain you have the proper security privileges to perform operations with Integrated Recovery, use the TS keyin, and boot the system.

The security administrator can use any of the following products. In this document, the term "security management product" refers to one of the following products:

- Apex Administration and Reporting is composed of two major components. An agent, APEXAG, is installed on the OS 2200 server, and a Web server service is installed on a Windows server. The user interface uses Web browser technology, thereby making any device that can access the Web server a potential system management workstation. A command line utility, @SECMGR, released with Apex, provides a batch/demand interface to perform those same administrator or end-user operations. For more information, see the *ClearPath OS 2200 Apex Getting Started Guide* (8207 4055), or *ClearPath OS 2200 Apex Help* (8207 4154).
- TeamQuest® Site Management Complex (SIMAN) provides a menu-driven interface for creating and maintaining security and resource control in an OS 2200 system. SIMAN can be run interactively in an OS 2200 demand session. SIMAN includes a batch processor whose command syntax is compatible with @SECMGR. For more information, see the *TeamQuest Site Management Complex (SIMAN) Administration and End Use Reference Manual* (TQ-01151).

1.4.8. Unit Duplexing

Unit Duplexing is an Exec feature that reduces system vulnerability to mass storage failures. It consists of a hardware configuration in which each mass storage device has an associated backup device that is transparent to all end-user software and to most system software.

If a user program issues a request to read or write to a file on a unit-duplexed device, the write request is automatically performed on both devices and the read request is performed on either device. From a software point of view, the unit-duplexed devices are bit-for-bit identical.

For additional information, refer to the *Unit Duplexing Planning, Installation, and Operations Overview* (7830 7592).

1.5. Operator Profile

System administrators have the option of setting up an operator profile for the system operators through Apex or TeamQuest SIMAN. This operator profile consists of an array of strings, called environmental variables, which contain information that defines operators, applications, and the system. Included in this profile is an environmental variable that lets operators receive messages from the Extended Language Message System (ELMS) in their native language, or another language, as determined by the site. Exec messages are currently displayed only in English.

The profile is set up at system initialization time using the security management product. After system initialization, profile changes are made through an Exec call. Refer to the *ClearPath OS 2200 Apex Help* (8207 4154), *TeamQuest Site Management Complex (SIMAN) Administration and End Use Reference Manual* (TQ-01151) or the *System Services Programming Reference Manual* (7833 4455) for more information.

Section 2

System Console

The system console is used to control and monitor system performance. Its configuration is governed by rules that are unique to each system.

On OS 2200 systems, workstations running Server Sentinel and Operations Sentinel provide the operations functions.

2.1. Console Operations Software

Unisys supports Server Sentinel and Operations Sentinel.

2.1.1. Server Sentinel

You perform many operations tasks on OS 2200 systems through the Server Sentinel interface. The Server Sentinel interface can run on the master Service Processor or on remote workstations.

The general functions you perform are as follows:

- Start and stop the partitions and their operating systems.
- Obtain information about the state of the server, partitions, and hardware units.
- Reconfigure partitions and their hardware units to add new units, remove failed units, restore units, or reconfigure hardware resources by assigning them to different partitions.

2.1.2. Operations Sentinel

Operations Sentinel enables an operator to monitor and control the following systems and servers:

- Dorado Series servers

These servers consist of OS 2200, UNIX, Linux, and Windows partitions.

- ClearPath NX, ClearPath LX, and ClearPath Enterprise Server Libra Series servers, and FS and EA series of MCP servers

These servers consist of MCP, UNIX, Linux, and Windows partitions or systems.

- ES7000 servers
These servers consist of Windows, UNIX, and Linux partitions.
- Windows systems
These include systems running Windows Server 2008 (Standard Edition), Windows 7 Professional, Windows 2008/R2, or Windows 8 Professional.
- UNIX systems
- Linux systems
- Other systems compliant with ANSI X3.64 protocol

Operations Sentinel also provides interfaces to Server Sentinel, NetIQ AppManager, and HP OpenView Network Node Manager including the Windows, Sun Solaris, and HP-UX versions. See the *Operations Sentinel Software Release Announcement* (7862 6512) for more information.

Operations Sentinel Console

Operations Sentinel Console is the principal client application of Operations Sentinel. It is the main user interface for operators to manage the various systems from a single workstation. Operations Sentinel Console provides the following capabilities:

- Topological displays of the systems you are managing. The status of each system, partition, and connection is indicated graphically.
- The means to open console windows, terminal windows, and remote desktops for the managed systems and partitions with simple mouse clicks.
- The means to launch site-defined external applications for the managed systems and partitions with simple mouse clicks.
- The console allows you to generate a system report of multiple or single OS 2200, MCP, Windows, and UNIX/Linux Systems. The report consists of bar charts where each bar chart is plotted for each monitored system for each system resource against a refresh interval.
- Alert icons that summarize the alert states of the managed systems and partitions. The appearance of these alert icons indicates the presence and severity of outstanding alerts.
- An Alerts window that lists all outstanding alerts. You select which alert attributes appear in this window and then you can clear and acknowledge the alerts and display detailed information about them.
- Tables that depict utilization properties for managed systems in real-time.
- The means to specify thresholds for system utilization properties, so that an alert is raised automatically when the value of a property goes out of acceptable range.
- The means to start other client applications of Operations Sentinel.

2.2. Keyin Format and Masking Rules

Masking characters are relational expressions with string attributes (the percent (%) character) that mask portions of the value of the expression.

ID fields in the following keyins can contain masking or wildcard characters, or both. The asterisk (*) is the defined wildcard character and matches any sequence of characters. The question mark (?) is the defined mask character and matches any character or a blank character. The ID fields also support embedded masking.

Examples:

T A*	Lists all runs starting with A.
T *A?	Lists all runs that end with A or have an A as the penultimate character.
T,B D?	Batch run D. Lists all runs starting with D and whose run ID is one or two characters long.
T,U C?X	Lists all runs executing under a user ID that starts with C, ends with X, and is three characters long.

If no arguments are specified, a full mask is implied.

Example:

"T" is same as "T *" which is also same as "T ??????".

If multiple fields are specified, an implied OR is assumed. If negation is specified through the N option, an implied AND is assumed.

T D SEC,PROD	Implied OR, that is run SEC, run PROD, or both, if both exist.
T,N D SEC,PROD	Implied AND, that is NOT run SEC AND NOT run PROD.

Note: The use of wildcard characters, masking characters, the user ID (U) and negation (N) options are controlled by Exec configuration parameters. If the Exec is not configured to enable some or all of these features, an error message is displayed.

The following table lists the keyins that support the masking feature.

Table 2–1. Keyins Supporting the Masking Feature

Keyin	Masking Rule	Limitations
BL	In some formats, accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	
CS	In some formats, accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	
DISPLVL	Accepts masking and wildcard characters in the operand field for account number, project ID, runid, and userid.	The wildcard character may not be in the first character position of the operand.
ERUNS	Accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	To prevent an operator from harming the system, ERUNS keyin has the following limitations: • The wildcard character is not permitted before the third character of an ID. • More than nine mask characters cannot occur.
RE	Accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	
RM	Accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs. The wildcard character is permitted in any position. The mask character can occur as desired.	To make it less easy to mistakenly remove the entire backlog with a mistyped keyin, the wildcard character cannot be used alone or just with mask characters. For example: 'RM *' and 'RM???' are illegal.
SP	Accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	

Table 2–1. Keyins Supporting the Masking Feature

Keyin	Masking Rule	Limitations
SQ	In some formats, accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs. For more information on the acceptable masking rules, see Section 11.70.3.	
SR	In some formats accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs. Note: "SR" keyin has a unique requirement as a result of the masking capability (need a "" around the "*" when NOT utilizing masking). For more information on the acceptable masking rules, see Section 11.71.4.	
SV	In some formats accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs. Note: "SV" keyin has a unique requirement as a result of the masking capability (need a "" around the "*" when NOT utilizing masking). For more information on the acceptable masking rules, see Section 11.75.5.	
SX	In some formats accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	

Table 2–1. Keyins Supporting the Masking Feature

Keyin	Masking Rule	Limitations
T	In some formats accepts one or more argument fields separated by commas. These fields refer to run IDs unless the U option is specified. In this case, the arguments refer to user IDs.	

2.3. Entering Unsolicited Keyins

Unsolicited keyins are commands that submit information to or make inquiries of the system. They are called unsolicited keyins because the system does not prompt you to enter information or respond to a message.

Unsolicited keyins are used to inform the system of changes in the external environment, request information about the internal system activity, or modify the internal system activity. Section 11 contains a list and description of all the unsolicited keyins. Use these keyins carefully—some slow down system performance.

Keyins are composed of the 1- to 8-character command (for example, AC or IOPHFL), and for some keyins, a single space, a comma and space, or a comma (depending on the keyin), and one or more parameters.

These keyins illustrate the various types of formats:

```
MS
CH H
FS,TAPES
FS ALL,component
MH FS,HOST
LG text
```

The maximum length of a keyin (including all parameters) is 76 characters.

To enter a keyin from a local console, press the MSG (message) WAIT key on the keyboard. This stops output to the console subsystem and allows input. The contents of the screen scroll upward, and you see the start-of-entry character in the bottom left corner of the console. When you have completed the keyin, press the transmit key. The console routine sets a 1-minute cycle during which you can make entries.

You can also enter a keyin by moving the cursor to any previous input line (a line that already has a start-of-entry character). When you enter new data, place the cursor immediately following the last existing character. When your entry is complete, press the transmit key.

2.4. Console Messages

Console messages generated by the Exec fall into one of four groups:

- System messages – General, informational, and system request
- Input/output messages – About peripheral device operations
- Communications messages – Concerning the communications complex
- Hardware confidence messages – About hardware status

There are also four additional message groups available for your site that are undefined and can be used by your site for specific system operations. Your site can define these additional console message groups during system generation.

Refer to the *System Console Messages Reference Manual* (7833 3275) for detailed information about message types and how to respond to them.

2.5. Controlling the System Console

The following keyins let you control the system console:

- The AC keyin sets or clears the message groups that is displayed on different console devices, and lists the consoles that are configured to display messages.
- The CMR keyin controls the console message redisplay feature that lets you redisplay messages that have scrolled off the screen.
- The MR keyin, from remote displays, recalls outstanding messages that have scrolled off the screen. Local displays keep the read-and-reply messages from scrolling off the screen.
- The TS keyin checks which terminals are disabled and restores terminals.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

2.6. Remote Consoles

Keyins can be entered at remote consoles under control of the security management product. Demand terminal users can receive console output, answer console messages, and enter certain unsolicited console keyins using the @@CONS statement.

Using the security management product, the site administrator gives each user one of five remote console mode capability levels, and gives the user the ability to perform either the maximum or a subset of the maximum of the keyins allowed in a particular level.

See the *Executive Control Language (ECL) and FURPUR Reference Manual* (7830 7949) for a description of the five console mode capability levels and a list of the keyins that can be performed by demand users, along with the console capability level required for each keyin.

When a remote console is in use, the status of the system can change without warning. You can have to respond to changes in unit status and print files that are deleted, as well as runs that terminate in error or are removed. You can also be required to respond to read-and-reply messages that have been routed to you from a terminated remote console.

If the system configuration parameter is set to echo remote console messages on the system console (echo_remote_console), the local system console receives messages sent to the remote console. If the message is a read-and-reply, then it only displays as read-and-reply on the remote console. On the local system console, it is displayed as a read-only message.

2.7. Console Faults

When a fault occurs on a console or console printer, an interlock message appears on all other available console devices. The message that is displayed depends on the type of fault. (Refer to the *System Console Messages Reference Manual* (7833 3275) to interpret these messages.)

If necessary, you can redirect messages to other console devices with the AC keyin.

Caution

Do not use the XB control page parameter, which changes the border color, when operating the console. It destroys the current screen.

After the fault, the system starts a 30-second cycle to retry output. Distribution of output to other devices on the console continues. When the condition is corrected, you can see backed-up messages on the corrected device, and normal output resumes.

Read-and-reply messages might appear with different message numbers, and only the last line of multiline read-and-reply messages is displayed. Also, read-only messages that were on display before the fault are lost. Use the MR or CMR keyin to redisplay these messages.

Section 3

System Start-Up and Recovery Procedures

This section describes system start-up and recovery procedures (called initial boot and recovery boot, respectively). Figure 3-1 illustrates the types of system boots you can perform, whether they are performed from tape or mass storage and whether they are performed manually or automatically.

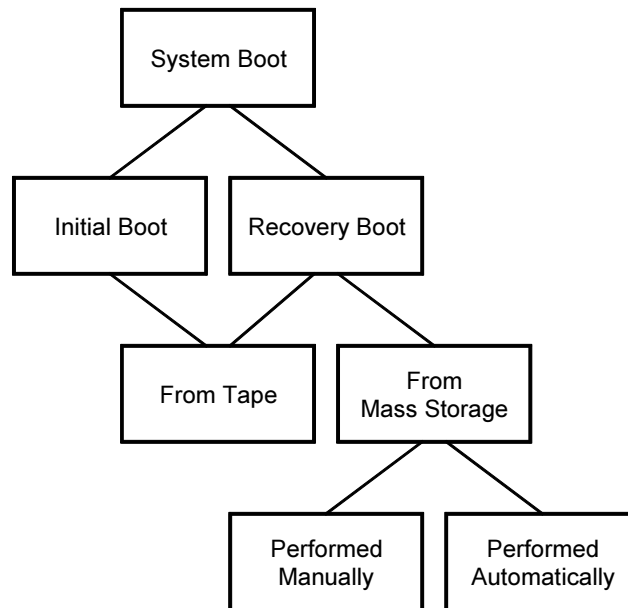


Figure 3–1. Types of System Boots

3.1. Loading System Software

The initial loading of software on the system, termed the *initial boot*, loads the Exec software into memory and starts the system running.

For an initial boot, you must load the Exec from a tape. If errors occur during this process, you can dump the contents of main memory. The dump is then used to help identify what caused the errors. See Section 6 for information on how to perform and process system dumps.

The Exec does not perform the entire initial boot procedure automatically. Several pauses occur, and you are given a chance to make various changes. During these pauses, the system displays several informational messages and several read-and-reply messages. For more information about system generation, see the *Exec System Software Installation and Configuration Guide* (7830 7915).

During the boot process, it might be necessary for you to identify the hardware that is available for use by the system if either of these situations occurs:

- The system that you want to load the Exec into does not contain all the components specified when your site generated the system (the configuration is different).
- You want to prohibit or restrict access to one or more components from the configuration to complete the load process.

Once the Exec has been loaded, it takes control of the system. If the Exec hangs, crashes, or must be brought down for some other reason, it is typically restarted from mass storage with a recovery boot to restart the system.

The boot and initialization elements of the Exec contain routines that do the following:

- Perform the initial load from tape
- Write the Exec on the system disk
- Load and execute the Exec from the system disk
- Take a system dump (in the event of failure)
- Perform an automatic recovery boot from mass storage ("automatic disk recovery") when the system dump is complete

Both the initial boot and the recovery procedures are described in this section. See Section 9 for the system security file initialization procedures.

Note: *Cross-booting to a system with fewer mass storage devices is allowed. See PLE 17809598 for more information on this functionality.*

3.1.1. Jump Keys and System Boot

Once the Exec system software is loaded, you can use a manual boot from disk or tape if required, but typically you can use an automatic boot when the system is already installed. This kind of boot performs an automatic disk recovery and quickly brings up the system.

Jump keys control the action of the Exec during a boot (both an initial boot and a recovery boot). Use the SJ and CJ keyins to set or clear jump keys, and the DJ keyin to display which jump keys are set (see Section 11 for more information on these keyins). A jump key is not set or cleared until you receive a confirming message on the system console.

There are 36 available jump keys in the system. Some are not used. Table 3–1 lists and explains the jump key settings:

- Jump keys 1 to 16 are reserved for Unisys.
- Jump keys 17 to 24 are available for users.
- Jump keys 25 to 36 are reserved for Unisys debugging.

Two jump keys, 9 and 13, can have undesirable consequences if you accidentally set them. For this reason, a message appears during the boot process asking you to confirm your action. Jump keys 9 and 13 are ignored during auto recovery.

Before booting your system, the following jump key settings are recommended:

- If this is an initial boot from tape, set jump keys 4 and 13.
- If this is a tape recovery with library load, set jump key 4.
- If you want a manual tape dump, set jump key 2. If you want a mass storage dump, set jump key 2. (See Section 6 for more information on performing system dumps.)

Table 3–1. Jump Key Settings

Key No.	Jump Key Name	Jump Key Task Description
1	MODIFY CONFIG	Solicits for configuration changes during autorecovery.
2	Manual Dump	If this is set on a manual boot, the Exec dumps main storage. By default, a selective dump is taken. If jump key 6 is set, a full dump is taken. Should be cleared when the dump is complete.
3	Auto Recovery Inhibit	Disables autorecovery and disables automatic dump feature after initialization.
4	Library Reload	Loads processors and libraries from the boot tape during a tape boot. See also jump key 13.
6	Full Dump	Performs a dump of all main storage, versus the selective dump that is the default dump type. The resulting dump is typically much larger when jump key 6 is set.
7	TIP and Step Control Initialization and Down Pack Initialization	Solicits a read-and-reply message for TIP initialization. During a recovery boot, it also lets you choose between recovering and initializing the pack status file.
9	Queue Recovery Inhibit	Inhibits recovery of all run scheduling information, backlog runs and print files. Ignored during autorecovery.
10	Debug dump	If set, allows any application that executes an SGNL 10 instruction to initiate an online dump. Useful when the Exec cannot detect an error but an application can.

Table 3–1. Jump Key Settings

Key No.	Jump Key Name	Jump Key Task Description
11	STPAUL only. Debug panic dump	If set and jump key 10 is also set, allows any application that executes an SGNL 10 instruction to trigger a system stop.
12	Enable breakpoint trace during boot	Acquires the breakpoint trace bank during boot, thus allowing the breakpoint trace mechanism to be used during the boot process.
13	Mass Storage Initialization	The I-boot jump key. Initializes the non-shared, fixed mass storage directory. All files on non-shared fixed disks are lost. Jump key 4 should be set with this jump key.
16	Halt System	Similar to jump key 3 but the system halts in a manner that bypasses all hardware initialization so that component state information can be harvested. Jump key 16 results in an “unclean halt” and can trigger heartbeat loss and have other undesirable side effects in a multi-host environment. For these reasons, it is used in a diagnostic capacity only.
25	STPAUL only. Partitioning checks.	If set, allows non-CEC components to be partitioned.
26	Allow tape error message display	If set, allows the display of certain tape I/O error messages that are normally suppressed. These messages may contain useful diagnostic information.
28	STPAUL only. I/O debug.	CLOADI: If set, does not require maintenance mode when loading XIOP microcode. IOCHAN: NIOP Nonsense Interrupt streaming trap.
30	STPAUL only. SCP data error.	If set, stops the system such that the cause of suspect SCP data can be determined.
33	Bypass Configuration Checks.	If set, bypasses configuration checks made during booting that ensure that the Exec contains all feature code required to handle the configuration as defined in the PDB, on the assumption that the hardware requiring the feature code is unavailable. Use only as directed.

3.1.2. Initial Boot

The procedure for booting the Exec depends on the type of system and whether or not there are additional hardware components. The following information is intended only as a guide to booting the system.

The instructions take you only to the point at which the Exec displays the first console message. Once you get to that point, the booting process is not yet complete; however, in most cases you must be able to weave your way through the console messages to complete it.

When you load the Exec, the system displays messages that you must respond to. For example, the Exec can ask for the name of a device or request that you confirm your intent to perform some action. If you use products such as Transaction Processing (TIP) or the Integrated Recovery Utility (IRU), you can see messages that apply to these products.

During a tape boot, the following steps occur for ClearPath IX systems:

1. The hardware loads the bootstrap code from the tape.
2. The bootstrap code loads the Exec from the tape into main storage.
3. The initialization portion of the Exec starts.
4. The initialization portion initializes or recovers the mass storage file structure.
5. The Exec copies itself from the tape to the designated mass storage device.
6. A runstream starts that loads the system libraries.

The System Control Facility (SCF) is used for start-up and control of system partitions, which are defined in the Site Configuration Management System (SCMS II). To boot the system, you enter SCF commands from a system control display session; once the system is up, you can enter console keyins from a system operation display session to manage daily system activity, such as run scheduling and facility control.

The procedure described here is the simplest one for performing an initial boot. Refer to the *SCF Operations Guide* for complete information on initial boot procedures and tape and disk recovery boot procedures and for a description of the SCF error messages that you might see during the system boot. Other boot messages are displayed by the Exec; see the *System Console Messages Reference Manual* (7833 3275) for a description of those messages.

1. Press the **F5** key to get to a system control display session.
2. From the system control menu, enter **A** to activate a partition and then select the partition. You are now in a Partition Control display session for that partition. The display shows the currently selected boot type, load path, and boot options for the partition.
3. Mount the boot tape on the tape drive.

4. If you need to change the boot type, enter **SET-TYPE TAPE** and transmit. If you need to change the tape device name, enter **SET-TYPE *device-name*** and transmit.
5. If jump keys 4 (library load) and 13 (initialization) are not set, enter **SJ 4, 13** to set them, and transmit. Enter **destroy files** to set the reinitialize option, and transmit.
6. Enter **SP** to start the partition. The Start Partition (SP) command initializes the partition and performs an initial program load of the partition.
7. During initialization, the MODIFY CONFIG console message gives you a chance to modify the configuration before the system starts running.

3.2. Recovery Boot Procedures

You can perform a recovery boot from tape or mass storage, and you can initiate it either manually or automatically.

- Manual recovery

Manual recovery never initializes the mass storage file area except when the file description on a particular device is destroyed. You can reload the libraries if the recovery boot is from tape. You can write a dump of main storage to mass storage on a mass storage recovery.

- Automatic recovery

If the system has automatic recovery enabled (jump key 3 is clear), it can automatically recover from most errors.

Both hardware and software can cause a recovery boot to be automatically initiated, which is similar to the manual mass storage recovery boot. A dump of main storage can be written to mass storage and saved for analysis.

The process of recovering a machine following a system hang, system stop, hardware autorecovery (on some systems), \$!, and so on, is governed by the jump keys set in the partition at the time the problem occurs. If a system stop, \$!, or hardware autorecovery occurs and jump key 3 is clear, the system attempts to recover itself. The only operator intervention is to respond to messages that the system displays during the recovery phase. The \$! keyin is described in Section 11.

If you have to deal with a system hang or you manually recover the system, then more intervention is required. Many of those actions depend on the state of the machine at the time of the stop and the state of the machine upon recovery. In some cases, you might need to stop autorecovery because the system cannot save the dump and you need to save the dump to tape or mass storage.

When a system error occurs in the Exec, autorecovery reloads the Exec without destroying mass storage files. During autorecovery, the console displays messages to inform you

- Whether a system dump was taken
- The type of boot and session number
- The date and time of recovery

For more information on performing and processing system dumps, see Section 6.

For additional information on recovery concepts, see the *Integrated Recovery Conceptual Overview* (7830 8186) and the *Integrated Recovery Utility Operations Guide* (7830 8194).

Section 4

General System Operations

This section covers the following general system operation procedures as they relate to the Exec:

- Monitoring system status
- Controlling job flow
- Controlling system components
- Logging system events
- Control unit operation
- Controlling symbiont devices and files
- Manually terminating and recovering the ASIS run

See Sections 5 through 9 for additional operations procedures.

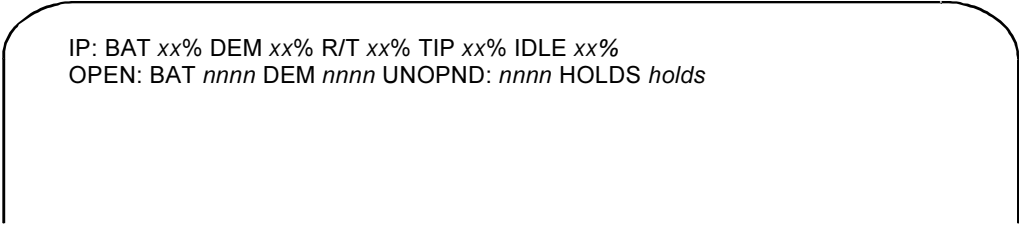
4.1. Monitoring System Status

System status is continuously monitored by the Exec and is displayed by four lines at the top of the system console. These four lines of information are actually displayed as two lines in a rotating sequence. Each sequence is updated twice (6 seconds apart) before the other sequence appears.

The appearance of these lines for 2200/XPA systems is shown in Figure 4-1 and Figure 4-2. See the SS keyin description for the 2200/XPA continuous console display and for explanations of the continuous console display parameters.

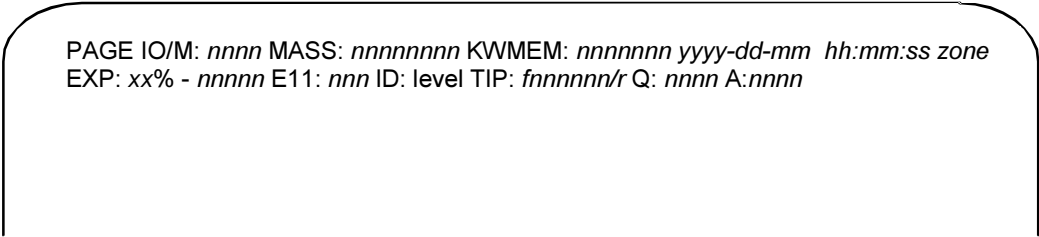
The following keyins let you monitor other aspects of system operation:

- The BL keyin displays the run-IDs of all runs in backlog. Included in the information provided by this keyin are the number of runs in backlog and the number of runs that are candidates for opening.
- The EU keyin displays EXPOOL use. Information provided includes the current size of EXPOOL, the current amount of EXPOOL in use, and other parameters.
- The FF keyin displays information about files. The keyin is very useful to determine which user has a file assigned when a hold condition occurs. The information returned includes the name of the common name section that has the file assigned, the run-ID or TIP run that has the file assigned, and the host system that has the file assigned, if applicable.



```
IP: BAT xx% DEM xx% R/T xx% TIP xx% IDLE xx%  
OPEN: BAT nnnn DEM nnnn UNOPND: nnnn HOLDS holds
```

Figure 4-1. System Status Information, Initial Two Lines



```
PAGE IO/M: nnnn MASS: nnnnnnnn KWMEM: nnnnnnnn yyyy-dd-mm hh:mm:ss zone  
EXP: xx% - nnnnn E11: nnn ID: level TIP: fnnnnnn/r Q: nnnn A:nnnn
```

Figure 4-2. System Status Information, Second Two Lines

- The FS keyin displays the status of components, interfaces, or other partitionable devices. The keyin provides information such as its current state (up, down, reserved, suspended), its availability, assignments, and other parameters relevant to the device.
- The HU keyin displays a list of up to 10 run-ids and the percentage of total available IP SUPs each of those run-IDs accumulated within 1 second after entry of the HU keyin. It also includes the statistics for application level test and set queued activities.
- The MS keyin displays the number of mass storage tracks currently available in the up and suspended states for fixed and removable mass storage devices.
- The MU keyin displays central complex memory use. The information is provided on a global basis, for a specific run, or for all runs, and is displayed by category (for example, Resident Exec and Down Memory).
- The RC keyin displays detailed information on backlogged runs, active runs, and runs terminated with output files remaining.
- The SS keyin displays expanded information describing system status. The information provided by an SS keyin includes the continuous display information, as well as information such as the number of read-and-reply messages sent to all consoles, the number of unopened runs waiting for start time, and the number of runs being held by the CS H run-ID keyin.
- The T keyin displays information about active demand runs, active batch runs, and runs that are being held. Information includes the name of the run and the run state.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

4.2. Job Control

Job control is one of the main functions of computer operations. It involves controlling jobs according to site requirements, user requests, and system performance. Your site requirements are typically set up when the system is configured. User requests occur when a run hangs or otherwise does not complete normally. For maximum system performance, you need to control the mix of jobs that enter the system, remove jobs that demand too many system resources during peak loads, and perform other control operations.

The following keyins let you control jobs in the system:

- The B keyin sets the maximum number of open batch runs allowed in the system.
- The CS keyin sets and releases various system- and run-specific hold conditions. Hold conditions include all runs in backlog (master hold), specific runs in backlog, demand scheduling runs, and demand terminals.
- The DISPLVL keyin changes the switching priority of batch or demand jobs. The keyin can be used to change the switching priority of a single job or multiple jobs. Jobs can be selected by categories such as account number, project-ID, user-ID, run-ID, and type (batch, deadline, or demand). Once the new priority has been activated, it can be reset to the site's normal priority using another format of the keyin.

All activities in the Exec are associated with switching levels (defined during system configuration) that define the activity's instruction processor switching priority.

- The E keyin terminates a run in error. If the keyin specifies a batch run, the run terminates with an error and allows a postmortem dump (PMD). If the keyin specifies a demand run, the current task is terminated, but the run continues with the next control image. The N option of the E keyin prevents the PMD for both the batch run and the current demand task that were terminated. The E keyin is preferred to the X keyin to terminate runs because it lets shared software subsystems and common banks (especially Checkpoint/Restart) remove the run from internal tables.

Note that after an E keyin terminates the current task, a termination task can then start up. The dynamic configuration parameter `run_termination_task` being nonblank enables the running of the termination task.

- The RM keyin removes a run from the system that is in the backlog queue. The run is removed without regard for any console hold or start time that is in effect on the run. When the run is removed, it is opened and then terminates in error.
- The SP keyin suspends a run and the RE keyin resumes a run that has been suspended. Jobs that require a long time to complete or are resource-intensive are good candidates for suspension, keeping them off the system during peak times and letting them back on during off-peak times. When the suspended run is resumed, it resumes where it left off. The keyin is also useful when a job gets hung in the IP and cannot be terminated with an X or E keyin. Using the SP keyin, you can suspend the run until an off-peak time, when a system halt affects fewer users and a dump can be issued.

- The ST keyin starts a runstream that is stored in a file or element that can be scheduled and executed as an independent run. Also, use this keyin to start runstreams for other software products.
- The X keyin terminates a run if the E keyin does not terminate the run. Using the X keyin on batch runs does not allow a postmortem dump; on a demand run, the keyin terminates the current task, but continues on the next control image. The X keyin does not notify shared software subsystems or common banks, and eventually new checkpoints or restarts cannot be processed because they are queued.

Note that after the X keyin terminates the current task, a termination task can then start up. The dynamic configuration parameter `run_termination_task` being nonblank enables the running of the termination task.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

4.3. Controlling System Components

Status and availability are the attributes of a system component that determine whether you can use it. A component can be up, but not available, because something in all paths to the component is down. Keyins are used to bring devices down (DN keyin), reserve them for future use (RV keyin), and bring them up (UP keyin), as well as modify (MD keyin) and remove (RD keyin) disk packs.

If an attempt to bring a component online fails, the UP keyin aborts, and you must correct the failure in order to use the component. If an attempt to take a component offline fails, the component is marked down, and you are warned that the component is down but online.

These keyins let you control system hardware status:

- The DN keyin brings down a component or interface. This keyin is most commonly used to bring down a component that is not operating properly for repairs or diagnostics. The keyin applies to a component or interface that has an up (UP) or reserved (RV) status, and any disk device that has a suspended (SU) status.
- The RV keyin reserves a component or a block of main storage.
- The SU keyin suspends a disk device from being allocated.
- The UD keyin establishes a unit-duplexing environment.
- The UP keyin brings a component or interface up (online). This keyin is most commonly used to bring up a component that was taken down because it was not operating properly. The keyin works with a component or interface that has a down (DN) or reserved (RV) status, any disk device that has a suspended (SU) status, and any printer that has been locked out or suspended with an SM *device* L keyin.
- The \$! keyin stops the system.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

The UP, DN, RV, and SU keyins are called facility control keyins. Table 4-1 shows the consequences of using the facility control keyins on different types of components and devices.

The following symbols are used to explain the changes for each device.

–	Keyin not allowed, not applicable, or not available.
S	Keyin forces system stop; change of status is effective following disk boot.
Sc	Keyin forces system stop when assignments exist; otherwise, status changes immediately.
C	Status changes immediately.
Ca	Status changes immediately.
Ci	Status changes immediately and unit is initialized.
Cn	Status changes if no assignments exist; otherwise, keyin is not allowed.

Table 4–1. Summary of Status Changes for Facility Control Keyins

Device	Current Status	Unsolicited Keyin			
		UP	DN	RV	SU
IP	Up	–	C	C	–
	Down	C i	–	Ci	–
	Reserved	C	C	–	–
IOP	Up	–	C	–	–
	Down	C	–	–	–
XIIP (data mover)	Up	–	C	–	–
	Down	C	–	–	–
XPC	Up	–	C	–	–
	Down	C	–	–	–
Channel module	Up	–	C	–	–
	Down	C	–	–	–
Control unit	Up	–	C	–	–
	Down	C	–	–	–

Table 4–1. Summary of Status Changes for Facility Control Keyins

Device	Current Status	Unsolicited Keyin			
		UP	DN	RV	SU
Fixed mass storage	Up	–	S	–	C
	Down	Ci	–	C	Ci
	Reserved	C	Sc	–	–
	Suspended	C	S	–	–
Removable mass storage	Up	–	Sc	Ca	C
	Down	C	–	C	C
	Reserved	Cn	Sc	–	Cn
	Suspended	C	Sc	Ca	–
Nonmass storage	Up	–	C	C	–
	Down	C	–	C	–
	Reserved	C	C	–	–
Main storage	Up	–	C	C	–
	Down	C	–	C	–
	Reserved	C	C	–	–

Control units are added to or removed from a partition by the Exec during UP or DN control unit keyins. If the keyin attempt fails, a message is displayed, the keyin aborts, and you must correct the cause of the failure.

4.4. Logging System Events

Use the LG keyin to make text entries in the system log file. The system log file contains information about such things as run starting and completion times, console messages, checkpoints, device assignments and releases, and hardware fault information.

Your site administrator must advise you of the type of system events that must be logged. Refer to the LG keyin module in Section 11 for detailed information about the keyin. For more information about the system log file, refer to the *System Log Operations and Support Reference Manual Volume 1* (7831 0315).

4.5. Controlling Symbiont Queues and Files

Use the SM keyin to display the status of symbiont devices and control symbiont operation. Symbiont devices include card readers, card punches, printers, and remote terminals.

These keyins are related to symbiont files:

- The SR keyin restores symbiont files that were previously saved to mass storage so you can queue them for printing (or punching).
- The SV keyin saves queued print and punch files to magnetic tape.
- The SX keyin deletes queued print and punch files from the system.
- The SQ keyin displays the status of the print files in the print queue.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

4.6. Manually Terminating and Recovering the ASIS Run

The ASIS (Authentication and Session Initiation Subsystem) background run normally has the run-id ASIS. Normally, the background run automatically restarts on every Exec recovery boot; however, SOLAR can install ASIS in a *noauto* mode, which prevents it from automatically restarting. Under some circumstances, an operator can have to kill the ASIS run, and manual recovery of the ASIS run can also be required in some circumstances. Always consult with the appropriate system administrator or security officer before performing either of these procedures.

4.6.1. Manually Terminating the ASIS Run

The normal method of shutting down the ASIS subsystem is with the ASISUT SHUTDOWN command; see the *User Authentication Administration Guide* (7850 4586) for more information about this command. If the administrator is unable to access the ASISUT, or the ASISUT SHUTDOWN command fails to work, then manual termination is required.

Always try the E keyin first to manually terminate the ASIS background run (E ASIS), and use the E keyin against any administrative activities that are processing within ASIS. This normally causes ASIS to perform a normal shutdown and to deactivate itself, with a dump that reflects the E keyin as the cause of the failure.

If the E keyin fails to deactivate the ASIS background run, the system administrator might want to try to deactivate ASIS with SOLAR's DEACT processor before proceeding to the last resort of using the X keyin.

The X keyin stops the ASIS subsystem from running, but does not deactivate it. The X keyin displays a warning message on the console indicating that this action affects the ASIS subsystem and asking the operator for permission to proceed. Because this action does not deactivate the ASIS subsystem, the Exec soon displays a message on the console reporting that ASIS is hung and asking if ASIS must be deregistered.

Always respond YES to the prompt to deregister ASIS. (This procedure also applies if the administrator used SOLAR's DEACT processor.)

Always try the E keyin first against any administrative activities that are processing within ASIS. If the E keyin fails to terminate the activities, then try the X keyin against them.

4.6.2. Manually Recovering the ASIS Run

The only time manual recovery of the ASIS background run is required is after an ASIS shutdown if the site wishes to restart ASIS authentications during the same Exec session. Use the ST keyin as follows to restart ASIS (assuming the run-ID is ASIS):

```
ST,/Z SYS$LIB$*RUN$.ASIS, [setc],,acct-id/ASIS
```

The ASIS background run is unable to distinguish between the following two error conditions and thus displays a message on the console stating that the ASIS is already initialized and asking whether it must be deactivated:

- Two or more background runs are inadvertently started; in this case, only the first one initializes the ASIS and is used for authentications.
- The background run was terminated with the X keyin, but the ASIS was never deactivated.

If possible, contact the system administrator before responding to the message about deactivating the ASIS. Otherwise, respond NO to the message if the cause is the first condition above (ASIS is up and running again) and respond YES if the cause is the second condition above (ASIS was terminated with the X keyin).

When the response to the message about deactivating the ASIS is NO, the background runs FIN.

When the response to the message about deactivating the ASIS is YES, the background run attempts to deactivate the ASIS. If the attempt fails, the background run displays a message to that effect on the console and then runs FIN. If the attempt to deactivate the ASIS succeeds, the background run displays a message on the console that indicates success and asks if the ASIS must be restarted. Respond YES to start normal initialization of the ASIS. (Responding NO causes the background run to FIN.)

Section 5

Tape and Disk Operations

This section presents operations information related to tapes and disks.

5.1. Tape and Tape Drive Operations

This section provides information about Tape Automatic Volume Recognition (TAVR), cartridge tape subsystems, and the Virtual Tape Handler (VTH).

TAVR provides great flexibility in mounting tape reels by allowing operators to premount tapes before user assignment and by allowing any compatible drive to be used to mount a tape after user assignment.

Half-inch cartridge tape subsystems, such as the 4780, OST4890, OST5136, and CTS5236, have many features that enhance system performance and reliability. This section contains information on how to use these tape subsystems in an auditing environment, and includes an explanation of how the devices operate, operational considerations, and error recovery implications for auditing with buffering on.

The DLT 7000 is a half-inch cartridge tape subsystem, but it cannot be used as an audit trail device. It is supported as a boot device and system dump device. You must also note that the operator clean message to the system console is not supported on the DLT 7000. An LED on the device indicates when the drive needs cleaning or that a bad tape has been loaded. Because of a DLT 7000 hardware limitation, the LED display is not supported. Therefore, the Exec does not display either the system-ID or the reel-ID on the drive.

The Fibre SCSI tape subsystem (for example, CTS9840) implements Persistent Reservation commands to exclusively acquire a tape unit. To release a current reservation, the unit must be released by the host currently reserving it by using the customary DN keyin.

A Fibre SCSI unit holding an unwanted reservation can be cleared by use of the existing TU keyin or by powering it down. Both methods, however, force another host to lose exclusive use of the device as explained in the following bulleted paragraph on the TU keyin. An important difference of the Fibre SCSI device is that channel initialization does not release the persistent reservation as stated in the TU explanation on exclusive use.

The following keyins let you control tape drives and handle tape labels:

- Use the TU keyin to release a tape unit from exclusive use. The keyin must be used only when the host that has exclusive use is down and is unable to perform channel initialization. If channel initialization has been performed, then other hosts in the tape unit environment can acquire the devices exclusively without using the TU keyin. This keyin unconditionally releases the unit and can force another host to lose exclusive use of the device.
- The unload (UL) keyin rewinds the unit or the units of the control unit (with interlock), and clears the reel-ID from the unit status tables. To use this keyin, the unit must be up (UP) or reserved (RV). The keyin does not cause a change in the status of a unit, but makes a unit containing a premounted reel available to you for a subsequent reel mount. Use this keyin instead of manually rewinding and removing a premounted reel at the tape unit. It ensures that the Exec is aware that the reel is no longer premounted. Also use the keyin in these circumstances.
 - When you need to unload premounted reels that you expect cannot be assigned in the near future.
 - When a job terminates in error before all anticipated reels have been used.
 - When the run scheduler changes jobs to a later time.
- The premount (PM) keyin designates a unit as premount-only. When a unit is designated as premount-only, it can only be used to premount tapes, or to handle tapes that are in the facilities pool of the user that is requesting the tape. An absolute device/control unit assignment overrides the premount-only designation.
- The Premount Reset (PR) keyin removes a premount-only designation and can be required if runs are waiting for tape facilities. Entering this keyin allows a run that is waiting for a tape unit to be assigned one.
- The P option on the Facilities Status (FS) keyin lets you see which units are designated as premount-only units.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

5.1.1. Compatibility Issues of 18- and 36-Track Tapes and Tape Drives

The following compatibility issues with regard to 18- and 36-track tapes and tape drives are important to keep in mind to protect tapes, tape drives, and data integrity.

- Although a tape written on an 18-track cartridge drive can be read on a 36-track cartridge drive, a tape written on a 36-track cartridge drive cannot be read on an 18-track cartridge drive.
- A labeled cartridge tape written in 36-track format appears as an unlabeled cartridge tape when mounted on an 18-track cartridge drive. If you answer N to the following message, the tape is rewritten without being checked for a label by the Exec:

Is Tape Labeled? Y,N

- The 36-track cartridge drives support the use of E-cart cartridge tapes, which are twice as long as standard cartridge tapes. Standard tapes can be mounted on either 18- or 36-track cartridge drives and written in either 18- or 36-track format. When using E-cart tapes, however, be aware of the following restrictions:
 - E-cart cartridge tapes must not be mounted on 18-track cartridge drives. This could damage the tape cartridge or the tape drive.
 - E-cart tapes can be written only in 36-track format.

Cautions

A labeled cartridge tape written in 36-track format appears as an unlabeled cartridge tape when mounted on an 18-track cartridge drive. If you answer N to the "Is Tape Labeled? Y,N" message, the tape is rewritten without being checked for a label by the Exec.

Do not mount E-cart cartridge tapes on 18-track cartridge drives. This can damage the tape cartridge or the tape drive.

5.1.2. Compatibility Issues of Low-Density (9840A-B) and High-Density (9840C) Tapes and Tape Drives

The following compatibility issues with regard to low-density and high-density tapes and tape drives are important to keep in mind to protect tapes, tape drives, and data integrity:

- Although a tape written on a low-density drive can be read on a high-density drive, a tape written on a high-density drive cannot be read on a low-density drive.
- A tape written on a low-density drive can be rewritten on a high-density drive, assuming the write starts at the Beginning Of Tape. However, once past the Beginning Of Tape, new data cannot be appended to the data written by the low-density drive.

5.1.3. Compatibility Issues of EDRC and Unisys Proprietary Compression

The following list summarizes the compression issues between enhanced data recording capability (EDRC) and Unisys proprietary compression.

- Devices with EDRC compression cannot read tapes made with the Unisys proprietary compression.
- Devices with Unisys proprietary compression cannot read tapes made with the EDRC compression.

5.1.4. Working with a TAVR System

TAVR simplifies the interfaces between a user requesting a tape, the operating system, and the system operator to provide more flexibility in mounting tape reels.

The principal advantages of TAVR are

- You can select the unit on which to mount the reel either before (PREMOUNT) or after (AFTERMOUNT) the load message is displayed. This lets you choose where and when to mount tapes.
- TAVR eliminates the relationship between the tape file assigned and the unit selected, and uses reel-IDs to supervise and control the assignment and release of tape reels.

TAVR Terminology

You must know the following TAVR terms:

Actually assigned

means that when a unit is actually assigned, aftermounting of the requested reel is not allowed.

Aftermount

means that you override the unit specified in the LOAD message and mount the reel on any unassigned unit that meets the assignment criteria.

Available unit

is an unassigned unit that is up (UP) or reserved (RV) and is compatible with the type of unit requested by the user.

Compatible

means that the unit equipment code is contained in the CONV selection sequence for the requested assign mnemonic. Also, the modes and translator/data converter are valid for the assign request. The unit must be within the user quota limits.

Interlock

means that a tape has been rewound and unloaded. This tape cannot be used until another LOAD message is issued for it and it is reloaded.

Labeled tape

is a tape that has a VOL1 label as the first block of data.

LOAD message

is a console message requesting that you mount a reel on a specific unit. When it is issued with a corresponding LOAD ANY message, you can override the unit selected in the LOAD message by mounting the requested reel on one of the compatible units in the LOAD ANY message.

LOAD ANY message

is a console message telling you the device types that are compatible with the user's request.

Premount

means that you mount a tape on a compatible unit before a request for the tape is processed. The reel-ID of the tape is now known to the Exec (you can display it by using the FS keyin).

Reel-ID conflict

means that the requested tape reel is being used by another run or another activity of the run that requested it. There are two basic conflict times: during file assignment and during tape swapping or initializing.

During file assignment

- If the reel is being used by another run, the request is put in the facilities hold queue until the other run finishes with it.
- If the reel is being used by this run and it is a single activity run, the request is rejected.
- If the reel is being used by this run and it is a multiactivity run, the request is put in the facilities hold queue until it is available.

During tape swapping or initializing, if the next reel is in use by another run or another activity in this run (multiactivity runs), the request is held in the reel-ID conflict queue.

SCRTCH tape

is a tape that is assigned to a user for the duration of the assignment and then made available to the system again for the next SCRTCH tape request. SCRTCH tapes have the following conditions:

Note: The industry use of the word "scratch tape" is different from the Unisys term: SCRTCH. To scratch a tape means to make the tape available for anyone else to use, that is, scratch it off a personal list of tapes.

- The tapes must be assigned with the reel-ID SCRTCH for the Exec to consider them SCRTCH tapes.
- The tape is rewound to the load point and reused when initialization or swapping of SCRTCH tape occurs. This rewrites tape 1 for both one-unit assignments and two-unit assignments when an attempt is made to swap beyond the first reel and the second reel.
- A tape TSWAP function 2 request is not allowed for SCRTCH tapes.
- SCRTCH tapes are not allowed for blank tape assignments.
- SCRTCH tapes are not allowed for a cataloged file.
- Reel substitution is not allowed (other than SCRTCH assignment).

- SCRTCH tapes are rewound with interlock only when the tapes are freed. The tapes are rewound to the load point and reused when they are initialized or swapped.

- SCRTCH tapes can be premounted if they are labeled.

Tape labeling system (TLS) checks only the VOL1 header label.

You cannot use reel-ID SCRTCH in response to either of the two possible ANSWER WITH messages for unlabeled tapes. The ANSWER WITH messages are

```
ANSWER WITH REEL NO. ON devnam  
ANSWER WITH REEL NO. ON devnam E, REEL
```

- The following types of assign requests are specific assignments of a tape unit (the tape unit is also considered to be assigned):
 - An assign request from a privileged account run (a run executing under the tape labeling privileged account)
 - An absolute device, control unit, or logical control unit assign request
 - An assign request with translator/data converter specifications

Note: If an @MODE facility request or a set-mode I/O request specifies a translator or data converter, or asks for block numbering to be turned on, the unit is specifically assigned (see *Actually assigned in preceding text*).

Tape labeling system (TLS)

is the area of the Exec that handles aspects of tape labeling, for example, tape label creation and tape label verification.

Temporarily assigned

means that a unit is tentatively assigned and the operator might mount the requested tape on any compatible unit (aftermount); see *Actually assigned* (defined earlier).

Unlabeled tape

is a tape that does not have a VOL1 label as the first block of data.

User pool unit

is a tape that was assigned by the user and subsequently freed by using the S option. The unit is part of the user facility pool until it is assigned again and subsequently freed without using the S option.

Premount Procedures for TAVR

Premounting is the process of mounting a reel on an unassigned unit in the up (UP) or reserved (RV) state before the reel is requested by a tape assignment, initialization, or swap.

When the assignment request is processed, the following events occur:

1. Tape assign processing checks to see whether the tape is already premounted on a compatible unit. It does this by assigning processing checks:
 - If the tape is premounted, the assignment is completed without soliciting (with LOAD messages) for intervention.
 - If the tape is premounted on an incompatible tape unit, it is rewound with interlock and a LOAD message is sent that requests mounting the reel on a compatible unit. You can mount the tape on any device type that TAVR has specified.
2. At tape initialization or swap time (TINTL/TSWAP requests), the Exec recognizes that the requested reel is premounted on a compatible unit and completes the user's request without operator intervention.
3. TAVR ensures that reel-IDs are unique within the system. Thus, you cannot premount the same reel-ID twice.

Aftermount Procedures for TAVR

Aftermounting is the process of overriding the unit specified in the LOAD message after the assign request has been processed. Aftermounting applies to any request except ones in which a specific device type is requested.

When the Exec processes the assign request, if the requested tape is not premounted on a compatible unit, the Exec selects the most compatible tape drive available and requests that you mount the tape. The Exec also informs you of other devices the tape can be mounted on. You can then override the tape unit that the Exec has selected.

If aftermount is allowed, the Exec sends a LOAD ANY message to you in addition to the LOAD message. The formats of the messages are

```
LOAD reel-id devnam [RING|NORING] filename - index run-id
(Load reel-id) (ANY UNASSIGNED dev-type1 [,dev-type2,...] )
(Load reel-id) (ANY UNASSIGNED cuname [,cuname,...] )
```

Special Considerations for TAVR

When using TAVR, consider the following:

- Users need to know that reel-IDs must be specified on @ASG statements. This lets you premount or aftermount the requested reels. If a tape is assigned without specifying a reel-ID, TAVR assumes there are no premounted units for the request.
- Each tape must have a unique reel-ID. All reel-IDs are unique, except for SCRTCH, BLANK, or reel-IDs specified by using the B option on the @ASG statement. With the B option, the Exec does not check for assigned reel-IDs.
- To use TAVR most efficiently, all tapes must be labeled. You can premount unlabeled tapes, but increased intervention is required because you can be solicited for a reel-ID for the tape.
- TAVR does not let you premount tapes on noninterrupting units.
- For SCRTCH tapes: You can use the SCRTCH reel-ID many times within a run or runs. The tape labeling system (TLS) does not check tape file header labels for SCRTCH tapes (except for the VOL1 volume header label). The tape is rewound to the load point and reused when initialization or swapping of a SCRTCH tape occurs.
- If the TAVR feature is installed, the media-ID automatically updates a load time for labeled tapes. The media-ID for a SCRTCH tape is not updated until access has been made to that tape.
- Reel number substitution is not allowed with TAVR, except for tape file assignments that are considered TAVR-specific requests, such as
 - Absolute device assignments
 - Control unit or logical control unit assignments
 - Assignments specifying the reel-ID SCRTCH
 - Assignments with data translation/conversion specifications
 - Tape file assignments for runs executing under the tape labeling privileged account

The recommended method for reel-ID substitution on a tape file assignment is to use an absolute device, control unit, or logical control unit assignment.

- If you premount or aftermount an 18-track tape on a 36-track drive and the user has assigned an 18-track device or if you premount or aftermount a 36-track tape on an 18-track drive and the user has assigned a 36-track device, an error message "Reel *reelid* cannot be mounted on *devicename*" is displayed and you have to remount the tape on an appropriate tape drive before continuing. See warnings in Section 5.1.1.

Special Considerations When Using TAVR with a CTL

When using a Cartridge Tape Library (CTL), consider the following:

- Using TAVR with a CTL allows tape swaps between drives belonging to Cartridge Library Units (CLU) and freestanding (non-CLU) drives. For example: A user assigns drive A and cartridge reel A in CLU A. The user swaps to drive B and cartridge reel B in CLU B. The tape drive in CLU A is released, a new tape drive is assigned to the user (drive B), and reel B is mounted on drive B from CLU B. The user then swaps to reel C, which does not reside in any CLU. Tape drive B is released and a freestanding drive, drive C, is assigned to the user. An operator LOAD message is then issued, and you mount reel C on the freestanding drive (drive C).
- To allow the assignment or swapping of cartridge tapes between two different CLUs or between a CLU and a freestanding device, two drives are necessary for tape drive switching. Consequently, tape drive contentions are more likely to occur than if tape assignments and swaps were confined to a single CLU. You can use one of the following procedures to avoid or resolve contention among runs that are trying to switch each other's tape drives:
 - When you see the message


```
run-id NEEDS reel-id CURRENT DEVICE device - CURRENT, HOLD, RETRY?
```

respond with CURRENT if you want to have the drive that is currently assigned by the run to satisfy that run's request, thus avoiding tape drive contention. You must then mount cartridge reel-ID on the current drive to satisfy the request.
 - To make the run wait for a drive to become available, respond with HOLD instead of CURRENT. If you then enter an RC run-ID keyin, the hold condition is indicated by OPEN/FACIL/TSWAP in the run status field. This hold condition indicates the run waits until a compatible tape unit becomes available. If another run is also attempting to select a busy tape unit, a deadly embrace situation can result. You can resolve the deadly embrace by mounting the tape requested by one of the runs on either the drive that is currently assigned by that run or any compatible drive.
 - You can also resolve a deadly embrace by entering an E run-ID keyin to cause one of the runs to terminate in error.

5.1.5. Half-Inch Cartridge Tape Subsystems

Half-inch cartridge tape subsystems have many features that enhance performance and reliability. This subsection contains background information on how these tape subsystems can be used in an auditing environment. It includes an explanation of how these devices operate, their operational considerations, and error recovery implications for auditing with buffering on.

Note: Although the DLT 7000 also is a half-inch cartridge tape subsystem, it cannot be used as an audit trail device.

To understand the buffered audit environment of the half-inch cartridge tape subsystem, you must also understand the basic operating and performance characteristics of the subsystem.

Note: *The hardware extended recovery mode is disabled by the Exec for 4780 drives used as audit trails to ensure that no I/O errors exceed 6 seconds in error recovery. If the attempt to turn off extended recovery mode fails, informational messages such as AT n leg device REEL reelid *DXR STATUS:024* are displayed. The messages are informational only. The audit trail continues to use the drive and tape, but with extended recovery enabled. If a hardware error occurs, it might take up to 12 seconds for the Exec to detect a problem and issue the timeout message. To avoid this problem, swap the audit trail to another control unit.*

Buffering

The half-inch cartridge tape subsystem operates in buffering-on (BUFON), buffering-off (BUFOFF), buffering-file (BUFFIL), or buffering-tape (BUFTAP) mode. BUFFIL is supported for CTS5236, CTS9840, CTS9940, T10000, DLT, and LTO tape subsystems. BUFTAP is supported for CTS9840, CTS9940, T10000, DLT, and LTO tape subsystems.

BUFON

enables buffered write mode for user data blocks. Because user data blocks do not have to wait for physical tape motion, the BUFON mode of operation is faster than the buffering-off mode.

BUFOFF

disables any buffered write mode.

BUFFIL

enables buffered write mode for user data blocks and file marks. The system ensures that writing of the data to tape occurs at the close of a file. Because BUFFIL mode enables buffered write mode for user data blocks and file marks, it is faster than BUFON mode.

BUFTAP

enables buffered write mode for user data blocks and file marks for the entire tape. Because BUFTAP mode enables buffered write mode for user data blocks and file marks for the entire volume, it is faster than BUFFIL mode.

Note: *BUFON and BUFOFF both turn off buffering of tape marks.*

BUFON is a user-selectable write mode that can be requested for a tape file on the assignment (@ASG) statement or by using an @MODE statement after the file is assigned. BUFON is the system default when the configuration tag tape_buffering is 0.

When BUFON is enabled, user data blocks are transferred from the host computer to the buffer of the control unit, where they are temporarily stored. The control unit releases the channel and notifies the host that the write is complete (this occurs before the data is actually written to the tape). Data from several write operations is collected in the buffer until the buffer is full or until a short timer expires. Then the control unit starts the tape drive and physically writes the data from the buffer to the tape.

In the BUFOFF mode, data is transferred from the host to the control unit to the tape. An ending status is not returned to the host until the tape drive is started and the data is transferred to the tape. Because the host computer must wait for physical tape motion, BUFOFF mode is slower.

When using a cartridge tape subsystem for audit tapes, you must consider the effects of buffering on the integrity of the audit tape. If buffering is turned off (BUFOFF mode is selected) when a write is made to tape, status is not returned to the Exec until the data is physically on the tape. If a drive or control unit error occurs in those circumstances, the integrity of the data can be guaranteed. With buffering turned off, however, performance is lower than with buffering on. For example, an audit write to a CTS5236 tape with buffering off takes an average of 68 milliseconds to complete, as compared to only 3 milliseconds with buffering on.

BUFFIL in the buffered-write subfield of the @ASG, @CAT, and @MODE statements enables tape marks, in addition to labels and data blocks, to be buffered in the tape subsystem until a synchronization point is reached. Synchronizing of the buffer to tape is done after the last EOF or EOV of a file or volume.

If BUFFIL is not set, each Write Tape Mark request causes a synchronization. These synchronizations can slow performance when many small files are written to a labeled tape.

When using TMB, it is your responsibility to retain the original data for the user to perform error recovery, if that is your requirement. The Exec provides no means for you to recover this data. With the use of the BUFTAP option, it is possible for you to have received good status on previous writes of thousands of files, when in fact none of the files made it to tape. If an error occurs while the files are being written to tape, the remaining blocks in the tape subsystem buffer cannot be written onto the tape. The Block ID SAFE (BSAFES) I/O function provides a means for the users to determine what block has been actually written to tape by the tape subsystem. A FSAFES I/O function also provides for synchronization of tape, but this impacts tape performance.

Note: *If an SBCON CTS9840 or SBCON CTS5236 is to be reset or powered down, you must down (DN) the tape unit first. If you reset or power it down while it is in the UP state, and a user assigns a tape with the BUFFIL or BUFTAP option enabled, the option causes a command reject with the fault symptom code "1A 1D" in sense bytes 10 and 11.*

Buffering Example

To achieve maximum performance, I/O must be done with buffering on. However, writing to the tape with buffering on can jeopardize the integrity of the audit trail, as illustrated by the following example.

In a half-inch cartridge tape subsystem (BUFOFF) environment, a block of data is physically written to tape before status is returned to the host. If an error occurs, the block is sent again. If the retry is unsuccessful, the audit reel is closed and the block is written to the next reel.

In the tape subsystem (BUFOFF) environment, a block of data is physically written to tape before status is returned to the host. If an error occurs, the block is sent again. If the retry is unsuccessful, the audit reel is closed and the block is written to the next reel.

To illustrate further, assume the control unit attempts to write block 50 to reel 1, but an error occurs. With buffering off, there is no problem. The audit reel is closed and block 50 is written to reel 2.

In the tape subsystem audit environment with buffering on, however, data is written to the control unit buffer and later to the tape. Using a similar illustration, the host writes block 50 to the buffer and continues to write blocks 51, 52, and 53.

If an error occurs at this point, before the control unit writes the blocks to tape, the host is notified on the next write to the buffer (in this example, block 54).

The Exec then writes ending system labels to reel 1 and attempts recovery by writing block 54 to the next reel. Notice that blocks 50, 51, 52, and 53 are not recovered and they are not on the reel 1 or the next reel, reel 2. This leg of the audit trail is now missing four audit blocks and cannot be used for recovery purposes.

Similar problems could happen if one or both control units lose power or have catastrophic hardware errors that cause buffer loss.

To maintain the integrity of the audit trail with buffering on, you must configure dual Exec audit trails, place each leg on separate subsystems (with either dual or single control unit), and protect each tape subsystem with a reliable Uninterruptible Power Source (UPS).

Caution

Using a single control unit can degrade system performance while a spare is being mounted. If only one control unit has access to the primary and spare tapes, I/O is locked out to the primary tape for 1 to 2 seconds after the spare tape is mounted. This delay reflects the time it takes the drive to read the ANSI-standard hardware density identifier and reposition itself for use.

Audit Trail Legs

In the buffered audit environment of the half-inch cartridge tape subsystem, the audit trail must be opened by using operator keyins which ensure that each leg of the audit trail is maintained on a separate subsystem. These keyins are a minor variation of the usual keyins used to open the audit and must be documented at your site. An example of the keyin and the instructions for keeping the audit legs on separate subsystems are given in Section 11. Once the audit is opened on separate subsystems, the Exec automatically maintains the separation of the audit trail tapes as long as no errors force the Exec to swap off the originally designated subsystem. In that case, you must manually swap the audit legs back to their original subsystems. The procedure to return the audit trails to the correct subsystem is very simple.

The only other significant error is a datacheck. A datacheck error can occur when the control unit attempts to write a buffered data block to the tape and the write fails due to drive or media errors. This error causes the Exec to enter error recovery and swap to the next reel, which results in the audit leg missing several blocks of data.

The simplest solution to this problem is to duplicate the part of the audit leg that did not have the datacheck. This approach can be implemented easily with an Executive Control Language (ECL) runstream. The runstream requires you to indicate which tape was mounted on the unaffected leg of the audit when the error on the damaged leg occurred. The runstream duplicates part of the good leg, ensuring that if the audit tapes are needed due to a subsequent system failure, two copies of the data necessary to recover the system exists. Sites can choose not to use this runstream or enhance it to make multiple copies of the good audit leg. A sample runstream is presented later in this section.

Even with multiple backup and fallback capability, it is still possible for multiple error conditions to combine and result in simultaneous failure of both audit legs and loss of some amount of audit data. These errors are extremely unlikely. The recovery for these types of errors requires manual procedures at the site.

The site needs to perform an IRU long recovery to recover as much of the audit data as possible between the two audit legs. After long recovery, the database is usable, although some transactions can be lost.

Automatic Cartridge Loader

The automatic cartridge loader (autoloader) is an optional device that can automatically load tapes into the drive one at a time and accepts unloaded tapes into a discharge tray.

The autoloader has three modes of operation:

- Auto Mode

The autoloader loads a cartridge into the drive whenever the drive is empty; the cartridges remain in the loader. No system intervention is necessary on a system with Tape Automatic Volume Recognition (TAVR). Each new tape is considered a premount. On a system without TAVR, the cartridge is unloaded immediately if no outstanding request exists for that tape drive.

You can use the autoloader in auto mode to premount a series of cartridges that is requested in a given order. As soon as one cartridge is unloaded, the next is loaded into the drive and the system considers the cartridge as premounted. This method of operation is best used only on drives that have been placed in the premount-only state by a PM keyin.

If this keyin is not used, as with any form of premount on a TAVR system, and if any other requestor needs the drive while it is in a premounted state, the premounted cartridge is unloaded and the operator is requested to load the requested cartridge. This can result in a sequence of messages as the autoloader loads each cartridge in the loader and then rejects it as the wrong cartridge. If this happens, put the autoloader in either system or manual mode and load the correct cartridge.

- Manual Mode

The autoloader works only when you press the START button and manually load a cartridge. Once the cartridge is unloaded, you must repeat the same process.

- System Mode

Notes:

- *System mode is not supported on OST5136, CTS5236 (SCSI or SCSI2W), or LTO tape subsystems. System mode is supported for the SBCON version of CTS5236.*
- *For the 4780 tape subsystem to detect system mode, the SYSTEM mode indicator must be lit.*

Multivolume file assignments or blank/scratch (BLANK/SCRTCH) pools. Priority is given to BLANK/SCRTCH tape assignments when a cartridge is in the autoloader; in this case, a drive is selected and a SELECTED message is displayed on the console, not a LOAD message. File assignments requesting more than one media-ID attempt to use a drive with an empty autoloader. LOAD messages are issued, requesting that up to the first 6, 10, or 24 (depending on the type of autoloader) cartridges be loaded.

When a TSWAP\$ function 0 or 1 request comes in (see the description of TSWAP\$ in the *Exec System Software Executive Requests Programming Reference Manual* (7830 7899)) and the autoloader contains cartridges, the autoloader loads the next cartridge into the drive to determine if the cartridge is usable. (For specific requests, the autoloader loads the next cartridge and determines whether the cartridge is correct.) When the loaded cartridge is not usable (correct), it is unloaded and a LOAD message is issued.

The autoloader does not give any indication to the software of the mode in which it is operating. If discrepancies arise between what is expected and what actually occurs on your drives, check the mode setting on the autoloader. Messages concerning the state of the autoloader and commands to the autoloader might not work properly in auto or manual mode.

The Exec displays three types of messages on the half-inch cartridge tape unit display:

system-id

appears when the unit has been brought up (UP) or reserved (RV) and no cartridge is requested or mounted. The system-ID is determined by the configuration statement `IDS IS SITEID 'site-id-name'` or by the `partition_name` field in the PDB.

media-id (flashing)

appears when a request is issued for a cartridge to be mounted on the unit.

Note: *The media-ID continues to flash on the display until the requested cartridge is loaded.*

media-id (nonflashing)

appears when a cartridge is in the tape drive.

Notes:

- *If the TAVR feature is not installed, labeled and scratch tapes cannot be updated until access is made to the tapes.*
- *Any cartridges remaining in the autoloader are assumed to be BLANK by the system and are loaded into the drive on a request for a BLANK or SCRATCH cartridge. If the cartridges are labeled, this results in the cartridge being unloaded, and the operator is requested to load a BLANK cartridge. If the cartridges are unlabeled, they can be overwritten by this request.*

When a multivolume file is assigned to a device with an autoloader configured, the following messages are issued to the console (the actual number of messages depends on the type of autoloader and the number of volumes for the file):

```
LOAD 1 vol11/vol2/vol3 devnam [RING|NORING] filename index runid
LOAD 2 vol4/vol5/vol6 devnam [RING|NORING] filename index runid
.
.
LOAD 8 vol22/vol23/vol24 devnam [RING|NORING] filename index runid
SERVICE 1 vol11/vol2/vol3 devnam [RING|NORING] filename index runid
SERVICE 2 vol4/vol5/vol6 devnam [RING|NORING] filename index runid
.
.
SERVICE 8 vol22/vol23/vol24 devnam [RING|NORING] filename index runid
```

Note: *Cartridges must be loaded into the autoloader in the correct order; if not, you require an operator intervention.*

Information Displayed by Subsystem LEDs

The LEDs on half-inch cartridge tape subsystems provide status and loading information. The LEDs can indicate any of the following:

- Load User-Defined Tape

When a user requests a specific tape, the media-ID of the requested tape is displayed by the LEDs. When the tape is requested, but not yet loaded, the media-ID flashes. Once the Exec recognizes the tape has been successfully loaded, the media-ID continues to be displayed but it no longer flashes.

- Load Blank Tape

When a blank tape is requested by a user, the LEDs indicate whether the tape must be labeled or unlabeled and if write protection has been specified.

When a labeled blank tape is requested, the LEDs display one of the following messages:

- x-BLNK F if write-protection was specified by the user
- x-BLNK U if write-enable was specified by the user
- x-BLNK if neither write-protection nor write-enable was specified by the user

where:

x is an L if the tape must be labeled and a U if the tape must be unlabeled.

- Cleaning Required

The cartridge tape drives maintain a record of the amount of tape that has passed through the read and write heads since the drive was last cleaned. When this number reaches a predetermined point and a tape is unloaded from the drive, a *CLEAN* message appears on the LED display of most cartridge tape subsystems. The *CLEAN* message does not disable use of any of the drives; you can continue to mount cartridges and use the drives even if they are not immediately cleaned.

- On 4780, SCSI 5136, SBCON CTS5236, CTS9840, U9840D, U9840C, T10000A, and LTO subsystems, the following operator message is displayed on the console: XXXX REQUIRES CLEANING, where xxxx is the name of the drive requiring cleaning. This operator message is displayed following each unload until the drive is cleaned.
- On SCSI 5136 subsystems, the *CLEAN* LED message does not appear while the drive is up even if it needs cleaning. The *CLEAN* LED message is shown only when the drive is brought down and needs cleaning.
- On SCSI CTS5236 subsystems, no message is sent to the console.
- On CTS5236 subsystems, the *CLEAN* LED message is not supported by the subsystem. The indication that cleaning is required differs among the CTS5236 models; refer to the hardware manual to find out the method of indication.
- On OST4890 subsystems, an operator message is displayed on the console and the LED displays "CLEAN ME."
- On the DLT 7000, no message is sent to the operator console. An LED on the device lights when the drive needs cleaning.

- System-ID

When no cleaning is required and no tape is loaded or requested, the LEDs display the system-ID number.

- Unit number

If the unit is down, the LEDs display the unit number of the drive.

Note: Due to hardware limitations, the Exec does not display either the system-ID or reel-ID on the DLT 7000 or LTO tape drive.

Duplex Audit Tape Open Procedure in

For modern tape devices (that is 9x40s), there is a one-to-one correspondence between subsystems and drives. However, for earlier tape device (that is 5136, 5236s), there were multiple drives for a subsystem. To protect from drive, media and subsystem failures for these earlier devices, the duplex audit trail legs need to be located on different subsystems.

To locate the legs of a duplex audit trail on separate subsystems, use the following example procedure. The example shows a duplex audit with two spares configured:

- Subsystem 1, audit leg 1
 - Tape CU TCN0,TCN1
 - Active unit TN00
 - Spare unit TN01
- Subsystem 2, audit leg 2
 - Tape CU TCN2,TCN3
 - Active unit TN20
 - Spare unit TN21

Determine which tapes to use. You need four tape units, two units on each of two available subsystems. The objective is to have one primary unit and one spare unit on each tape subsystem. Always use the following keyin to open the audits.

```
AT 1 OP primary1/primary2,spare1/spare2
```

where:

primary1

is the primary unit for leg 1 on subsystem 1.

primary2

is the primary unit for leg 2 on subsystem 2.

spare1

is the spare unit for leg 1 on subsystem 1.

spare2

is the spare unit for leg 2 on subsystem 2.

Using the example configuration, this keyin is

```
AT 1 OP TN00/TN20,TN01/TN21
```

Units TN01 and TN21 are spares. The Exec always picks a spare unit on the same string as the active audit tape. If the active audit tapes ever end up on the same subsystem due to error, you have to manually swap one of the active audit tapes to a spare on the other subsystem. Once the swap is done, no further action is required.

Audit Tape Copying

To recover from drive or media failures, the buffered audit tape backup program on the half-inch cartridge tape subsystem must be available on your system at all times. This program protects your site from data loss due to failure in one audit leg.

The program relies on the requirement that, on half-inch cartridge tape subsystems, all buffered audit environments be duplex. Duplexing means that at any time there are two identical copies of any particular audit record, and this implies that the typical duplex environment always has four tape drives dedicated to audit: two active primary drives or legs, and two spare drives.

For example, consider the following duplex audit environment:

- Subsystem 1, audit leg 1
 - Tape CU TCN0,TCN1
 - Active unit TN00 tape reel C001
 - Spare unit TN01 tape reel C002
- Subsystem 2, audit leg 2
 - Tape CU TCN2,TCN3
 - Active unit TN20 tape reel C003
 - Spare unit TN21 tape reel C004

If an error occurs on TN00, reel C001, it is likely that some of the data that must be on reel C001 is missing. The data that is missing on reel C001 is present on reel C003, and reel C003 is the only copy of the data. It can be unlikely that problems occur with reel C003, but it is necessary to protect the data as much as possible by using the BACKUP runstream.

This runstream protects the audit data by making a duplicate copy of the data on the good audit leg (reel C003).

You need to back up an audit tape whenever you suspect an error on the tape has caused data to be missing from one of the duplexed legs. These cases include (but are not limited to) occurrences of the DATCHK, INTREQ, or EQPCHK I/O error messages; or any time an unusual error occurs on the drive. If you are in any doubt about whether to back up a tape, it is always best to do the backup.

To perform the backup, proceed as follows:

1. Identify the tape to be backed up. Remember, you are backing up the good leg of the audit. Examine the printed output to determine which reel was active on the audit trail leg that is free of errors (the good reel) when the error occurred on the other leg (the bad reel).
2. Determine whether the good reel is still active (mounted); use the AT 1 FS keyin to do this. If the tape is still mounted, swap off the reel by using the AT 1 SW keyin.
3. If you are not sure which tape was active at the time of the error on the other leg, make backup copies of several of the tapes that were active around the time of the error. You might encounter this case if the good leg happened to swap to the spare at or near the time of the error on the bad leg.

The following example shows how the BACKUP utility could be used. It also shows a typical sequence of system console messages; the reel numbers and drive names used are for example only. In addition, the example is annotated to explain the actions that can be executed at each step of the program (comments are enclosed in parentheses).

```
ST qualifier*filename.BACKUP,,,OPUTIL/0
```

```
BACKUP START
```

```
1-BACKUP*ENTER REEL NUMBER OF ... AUDIT TAPE TO COPY OR X TO END
```

Reply X to end the program without a backup or enter the reel number of the AUDIT tape you want to back up.

```
1 C003
```

```
1-BACKUP*ENTER BLANK REEL NUMBER FOR BACKUP REEL.
```

Reply with the number of a blank reel you want to become the backup of the AUDIT tape previously entered.

```
1 C006
```

```
1-BACKUP*ARE YOU SURE REEL C006 IS A BLANK? ACK=YES, X= ABORT
```

Reply X if you have entered the incorrect blank reel number; otherwise, transmit to continue the program.

```
1 YES
```

```
LOAD C003 TN00 NORING AUDIT$01 -1 BACKUP
```

Mount reel C003. Note that this reel must be write-protected (NORING).

```
LOAD C006 TN01 RING BACKUPREEL -1 BACKUP
```

Mount reel C006. Note that this reel must be write-enabled (RING).

```
BACKUP*MSG: BACKUP OF TAPE C003 DONE.
```

```
1-BACKUP *ENTER REEL NUMBER OF AUDIT TAPE TO COPY OR X TO END
```

If additional backup copies are needed, enter another blank reel to continue the program; otherwise, enter X to end.

```
1 X
```

```
BACKUP*TAPE BACKUP DONE.
```

In this example, reel C006 is now a backup tape for reel C003 and can be used by Integrated Recovery Utility (IRU) during recovery if there is an error on reel C003.

Remember that the backup reel is needed only if the audit tapes are required for recovery and if the original audit tape is unreadable. Although this situation is not likely to occur, the following example shows how the backup reel is used. Once again, the example is annotated to explain actions to be taken by the operations personnel during recovery.

Note: *It is imperative that your system have the tape configuration tag `TLSUB (reel_nbr_substitution_allowed)` set to 1 or TRUE. This allows the substitution of the backup reel for the failing recovery tape.*

IRU displays a message with the following format during recovery:

```
IRU*7015 Audit trail read error - 012- OPER B REPLY
IRU* on audit reel C003
IRU*AT3 Read TBSN: xxx TIME: yy/mm/dd hh:mm:ss zzz
1-IRU*7234 Enter DEVICE, DUPLEX, SKIP, DUMP, END, or ?
```

If the reel just listed has a backup, you can reply as follows:

```
1 DEVICE
```

```
LOAD C003 TN00 NORING AUDIT$01 -1 IRU
```

Do not mount C003; instead, mount the backup reel C006. You can expect to get the following message:

```
1-TN00 WRONG REEL, LOAD C003 A,E,REEL
```

Reply with the number of the backup reel.

```
1 C006
```

```
LOAD C006 TN00 NORING AUDIT$01 -1 IRU
```

Remount reel C006 if necessary. The IRU now uses reel C006 as a substitute for reel C003; this is acceptable because reel C006 is an exact copy of C003.

Following is an ECL runstream that duplicates a part of the audit leg that did not have the deferred datacheck. You must indicate which tape is mounted on the unaffected leg of the audit when the error on the damaged leg occurred. The runstream duplicates part of the good leg, ensuring that if the audit tapes are needed due to a subsequent system failure, two copies of the data necessary to recover the system exists.

```
@run backup,0,sys$,999,999
@elt,iq tpf$.sgs,,stop
* .
* . This is a sample runstream that can be used as a guide for sites
* . to make their own audit U47M tape backup run. Most sites will want to
* . make modifications to accommodate local programming standards.
* . All sites should test the backup runstream to be sure it works at
* . their site.
* .
* . There must be at least one DEVICE SGS and the first one
* . will be the only one used; other DEVICE SGSs are ignored.
* . The DEVICE SGS has the following fields.
* . Field 1 = the assign mnemonic to use for tape @asg
* .
* . sgs
device hicm
device u9s
device t
* .
* . The EXPIN and EXPOUT SGSs are used to supply appropriate
* . expiration fields for the input and output tape assignments.
* .
expin 31
expout 31
* . Define the name of the audit file.
* . This is necessary since the Exec assigns each audit tape without
* . the 'F' option so the file name here must match the file name
* . the Exec used to create the tape.
file sys$*audit$01
* .
@end stop
```

```
@ssg,ike ,tpf$.sgs,tpf$.addstream,,tpf$.sysstream
skel
* . SSG runstream to make a backup tape of a given audit tape.
* .
*accept,blo audit 'Enter reel number of original audit tape to copy or "x"
to end'
* .
*if [audit,1,1,1,ucstr$] <>X
*accept,blo backupreel 'Enter blank reel number for backup reel.'
* .
* . Give operator a chance to abort run if input is no good.
#msg copy original audit reel [audit,1,1,1] to [backupreel,1,1,1]
#msg,w are you sure reel [backupreel,1,1,1] is a blank? ack=yes, x=abort
* .
#asg,tf [file,1,1,1].,[device,1,1,1],[audit,1,1,1],[expin,1,1,1],noring
* .
* . note the 'F' option is used here to make backup verification easier.
#asg,tf backupreel.,[device,1,1,1],[backupreel,1,1,1],[expout,1,1,1],ring
* .
* . Now copy the good reel to the backup reel.
* .
copy,m [file,1,1,1].,backupreel.
#msg backup of tape [audit,1,1,1] done.
#free [file,1,1,1].
#free backupreel.
* .
#ssg,ike tpf$.sysstream,tpf$.sgs,tpf$.addstream
sgs
#add,l tpf$.sysstream
#eof
#eof
*else
*display,o 'runstream terminated by operator X keyin'
*endif
@eof
@eof
@fin
```

5.1.6. Virtual Tape Handler

The virtual tape handler (VTH) simulates tape drives on mass storage, which can be fixed or removable packs. A tape assign mnemonic mapped to VTH indicates mass storage is simulating that type of tape drive. There is a limit of 90 types of devices that can be defined as virtual tape devices at any given time. Files with qualifiers and filenames of SYSS*\$VTHR\$xxxxxx are considered virtual tapes, with xxxxxx representing the reel-ID.

Programs cannot differentiate between a virtual tape and a real tape except on a virtual tape

- Fast tape access (FTA) is not supported.
- The capacity of newer cartridge drives is not simulated.
- A program that reads backwards can get a different word returned than from a real tape.
- Audit trails are not supported on VTH.

See the *Exec System Software Executive Requests Programming Reference Manual* (7830 7899) for more specific information about the preceding exceptions.

An area of mass storage is designated as a pool (SYSS*\$VTH\$POOL) of reel-IDs to be used when programs require a blank or scratch tape. When a program no longer needs a scratch tape, the reel-ID is returned to the pool and is available for use as a blank or scratch tape by another program. The system administrator defines how many IDs must be in the pool and has to add more if the pool becomes depleted.

This pool also identifies the types of devices that can and cannot be defined as virtual tape devices. The type of device used for the default system tape must not be allowed as a virtual device because this would prevent rollbacks.

5.1.7. DVD Tape

The DVD tape is a DVD that simulates a read-only tape. You cannot write to it. For example, it cannot be an Audit Trail device or Dump device. It is supported as a read-only tape and also a boot device. DVD tapes have the same characteristics of most open reel types and therefore do not support fast tape access (FTA), block numbering, data compression, buffered writes, or expanded buffer. DVD tapes do not support ACS or CTL; therefore assignments with CTL-pool are rejected. Because the DVD-tape is read-only, the use of the W option, the indication of RING or WRITEENABLE in the ring field, and the indication of BLANK or SCRTCH in the reel-ID field are all rejected.

5.2. Disk and Disk Drive Operations

You must use DPREP to prepare disks for use. See the *Exec System Software Installation and Configuration Guide* (7830 7915) for more information.

Each disk pack is labeled as either a fixed disk or a removable disk. Before using a disk pack, the system reads the label of the pack to identify it as fixed or removable, and to make sure that it is correctly prepped. If it is not correctly prepped, the system displays a series of messages that you must respond to.

Once a fixed disk is used by the system, it cannot be removed from the system without rebooting the Exec. If a fixed disk is removed from the system, all the files on that fixed disk is lost or disabled. To avoid any loss of files, the same fixed disks that were used during a system session must be available when a manual or automatic recovery is performed.

During either the initial load or a recovery, the Exec tries to read a label record from each disk pack, either online or suspended. If the pack is not prepped or there is no label, the system displays I/O error messages on the console. The system displays a message that prompts you to modify the configuration. At that time, you can change the status of any units in the configuration.

During an initial load or a recovery, the Exec counts the number of fixed mass storage devices and displays this message:

```
0-FIXED MS DEVICES=nn-CONTINUE? YN
```

If the number, *nn*, is correct, respond by entering a Y, and the boot continues. If the number is not correct, respond by entering an N. After a short delay, the system stops with an error. If the system does not display the correct number of devices in the message, enter an FS,MS keyin before you answer the message. The keyin produces a list of devices found, and you can see which ones are fixed.

These keyins let you control disk drives and disk packs:

- The DC keyin copies the contents of one disk pack to another disk pack. All disk types, except those that are physically removable, are valid. The DC keyin can be used for local and shared disks.
- The IN keyin initializes a fixed mass storage device (disk drive). Use caution, since this keyin destroys all information currently on the device. Do not use this keyin on removable mass storage devices.
- If DN,PACK (down pack) keyins are configured, you can use the MD keyin to place inhibits on or to modify the current inhibits of a disk pack in the UP, DOWN, or PRESENT state. If you use the MD keyin on a disk pack that does not currently exist, the pack is added to the list of packs in the PRESENT state. The benefit of adding disk packs in this manner is that it allows you to place inhibits on a pack currently not on the system so that when it is put on the system its access is restricted as needed. (See 5.2.1 for more information on the down pack keyins.)

- The MS keyin displays information about the availability of mass storage on the system. The keyin displays, for each device type, the number of mass storage tracks currently available.
- The MV keyin moves a fixed disk pack or an assigned removable disk pack from a failing drive to another drive. The keyin must be entered before the pack is moved. If the disk pack is moved first, I/O errors can occur. The keyin is valid only for physically removable devices.
- The RD keyin removes disk packs in the DN or PRESENT state from the pack status table to make room for other packs. The maximum number allowed is defined in the system configuration. This keyin cannot be used on a unit-duplexed device. (See 5.2.1 for more information on the down pack keyins.)
- Use the RP keyin to remove all pointers to files that reside on removable disk packs from the STD or SHARED master file directory (MFD).
- The EXPAND keyin expands the internal OS data structures of a pack to make use of the larger disk capacities. It can be used to expand local and shared disks.

Refer to the keyin modules in Section 11 for a complete description of each keyin and its operation.

5.2.1. Controlling Removable Disk Packs with Down Pack

The Down Pack feature lets you bring down or restrict access to a pack prepped as removable (without causing a system stop). You can bring down or restrict access to the removable pack even though files are assigned on the pack, user I/O requests are in progress, or the pack is unmounted.

Down pack keyins are most valuable in an environment where all files on removable packs are TIP duplex files. See the *Transaction Processing Administration and Operations Reference Manual* (7830 7881) for examples on the use of down pack keyins when the removable packs contain TIP duplex files.

Notes:

- *The NEWCON configuration parameters REQNBPRST0 and REQNBPRST1 control the availability of the Down Pack feature. See the Exec System Software Installation and Configuration Guide (7830 7915) for more information.*
- *When the TIPDUPRSL configuration parameter is TRUE, it will initiate the DN,PACK keyin and suppress I/O messages and provides a forced "G" response to the failing removable pack. See the Exec System Software Installation and Configuration Guide (7830 7915) for more information.*

Down Pack Keyins

The down pack keyins are as follows:

DN,PACK

Changes the status of a removable pack to down (DN).

UP,PACK

Changes the status of a removable pack to up (UP).

FS,PACK

Reports the status of a removable pack.

MD (modify disk)

Restricts read, write, or assign access to a removable pack.

RD (remove disk)

Removes a pack from the system pack list.

See Section 11 for more information on these keyins.

Changing the Status of a Removable Pack

The status of a removable pack is distinct from the status of the disk device that contains it. By manipulating the pack status, you might be able to bring down an assigned disk device and make repairs without a system reboot.

The status of a removable pack can be: UP, DN, PRESENT, or NOT PRESENT.

UP Status

When they are assigned, removable packs receive an initial status of UP, and access to an UP pack is unrestricted.

DN Status

Removable packs receive a status of down (DN) when a DN,PACK keyin is performed on the pack. A down pack is disassociated from the disk device where it resides. Once this logical separation occurs, the device status can change without regard to assignments on the pack.

Any attempt to access a file on a down pack is rejected. If the file is TIP duplexed, an I/O request receives do-not-read/do-not-write (DNR/DNW) status for the leg on the down pack, but the request proceeds on the other file leg. If the file is TIP duplexed and the TIPDUPRSL configuration parameter is TRUE, then I/O to the pack is given an immediate I/O status 015 without setting the DNR/DNW status for the leg. This improves the TIP Duplex recovery time and prevents undesirable I/O console messages from flooding the system. If the file is non-TIP or TIP simplex, an I/O request fails with an I/O status 015. Files might be cataloged, assigned, and freed on a

pack while it has a status of DN. However, any attempt to expand an existing file on a down pack is rejected.

The Exec is unable to update directory information on a down pack. Therefore, when one of the following events occurs, an UP,PACK keyin is not allowed until all files on the pack are freed or the system is rebooted:

- A cataloged file is assigned or freed on a down pack.
- An RP keyin is successfully performed on a down pack.
- A down pack is reprepiped.

Note: *If the pack contains only TIP files, you might release all file space on the pack instead of freeing all files.*

PRESENT Status

Removable packs receive a status of PRESENT when a modify disk (MD) keyin is used to restrict access to an unassigned pack (that is, the pack has not yet been placed in the system's list of UP, DN, or PRESENT packs). This status allows restrictions to be placed on a pack before it enters the system.

NOT PRESENT Status

The NOT PRESENT status implies the removable disk is not assigned, has no inhibits set, and has not been brought down by the DN,PACK keyin (that is, it was not found by the system in the list of UP, DN, or PRESENT packs). By definition, removable packs have a NOT PRESENT status if they do not have a status of DN, UP, or PRESENT.

Determining the Status of a Removable Pack

Use the FS,PACK keyin to display the status, inhibits, assign count, and directory-ID of a removable pack. See Section 11 for more information on this keyin.

Restricting Access to a Removable Pack

Use the MD keyin to establish or clear read (R), write (W), or assign (A) inhibits. Read and write inhibits correspond to the DNR and DNW I/O inhibits for TIP files. The creation of global DNR/DNW attributes for all files on a pack allows TIP recovery procedures to be implemented more easily.

The following apply to disk pack inhibits:

- Any attempt to assign a file on an assign-inhibited (A) pack is rejected.
- Any attempt to read from a file on a read-inhibited (R) pack is rejected. If the file is TIP duplexed, a read request receives DNR status for the file leg on the down pack, and the request is redirected to the other file leg. If the file is non-TIP or TIP simplex, the request fails with an I/O status 015.

- Any attempt to write to a file on a write-inhibited (W) pack is rejected. If the file is TIP duplexed, a write request receives DNW status for the file leg on the write-inhibited pack, but the write proceeds on the other file leg. If the file is non-TIP or TIP simplex, the request fails with an I/O status 015.
- TIP maintenance I/O can penetrate read and write inhibits.
- Combinations of inhibits are allowed. However, a write inhibit (W) can be established only in combination with a read inhibit (R). If the pack is UP and unassigned, inhibits cannot be established unless the assign inhibit (A) is included.

Removing a Pack from the System Pack List

Use the RD keyin to delete a DN or PRESENT removable pack from the system's list of packs. The RD keyin can be used to make room for new packs when the system pack list is full. A pack cannot be removed until all files on the pack have been freed.

Down Pack Cautions and Limitations

Use the Down Pack feature cautiously in a non-TIP or TIP simplex environment. I/O requests to down packs fail with an I/O error 015. If an Exec I/O request receives an I/O error 015 (for example, a down pack contains system files), the system can stop. If a UDS I/O request receives an I/O error 015 (for example, a down pack contains database files), UDS can bring down the entire software application. In this environment, you must carefully consider whether using the Down Pack feature, and the data recovery procedures it requires, is preferable to a system stop and autorecovery.

A DN,PACK keyin cannot prevent the system stop that can occur when an Exec I/O request receives an error status.

A removable pack is not placed in the list of UP, DN, or PRESENT packs until it has been assigned, brought down (using a DN,PACK keyin), or had inhibits placed on it (using an MD keyin). Therefore, such a pack does not appear in the output of an FS,PACK keyin.

Since a down pack is disassociated from the disk device that contains it, a device containing a down pack does not appear in the output of an FS,RDISKS keyin. An FS devnam keyin reports the status of the device, but no information about the down pack.

5.2.2. Adding a New Fixed Disk During a Boot

You can add a new fixed disk to the system at MODIFY CONFIG time, but to be up and usable it has to be the current DRS disk. This implies that you are doing a tape boot, and the new device is specified as the system disk in SCF. If the new disk is not used as the current DRS disk, the boot proceeds, but the following messages are displayed, and the disk is internally downed:

```
DEVICE nnn, FRESHLY PREPPED  
DN OF nnn INTERNAL REQUEST WAS RECEIVED  
nnn DN
```

where *nnn* is the new disk device.

5.2.3. Unit Duplexing

Unit Duplexing reduces system vulnerability to mass storage failures. It consists of a hardware configuration in which each mass storage device has an associated backup device that is transparent to all end-user software and to most system software.

If a user program issues a request to read or write to a file on a unit-duplexed device, the write request is automatically performed on both devices and the read request is performed on either device. From a software point of view, the unit-duplexed devices are bit-for-bit identical.

TIP files on unit-duplexed devices must be simplex TIP files.

For additional information, refer to the *Unit Duplexing Planning, Installation, and Operations Overview* (7830 7592).

Section 6

Performing and Processing System Dumps

This section describes how to perform system dumps (also referred to as Exec dumps, panic dumps, or Exec panic dumps), how to use runstreams from the Dump Analysis Library (DMPLIB) to process the dumps, and how to register and process dumps through the RSS interface. Information on dump file management is provided in the *Exec System Software Administration Reference Manual* (7831 0323).

This section also describes how to download the WinDump utility and access the *WinDump User's Guide*. WinDump provides a GUI interface to DMPLIB and an efficient alternative to RSS.

6.1. Dumping System Memory

Note: The Exec supports only labeled dump tapes. All Exec dump tapes must be labeled.

System dumps are used for diagnosing Exec software problems. The dump process, which copies the contents of registers and main memory to either tape or mass storage, can be either automatic or manual.

An automatic dump (part of autorecovery) is generally more efficient than a manual dump. A manual dump is an option during a mass storage or tape recovery. The system dump file that the Exec writes is SY\$\$*PANIC\$DUMP\$\$\$. If this file is copied from tape to mass storage, the result is a one-file dump file. See the *Exec System Software Administration Reference Manual* (7831 0323) for information on using the COPYDUMP program and other means of managing system dump files, and details on system dump file names and formats.

Mass storage dumps rather than tape dumps are recommended because of the large number of tapes required for tape dumps. A consideration for mass storage dumps, however, is that you cannot access user files on mass storage while booting the system. Thus, you cannot copy the contents of those files into a dump file, and you cannot obtain a dump listing of those files.

6.1.1. Automatic Dumps

These are the steps for performing an automatic dump (to mass storage):

1. The system reloads the Exec from disk.
2. Answer M to the message

```
SAVE PANIC DUMP TO MASS STORAGE M,NO?
```
3. The system automatically assigns mass storage files and writes to them.
4. When the dump is complete, the message PANIC DUMP TAKEN is displayed on the system console.

If an error occurs so that the dump cannot be written to mass storage, you receive an I/O error message, the system frees and deletes the dump file, then asks you again if you want to save the dump to mass storage. If jump key 3 is set, autorecovery is suppressed.

To dump to tape, you must do a manual dump (see Section 6.1.2).

6.1.2. Manual Dumps for ClearPath IX Systems

To perform a manual dump to tape on a ClearPath IX system

1. Load the boot tape on the tape drive.
The tape drive has to correspond to the drive specified in the load path of the partition definition.
2. Set the boot type to tape by entering **SET-TYPE TAPE** (or **ST TAPE**).
3. Enter **SET-TAPE <device>** (or **STD <device>**) to specify the name of the boot device.

Note: The *SET-TAPE <device>* command needs to be entered only once on your system. This setting stays in effect for future boots. However, if you set the type to tape, the setting does not stay in effect for future boots; it defaults to disk.

4. Enter **SJ 2** (for jump key 2).
5. Enter **SP** (for start partition).
6. When you receive the following message, answer with the device name:

```
Enter name of tape drive for manual tape dump
```

7. When you receive the following message, answer with the volume serial number (transmit to request a labeled blank):

```
Enter Volume Serial Number for manual tape dump
```


8. Mount the tape and answer any messages that appear.

The message

```
Panic dump saved...
```

signifies that the dump has been successfully written to tape.

If an error occurs and the dump cannot be written to the specified tape device, an I/O error is issued and you are prompted again for a dump device name.

To perform a manual dump to disk:

1. Set the boot type to disk by entering **SET-TYPE DISK** (or **ST DISK**).
2. Enter **SJ 2** (for jump key 2).
3. Enter **SP** (for start partition).

You receive the following message:

```
Save panic dump to mass storage - M,NO?
```

4. Respond with **M** to save the dump to disk or **NO** to abort the dump.

If you respond with M, the message

```
Panic dump saved...
```

signifies the dump has been successfully written to disk.

If an error occurs and the dump cannot be written to the specified disk device, an I/O error is issued and you are prompted again for a dump device name.

6.1.3. Online Dumps

If you want to obtain an Exec dump without stopping the system, or if you want to look at some data structure at a specific time, take the steps below to perform an online dump.

Note: See the *DU MP* keyin in Section 11 for information about specifying a run-ID to get output on a desired run.

1. Enter the online dump keyin as follows:

```
DU MP
```

The keyin produces the following messages:

```
Panic dump taken  
State the reason for the online dump
```

2. Provide a 60-character, free-format response to be placed in the dump file reader for future reference.

The following message appears:

```
Save online dump to tape - T,NO?
```

3. Respond with **T** to write the dump to tape; respond **NO** to abort the online dump and terminate all further DU MP keyin processing.
4. If you respond with T, the following message is generated in preparation for writing the online dump:

```
Enter name of tape drive for online tape dump
```

5. Enter the name of an up and available tape drive.
6. When you receive the following message, answer with the volume serial number (transmit to request a labeled blank):

```
Enter Volume Serial Number for online tape dump
```

7. Mount a tape and answer any messages that appear.

The message

```
Panic dump saved...
```

signifies that the online dump has been successfully written to tape.

Performing an online dump may produce invalid data, which could lead to unacceptable results from DMPLIB processing. Although most DMPLIB functions perform general checking, this checking is successful only if the structure being processed is valid and correct.

6.2. Processing Dumps

DMPLIB, which includes Dump Analysis Procedures (DAP) functions, provides automated analysis and display of Exec data structures to aid in problem diagnosis.

The input used in dump analysis is either a dump tape or a one-file mass storage dump. DMPLIB output is an edited display or printout of Exec main and mass storage structures that vary in format, depending on which type of dump processing you select.

An alternative to the methods described in this subsection is the WinDump utility, which lets you use DMPLIB from a PC workstation. See Section 6.3 for more information.

6.2.1. Dump Processing in Batch Mode

You can use DMPLIB in batch mode at the system console when a demand terminal is unavailable. You can select several different dump types and input formats with the ST keyin. The DMPLIB batch runstreams are generated by elements found in file EXEC*DUMPLIBS. as follows:

```
ST EXEC*DUMPLIBS.type,set,runid,account/user-id
```

where:

type

specifies the type of analysis to be performed. The following are possible choices:

ILSDMP	This is an ILES dump. See the special condition word values for ILES dump processing under the description of the set parameter.
MICRO	This is a MICRO dump. This analysis produces the MICRO dump output as in MICRO/RSS, but does not register the output file with Remote System Support (RSS). This dump can be used if RSS is not installed at your site.
MICRO/RSS	This is a MICRO dump. This is a summary of data structures that pertain to the problem. The MICRO dump format varies with the problem. Using the RSS program, DMPLIB registers the dump with RSS so it can be transmitted to the Unisys Customer Support Center.

Caution

You should use MICRO/RSS for tape dumps only. If you use MICRO/RSS to process a mass storage dump, the dump file is deleted from the system. This prevents further analysis of the problem. To process mass storage dumps through RSS, use the RSS-REG runstream described in Section 6.4.

RSDUMP	This is a SHORT dump. The output is the same as SDUMP, but the DIAG\$ file is recovered from the system boot tape that is required as additional input. Note: <i>RSDUMP is supported (and accepts input) only for dumps that have been written to tape. Mass storage dumps cannot be processed with RSDUMP. That is, do not specify 2, 4, 5, or 6 as the set value.</i>
RTSDUMP	This is a Communications Simulator (CS 2200) dump.
SDUMP	This is a SHORT dump. The edited output contains a detailed analysis of all Exec data structures.

set

This is a number specifying the input format of the Exec system dump or dumps to be processed. The following are possible choices for Exec system dumps:

- 0 processes all existing cycles of SYSS\$*PANIC\$DUMP\$\$ and then asks you for the reel number and equipment type of dump tapes to be processed.
- 1 processes several dump tapes. Asks you for the reel number and equipment type of each dump tape to be processed.
- 2 processes only the latest cycle of SYSS\$*PANIC\$DUMP\$.
- 3 processes only one dump tape. Asks you for the reel number and equipment type of the tape to be processed.
- 4 processes all existing cycles of SYSS\$*PANIC\$DUMP\$.
- 5 asks you for the mass storage name of the dump file.
- 6 processes only the oldest cycle of SYSS\$*PANIC\$DUMP\$.

For ILES dumps (ILSDMP as type), use one of the following values for the set (or condition word) parameter to control the dump processing:

- 0 processes the latest cycle of the ILES dump. Do not delete the dump file.
- 1 processes the latest cycle of the ILES dump. Delete the dump file.
- 2 processes all cycles of the ILES dump. Do not delete the dump files.
- 3 processes all cycles of the ILES dump. Delete the dump files.
- 4 processes the oldest cycle of the ILES dump. Do not delete the dump file.
- 5 processes the oldest cycle of the ILES dump. Delete the dump file.
- 6 asks you for the ILES dump file cycle number and delete option.

Note: To perform an ILES dump, user-ID must be executing with system high clearance level, all compartments, and the bypass ownership privilege.

runid

is the run ID number for the dump analysis run.

account

is the account number for the dump analysis run.

Examples of Dump Editing Options

Following are some examples of the dump editing options and input formats you can select with the ST keyin.

The following console keyin produces an SDUMP, with run-ID SHORT, from a dump tape. A query requests the reel number and equipment type of the dump tape to be processed. The SDUMP is processed under account 123456 for user-ID ABC.

```
ST EXEC*DUMPLIBS.SDUMP,3,SHORT,123456/ABC
```

The following keyin produces an SDUMP, with run-ID XER070, from one or three user-specified mass storage files. A system console query requests mass storage file name or names.

```
ST EXEC*DUMPLIBS.SDUMP,5,XER070,123456/ABC
```

The following keyin produces an SDUMP recovery dump with the DIAG\$ file recovered from the system boot tape. A system console query requests the reel number and equipment type of both the dump tape and the system boot tape.

```
ST EXEC*DUMPLIBS.RSDUMP,3,RECDMP,123456/ABC
```

The following keyin produces an SDUMP from the latest cycle of file SYSS\$*PANIC\$DUMP\$\$.

```
ST EXEC*DUMPLIBS.SDUMP,2,PNCDDMP,123456/ABC
```

The following keyin produces a MICRO dump, with run-ID DUMP01, from a dump tape, and then registers the output file with RSS. The file can then be transmitted to the Unisys Support Center.

```
ST EXEC*DUMPLIBS.MICRO/RSS,3,DUMP01,123456/ABC
```

Additional Operator Queries

The DUMPLIBS batch runstreams can query you for additional information about the dumps to be processed.

If the set value specified in the DUMPLIBS start statement is 5, the following query appears. Enter only one file name and transmit. To terminate the query, enter nothing and transmit.

```
Enter one dump file name
```

For mass storage dumps involving one or more cycles of SYSS\$*PANIC\$DUMP\$\$ (for example, if the set value is 0, 2, 4, or 6), the following query appears. Enter Y (yes) to delete the dump cycles or N (no), which is the default, to save them.

```
Should cycle(s) of SYSS$*PANIC$DUMP$$ be deleted after processing? Y/<N>
```

An N or NO response to the following panic dump tape query terminates the query and runstream generation.

```
Enter panic dump tape REELNO,EQP or NO
```

or

```
Enter boot (recovery) tape REELNO,EQP
```

To specify a reel number for the panic dump tape or boot tape, include the equipment type after a comma, as in the following example:

```
reel,equip-type
```

This assigns the reel on the specified equipment. If you do not specify an equipment type, the default is T.

6.2.2. Dump Processing in Demand Mode

You can process any dump from a demand terminal by entering the following @ADD statement:

```
@ADD EXEC*DUMPLIBS.IADUMP
```

IADUMP queries you for dump information. It then generates a runstream to process the dump. IADUMP prints a message stating the name of the file and element that the runstream is placed into.

For example:

```
Dump runstream is in file DUMP-hhmmss.DUMPRUN
```

Queries encountered in IADUMP are as follows. You can enter EXIT in response to any query to terminate IADUMP processing.

Query 1

This query lets you select the type of dump to be performed. Dump types are described in Section 6.2.1:

```
Enter the corresponding number to select one of the following dump types:
```

- 1: SDUMP - a normal SHORT dump
 - 2: RSDUMP - a normal SHORT dump with recovered DIAG\$ file
 - 3: MICRO - a MICRO dump
 - 4: MICRO/RSS - a MICRO dump registered with RSS
 - 5: RTSDUMP - a CS 1100 dump
- ```
[n]/<EXIT>
```

## Query 2

This query lets you select the type of input media the dump to be processed resides on:

Enter the corresponding number to select the number and type of dump files to be processed:

- 1: process all existing cycles of SYSS\*PANIC\$DUMP\$\$, and then query terminal for reel numbers of tape dumps to process.
- 2: query for and process several tape dumps.
- 3: process only latest cycle of SYSS\*PANIC\$DUMP\$.
- 4: query for and process only one tape dump.
- 5: process all existing cycles of SYSS\*PANIC\$DUMP\$.
- 6: query for and process dump in one mass storage file.
- 7: process only the oldest cycle of SYSS\*PANIC\$DUMP\$.

[n]/<EXIT>

## Query 3

This query lets you indicate whether you want the dump analysis runstream added (A), started (S, the default), or held (H):

Add, Start, or Hold the runstream? A/<S>/H/EXIT

The possible responses have the following effects:

- |           |                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| A (add)   | The generated runstream is dynamically added to the demand session and the dump is processed in demand mode with output directed to the demand terminal. |
| S (start) | The generated runstream is submitted as a batch run. This is the default option.                                                                         |
| H (hold)  | The generated runstream is placed in an element for future submission. The runstream can be edited in this element prior to submission.                  |

## Query 4

This query appears only if you have chosen to start (S) or hold (H) the generated runstream:

Default Run-Id = *rid*, Project-Id = *pid*, Priority = J  
Enter Run-Id, Project-Id, and Run Priority (A-Z) separated by commas:  
*rid,pid,pri*/EXIT

## Performing and Processing System Dumps

---

The run-ID project-ID, and run priority defaults from the current demand session are displayed so you can change them. If you do not specify any parameters, the defaults are used. Enter each parameter in its corresponding positional field, separated from the next field by a comma. Some examples of responses to the query and the effects of the responses are as follows.

| Response        | Run-ID  | Project-ID | Run Priority |
|-----------------|---------|------------|--------------|
| DMP008,6K9100,F | DMP008  | 6K9100     | F            |
| DMP008          | DMP008  | default    | default      |
| ,6K9100,D       | default | 6K9100     | D            |
| „A              | default | default    | A            |
| EXSTOP„C        | EXSTOP  | default    | C            |
| TX1040,W13A35   | TX1040  | W13A35     | default      |

### Query 5

This query asks whether you want to enter additional stream generation statements (SGS), which are described in Section 6.2.3, to specify certain processing options for runstream generation:

```
Additional SGSs? Y/<N>/EXIT
```

If you enter Y, you see the following query:

```
Enter SGSs. XMIT blank line when done.
```

Enter one SGS per input line. Transmit a blank line when you have finished.

Additional queries appear depending on the type of input media the dump resides on. These queries are the same as those seen on the system console when you start a batch run and are explained in Section 6.2.1 under "Additional Operator Queries."

### Query 6

If the dump to be processed resides on mass storage, you receive the following query:

```
Do you wish to save dumpfile on tape? Y/<N>
```

Answer Y to instruct IADUMP to generate ECL, which copies the dumpfile to tape prior to processing. You are then asked to provide the tape reel-ID and equipment type.



### 6.2.3. Optional Runstream SGSs

Optional runstream SGSs can be entered during demand mode runstream generation in response to the query "Additional SGSs?" Enter the SGSs in any order.

#### Using a DMPLIB Library from Another File

In certain cases (such as testing a new DMPLIB), it can be desirable to use a DMPLIB library that resides in a file other than EXEC\*DUMPLIBS. To specify a different DMPLIB, use the following SGS:

```
USE DUMP LIBRARY FROM FILE file
```

where *file* is the name of a file containing an alternate DMPLIB library. Enter the file name without the terminating dot. This SGS generates the \*RESTORE directive with the specified file name. When processing the dump, the dump analysis run uses the functions found in this file.

#### Calling Functions Individually in Demand Mode

If you specified that the runstream must be added (see query 3 under Section 6.2.2), you can use the following SGS to call individual functions:

```
CALL FUNCTIONS INDIVIDUALLY IN DEMAND MODE
```

This SGS assigns and loads the dump file (or files), calls the DAP processor, and restores the DMPLIB library. The initial DMPLIB function EXECBDRS is then called. EXECBDRS sets up the environment for subsequent DMPLIB function calls. The demand session is ready for the user to call other DMPLIB functions.

### 6.2.4. DMPLIBs

This subsection describes how to maintain alternate levels of DMPLIBs. It also identifies a procedure for submitting raw dump files with WinDump or RSS.

If your site wants to maintain more than one level of DMPLIB on a system (to process dumps with different Exec levels), you can do so by storing other DMPLIB levels in alternate files. While processing the dump, DMPLIB restores a library file that corresponds to the Exec level it finds in the dump. (The runstream must be generated by an ST keyin or an IADUMP.)

The default file used to process dumps is EXEC\*DUMPLIBS. If the Exec level that DMPLIB detects in the dump matches the installed DMPLIB level, or if the level is not found in the table, then the dump is processed using EXEC\*DUMPLIBS. If you need to process a dump taken from an earlier level of EXEC, you must catalog an alternate DMPLIB file and copy the DMPLIB level required for that Exec level into that alternate file. See the table below to determine the file names to catalog for alternate DMPLIBs.

After the dump is processed, you can @START the SDUMP runstream from the EXEC\*DUMPLIBS file. SDUMP determines the Exec level from the dump and processes the dump using the file corresponding file from the table below. The rows in the table show the installed DMPLIB level, and the columns show the Dump Exec level to process. Use the appropriate row and column to find the corresponding DMPLIB file qualifier to use for Alternate Exec level processing. For example, if DMPLIB level 9Ry is installed and SDUMP detects the dump originated on a level 48Ry Exec, it uses file EXEC48\*DUMPLIBS (DMPLIB 8Ry) for the remainder of the processing. Note that all DMPLIB files end with the filename DUMPLIBS.

|              | Dump Exec Level |       |         |
|--------------|-----------------|-------|---------|
| DMPLIB Level |                 | 49Ry  | 48Ry    |
|              | 9Ry             | EXEC* | EXEC48* |
|              | 8Ry             |       | EXEC*   |

If you do not know which level of DMPLIB is currently installed, refer to the updated source for function/element SELDUMPLIBS in the REL or NEWSO file. It shows the DMPLIB level currently installed in file EXEC\*DUMPLIBS (the file referenced by ST keyins and used by IADUMP) and the file names it uses when switching to different levels of DMPLIB.

### Submitting Raw Dump Files

You can submit raw dump files using either WinDump or RSS. WinDump is the recommended method.

#### WinDump

WinDump is a Windows application that can submit dump files electronically to the Support Center Remote Server using FTP. Dumps sent directly to the Support Center Remote Server with WinDump can generally be processed more efficiently, because of the fewer steps required to prepare a dump for analysis.

To use WinDump, you must send dumps from an OS 2200 host that allows Telnet access and has either TAS or cpFTP installed.

See Section 6.3 for information about downloading and installing WinDump.

#### RSS

The traditional method for submitting dumps that accompany Contacts and User Communication Forms (UCFs) is to use the following statement to call an SSG skeleton, which submits raw dump files with RSS:

```
@ADD EXEC*DUMPLIBS.RSS-DUMP
```

The SSG skeleton does the following:

- Queries you for information pertaining to the dump.
- Builds and executes a runstream that facilitates the transfer of the dump files.

**Note:** *The building of the runstream with this skeleton and the subsequent execution of this runstream in DEMAND or BATCH modes requires the appropriate privileges based on the security level of the system.*

## **6.3. WinDump Interface—Downloading and Installing WinDump**

WinDump is a full-service dump-handling and analysis program, a Windows application that allows you to work with dumps on a PC workstation through a GUI interface. It is an alternative to both RSS and the direct use of DMPLIBs. Using WinDump, you can process a dump, manage the dump files, and transmit dumps to the Unisys Support Center Remote Server using FTP. Analysts also use WinDump to run DMPLIBs functions through the GUI interface.

Dumps sent directly to Unisys by WinDump generally arrive faster and can be processed more quickly than those sent by RSS. Fewer steps are needed to prepare the dumps for analysis because WinDump handles the formatting of dumps automatically, correctly, and reliably. For this reason resubmission is rarely necessary—an additional savings.

To allow the use of WinDump, the OS 2200 host from which you wish to send dumps must allow Telnet access and have either TAS or cpFTP installed.

WinDump is supplied with both context-sensitive online help and two printable documents: an installation guide and a user's guide. This subsection describes two methods for obtaining WinDump and accessing its printable help.

### **6.3.1. Downloading WinDump from the Support Center Remote Server**

You can download WinDump from the Support Center Remote Server by supplying the following URL to your browser:

`https://www.support.unisys.com/pub/2200/utilities/windump/`

This location has the latest downloadable version of WinDump in a self-extracting .EXE file. The installation instructions are in a file called INSTALL.DOC.

After you have installed WinDump, you can find the *WinDump User's Guide* in the file WINDUMP.DOC in the installed directory. Follow the instructions in Section 4.6 of the *User's Guide*, "Transferring Dump Files," to prepare dumps and transfer them to the Unisys Support Center.

### 6.3.2. Downloading WinDump from DMPLIBs

WinDump is also available as part of DMPLIBs. After you install DMPLIBs, look for the elements *Wd/level/EXE* and *WDINSTALL/DOC* in the file *EXEC\*DUMPLIBS*, where *level* is the current level of WinDump (for example, 36 for 3.6, 37 for 3.7, and so on). Element *Wd/level/EXE* contains a self-extracting .EXE, which when executed creates the installation files. *WDINSTALL/DOC* is a Microsoft Word document that describes how to install WinDump from *Wd/level/EXE*.

To obtain and install WinDump from *EXEC\*DUMPLIBS*, you must download *Wd/level/EXE* and *WDINSTALL/DOC* to the workstation on which you wish to install it. Open an MS-DOS window on the workstation as follows:

1. Click **Start**.
2. Click **Run**.
3. Type **command** in the **Run** dialog text box.
4. Click **OK**.

Enter the following commands:

```
>ftp.exe
```

```
>open [IP-address or host-name]
```

(where *IP-address or host-name* is the IP address or symbolic Internet host name for FTP of the OS 2200 system from which you are downloading)

```
>[userid]
```

```
>[password]
```

```
>[project]
```

```
>type binary
```

```
>quote site ftype omn
```

(Not needed if using TAS; required if using cpFTP)

```
>get exec*dumplib.wd[level]/exe c:\temp\wd[level].exe
```

```
>get exec*dumplib.wdinstall/doc c:\temp\wdinstall.doc
```

```
>quit
```

```
>exit
```

After downloading the two files, open the file *C:\TEMP\WDINSTALL.DOC* in Microsoft Word and follow the instructions for installing WinDump.

When you have finished the installation, you can find the *WinDump User's Guide* in the file WINDUMP.DOC in the installed directory. Follow the instructions in the *WinDump User's Guide* to prepare dumps and transfer them to the Unisys Support Center.

## 6.4. RSS Interface

This subsection explains how to register, process, and submit dumps through the Remote System Support (RSS) interface.

### 6.4.1. Registering Dumps

Mass storage dump files can be registered with RSS for future analysis by using the RSS-REG runstream. RSS-REG takes all existing mass storage dump files and copies their contents to RSS files. After the dump data is given to RSS, the mass storage dump files (all SYS\$\*PANIC\$DUMP\$\$ F-cycles) are deleted.

The ST keyin that starts RSS-REG from the system console is

```
ST EXEC*DUMPLIBS.RSS-REG,,,acct-id/user-id
```

### 6.4.2. Processing Dumps with RSS

Every mass storage dump registered by RSS-REG can be analyzed by using the RSS PROCESS task. System dump files are registered under the name EXEC. The following three process names are associated with each dump file:

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MICRO | This process name creates a brief synopsis of the problem, containing only pertinent information. This summary file is registered with RSS and can be transmitted to the Unisys Support Center.                                                                                                                                                                                                                                                                                                             |
| SHORT | This process name creates and prints a complete listing of all Exec internal structures. The SHORT process name can be used when the support center requests information that you cannot transmit electronically in a MICRO dump.                                                                                                                                                                                                                                                                           |
| TAPE  | This process name creates a tape containing the unprocessed mass storage dump data. The tape file is in @COPY,G format. The TAPE process name can be used on the rare occasions when a SHORT dump listing is insufficient to diagnose the problem and all dump data must be examined at the support center. Note that the tape created here is different from a tape dump the Exec creates directly. Therefore, it cannot be used as input to the batch dump analysis runstream described in Section 6.2.1. |

### 6.4.3. Submitting Raw Dump Files with RSS

Before submitting dumps that accompany contacts and UCFs, enter the following:

```
@ADD EXEC*DUMPLIBS.RSS-DUMP
```

This statement calls an SSG skeleton which queries you for information pertaining to the dump and builds and executes a runstream that facilitates the transfer of the dump files.

## 6.5. Dumping XPC Storage

Selected nonvolatile storage (NVS) segments of the XPC can be dumped to tape or mass storage for analysis of specific XPC internal tables. Refer to the XPCDUMP keyin description in Section 11 for keyin format and other information. The dump keyin can be initiated anytime after the MODIFY CONFIG message.

Processing and analysis of XPC dumps cannot be performed with the same utilities available for system dumps. XPC dumps are typically used by Unisys engineering personnel as troubleshooting aids.

## Section 7

# Integrated Recovery and System Operation

This section provides operations information on Audit Control and Step Control, which are used in Integrated Recovery.

### 7.1. Audit Control and the Audit Trail

The audit trail is a log of recoverable database and message processing information for an application group. The number of the audit trail is also the number of the application group. You control an audit trail with the Audit Trail (AT) keyin.

There are five audit trail states:

- Open (OP)  
An open audit trail has audit data files and an Audit Control Interface (ACI) file assigned. Audit records are received and written to the data files while the trail is open.
- Closed (CL)  
A closed audit trail does not have audit data files assigned. When an audit request is received, a trail can be automatically opened, or you can be prompted to open a trail. The initial state of an audit trail is closed.
- Down (DN)  
An audit trail in the down state does not have main storage, audit data files, or an ACI file assigned. Audit requests are not allowed to a trail in a down state.
- Deferred (DF)  
An audit trail is in the deferred state when the audit recovery file (ARF) is not assigned to the Exec. This can occur if ARF recovery of the trail resulted in an error or has not occurred (recovery has been deferred). The system displays a message notifying you of the deferred state. Use the AT *n* IN, AT *n* UP, and AT *n* OP to recover.

- In transition

Normal auditing is temporarily suspended or is in transition when one of the following events occurs:

- You use certain operator keyins (for instance, IN, OP, or LC).
- There is an I/O error (for example, autoswap processing).
- The audit trail is being initialized and recovered during system recovery.

The initial state of an audit trail is closed (CL). During a mass storage recovery boot, each audit trail recovers its state from the previous session and completes any activity in process during the previous session.

The Master Configuration Table (MCT) copy of an audit trail database can be used to reinitialize the database for an audit trail. This can occur when:

- You perform a tape boot and there is no database to recover from.
- You perform a tape boot and the database is created, but a comparison of the database with MCT data indicates that the configuration parameters in the MCT have been modified since the database was created.

You can defer recovery of one or more audit trails by using the AT *n* NR keyin. This keyin must be entered before the MODIFY CONFIG (select jump key 1 boot) message is answered.

Table 7-1 defines and describes the AT keyin functions. Refer to Section 11 for complete information on the AT keyin.

**Table 7-1. AT Keyin Functions**

| Function | Definition      | Description                                                                                                                                                                                                                               |
|----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CL       | Close           | Closes the audit trail for the specified application group and frees its audit data files. This keyin can be restricted by system generation.                                                                                             |
| DF       | Deferred        | Puts an audit trail in the deferred (DF) state                                                                                                                                                                                            |
| DN       | Down            | Disables an inactive (closed) audit trail if not restricted by system generation                                                                                                                                                          |
| FS       | Facility status | Displays the status of audit trails                                                                                                                                                                                                       |
| IN       | Undefers        | Initiates recovery of a trail from a deferred (DF) state                                                                                                                                                                                  |
| LC       | Leg change      | Lets you change the mass storage file (leg) assignment parameters for the specified leg. The change takes effect when the leg is swapped to the next F-cycle, if the trail is already open, or if the trail is closed, when it is opened. |
| LD       | Leg display     | Displays media summary characteristics of each leg of audit trails                                                                                                                                                                        |



**Table 7-1. AT Keyin Functions**

| Function | Definition      | Description                                                                                                                                                                                                   |
|----------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MV       | Move            | Allows a spare audit trail to be moved from its current tape drive to another drive                                                                                                                           |
| NR       | No recovery     | Permitted only during system recovery before the MODIFY CONFIG message is answered. The specified audit trail is placed in a deferred (DF) state. This keyin is not permitted for the system log audit trail. |
| OP       | Open            | Opens the specified audit trail and makes it available for auditing                                                                                                                                           |
| RM       | Remove premount | Lets you remove a premount unit within system generation constraints                                                                                                                                          |
| SP       | Spare premount  | Lets you add a premount unit (spare) within system generation constraints                                                                                                                                     |
| SW       | Swap            | Closes the current tape reel or mass storage leg and then opens the next reel or F-cycle                                                                                                                      |
| UP       | Up              | Changes the state of an audit trail from down (DN) to closed (CL), thus permitting auditing to resume                                                                                                         |

### 7.1.1. Audit Trails Configured as Tape

All audit tapes have 8-bit ASCII labels, and you can assign one or more spares to the audit trail. (A spare is a premounted reel, prepared for a future need to change reels. When a swap is made to a premount, the system prompts you to mount the next premount.)

If you premount a reel with a specified expiration date and then fail to use it, you must initialize the expiration date before you can use the reel again.

There are special procedures to handle audit tapes, depending on the tape labeling system configuration options that follow:

|        |                                                          |
|--------|----------------------------------------------------------|
| TLAUTO | set for automatic tape labeling                          |
| TLSIMP | set to control degree of tape labeling                   |
| TLSUB  | set to allow reel number substitution during label check |
| TPOWN  | set to enable account number checking during label check |

Audit Control assigns blank tapes and controls tape labeling with an Executive request. Whether you mount a labeled tape or an unlabeled tape depends on how your system is configured. For systems with automatic tape labeling set false, you must mount a labeled tape. For systems with automatic tape labeling set true, unlabeled (as well as labeled) tapes are acceptable. After you mount the tape, each reel of an audit trail has the file name appropriate for the audit complex (Step Control, TPM, COD, log).

See the *Exec System Software Administration Reference Manual* (7831 0323) for details on the file names. Audit Control tries to self-swap when the self-swap option is on and no premount (spare) is available.

You have a range of options for controlling the audit trail. You can choose to specify audit devices by name, by type, by control unit, or with a default type. Audit Control first tries to use the type you specified (the preferred assignment). For example, if a choice of tapes to premount exists at swap, the premount is chosen on either the same control unit (if both swap-from and swap-to are specified by control unit) or on the same logical channel.

When you choose a subsystem, Audit Control maintains this choice over successive swaps (if premounts are also available on this subsystem).

You can use another approach if you want to maximize the use of tape units for other (nonaudit) uses. In this case, you control the swapping by anticipating end-of-reel and using the swap-specific keyin (such as AT *n* SW). When you swap to a new device, the closing device is freed for other uses, unless the swap is to a spare. In this case, the system prompts you to mount a blank labeled tape, and the closing device is used as a spare.

On tape, if the end-of-reel is found or if there is an unrecoverable I/O error, writing automatically continues on a new reel (probably on a different tape unit). If no spare is available and self-swap is not enabled, a new unit is assigned. This new unit has a preferred assignment according to the original assignment, if the terminating reel was assigned by control unit name or tape type. Otherwise, the audit trail's default tape type is used.

### 7.1.2. Audit Trails Configured as Mass Storage

On disk, if there is an end-of-file error or an unrecoverable I/O error, an autoswap is triggered. This allows auditing to continue on the next (+1) F-cycle of the mass storage audit leg. The current F-cycle of the file is freed, and the (+1) F-cycle is assigned and cataloged.

To catalog the (+1) F-cycle and continue auditing, you must allow the oldest F-cycle to be deleted after the maximum number of allowable F-cycles for a mass storage audit file has been reached. This might occur under the following circumstances:

- The audit file (leg) is opened.
- You use the AT *n* SW *Li* keyin.
- An I/O error occurs, causing an autoswap.
- An end-of-file occurs, causing an autoswap.

For the system log audit trail, F-cycle wraparound automatically occurs. The system notifies you by displaying the following message:

```
AT n Li *F-CYCLE WRAP-AROUND* MAX: xx
```

For all other audit trails, the system displays this message:

```
a- AT n Li *F-CYCLE WRAP-AROUND* MAX: xx CONTINUE YN?
```

where:

*a*

is the message number.

*n*

is the trail-ID.

*Li*

is the mass storage leg-ID (L1 or L2).

*xx*

is the maximum number of relative F-cycles allowed for the file.

Respond Y to allow F-cycle wraparound. This effectively destroys the oldest relative F-cycle data for the specified leg of the trail. Answer N to prevent an F-cycle wraparound.

These situations can arise:

- If the trail is being opened automatically when the wraparound occurs, the trail is automatically brought down.
- If the trail is being opened by an operator keyin when the wraparound occurs and the trail down (DN) keyin is disallowed, the system prompts you to try again, disallowing a down response.
- If the trail is being opened by operator keyin when the wraparound occurs and the trail down keyin is allowed, the trail is brought down.
- If you did a swap keyin to cause the wraparound, the swap does not occur; the trail remains open, using the current F-cycle.
- If an I/O error caused the wraparound, the trail is automatically placed in a down state. Any pending audit I/O, for all legs of the trail, and any queued audit requests are terminated. A trail unavailable status is returned to all requestors.
- If there is a wraparound when an end-of-file is being processed, the result is the same as for an I/O error.

The current time is recorded in each block written to an audit file.

## 7.2. Step Control Recovery and Initialization Options

This subsection describes the Exec Step Control recovery and initialization options. For information on other Integrated Recovery concepts, see the *Integrated Recovery Conceptual Overview* (7830 8186). For more information on the Integrated Recovery Utility, see the *Integrated Recovery Utility Operations Guide* (7830 8194). For Exec recovery status and error messages, see the *System Console Messages Reference Manual* (7833 3275).

Step Control tracks the progress of each message and transaction through an Integrated Recovery application group. Step Control uses queues and data structures, referred to as queued items, to identify each message and its associated database changes. When a failure occurs, Integrated Recovery uses the contents of the queue items to maintain the integrity of the database by ensuring that only changes requested by successfully completed transactions are applied.

**Note:** In the message definitions throughout this subsection, the field *n* refers to the application group number.

### 7.2.1. Introduction to the Options

The following are the Exec Step Control recovery and initialization options:

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PANIC   | This option uses information from the system dump (also referred to as panic dump) and the audit trail to recover the application group's queue items. The system dump is created during a disk recovery (or reboot). PANIC recovery ensures that the queue item state changes existing at system failure time match the queue item state changes that were successfully written to the audit trail.                                                                                                    |
| REBUILD | This option is the same as PANIC, except that it uses information from the periodic savefile and the audit trail rather than the system dump to recover the application group's queue items.                                                                                                                                                                                                                                                                                                            |
| SUTIL   | This option is the same as REBUILD, except that the IRU (rather than the Exec) reads queue items from the audit trail. SUTIL recovery closes the audit trail if the audit trail was open. To recover the application group's queue items, SUTIL uses information from the periodic savefile plus information passed in from the IRU short recovery command. The IRU short recovery command uses information from the periodic save audit records and queue item audits to provide information to SUTIL. |
| INIT    | This option closes the audit trail if the audit trail was open. It initializes the application group's queue item storage in memory and various files, including the periodic savefile. During INIT, the system does not attempt message and database recovery because both types of recovery use the recovered queue items, which are not available during INIT processing.                                                                                                                            |

|       |                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DOWN  | This option releases queue item storage in memory, closes the audit trail if the audit trail was open, and leaves the application group in a down state. You can perform application group recovery at a later time (SUTIL and INIT options are available later). |
| DEFER | This option is the same as DOWN except that it frees the audit recovery file (ARF) and leaves the application group in a deferred state.                                                                                                                          |

## Options Available After a System Failure

After a system failure, the system attempts to recover the application groups and can prompt you with console messages to select the initialization or recovery option you want. Two different recovery messages can appear. Which one you encounter depends on the highest application group number configured on your system and the setting of jump key 7.

Recovery Message 1 is displayed if the highest application group number configured on your system is greater than nine and jump key 7 is set. Recovery Message 2 is displayed in all other cases. Refer to the appropriate section below.

### Recovery Message 1

When the highest application group number configured on your system is greater than nine and jump key 7 is set, a multiline read-and-reply message is displayed to let you select the recovery option for each application group. The options for an application group and the corresponding console message depend on the type of boot, as follows:

- Initial boot

```
Specify Recovery Action for the following application groups:
Non-recoverable APPs specify (INIT/DOWN)
Non-recoverable APPs are: application group numbers
Recoverable APPs specify (INIT/DOWN/DEFER)
Recoverable APPs are: application group numbers
0-Specify Recovery Action with "AP x RA" keyin, then answer DONE.
```

- Tape recovery boot

```
Specify Recovery Action for the following application groups:
Non-recoverable APPs specify (INIT/DOWN)
Non-recoverable APPs are: application group numbers
Recoverable APPs specify (REBUILD/SUTIL/INIT/DOWN/DEFER)
Recoverable APPs are: application group numbers
0-Specify Recovery Action with "AP x RA" keyin, then answer DONE.
```

- Disk recovery boot

```
Specify Recovery Action for the following application groups:
Non-recoverable APPs specify (INIT/DOWN)
Non-recoverable APPs are: application group numbers
Recoverable APPs specify (PANIC/REBUILD/SUTIL/INIT/DOWN/DEFER)
Recoverable APPs are: application group numbers
0-Specify Recovery Action with "AP x RA" keyin, then answer DONE.
```

Use the AP x RA keyin to respond to the message (refer to the AP keyin information in Section 11 for a complete description of the AP x RA keyin). You can respond using one or more of the application group numbers or a range of application group numbers. Recovery of the specified application group or groups begins as soon as the keyin is entered.

After responding to the outstanding read-and-reply message and specifying the action for each application group, a final message is displayed to which you must respond with "done." If you answer with anything other than "done," the message is repeated.

If you answer "done" and there are application groups for which you have not yet specified a recovery action, the multiline read-and-reply message is repeated and displays the remaining application groups. You must specify a recovery action for each application group.

**Note:** The AP x RA keyin is valid only during system initialization (I-boot or recovery boot) while the read-and-reply message is outstanding. If the AP x RA keyin is entered at any other time, the system responds with an error message.

### Recovery Message 2

After a system failure, the system attempts to recover the application groups automatically. If automatic recovery fails, or if jump key 7 is set, the system prompts you to enter one of the initialization or recovery options. The options for a recoverable application group and the corresponding console message depend on the type of boot as follows:

- Initial boot

```
APP n INITIALIZE, DOWN, OR DEFER? ENTER: (INIT/DOWN/DEFER)
```

Specify INIT, DOWN, or DEFER.

If jump key 7 is not set, the system assumes INIT. If valid data exists on the periodic savefile, the system changes INIT to SUTIL.

- Tape recovery boot

APP *n* RECOVERY TYPE? ENTER: (REBUILD/SUTIL/INIT/DOWN/DEFER)

Specify REBUILD, SUTIL, INIT, DOWN, or DEFER. REBUILD and SUTIL recover the application group as it was previously configured. If you configured a new application group, enter INIT for the tape recovery boot. If an application group is not used immediately after the tape reboot, enter DOWN or DEFER.

If jump key 7 is not set, the system automatically tries REBUILD; if REBUILD fails, the system automatically performs the Exec portion of SUTIL recovery (SUTIL setup) and prompts you to start IRU short recovery.

- Disk recovery boot

APP *n* RECOVERY TYPE? ENTER: (PANIC/REBUILD/SUTIL/INIT/DOWN/DEFER)

Specify any of the six options.

If jump key 7 is not set, the system automatically tries PANIC recovery; if PANIC does not succeed, the system automatically tries REBUILD; if REBUILD does not succeed, the system automatically performs SUTIL setup.

For a nonrecoverable application group, the following message appears:

APP *n* NON-RECOVERABLE APPLICATION - ANSWER INIT OR DOWN

Answer INIT or DOWN. A nonrecoverable application group is an application group that does not have recoverable messages and database files. Specifically, the application group configuration does not specify a periodic savefile, Universal Data System (UDS) retention file or files, or rollback page file; only one message retention file (MRF) for nonrecoverable messages is configured.

### Options Available After an Application Group Failure

If a single application group fails, bring the application group up with an AP *n* UP keyin. Select a recovery or initialization option by answering the prompt that is produced, or include the recovery option on the AP *n* UP keyin (AP *n* UP SUTIL or AP *n* UP INIT, for example).

The following options are available:

- Recoverable application group - SUTIL, INIT, DOWN, or DEFER
- Nonrecoverable application group - INIT or DOWN

If the Message Control Bank (MCB) or UDS Control component of an application group aborts, no console message appears prompting you for a recovery or initialization option. Instead, just start IRU short recovery to perform recovery and to bring up MCB and UDS Control. The Step Control queues and audit trail remain intact after an MCB or UDS Control abort.

### 7.2.2. Short Recovery Options

PANIC, REBUILD, and SUTIL are Exec short recovery options. The following parts of recovery occur with these options:

- Queue item and audit trail recovery (method varies according to the recovery option)
- Message and database recovery
- Application group restart

The Exec short recovery procedures work together with the Integrated Recovery Utility short recovery procedure to recover queue items, messages, and databases, and to bring up an application group. After an Exec short recovery procedure is completed, Integrated Recovery Utility short recovery must generally be started to complete recovery processing. Integrated Recovery Utility short recovery cannot be necessary if an application group is configured without MCB or UDS Control and the Step Control application group is configured with the "Bypass IRU Short Recovery" SGS.

#### **Notes:**

- *If the "Bypass IRU Short Recovery" SGS is not configured, even application groups that are not MCB or UDS require Integrated Recovery Utility short recovery.*
- *If the "Bypass IRU Short Recovery" SGS is configured, you cannot have UDS or MCB configured and audit-only recovery processing is turned off.*

### PANIC Option

Figure 7-1 illustrates PANIC and Integrated Recovery Utility short recovery for a recoverable application group that uses TIP, Universal Database Control, and MCB. The following steps describe the combined PANIC and Integrated Recovery Utility short recovery:

1. Exec Audit Control uses the ARF to recover the audit trail. Step Control recovers queue items from the system dump and the audit trail, updates the information in the Step Control queues, and initiates TIP database recovery. Messages appear at the console, indicating progress of the recovery and the status of the audit trail; for example:

```
APP 3 SAVEFILE RECOVERY IS FROM LEG 1
AT 3 L1 TAPJ2 REEL PR5312 1202:34 CST
AT 3 L1 BLK 29 TBSN 27
```

When PANIC successfully recovers the Step Control queue items, the following message appears:

```
APP n PANIC COMPLETE - QUEUE ITEMS RECOVERED FROM PANIC DUMP FILE
```



2. TIP File Control recovers the TIP database files using the rollback page file and the Step Control queues. The following messages appear:

```
APP n {Host host-id} Rollback recovery started
APP n {Host host-id} Rollback recovery completed
```

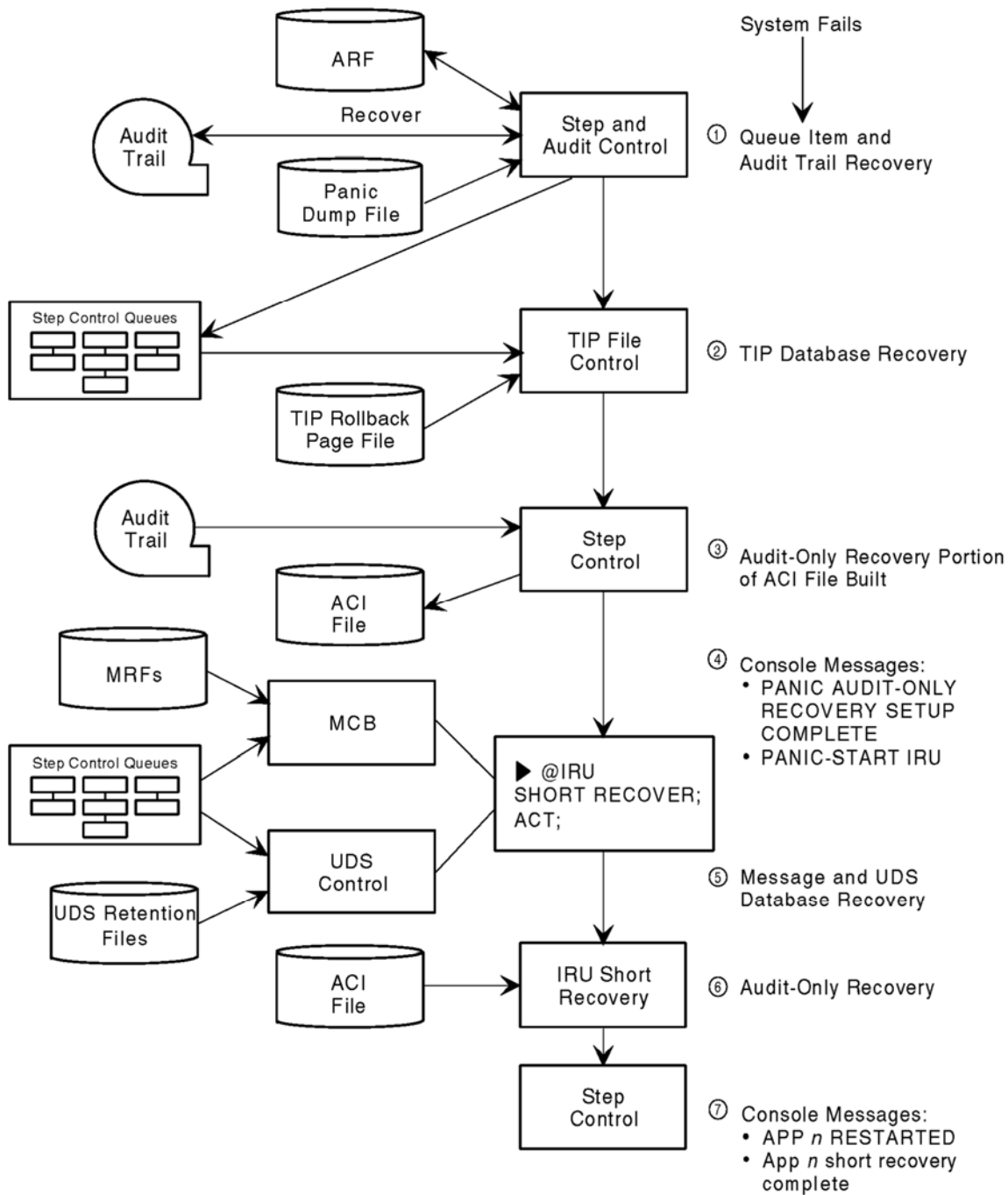
3. If any audit-only recoverable steps exist, Step Control uses the audit trail to build the audit-only recovery portion of the Audit Control interface (ACI) file. In doing so, Step Control locates the audit records that are required for audit-only recovery and copies them to the ACI file.
4. The following messages appear at the console:

```
APP n PANIC AUDIT-ONLY RECOVERY SETUP COMPLETE
APP n PANIC - START IRU
```

Start IRU short recovery at this time.

Messages appear at the console indicating application group recovery was successful.

**Note:** If the "Bypass IRU Short Recovery" SGS is configured, the START IRU message does not appear, Integrated Recovery Utility short recovery is bypassed (steps 5 and 6 are skipped), and the Exec, rather than the Integrated Recovery Utility, performs step 7.



**Figure 7-1. Short Recovery with the PANIC Option**

5. Integrated Recovery Utility short recovery queries Step Control about the application group state. For short recovery, IRU calls MCB to recover the MCB messages and Universal Database Control to recover the UDS database. This is accomplished by using the MRFs, Step Control queues, and UDS retention files.
6. If any audit-only recoverable steps existed in step 3, Step Control indicates to Integrated Recovery Utility that the audit-only recovery portion of IRU short recovery is required. Integrated Recovery Utility uses the ACI file to recover the database updates for the audit-only recoverable steps.
7. Integrated Recovery Utility requests Step Control to restart the application group. The following messages appear at the console:

```
APP n RESTARTED
run-id: App n short recovery complete
```

where *run-id* is the run-ID of the IRU run.

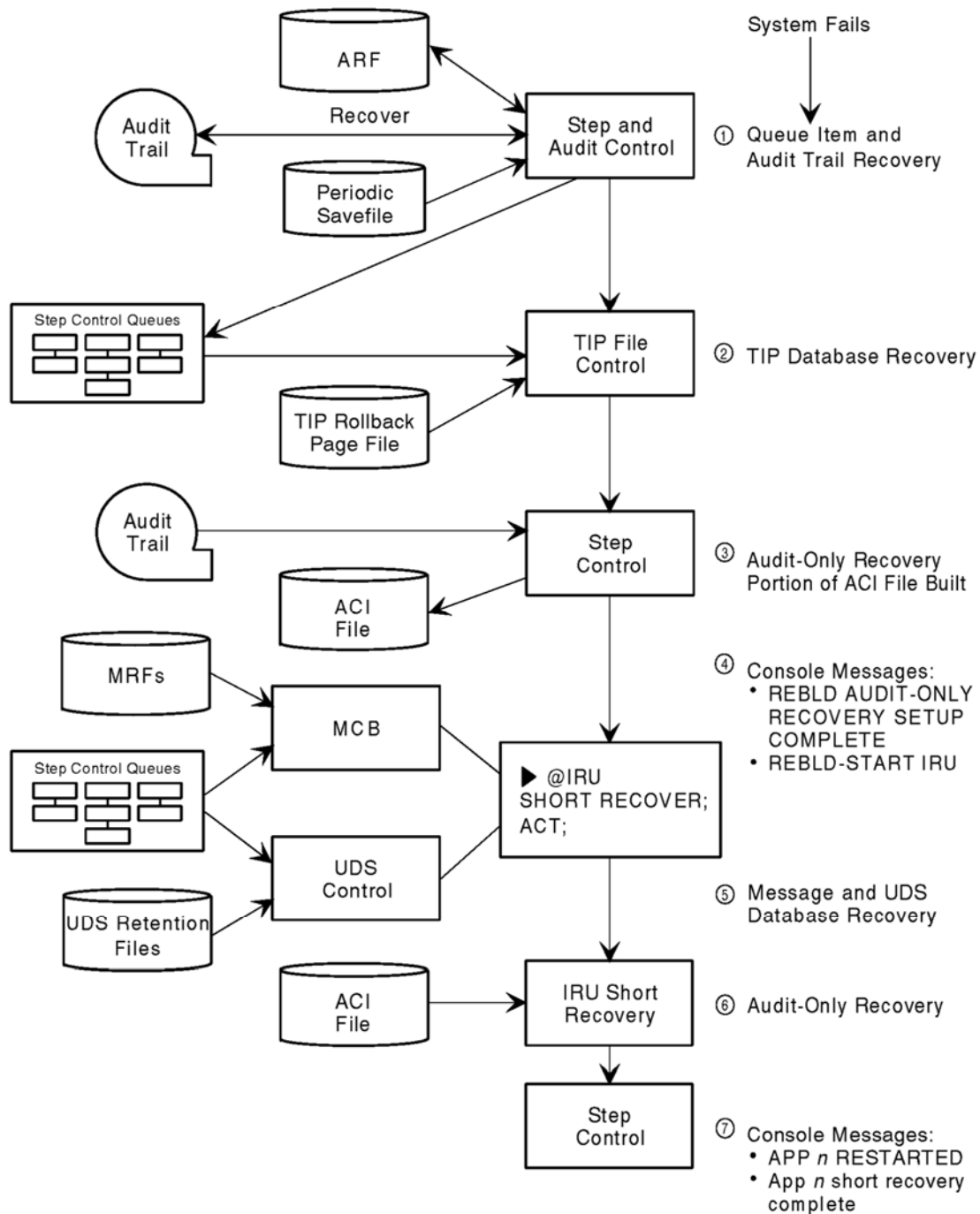
## REBUILD Option

Figure 7-2 illustrates REBUILD recovery for a recoverable application group that uses TIP, Universal Database Control, and MCB. The following steps describe REBUILD recovery:

1. Exec Audit Control uses the ARF to recover the audit trail. Step Control recovers queue items from the periodic savefile and the audit trail, updates the information in the Step Control queues, and initiates TIP database recovery. Messages appear at the console indicating progress of the recovery and the status of the audit trail; for example:
 

```
APP 3 SAVEFILE RECOVERY IS FROM LEG 1
APP 3 REBLD PERIODIC SAVE FILE VERIFIED
AT 3 L1 TAPJ2 REEL 536E 1030:31 CST
AT 3 L1 BLK 53 TBSN 50
APP 3 REBLD QUEUE ITEMS REBUILT FROM SAVEFILE AND AUDIT
```
2. TIP File Control recovers the TIP database files using the rollback page file and the Step Control queues. The following messages appear at the console:
 

```
APP n {Host host-id} Rollback recovery started
APP n {Host host-id} Rollback recovery completed
```
3. If any audit-only recoverable steps exist, Step Control uses the audit trail to build the audit-only recovery portion of the Audit Control interface (ACI) file. In doing so, Step Control locates the audit records that are required for audit-only recovery and copies them to the ACI file.



**Figure 7-2. Short Recovery with the REBUILD Option**

4. The following messages appear at the console:

```
APP n REBLD AUDIT-ONLY RECOVERY SETUP COMPLETE
APP n REBLD - START IRU
```

Start IRU short recovery at this time.

**Note:** If the "Bypass IRU Short Recovery" SGS is configured, the *START IRU* message does not appear, IRU short recovery is bypassed, and the Exec, rather than the IRU, performs step 7.

5. IRU short recovery queries Step Control about the application group state. For short recovery, IRU calls MCB to recover the MCB messages and UDS Control to recover the UDS database. This is accomplished by using the MRFs, Step Control queues, and UDS retention files.

Messages appear at the console indicating application group recovery was successful.

6. If any audit-only recoverable steps existed in step 3, Step Control indicates to IRU that the audit-only recovery portion of IRU short recovery is required. IRU uses the ACI file to recover the database updates for the audit-only recoverable steps.
7. IRU requests Step Control to restart the application group. The following message appears at the console:

```
APP n RESTARTED
run-id: App n short recovery complete
```

where *run-id* is the run-ID of the IRU run.

## SUTIL Option

Figure 7-3 illustrates SUTIL recovery for a recoverable application group that uses TIP, UDS Control, and MCB. The following procedure describes SUTIL combined with short IRU recovery:

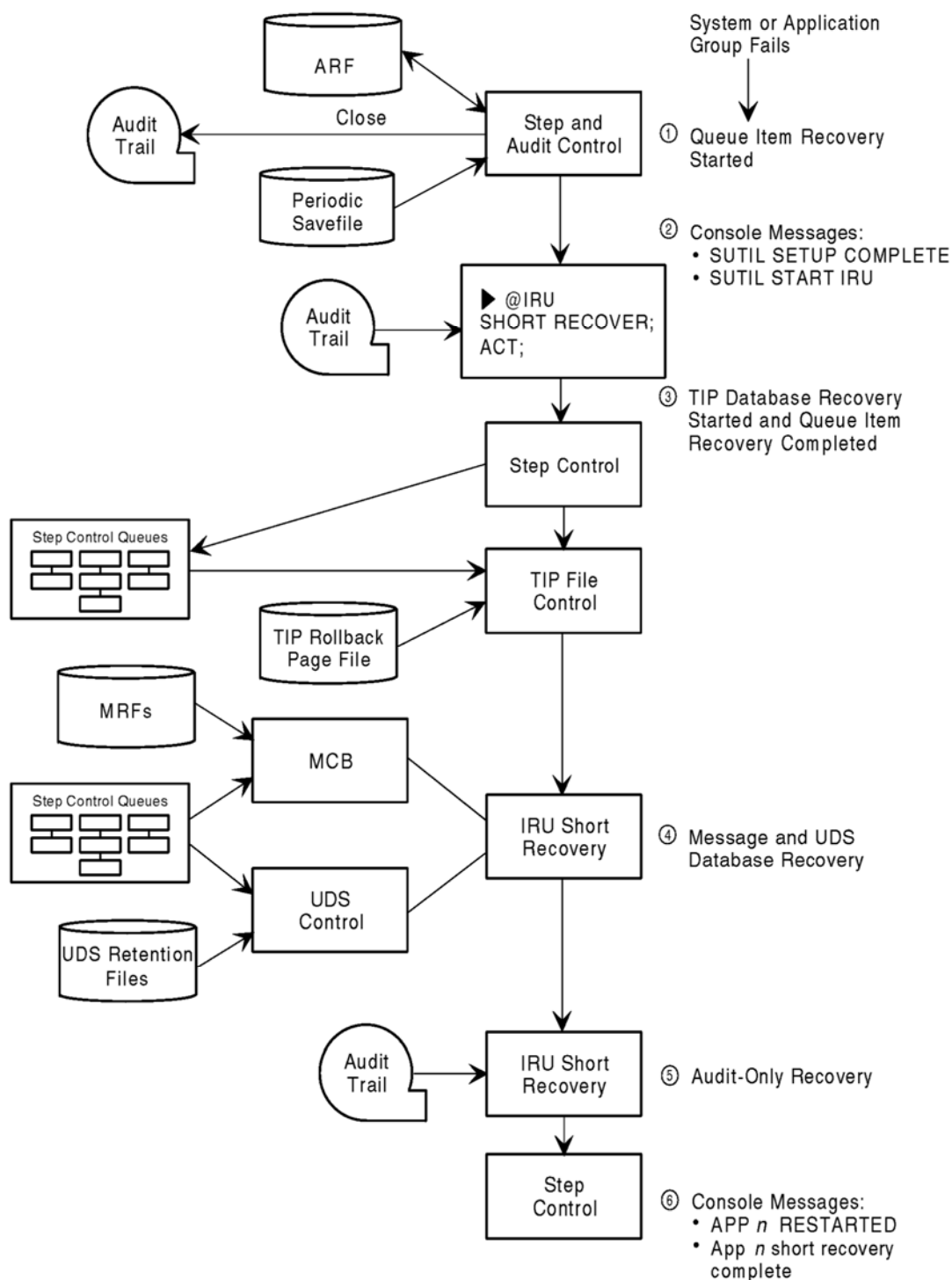
1. The system uses the ARF to close the audit trail if it is not already closed. Step Control obtains application group queue item information from the periodic savefile and uses the information to set up the Step Control queues in memory (that is, to start queue item recovery). Messages appear at the console, indicating progress of the recovery and the status of the audit trail; for example:

```
APP 3 SAVEFILE RECOVERY IS FROM LEG1
AT 3 ABNORMALLY CLOSED
APP 3 SUTIL PERIODIC SAVE FILE VERIFIED
```

2. The following messages appear at the console:

```
APP n SUTIL SETUP COMPLETE
APP n SUTIL START IRU
```

These indicate that SUTIL setup is complete and that the site must start IRU short recovery.



**Figure 7-3. Short Recovery with the SUTIL Option**

3. IRU queries Step Control about the application group state. IRU starts at the beginning of the audit trail reel/file obtained from the query (usually the last reel/file) and reads forward until it finds the periodic savefile audit record that matches the state of the application group's queue items from the periodic savefile. IRU reads the audit trail from the periodic savefile audit record to the end of the audit trail and passes each queue item audit record found in that portion of the audit trail to Step Control. Step Control then updates the encountered queue item state change in queue item storage. When IRU reaches the end of the audit trail, it informs Step Control that there are no more queue items. Step Control initiates TIP database recovery and completes queue item recovery.

The following messages appear at the console:

```
APP n {Host host-id} Rollback recovery started
APP n {Host host-id} Rollback recovery completed
```

```
run-id: Recovery of appl n queue items successful
```

4. When queue item and TIP database recovery are complete, IRU performs short recovery by calling MCB to recover the MCB messages and UDS Control to recover the UDS database. This is accomplished by using the MRFs, Step Control queues, and UDS retention files.

Messages appear at the console indicating application group recovery was successful.

5. If any audit-only recoverable steps were active when the system or application group failed, IRU performs the audit-only recovery portion of short recovery by reading the after-looks on the audit trail for commit-in-progress steps and applying them to the database.
6. IRU requests Step Control to restart the application group. The following messages appear at the console:

```
APP n RESTARTED
run-id: App n short recovery complete
```

where *run-id* is the run-ID of the IRU run.

### 7.2.3. INIT Option

Use the INIT option followed by IRU short recovery when creating an application group.

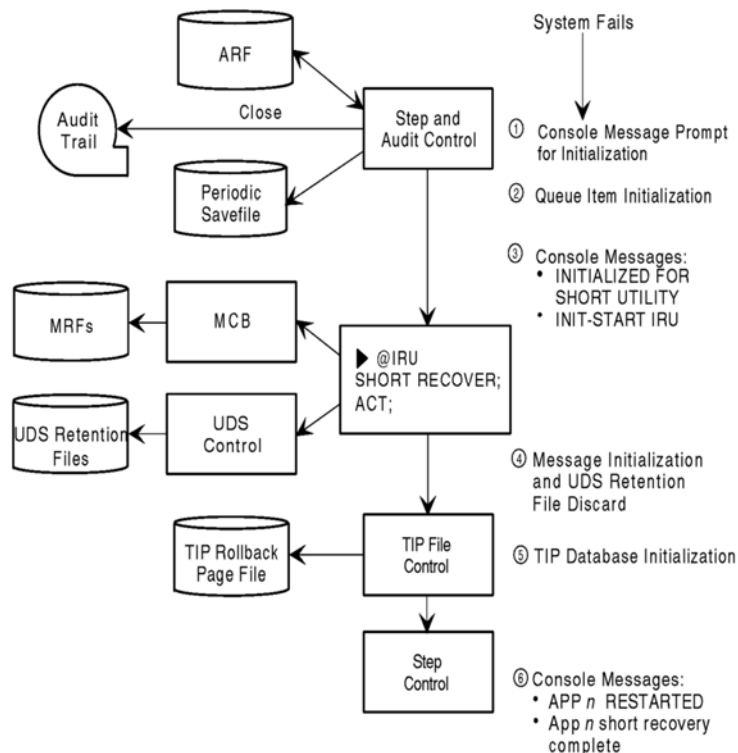
Use the INIT option followed by IRU medium or long recovery in the following situations:

- The periodic savefile does not contain recovery information (for example, it was lost).
- Before performing long queue item recovery (See the *Integrated Recovery Utility Operations Guide* (7830 8194) for more information.)

INIT does not perform message and database recovery. To recover messages, databases, queue items, or a combination, perform some form of recovery, such as IRU medium recovery or long database, message or queue item recovery.

Figure 7-4 illustrates the INIT procedure for a recoverable application group that uses TIP, UDS Control, and MCB.

In this example, the site initializes the application group by performing IRU short recovery after INIT. The following steps describe the INIT procedure followed by IRU short recovery.



**Figure 7-4. Creating an Application Group with the INIT Option**

1. The following messages appear when you select INIT following a recovery boot:

```

APP n INIT MODE - APPLICATION QUEUE ITEMS TO BE INITIALIZED
APP n INITIALIZE STEP CONTROL QUEUES? ANSWER NO OR GO

```

Answer GO to proceed with initialization.

2. The system uses the ARF to close the audit trail if it is not already closed. Exec Step Control initializes the queue items and the periodic savefile.
3. The following messages appear at the console:

```

APP n INITIALIZED FOR SHORT UTILITY
APP n INIT - START IRU

```

Start the IRU short recovery procedure.



**Note:** If the "Bypass IRU Short Recovery" SGS is configured, the START IRU message does not appear, IRU short recovery is bypassed, and the Exec, rather than the IRU, performs step 6.

4. IRU queries Step Control about the application group state and calls MCB and UDS Control. MCB initializes the message retention files, and UDS Control discards the information in the UDS retention files. IRU requests Step Control to restart the application group.

Messages appear on the console indicating application group initialization was successful.

5. TIP File Control initializes the TIP rollback page file. The following messages appear:

```
APP n {Host host-id} Rollback initialization started
APP n {Host host-id} Rollback initialization completed
```

6. The following messages appear at the console:

```
APP n RESTARTED
run-id: App n short recovery complete
```

where *run-id* is the run-ID of the IRU run.

## **7.2.4. DOWN Option**

Specify DOWN if the application group does not need to be active yet or if further diagnosis of the failure is needed before selecting another Exec option. In processing the DOWN option, the system

- Closes the audit trail, if it is open
- Releases queue item storage in memory
- Frees the periodic savefile if it is an Exec file
- Leaves the application group in a down state

When DOWN is successfully completed, the following message appears at the console:

```
APP n APPLICATION GROUP DOWN
```

After initiating the DOWN option, you can use the SUTIL or INIT option at a later time. Before specifying SUTIL or INIT, however, you must perform the AP *n* UP keyin to bring up the application group. Select a recovery or initialization option by answering the console message prompt that is produced, or alternatively, includes the recovery type on the AP *n* UP keyin (AP *n* UP SUTIL or AP *n* UP INIT). You cannot use PANIC and REBUILD recovery because the DOWN option closes the audit trail.

### 7.2.5. DEFER Option

DEFER is a Step Control option you can want to specify in the following situations:

- You want to run a concurrent application group in non-RLP mode (described in the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919)).
- Maintenance needs to be done on the ARF or periodic savefiles.

The DEFER option is similar to the DOWN option, except it frees the ARF from Step Control so the host to which you are switching the application group can access the ARF. In processing the DEFER option, the system

- Closes the audit trail, if it is open
- Releases queue item storage in memory
- Frees the ARF from Step Control

**Note:** *The ARF is not freed from Audit Control until the audit trail is deferred. Both Step Control and Audit Control for an application group must be deferred before the ARF is freed.*

- Leaves the application group in a deferred state

If the DEFER option is completed successfully, the system displays a message indicating the application group is deferred.

You can use the AP *n* UP keyin to bring up the deferred application group.

## 7.3. Local TIP Recovery Boot

Several local TIP configuration parameters can be changed dynamically but do not take effect until TIP is initialized again. Changing the values of these parameters sets a flag that generates messages during a local TIP recovery boot indicating that a local TIP initialization boot is required.

The messages offer you a bypass option to override the TIP initialization boot, and perform an Exec tape recovery boot.

```
TIP CONFIG has changed since last session. local TIP Initialization is
required.
```

```
To bypass this required local TIP Initialization, perform EXEC tape
recovery boot.
```

```
Answer: I (INIT TIP) OR D (DOWN TIP for this session):
```

## Section 8

# System Operations and Security

This section presents the following security-related operations information:

- The procedures for initialization of the system security files during initial and recovery boots
- A description of the printout labeling feature
- Security considerations when handling removable disks and tapes

### 8.1. Security Initial and Recovery Boots

During an initial boot, the operator is asked the following question:

```
n Specify Method to obtain Security Files--Load, Initialize (L,I)
```

Choose L to load the security files from a tape or tapes. Choose I to initialize the security files. The response applies to all security files. When security files are initialized, the header is cleared or created, links are established to the allocatable portion, and lookup tables are initialized. With SYSS\$SEC@USERID\$, user-ids and security records are created automatically for the Exec (EXEC8), the overhead user-id, and the security officer. With SYSS\$ACCOUNT\$R1, the system standard account and master account are created.

With Fundamental Security, the system always tries to recover the security files during a tape or autorecovery boot. If all security files cannot be recovered, they are all initialized with messages indicating that fact.

With Security Level 1 or higher, the operator is given the choice as to whether to recover the files as follows:

```
n Specify Method to obtain Security Files--Load, Init, Recover(L,I,R)
```

The response applies to all security files.

If the response is L, the operator is asked to load the SEC,RESTORE tape.

If security files are loaded or recovered and the configured security officer does not match the security officer in the SEC@USERID\$ file, the following message is displayed:

```
0 - The Security Officer configured in the Exec does not match the file [Recovered | Loaded].
```

```
Use the Configured or File Security Officer? (C, F)
```

If any of the security files cannot be assigned, fail integrity checks, or receive I/O errors, one of the following messages appear, as applicable:

```
Security file, < filename>, is corrupted.
```

```
n Specify Method to obtain Security Files--Load, Initialize (L,I)
```

```
Security file, < filename>, could not be cataloged. Facilities status = xxxxxx.
```

```
n Specify Method to obtain Security Files--Load, Initialize (L,I)
```

```
Security file, < filename>, could not be assigned. Facilities status = xxxxxx.
```

```
n Specify Method to obtain Security Files--Load, Initialize (L,I)
```

```
0 - Security file, < filename>, received I/O error xx; function = yy; sector addr = zz
System security may be impacted. Contact your Site Administrator.
```

```
Continue using a bad Security File or Stop the System? (Continue, Stop)
```

**Note:** If you restore ACCOUNT\$R1, accumulated file mass-storage quota values are incorrect. A recovery boot is required to recalculate the actual values from the current catalogued files.

## 8.2. Security Labeling for Printouts

Your system can be configured to label printouts with security information. Depending on how your system is configured, labels can be printed on every page or only on the banner and trailer pages of a printout. These labels (see Figure 8–1) appear at the top and bottom of each page. Labels are one to four lines long (up to 225 characters).

The label contains the mandatory attributes of a print file: the clearance level and the compartment set. The security officer (or site administrator) defines the compartments and clearance levels for the system and each of its users.

A clearance level is a numerical classification for objects, which ranges from 0 to 63, with 63 being the most restrictive. Clearance levels can also be associated with symbolic labels such as unclassified, classified, secret, and top secret. A compartment is a group based on interest or category, such as accounting, payroll, or personnel. In contrast to clearance levels, compartments are nonhierarchical classifications. A printout that has been identified as system high by the security system software is labeled with clearance level 63 and compartment set ALL.

Users might cause misprinting of the label through certain actions. For example, they can request a different print band or font or they can use ER APRTCN\$ in the middle of a print file and change the character set representation. After such changes, the system continues printing the original label, possibly obscuring the real attributes.

The printouts also have unique sequence numbers printed on the first and last pages. Sequence numbers are randomly generated 3-digit numbers printed on the banner and trailer pages of every printout to prevent spoofing. They are created when the print queue receives the first print file, and are increased by 1 each time another print file enters the queue. This process stops as soon as the print queue is empty. When the printer becomes active again, a new random sequence number is generated to start the next series of printouts.

When you are sorting print files to distribute to users, check the sequence numbers at the beginning and end of print files to make sure they match. Report any mismatches to your security officer. You can make an additional check by comparing printout labels on the banner and trailer pages, if labeling is configured for your site.

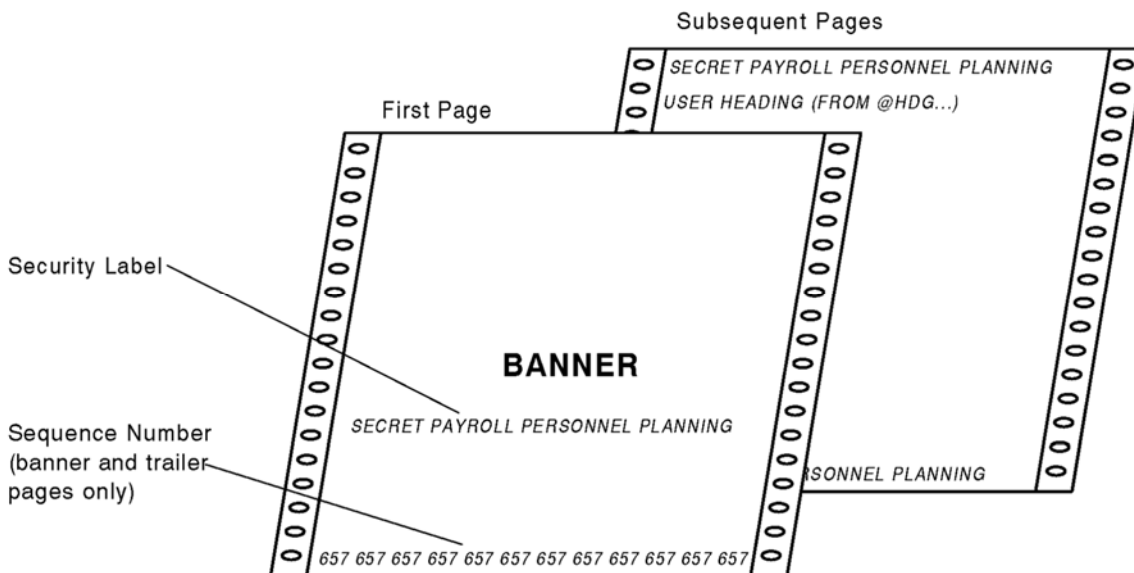
Compartment set names longer than 225 characters can be used internally, but cannot be printed out. An error message results and the printout is labeled system high (clearance level 63 and compartment set ALL). Handling of printouts labeled system high depends on local site procedures. For example, such printouts can be distributed to the security officer.

The security system creates a log entry in the system log file to show which files have a misleading output label.

Use the SR and SV keyins (see Section 11) to move labeled print files from system to system. These keyins also save the current print queue for a single host. If you plan to transport SV tapes, make sure the destination system has the identical security environment or labels cannot be handled properly.

The file's symbolic clearance level appears first in the label, followed by the compartment set. Compartment names are never split between lines, and only ASCII characters are used in labels.

Figure 8-1 shows the labeling on a sample output where the symbolic clearance level is SECRET, and the user's compartment set has the compartments PAYROLL, PERSONNEL, and PLANNING.



**Figure 8-1. Sample Output Label**

The label can appear slightly different if any of the following conditions exists:

- The ALL compartment bit is set, and the label has ALL in place of a list of compartments.
- No compartments are associated with the file and the label has NULL in place of a list of compartments.
- The label is longer than 225 characters and the label has ALL in place of a list of compartments and contains this message:

The security label is too long to be printed.

In this case, the system also makes a log entry.

### **8.3. Handling Removable Disk Files and Tapes in a Security Environment**

In a security system, removable disk files and tapes must be handled properly. Any removable object not labeled with a valid security label that contains a clearance level and compartment set is defined as nonlabeled. Removable objects are known to the system only when the pack is registered or the tape is mounted. They can remain nonlabeled until a user references them, possibly after a long time delay. The security system has detection mechanisms for nonlabeled removable objects.

If a removable object is labeled, its label has meaning only within a system's specific security environment. An object's security level (clearance level and compartment set) is interpreted relative to the definitions of the system symbolic clearance levels and system compartment set.

When objects are moved from one system to another, the label can then refer to the same attributes, but those attributes can have different meanings on each of the systems. Thus, the integrity of the label is endangered.

The security system defines specific requirements for maintaining security while transporting removable objects between systems. When removable objects are transported between systems, your security officer must check the definition of their original security environment, that is, how the system clearance level and system compartment set are defined. If they are different from the definitions on your destination system, the labels need to be changed immediately, so that the labels reflect the original level and compartment set definitions.





## Section 9

# **System Operations in a Multihost Environment**

This section describes the ways in which a multihost operations environment is different from a one-host operations environment. It presents an overview of the features as a system operator deals with them, tells which console keyins to use to determine the status of and control parts of the multihost environment, and tells where to look in this manual or in other documents for detailed information.

The section covers these topics:

- Types of multihost environments (Section 9.1)
- Hosts and shared devices (Section 9.2)
- Files and file directories (Section 9.3)
- Application groups (Section 9.4)
- Audit trails (Section 9.5)
- TIP File Control, scheduling, and automatic recovery (Section 9.6)
- Automatic Recovery of Components (ARC) and the monitor run (Section 9.7)
- XPC (Section 9.8)
- Shared unit duplexing (Section 9.9)

## **9.1. Types of Multihost Environments**

There are two basic types of multihost environments, depending on which features you have installed on your system

- A Multi-Host File Sharing (MHFS) environment
- An Extended Transaction Capacity (XTC) environment

MHFS provides the foundation for an XTC environment.

### 9.1.1. MHFS Environment

MHFS is a feature that is available by itself or as part of the XTC feature grouping. The acronym *MHFS* and the term *file sharing* are used interchangeably in Unisys documentation; for example, the MHFS environment and the file-sharing environment both refer to an operations environment in which two or more hosts can access Exec files on shared mass storage devices. You must use the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469) as the starting point for dealing with operations in a multihost environment.

In a file-sharing environment, there is a shared master file directory (MFD) that defines all files cataloged on shared mass storage devices; the directory-ID for these files is SHARED#. There is also a local (standard) MFD for each host that defines the host's local set of cataloged files; the directory-ID for a local file is STD#. For more information on files and file directories, see Section 9.3.

All shared devices in the file-sharing environment contain shared files only and are physically connected to shared mass storage control units, which allow all hosts to access the devices. A host can remain in the MHFS configuration as long as it has at least one path available to each shared subsystem. (It is not necessary for all hosts to be connected to all shared control units.) Host LAN Controller (HLC) connections link all hosts in a network MHFS configuration. An XPC links all hosts in an XPC MHFS configuration.

File sharing is set up and maintained host by host, not application group by application group. A file-sharing host is identified by an alphabetic host-ID—A, B, C... Each host has a set of internal tables indicating the identity and status of the other hosts in the environment. A host is not officially in the file-sharing environment until the other hosts have been informed of its presence; a host can be up to the Exec without being up and active in the file-sharing environment. When a host leaves the environment, other hosts also need to have their internal tables updated to indicate that the host is no longer present; a host can be down to the Exec without being down to file sharing (from the perspective of other MHFS hosts). If the tables are not correct, you cannot get accurate information in response to keyins about the status of other hosts or in other console messages. For more information on file-sharing hosts, see Section 9.2.

MHFS provides batch and demand access to shared Exec files, but XTC (Section 9.1.2) is needed to provide transaction access to shared databases belonging to TIP or UDS application groups.

### **9.1.2. XTC Environment**

XTC is an XTPA feature grouping that provides improved capacity for TIP and UDS systems. For an introduction to XTC, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).

The MHFS feature is included in the XTC feature grouping. Because of the XTCEXEC feature, multiple hosts can access shared TIP or UDS databases concurrently. An XPC-L controls access to the database files from all hosts, to preserve database integrity. All hosts are informed of changes to the databases and to the transaction environment. The end result is a system that can process many more transactions than a one-host system could.

The XPC-L also handles interhost broadcast messages as well as lock requests.

The UDS product release includes the Receiver of UDS Broadcast (RUB) program, which must be active on all XTC hosts to permit proper communication between application groups on the host and UDS. See the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919) for information on starting the RUB run.

The ARC feature is an optional part of the XTC feature grouping. Recovery runstreams registered through the ARC feature automatically recover and restart application groups and software components such as UDS and MCB. Heartbeat runstreams (one for each application group) detect any failure and start the appropriate recovery runstreams. If one host fails, runstreams can perform short recovery for local application groups and for concurrent application groups (see Section 9.4) on that host after the host recovers. In an XTC environment, ARC does not provide shared (host-to-host) short recovery. If a component fails but the host does not, the system automatically detects the failure and recovers the component.

The ARC feature consists of the application interface bank (AIB) and the monitor run, a real-time program that communicates with the AIB. See Section 9.8 for information on starting and controlling the monitor run. The heartbeat run and the monitor services control program (MSCP) are also real-time programs. For detailed operations information on all of these programs, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).

Concurrent TIP and UDS application groups are defined on all XTC hosts and can be active on all hosts at the same time. See Section 9.4 for more information on application groups and Section 9.6 for information on TIP File Control.

On an XTC system, time must be coordinated across hosts. Refer to the *Exec System Software Administration Reference Manual* (7831 0323) for more information on time management. You can use the D keyin (see Section 11) to display the date and time on a host, but you cannot use D keyins to alter the time.

### 9.1.3. Using the SPAIR Utility in an XPC or ARC Environment

You may use the Software Product Application Initialization and Recovery (SPAIR) utility if you have the Automatic Recovery of Components (ARC) software. ARC is included with the Extended Transaction Capacity (XTC) feature grouping.

If the audit trail is configured as removable mass storage, , then when the audit trail is opened or recovered, you can supply up to three pack-IDs for each leg (legs 1 and 2), or choose to execute the SPAIR utility to provide the initial equipment type and more or less than three pack-ids for each leg.

The *Exec System Software Administration Reference Manual* (7831 0323) explains how to use the SPAIR utility. The manual describes SPAIR security, the SPAIR processor call, how to execute SPAIR commands, how to use SPAIR in a runstream, and how to interpret SPAIR messages.

## 9.2. Hosts and Shared Devices

The Exec maintains the shared environment automatically unless you down Multi-Host File Sharing. After a host failure, or after one host is manually removed from file sharing, the remaining hosts stay in the MHFS environment and have access to TIP and UDS files on shared mass storage.

The MHFS status of a host is one of the following:

- The host is up and active in the file-sharing environment (UP).
- The host is down to MHFS (DN).
- MHFS is being brought down gracefully on this host (RV).

To determine the status of a single host in file-sharing environment, use this keyin:

```
MH,FS HOST,host-id
```

where *host-id* is A, B, C...

To determine the host-ID and status of all hosts in file-sharing environment, use this keyin:

```
MH,FS HOST,ALL
```

To see the host-ID and site-ID of this host and all hosts in the file-sharing environment, use this keyin:

```
MH,FS HOST
```

The RV status can be displayed only as the status of the host on which you enter the MH,FS HOST keyin. The status of all other hosts is displayed as either UP or DN to MHFS.

To determine whether file sharing is up at all, you can use either an MH,FS HOST or an MH,FS HOST,ALL keyin.

To determine the status of all devices active in the file-sharing environment, use this keyin:

```
FS,SHRD
```

To determine the status of all shared mass storage devices, use this keyin:

```
MS,SHRD
```

For detailed information on shared files, shared devices, and MHFS features, see the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469). Although MH keyins are documented in Section 11 of this manual, do not use these keyins for file-sharing control until you become familiar with the information in the *MHFS Guide*, which is summarized as follows:

- The MH,IN DIR,SHARED keyin causes shared file initialization and initialization of the shared MCT when you boot a local host and no other hosts are up in the file-sharing environment.
- The MH,RC DIR,SHARED keyin causes shared file reconfiguration and initialization of the shared MCT when no other hosts are active in the file-sharing environment.
- The MH,AD *directory-id*, *device-id* keyin changes the directory-ID of a device from shared to local or from local to shared. Note the specified device must be marked down.
- The MH,UP keyin adds a host dynamically to the file-sharing environment while the host is running and then notifies the other active hosts of the addition. If the host was previously in the MHFS environment, then this keyin causes a shared recovery if no other hosts are up in the MHFS environment or a shared reconfiguration if other hosts are up.
- The MH,DN keyin is coordinated across all hosts to physically and logically remove a host from file sharing. The removal of the host immediately terminates all runs with shared files assigned and prevents future shared file assignments. (This keyin results in an XER055 stop in an XTC environment.)
- The MH,RV keyin removes a host from the file-sharing environment by preventing new shared file assignments. The host is completely removed when existing shared file assignments are freed. (This keyin results in an XER055 stop in an XTC environment.)
- The MH,DN HOST,*host-id* keyin is executed from one host to mark another host down to file-sharing. Do not attempt to use the MH,DN HOST,host-ID keyin to remove another host from the file-sharing environment.

You can use the HK,*host-id* keyin to execute unsolicited keyins on other MHFS hosts from one MHFS host. The responses to the unsolicited keyins are displayed on the console of the host on which the HK keyin is executed. You can also use the HM,*host-id* keyin to send console messages to other hosts.

MV keyins are not allowed in a file-sharing environment.

MHFS console messages are described in the *System Console Messages Reference Manual* (7833 3275).

As mentioned in Section 9.1.2, in an XTC environment you can use the D keyin (see Section 11) to display the date and time on a host, but you cannot use D keyins to alter the time. For information on establishing time coordination among XTC hosts, see the *Exec System Software Administration Reference Manual* (7831 0323).

### 9.3. Files and File Directories

The directory-ID for a cataloged file depends on the type of storage on which it resides. If the file resides on shared mass storage, it has a directory-ID of SHARED# and is defined in the shared master file directory (MFD). If the file resides on a host's local mass storage, it has a directory-ID of STD# (standard, meaning local) and is defined in the local MFD for that host. For an Exec file registered with TIP, there is another set of TIP Directories: shared TIP directory and local TIP directory. There is no direct correspondence between the MFD and TIP directories. The local TIP directory might contain files cataloged on local and shared mass storage. However, the shared TIP directory can only contain files cataloged on shared mass storage. See Section 9.6 for more information on TIP directories.

Database files belonging to a concurrent (for XTC) TIP or UDS application group are on shared mass storage, and if TIP, in the shared TIP directory. System files that belong to concurrent and local application groups vary in regards to local or shared mass storage. To find out which system or Integrated Recovery files exist for each one of these types of application groups, and where they are located, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).

To determine whether a cataloged file is currently assigned or if it even exists, you can use this keyin:

```
FF [directory-id#]qualifier*file.
```

Because of the MHFS double-MFD search feature, if you do not specify the directory-ID, the system tries to locate the file by checking the local MFD first and then the shared MFD, unless a feature has been installed that changes this behavior. In the response to the keyin, the system tells you whether a common name section (CNS), such as EXEC8 (the Exec run-ID), a run, or a host has the file assigned. To protect shared files in an XTC environment, the MHFS application-private assignment feature allows the Exec to assign shared TIP/Exec files exclusively and UDS to assign shared UDS/Exec files exclusively, but this assignment is allowed on multiple hosts. For information on the FF keyin, see Section 11; for information on application-private assignment, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).

## 9.4. Application Groups

There are two types of application groups (AG)—local and concurrent

- A local application group is known to only one host and shares no resources (including its database) with other hosts.
- In an XTC environment (Section 9.1.2), a concurrent application group is defined on both hosts and can be active on both hosts at the same time. All XTC hosts can access the database for the application group concurrently.

An AG/host is the portion of a concurrent application group that exists on a particular XTC host. For example, concurrent TIP application group 4 exists on host A as AG4/A and on host B as AG4/B. The application group must be recovered separately on each host—that is, each AG/host must be recovered separately. Note that the term *AG/host* refers only to the portion of a concurrent application group that exists on a particular XTC host and does not refer to a local application group configured on a particular host.

The information presented here covers only concurrent application groups because they exist only in special multihost environments.

The status of concurrent AG/host can be one of the following:

- Up (UP)  
The AG/host has been initialized or recovered, and it is ready for processing. All database and associated system files are available and assigned.
- Down (DN)  
The AG/host is not available; queue-item storage has been released and the audit trail has been closed (if it was open). SUTIL or INIT recovery can be used to recover the AG/host. The ARF file remains assigned.
- Recovery in progress (RCV)  
The AG/host is being recovered; it is available for processing when recovery is complete.
- Deferred (DF)  
The AG/host is not available and all its files, including the ARF files have been freed from Step Control. If you have also deferred the audit trail, the ARF is then freed by Audit Control and by the Exec.

To determine the status of and other information about a concurrent AG/host, use this keyin:

```
AP n FS
```

The format of the console message displayed in response to the AP *nn* FS keyin is

```
APP n state qitem-usage PRG:steps REC:type MCB:MCB-status
APP n TYPE: ag-type DR: DR-flag READY STEPS: num-ready
```

This message describes the status of application group *nn* on this particular host. The fields in the message are explained in Section 11, in the description of the AP keyin. The second line of the message appears only in a multihost environment: TYPE is the application group type (LOCAL or CONCURRENT) and DR is the database recovery (DR) flag, which contains the status of database recovery.

These are some of the actions that you can perform to change the state of an application group:

- Bring up (enable) a concurrent AG/host with an AP *n* UP keyin.
- Bring down (disable) a concurrent AG/host with an AP *n* DN keyin.
- Defer a concurrent A/Ghost with an AP *n* DF keyin.

See the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919) for specific application group recovery information. For general application group recovery information, see Section 7, "Step Control Recovery and Initialization Options," in this manual and the *Integrated Recovery Utility Operations Guide* (7830 8194). The AP keyins are documented in Section 11 of this manual.

## 9.5. Audit Trails

A local application may be configured with an audit trail. If an audit trail is configured, then for each audit trail, there is one ARF and one Audit Control Interface (ACI) file.

In an XTC environment (Section 9.1.2), each concurrent AG/host has an audit trail, so each concurrent application group has multiple audit trails, one for each AG/host; the IRU blends all of the AG/host audit trails together when recovering the database for the concurrent application group. For each audit trail, there is one ARF and one ACI file.

The audit trail for an application group must be open or available to be opened (that is, closed) for recoverable transaction programs to run. AG/host or application group recovery also usually requires that the audit trail be available for auditing.

For a discussion of Audit Control, audit trail states, and AT keyins, see Section 7. For a description of all AT keyins, see Section 11.

## 9.6. TIP File Control, Scheduling, and Automatic Recovery

An XTC system has both local TIP File Control and shared TIP File Control. Shared and local TIP File Controls are handled separately. Local TIP File Control can be initialized or recovered. Shared TIP File Control can be initialized or reemployed. For information on messages that you might see while initializing or reemploying shared TIP File Control, see "Multihost Initialization, Recovery, and Reemployment" in the *Transaction Processing Administration and Operations Reference Manual* (7830 7881). For specific operations procedures and scenarios in an XTC environment, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).



Only TIP/Exec files cataloged on shared mass storage can be registered in the shared TIP directory. TIP files reserved in TIP/Exec files that are registered in the shared TIP directory are automatically reserved in the shared TIP directory. TIP/Exec files cataloged on local mass storage are registered in a local TIP directory, but TIP/Exec files cataloged on shared mass storage can also be registered in a local TIP directory; the TIP files reserved in those TIP/Exec files are automatically reserved in the local TIP directory. For a TIP/Exec file, the directory-ID refers to the TIP directory in which it is registered, not to the MFD indicating the type of mass storage on which it resides; a local TIP/Exec file can reside on shared mass storage. The same is true of the TIP files reserved in the TIP/Exec file.

On a one-host TIP system, there is one validation table (VALTAB) with entries defining all transaction programs, in TIP system file 4. On an XTC system, there is only VALTAB, so TIP system file 4 is shared. There is also a single set of TIP program files (SUPUR files for TIP programs, TPUR files for HVTIP program libraries, or standard online program files). TIP scheduling is a system-wide resource on an XTC system; however, you enable or disable it host by host. Some flagbox settings are broadcast to all hosts, while others affect only the local host. For details on the effects of flagbox bits in an XTC environment, see the flagbox section in the *Transaction Processing Administration and Operations Reference Manual* (7830 7881) (the FB keyin is documented in that section and also in Section 11 of this manual).

The TIP automatic recovery feature is required for XTC. This feature automatically records the state of the TIP environment as it changes, so that TIP recovery from a system failure can be accomplished without manual procedures. TIP automatic recovery recovers the TIP scheduling and program loading environment (such as the flagbox and HVTIP libraries that were loaded).

On a one-host TIP system, there is one TIP automatic recovery file (ARF). On an XTC system, there is also one TIP ARF, which is located on shared mass storage. Each time a host enters the XTC environment, you must enable TIP automatic recovery manually with a TIP UP keyin. From that point on, the feature functions automatically.

For detailed information on TIP automatic recovery and on the use of TIP UP, TIP DN, and TIP RS keyins, see "TIP Automatic Recovery and TIP Keyins" in the *Transaction Processing Administration and Operations Reference Manual* (7830 7881). The TIP keyins are also documented in Section 11 of this manual.

### Caution

The configuration parameter TPLIB contains the TIP file number that contains the first HVTIP library. This parameter can be changed dynamically. In an XTC environment, failing to keep TPLIB set to the same value on each host in the XTC environment can result in TIP scheduling errors, TIP program aborts, or TIP Automatic Recovery failure.

### 9.7. ARC and the Monitor Run

ARC, an optional feature of XTC, uses a monitor run to monitor the heartbeat of application group components and to start the appropriate recovery runstream if a component fails. The monitor run includes a primary run and a backup run. If the primary run fails, the backup run becomes the primary run and another backup run is created.

To start the monitor run, you use an @START statement or an ST keyin. Once the monitor run is active, you can use ARCMR keyins to determine the status of the monitor run and of application group components, and to terminate the run. See the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919) for detailed operations information on the monitor run. The ARCMR keyin is also documented in Section 11 of this manual.

### 9.8. Extended Transaction Processing

In an XTC system, an extended transaction processor (XPC) controls access to the database files from all hosts. For XTC, the XPC is configured as a control unit and is in the UP or the UP NA state. Each host has two or more operational paths to the XPC. To bring the XPC up or down, you use UP or DN keyins.

If the XPC is unavailable, the system must run in non-XPC mode with one host designated as the non-XPC host. For a description of non-XPC mode and procedures to follow on all hosts, see the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919).

See the *Integrated Recovery Reference and Administration Guide for Multihost Environments* (7831 0919) for information on XPC preparation. The guide also describes XTC operations scenarios involving the XPC. See Section 11 of this manual for descriptions of UP and DN keyins and the *System Console Messages Reference Manual* (7833 3275) for descriptions of XPC messages.

### 9.9. Shared Unit Duplexing

Unit Duplexing can be used as a stand-alone feature with MHFS, with MHFS and XTC Applications feature grouping. Duplex associations between local devices and shared devices can only be established dynamically.

Keyins used with Unit Duplexing are described in Section 11, and messages are described in the *System Console Messages Reference Manual* (7833 3275). For detailed information on Unit Duplexing, see the *Unit Duplexing Planning, Installation, and Operations Overview* (7830 7592).

If you use an FS *device* keyin to determine the status of a shared device that is unit-duplexed, the response to the keyin has this format:

```
device1 -UD- device2 S status [*] F PACKID packed [EXEC|SHARED DRS]
```

Since the devices are unit-duplexed, they both have the same status, which can be UP or SU; a unit-duplexed device cannot be RV. S indicates that the devices are shared, and F indicates that they are prepped as fixed.

To establish or reestablish a duplex association between devices that are configured as shared, use the UD keyin:

```
UD device1,device2
```

where *device2* is the device that you want to be an exact copy of *device1*.



## Section 10

# Keyins by Group

This section lists the unsolicited keyins by functional group. Not all of these keyins can be performed on every supported OS 2200 system; see Section 11 for detailed descriptions of each keyin, including the restrictions that apply to that keyin.

Table 10-1 shows the keyins according to functional group.

**Table 10–1. Keyins by Functional Group**

| Keyin                            | Function                                                                                  |
|----------------------------------|-------------------------------------------------------------------------------------------|
| <b>Checkpoint/Restart Keyins</b> |                                                                                           |
| CK                               | Starts a checkpoint                                                                       |
| RS                               | Restarts a run that was checkpointed                                                      |
| <b>Communications Keyins</b>     |                                                                                           |
| HK                               | Sends unsolicited keyins to other hosts                                                   |
| HM                               | Sends messages to other hosts                                                             |
| TB                               | Sends messages to demand terminals                                                        |
| TM                               | Sends messages to other site-IDs, run-IDs, or user-IDs                                    |
| <b>Console Control Keyins</b>    |                                                                                           |
| AC                               | Console control                                                                           |
| CMR                              | Redisplays console messages                                                               |
| MR                               | Displays information about active remote consoles and outstanding read-and-reply messages |
| OM                               | Displays or changes operating mode                                                        |
| <b>Component Control Keyins</b>  |                                                                                           |
| ABORT                            | Halts processing of DC, UD, and XPCDUMP keyins                                            |
| CTL                              | Changes state or displays status of a cartridge tape library (CTL)                        |
| DC                               | Copies a disk                                                                             |
| DN                               | Brings down components                                                                    |

**Table 10–1. Keyins by Functional Group**

| <b>Keyin</b>                          | <b>Function</b>                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------|
| EXPAND                                | Expand local and shared disks                                                                               |
| FA                                    | Simulates an attention interrupt                                                                            |
| IN                                    | Initializes fixed mass storage at "Modify Config" time                                                      |
| MD                                    | Modifies a disk pack                                                                                        |
| MH                                    | Performs Multi-Host File Sharing facility control                                                           |
| MS                                    | Displays information about mass storage                                                                     |
| MV                                    | Moves disks                                                                                                 |
| PM                                    | Designates a unit as a premount-only unit (TAVR only)                                                       |
| PR                                    | Removes the premount-only designation on a unit (TAVR only)                                                 |
| RD                                    | Removes disk packs                                                                                          |
| RP                                    | Removes pointers to files on a pack                                                                         |
| RV                                    | Reserves components                                                                                         |
| SM                                    | Handles symbiont devices                                                                                    |
| SU                                    | Suspends a component                                                                                        |
| TU                                    | Releases a tape unit (cartridge tape only)                                                                  |
| UD                                    | Establishes Unit Duplexing                                                                                  |
| UL                                    | Rewinds with interlock tape units (TAVR only)                                                               |
| UP                                    | Brings up components                                                                                        |
| <b>Date/Time Keyin</b>                |                                                                                                             |
| D                                     | Displays or controls the date and time                                                                      |
| <b>General Status, Display Keyins</b> |                                                                                                             |
| EU                                    | Displays EXPOOL usage                                                                                       |
| FF                                    | Displays information on a file                                                                              |
| FS                                    | Displays information on components or interfaces                                                            |
| HU                                    | Displays host resource (IP SUPs) usage, and statistics for application level test and set queued activities |
| MODELNUM                              | Displays information on the model number and performance level                                              |
| MS                                    | Displays information about mass storage                                                                     |
| MU                                    | Displays central complex memory usage                                                                       |
| SS                                    | Displays system status                                                                                      |
| KEYINS                                | Displays registered user keyins                                                                             |
| PATHS                                 | Displays all paths to a device and their states                                                             |

**Table 10–1. Keyins by Functional Group**

| Keyin                             | Function                                                                               |
|-----------------------------------|----------------------------------------------------------------------------------------|
| <b>Logging (System) Keyins</b>    |                                                                                        |
| LG                                | Makes entries in the system log file                                                   |
| AT                                | Controls operation of the audit trails (system log, Integrated Recovery, COD, and TPM) |
| <b>Reload Common Banks Keyin</b>  |                                                                                        |
| RL                                | Reloads a common bank from a library file                                              |
| <b>Performance Analysis Keyin</b> |                                                                                        |
| SB                                | Controls the collection of performance data                                            |
| <b>Run Process Control Keyins</b> |                                                                                        |
| B                                 | Sets the number of batch runs                                                          |
| CS                                | Sets and releases the hold conditions                                                  |
| DISPLVL                           | Changes the switching priority of demand or batch jobs                                 |
| E                                 | Terminates a run in error                                                              |
| ERUNS                             | Terminates in error one or more runs                                                   |
| II                                | Causes an interrupt                                                                    |
| RM                                | Removes a backlogged run                                                               |
| ST                                | Starts a stored runstream                                                              |
| SP                                | Suspends a run                                                                         |
| RE                                | Restarts a run previously suspended                                                    |
| X                                 | Aborts a run                                                                           |
| <b>Run Status, Display Keyins</b> |                                                                                        |
| BL                                | Displays runs in backlog                                                               |
| DISPLVL                           | Displays active DISPLVL switching priorities by category                               |
| RC                                | Displays information on a specific run                                                 |
| T                                 | Displays information on demand and batch runs                                          |
| <b>Security Keyins</b>            |                                                                                        |
| SEC                               | Saves and restores security files on tape                                              |
| TS                                | Lists disabled terminals or enables terminals                                          |

**Table 10–1. Keyins by Functional Group**

| Keyin                                | Function                                                                               |
|--------------------------------------|----------------------------------------------------------------------------------------|
| <b>Symbiont Control Keyins</b>       |                                                                                        |
| SM                                   | Handles symbiont devices                                                               |
| SQ                                   | Displays the status of print files                                                     |
| SR                                   | Restores saved symbiont files                                                          |
| SV                                   | Saves queued output symbiont files on tape                                             |
| SX                                   | Deletes queued files from the system                                                   |
| <b>System Debug Keyins</b>           |                                                                                        |
| BP                                   | Controls the breakpoint setting                                                        |
| C                                    | Displays and modifies contents of Exec banks                                           |
| CJ                                   | Clears jump keys                                                                       |
| DJ                                   | Displays jump keys                                                                     |
| SJ                                   | Sets jump keys                                                                         |
| JH                                   | Controls jump history tracing                                                          |
| DU MP                                | Obtains an online dump                                                                 |
| FC                                   | Initiates and terminates internal Exec traces                                          |
| IODELAY                              | Controls artificial time delays for secondary units of duplexed pairs.                 |
| IT                                   | Controls I/O traces and timeout value for mass storage I/O                             |
| PURGE                                | Removes residue data from an XPC. An XPC-L device does not support this keyin.         |
| XPCDUMP                              | Dumps XPC storage                                                                      |
| \$!                                  | Stops the system                                                                       |
| <b>Transaction Processing Keyins</b> |                                                                                        |
| AP                                   | Controls application group processing                                                  |
| ARCMR                                | Displays monitor run status or terminates monitor run                                  |
| AT                                   | Controls operation of the audit trails (system log, Integrated Recovery, COD, and TPM) |
| FB                                   | Displays or changes the flagbox settings                                               |
| LB                                   | Controls the TPM mask settings                                                         |
| LC                                   | Controls setting of TPM (ON or OFF)                                                    |
| TF                                   | Initializes TIP File Control                                                           |
| TIP                                  | Displays or changes the status of TIP automatic recovery                               |



**Table 10–1. Keyins by Functional Group**

| <b>Keyin</b> | <b>Function</b>                         |
|--------------|-----------------------------------------|
| TIPSS        | Displays TIP Session Control statistics |
| TP           | Controls Transaction Processing         |



## Section 11

# Keyin Reference

This section describes the unsolicited keyins that are available on OS 2200 systems. The following information is included as applicable for the different operator keyins:

- Purpose
- Keyin formats
- System response
- Restrictions and other considerations
- Examples of correct use

See Section 2 for general information about entering unsolicited keyins.

### 11.1. \$!—Stop the System

The \$! keyin stops the system for diagnostic purposes and is also used to recover from a hung system (assuming there is response to console input or user terminals).

#### 11.1.1. Format

\$!

#### 11.1.2. System Response

The system is rebooted with an XER 0150.

### 11.2. ABORT—Halt Processing of DC, UD, and XPCDUMP Keyins

The ABORT keyin halts the processing of DC, UD, and XPCDUMP keyins. You can enter the keyin at any host, not just the host from which the original keyin was entered.

#### 11.2.1. Format

ABORT *keyin*

where *keyin* is the entire string of the keyin to be halted.

### 11.2.2. Examples

```
ABORT DC DISKA,DISKB
```

```
Halts DC DISKA,DISKB.
```

```
ABORT UD DISK1,DISK2
```

```
Halts UD DISK1,DISK2.
```

```
ABORT XPCDUMP XPC0
```

```
Halts XPCDUMP XPC0.
```

### 11.2.3. System Response

Response messages include "ABORT" followed by the first 12 characters of the ABORT keyin (the 12 characters include the keyword ABORT). The ABORT keyin can be rejected because the user

- Specified a device name that is not a disk
- Specified a disk pack not involved in a disk copy operation (when trying to halt a DC)
- Is trying to halt a DC or UD just as a copy is completing
- Issued the ABORT against a keyin other than DC, UD, or XPCDUMP

A UD keyin that receives the following message is a special case:

```
Disk Copy will destroy files on destination-device Continue Copy Y, <N>
```

If you answer Y to this message, you must prep the destination device again before it can be used.

## 11.3. AC—Console Control

The AC keyin sets and clears the display of message groups on console devices, lists the associations between control units, console devices, and message groups, and controls the use of the backup console.

### 11.3.1. Formats

#### AC C and AC S

```
AC C/ALL/devname1, devname2, ...,devnameN
```

```
AC C/groupname/devname1, devname2, ...,devnameN
```

```
AC S/ALL/devname1, devname2, ...,devnameN
```

```
AC S/groupname/devname1, devname2, ...,devnameN
```

AC C clears the message groups you specify from being displayed on the console devices you specify. AC S sets the message groups you specify to be displayed on the console device you specify.

### **AC I**

AC I or AC I/ALL  
AC I/*groupname*

The AC I keyin displays the associations between console devices and message groups.

AC I (or AC I/ALL) lists all the console devices for all message groups and the response console for each message group.

AC I/*groupname* lists all the console devices that display the specified message group and gives the response console for that message group.

### **AC R**

AC R/ALL/*devname*  
AC R/*groupname*/*devname*

These AC R keyins change the response console for read-and-reply messages (for systems with multiple consoles only).

where:

ALL

indicates all message groups.

*devname*

is the console name.

*groupname*

is the message group name.

## **11.3.2. System Response**

When you enter an AC keyin, a system message verifies completion.

If you perform an AC C keyin to clear a console for read-and-reply messages and receive either of the following messages, perform an AC R keyin directing the message group to another device and retry the AC C keyin:

CLEAR OF *group-name* RESPONSE CONSOLE NOT ALLOWED

AC C/ALL/COMPLETED BUT RESPONSE CONSOLE NOT CLEARED

The system response to an AC *I/groupname* keyin is

```
group-name RESPONSE CONSOLE IS devname
```

If the response console is a remote console, the messages are

```
group-name RESPONSE CONSOLE IS CURRENTLY SITEID site-id
```

```
DEFAULT group-name RESPONSE CONSOLE IS devname
```

### 11.3.3. Restrictions

Messages entered through the main storage SIZE message are displayed only on the bootstrap console. Messages following the MASS STORAGE message are routed to the console devices as specified during system generation and modified by AC keyins.

If you use the AC keyin to route system messages to a remote demand console in addition to a PC console, note the following: If you enter any of the jump key keyins (DJ, CJ, and SJ) at the demand terminal, the keyin response is displayed only on the demand terminal, not on the PC console.

The AC keyin cannot be entered from a remote console (@@CONS), nor do AC R keyins affect remote-response consoles. Read-and-reply messages issued at a remote-response console are not rerouted. Any message group specified on an AC R keyin remains assigned to its @@CONS remote-response console.

Changes to the console message group that result from an AC keyin are saved over a disk or autorecovery bootstrap.

### 11.3.4. Examples

```
AC S/SYSMSG/CNSOL3,CNSOL4
```

Allows message group SYSMSG to be displayed on devices CNSOL3 and CNSOL4, as well as the system console.

```
AC I/IOMSG/CNSOL1
```

Causes the system to list the devices on console CNSOL1 that display message group IOMSG.

```
AC R/ALL/CNSOL2
```

Makes the CRT on console CNSOL2 the response device for all read-and-reply messages.

## 11.4. AP—Application Group Status and Control

The AP keyin displays the status of application groups and brings up or shuts down an application group.

### 11.4.1. Formats

#### AP FS

AP ALL FS

Display the status of all application groups.

AP *n* FS

Display the status of application group *n*.

#### AP DN

AP *n* DN

Bring down application group *n*.

AP *n* DN RECOVER, *option*

Bring down application group *n*, and initialize or recover the application group. *option* is REBUILD, SUTIL, INIT, or DOWN.

#### AP DF

AP *n* DF

Defer the application group. The application must be in the DN state. All system files assigned by Step Control for the application group are freed.

#### AP UP

AP *n* UP

Start up application group *n*.

AP *n* UP *option*

Bring up application group *n*, and initialize or recover the application group. The *option* can be REBUILD, SUTIL, or INIT.

**Note:** The INIT option of this keyin, followed by IRU short recovery, discards all queue items, database updates, and XPC-L locks.

### AP RA

*AP x RA option*

Recover application groups across the boot. It is valid only when the highest application group number configured in the system is 10 or higher and jump key 7 is set, and while the read-and-reply application group recovery message is outstanding (see Section 7).

where:

*x*

is any of the following:

- A single number
- A series of numbers separated by commas
- A range of numbers connected with a hyphen
- A combination of any of the preceding
- ALL (for all application groups)
- An asterisk (\*) to represent the remaining application groups numbers

*option*

can be any valid recovery option: PANIC, REBUILD, SUTIL, INIT, DOWN, or DEFER.

### Example

*AP 1,3-5,7,9-11 RA PANIC*

### AP RS

*AP n RS*

Resets recovery coordination flag on all hosts (for this application) to Database Recovery Complete With Unlock (DCRWU). Application on executing host must be in a DN state. On all other hosts, the application (and audit trail) must be DEFER'ed or the host itself must be down. Applies to concurrent application groups only.



## 11.4.2. System Response

The response to the AP ALL FS keyin displays status information starting with the smallest numbered application group and ending with the highest configured application group. Information is displayed in the following format:

```
APP 3 state qitem-usage PRG:steps REC:type MCB:MCB-status
APP 3 TYPE: ag-type DR: DR-flag READY STEPS: num-ready
APP 7 state qitem-usage PRG:steps REC:type MCB:MCB-status
APP 7 TYPE: ag-type DR: DR-flag READY STEPS: num-ready
```

where:

*state*

is the state of the application group: UP, DN, recovery-in-progress (RCV), DF, or not configured. (If no application groups are configured, no other information is displayed.)

*qitem-usage*

is the percentage of queue items in use.

*PRG:steps*

is the total number of queue items on the active, ready, and commit-in-progress lists if the application group is up, where *steps* is a 1- to 6-digit decimal number that corresponds to the number of application group program activities executing within a step. If the application state is still RCV, *PRG:steps* is replaced with the IRU short recovery status—Either IRU:PND for IRU short recovery pending or IRU:INP for IRU short recovery in progress.

*REC:type*

is the type of application group recovery executed, where *type* can be one of the following:

|         |                                         |
|---------|-----------------------------------------|
| INIT    | Initialized                             |
| SUTIL   | Recovered by short utility processing   |
| REBLD   | Recovered by rebuilding the queue items |
| PANIC   | Recovered from the system (panic) dump  |
| (blank) | Down or has recovery-in-progress        |

*MCB:MCB-status*

is the status of MCB: up, down or abort-in-progress (AIP). The field is omitted if the application group is not using MCB.

### *ag-type*

is the application group type configured at system generation. This field can be set to one of the following:

LOCAL

CONCURRENT

### DR: *DR-flag*

is the status of medium database recovery. This consists of one of the following values:

|              |                                                                                                                                                                                                                                            |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COMPLETELK   | Medium database recovery is complete, but the locks for this AG or AG/host still exist at this point.                                                                                                                                      |
| COMPLETENOLK | Medium database recovery is complete and the locks have been unlocked for this AG or AG/host.                                                                                                                                              |
| IN-PROGRESS  | Medium database recovery is in progress.                                                                                                                                                                                                   |
| NONE         | If the Ag or AG/host is up (UP), it signifies that short recovery is complete. If the Ag or AG/host is down (DN) or recovery is in progress, it signifies that no short or medium recoveries have occurred since the last AG/host failure. |
| READYSREMAIN | Medium database recovery is complete, but some ready steps remain in the Ag or AG/host, awaiting completion of the distributed transactions.                                                                                               |
| REMOTERDYINP | Medium database recovery is complete, but some ready steps remained. Interhost resolution of at least one of the ready steps was started for this AG/host. (This status applies only to concurrent application groups.)                    |
| REMOTERDYCMP | Medium database recovery is complete. Interhost resolution of the ready steps was also completed for the AG/host. (This status applies only to concurrent application groups.)                                                             |
| UNKNOWN      | The status of the database is unknown. Operator intervention is required to continue with recovery.                                                                                                                                        |

### READY STEPS:*num-ready*

is the number of ready steps present in the AG or AG/host.

The system responds to an AP *n* UP keyin with the following message:

```
APP n RECOVERY TYPE - ENTER: (REBUILD/SUTIL/INIT/DOWN/DEFER)
```

See Section 7 for information on the recovery options.

The system responds to an AP *x* RA keyin with a message indicating whether the keyin was successful or unsuccessful.

If you try to select an invalid option, enter unconfigured application group numbers, or make other syntax errors when entering an AP x RA keyin, system messages appear to inform you of the error. Refer to the *System Console Messages Reference Manual* (7833 3275) for explanations.

The system responds to an AP x RS keyin with the following type and read message:

```
AP RS keyin will disable coordination of application recovery. All
updates in progress will be lost. Once a response of RESET is entered,
application n must be INITIALIZED on ALL hosts, even if the keyin does
not complete successfully. Failure to do so could result in database
corruption on subsequent recoveries. Answer RESET to continue.
```

The user must enter RESET in order for the keyin to proceed. If the user transmits or enters any other data, the AP RS keyin is ignored. If you try to select an invalid option, enter an unconfigured application group number, or make other syntax errors when entering an AP x RS keyin, system messages appear to inform you of the error. Refer to the *System Console Messages Reference Manual* (7833 3275) for explanations.

### 11.4.3. Examples

#### Example 1

The system displays the following read-and-reply message during recovery:

```
Specify Recovery Action for the following application groups:
Non-recoverable APPs specify (INIT/DOWN)
Non-recoverable APPs are: 1-3,6,8
Recoverable APPs specify (PANIC/REBUILD/SUTIL/INIT/DOWN/DEFER)
Recoverable APPs are: 4-5,12,15
0-Specify Recovery Action with "AP x RA" keyin, then answer DONE.
```

Operator response is as follows:

1. The operator first wants to perform an SUTIL recovery of application groups 4, 12, and 15, so the following keyin is entered:

```
AP 4,12,15 RA SUTIL
```

This results in application group 4 beginning SUTIL recovery followed by applications 12 and 15.

2. The operator then wants to initialize application groups 1, 2, 3, and 8, so the following keyin is entered:

```
AP 1-3,8 RA INIT
```

This results in application groups 1, 2, 3, and 8 beginning initialization.

3. Next, the operator wants to bring down the remaining application groups 5 and 6, so the following keyin is entered:

```
AP * RA DOWN
```

This results in application group 5 and 6 being marked DOWN.

4. Finally, the operator must tell the system that the recovery is complete, so the following keyin is entered:

```
0 DOWN
```

This terminates the outstanding operator message.

### Example 2

The system displays the following read-and-reply message during recovery:

```
Specify Recovery Action for the following application groups:
Non-recoverable APPs specify (INIT/DOWN)
Non-recoverable APPs are: 1-3,6,8
Recoverable APPs specify (PANIC/REBUILD/SUTIL/INIT/DOWN/DEFER)
Recoverable APPs are: 4-5,12,15
0-Specify Recovery Action with "AP x RA" keyin, then answer DONE.
```

Operator response is as follows:

1. The operator wants to initialize all application groups, so the following keyin is entered:

```
AP ALL RA INIT
```

This results in the following messages:

```
APP 1 INIT - START IRU
APP 2 INIT - START IRU
APP 3 INIT - START IRU
APP 6 INIT - START IRU
APP 8 INIT - START IRU
1-APP 4 INITIALIZE STEP CONTROL QUEUES- ANSWER NO OR GO
2-APP 5 INITIALIZE STEP CONTROL QUEUES- ANSWER NO OR GO
3-APP 12 INITIALIZE STEP CONTROL QUEUES- ANSWER NO OR GO
4-APP 15 INITIALIZE STEP CONTROL QUEUES- ANSWER NO OR GO
```

All nonrecoverable application groups are initialized. The operator has to confirm initialization of each of the recoverable application groups, because recoverable information exists from the previous session. The operator responds GO to proceed with initialization, NO to stop initialization.

2. After recovery options are selected for all application groups, the following keyin is entered:

```
0 DOWN
```

This terminates the outstanding operator message.

## 11.5. ARCMR—Monitor Run Control

The ARCMR keyin displays information about the monitor run and terminates the monitor run with or without a dump.

### 11.5.1. Formats

#### ARCMR TERM

ARCMR TERM

Terminate the monitor run without producing any dumps. The termination process involves deregistering components that were registered with monitor run.

If any components are registered, the following message is displayed:

```
ARCMR: n COMPONENTS REGISTERED WITH MONITOR RUN...CONTINUE TERM Y/N?
```

If you answer Y, the components are deregistered, and both the monitor run and the backup monitor run completes (FIN) normally. If you answer N, the run is not terminated.

#### ARCMR ABORT

ARCMR ABORT

Terminate the monitor run with an ER EABT\$. A postmortem dump and a dump of the common data bank are produced for both the primary and backup monitor runs. Aborting the monitor run does not deregister components. If a system recovery occurs, these components are still registered and is restarted. To avoid unwanted components from restarting, initialize monitor run. Only those components reregister with monitor run will restart.

#### ARCMR STATUS

ARCMR STATUS

Display information about the monitor run in the following format:

```
ARCMR: PRIMARY RUNID: prunid BACKUP RUNID: brunid
ARCMR: RECOVERY MODE: mode yymmdd/hhmm:ss
{ARCMR: NUMBER OF COMPONENTS REGISTERED: n} or
{ARCMR: NO COMPONENTS REGISTERED}
ARCMR: RUNIDS OF REGISTERED COMPONENTS: runid1 runid2 ...
ARCMR: runid3 runid4 ...
.
.
```

where:

*prunid*

is the run-ID for the primary monitor run.

*brunid*

is the run-ID for the backup monitor run. If the backup run's run-ID is not available, the brunid field displays the following:

NOT AVAILABLE

*mode*

is the recovery mode: either SHARED or LOCAL.

*yymmdd/hhmm:ss*

is the time of the keyin.

*n*

is the total number of components registered with the monitor run. If *n* is zero, the message

ARCMR: NO COMPONENTS REGISTERED

is displayed.

*runid1 runid2 ...*

The run-IDs of the components registered with the monitor run. If no components are registered, NONE is displayed.

### ARCMR RUNSTATUS

ARCMR RUNSTATUS *runid*

Displays status information for a monitor run component in the following format:

```
ARCMR: RUNID: runid yymmdd/hhmm:ss
ARCMR: HEARTBEAT ID: id HEARTBEAT INTERVAL: n MS
ARCMR: TERMINATE BEFORE RESTART: yes/no
ARCMR: RECOVERY RUNSTREAMS:
ARCMR: COMPONENT: cfile
ARCMR: LOCAL: lfile
ARCMR: SHARED: shfile
```

where:

*runid*

is the run-ID of the component on the ARCMR keyin.

*yymmdd/hhmm:ss*

is the time of the keyin

*id*

is the heartbeat-ID of the component (in decimal).

*n*

is the heartbeat interval (5,000 to 30,000 milliseconds).

*yes/no*

indicates whether the run should be terminated before starting a recovery runstream for the component. If set to yes, terminate the current run and start the recovery runstream. If set to no, start the recovery runstream without terminating the run.

*cfile*

is the file or *file.element* that contains the component recovery runstream. The component recovery runstream is started (@START) to recover a component when the component fails, but the system is still active. If a runstream is not registered, NONE appears in the field.

*lfile*

is the file or *file.element* that contains the local recovery runstream. The local recovery runstream is used to recover components after a host failure and reboot. If a runstream is not registered, NONE appears in the field.

*shfile*

is the file or *file.element* that contains the shared recovery runstream. The shared recovery runstream is used to recover a component on the backup host when the production host fails. If a runstream is not registered, NONE appears in the field.

## **11.5.2. Restrictions**

The ARCMR keyins are available only when monitor run is active on the system.

## 11.6. AT—Controlling the Audit Trail

The AT keyins are used to perform various manual functions to audit trails of all types. There are currently four supported types of audits trails: Step Control, System Log (LOG), Transaction Processing Monitor (TPM) and Capacity On Demand (COD). Although each type of trail collects a unique set of data, all audit trails are constructed, configured, and manipulated exactly the same.

The parameters for each function on the AT keyin vary according to trail configuration. *L<sub>i</sub>* refers to the audit trail leg, leg 1 or 2. The variable *n* refers to the audit trail name. For Step Control type trails, *n* is the application number associated with this trail. For the system log trail, the default name of the trail is LOG. For TPM type trails, the default trail name is TPM. The system COD default trail name is COD.

### 11.6.1. Formats

#### **AT CL**

AT *n* CL

Closes the audit trail for the application group specified by *n* and frees its audit data files. This keyin can be restricted by system generation.

#### **AT DF**

AT *n* DF

Puts the audit trail of the application group specified by *n* in a deferred state. The trail must be in the down (DN) state before using this keyin.

#### **AT DN**

AT *n* DN

Places the inactive (closed) audit trail specified by *n* in a down state, if not restricted by system generation.

#### **AT FS**

AT *n* FS

Displays the status of the audit trail specified by *n*. When *n* = ALL, it displays the status of all configured audit trails. When *n* = LOG, it displays the status of the audit trail for the system log. When *n* = COD, it displays the status of the audit trail for COD. When *n* = TPM, it displays the status of the audit trail for TPM.

#### **AT IN**

AT *n* IN

Starts recovery of the audit trail specified by *n*. The audit trail must be in the deferred (DF) state before this keyin is used.



**AT LC**

AT *n* LC *Li*

Changes the audit trail leg specified by *i*. If the trail is open, the change takes effect when the leg is swapped to the next F-cycle. If the trail is closed, the change takes effect when the trail is opened. See Section 11.6.2 for details on what can be changed and how it is done.

**Note:** You cannot use the AT *n* LC *Li* keyin to switch the audit trail from a fixed disk to a removable disk or conversely. A system generation is required to switch an audit trail from a fixed to a removable disk or from a removable to a fixed disk.

**AT LD**

AT *n* LD

Displays the media summary for audit trail *n*. When *n* = ALL, it displays the media summary for all audit trail legs.

**AT MV**

AT *n* MV *unit1* [, *unit2*]

Allows an audit trail to be moved from its current tape drive to another, spare, drive.

Parameter *unit-1* specifies a tape drive mnemonic for a "spare" currently assigned to Exec audit trail *n*. Parameter *unit-2* specifies a tape unit to which to move *unit1* and can be any one of the following:

- Assign mnemonic (HIC or T, for example)
- Tape control unit mnemonic (T34, for example)
- *unit1* mnemonic, for instance (T34A0, for example)
- Void, which allows Audit Control to select the most compatible assign device to *unit1*

**AT NR**

AT { *n* | ALL } NR

Bypasses the recovery of the specific audit trail *n*, or ALL configured audit trails except for the system LOG audit trail. This action leaves each target audit trail in a deferred (DF) state.

The NR keyin is only permitted during system recovery prior to the start of audit trail recoveries. You can assure this by setting jump key 1 and entering the NR keyin before you respond to the MODIFY CONFIG read/reply message.

The NR keyin is not permitted for the system LOG audit trail.

### AT OP

`AT n OP [name-1/name-2, spare-unit-1/spare-unit]`

Opens the audit trail specified by *n* so that it is available for auditing.

Parameters *name-1* and *name-2* specify which legs of the trails to open, and whether the trail is open for simplex or duplex auditing. They also specify a tape unit for tape audit trails.

Parameters *spare-unit-1* and *spare-unit-2* specify spare or premount tape file assignments.

### AT RM

`AT n RM unit`

Allows you to remove the premount unit specified by *unit* from the audit trail specified by *n*.

### AT SP

`AT n SP unit`

Adds a spare premount *unit* to the audit trail specified by *n*.

### AT SW

`AT n SW unit1[,unit2] [reel-id]`

For manual control of tape swapping, this keyin closes the current tape reel (*reel-id*) for the audit trail specified by *n*, then opens the next reel.

where:

*unit1*

is the name of a tape unit device currently assigned and ready to receive audit data.

*unit2*

is a tape assign mnemonic, such as

- Tape device name (for example, T40A0)
- Tape assign type (for example, HICL)
- Tape control unit name (for example, CU42A)

*reel-id*

is the identifier of the *unit2* tape that is assigned to complete the tape swap. The *reel-ID* is written in the close record of the tape assigned to the *unit1* device before the unit is freed. This provides a forward link to the swap-to device.

**Notes:**

- *Minimum spares are ensured by using unit1 as a spare when appropriate.*
- *Spare unit selection, if called for, is based on the same logical channel assign, the same control unit assign, the same assign type, or the default assign.*
- *If assigned, the unselected spare is placed so that it is selected as the next choice of a swap-to spare (to rotate spares).*

AT *n* SW *Li*

Closes the current mass storage leg, *Li*, for the audit trail specified by *n*; then opens the next generation (file cycle or F-cycle).

The *Li* (leg) parameter can be L1 or L2, and is order-independent.

**AT TI**

AT { *n* | ALL } TI

Reinitializes the specified audit trail, or all audit trails, to their configured values. Each target audit trail is placed in a closed or down state, depending on the RECOVERY system generation option.

You can use the TI keyin to recover from a software problem, or to ensure that parameters that were changed during system generation are applied or reapplied. The TI keyin copies the configuration source of the Audit Control structure to the audit recovery file (ARF). The target trail must be in a not-recovered state (during system boot), or in a deferred (DF) state (after trail recovery completes).

The results of the TI keyin differ, depending on whether the keyin is issued at boot time or after system recovery:

- If you issue the keyin at system boot time, the keyin must take effect prior to the start of audit trail recoveries. To assure this, set jump key 1 and enter the keyin before responding to the MODIFY CONFIG read/reply message.

Even if TI action is set for a trail, the trail is not initialized until the trail is actually recovered. Recovery of trails configured with the RECOVERY,DEFER attribute bypass recovery at boot time; they are not initialized until they are acted upon by an operator AT *n* IN keyin. Likewise, trails targeted by an NR keyin bypass recovery, and initialization of those trails does not occur until after IN processing.

TI action bypasses recovery of open audit trails. Close records are not written, and duplex legs are not resynchronized. PANIC and REBUILD fail if a trail is related to Step Control. The state of a trail upon completion of TI action is closed or down.

- If the TI keyin is issued after audit trail recoveries begin, a trail must be in a deferred state for TI action to be applied. TI action is skipped for trails that are not in DF state. The keyword ALL is not allowed at this time.

Audit close records are written normally and duplex files are synchronized.

The system LOG and COD trails are never in a deferred state, so TI keyins are only allowed for these trails at system boot time (AT LOG TI, AT COD TI, or AT ALL TI keyins).

To initialize an open audit trail after audit trail recoveries are complete, the operator keyin sequence is as follows:

```
AT n CL
AT n DN
AT n DF
AT n TI
```

### AT UP

```
AT n UP
```

Changes the state of the audit trail, specified by *n*, from down (DN) to closed (CL).

## 11.6.2. System Response

The system response to the AT *n* LC *Li* keyin is a message that lets you change parameters:

```
a - AT n Li FIX/REM[,cache] parameters
```

To change one parameter, you must reenter all of them. If you do not reenter a parameter, the system uses a default value. The parameters are

```
[,cache] type/res/gran/max/rfp[/place][,pack-id/...]
```

where:

*a*

is the message number.

*cache*

Currently, cache'ing is not supported. Therefore, only a cache value of DNC is allowed.

*type*

is the media type of mass storage, either FIX or REM. This cannot be changed and must be resupplied as is.

*res*

is the initial memory reserve. The value of *res* must be greater than 0 and less than 262143. It must also be equal to the value for *max*.

*gran*

is the physical granularity used for input/output, which can be TRK (track) or POS (position).

*max*

is the maximum memory reserve. The value of *max* must be greater than 0 and less than 262143. It must also be equal to the value for *res*.

*rfp*

is the removable mass storage audit F-cycle pack-ID placement mnemonic, which can have the following values:

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| RFPON  | F-cycle to pack-ID placement is enabled for this removable file pack set.  |
| RFPOFF | F-cycle to pack-ID placement is disabled for this removable file pack set. |

*place*

is the placement on a controller or on a device.

*pack-id*

is the name of the disk pack when the file is maintained on removable mass storage.

**Notes:**

- *The AT n LC keyin provides flexibility in manipulating files, but use it cautiously. Only someone who understands the effects of modifying the audit trail parameters should use this keyin. For example, the pack-ID can be changed only if a file has no previous F-cycles. When you are changing pack-IDs, the new pack-ID must have the same directory type (standard or shared) as that of the old pack-ID. Also, a small value for max can cause frequent swaps to occur.*
- *The parameters for the audit trail changed with the AT n LC keyin are saved in the audit recovery file (ARF) for the target audit trail. They stay in effect until a system tape boot occurs, which results in the ARF being rewritten due to a change in the static configuration of the audit trail. When this occurs, the parameters established by system generation replace those previously set by the keyin, except for the cache parameters as described above.*

### 11.6.3. Restrictions

When changing the parameters of the system log audit trail, always set the initial reserve and maximum granules to the same value.

The AT *n* NR and AT ALL NR keyins are permitted only during system recovery before all audit trail recoveries begin (for example, before the MODIFY CONFIG message is answered). The NR keyins are not permitted for the system LOG audit trail.

### 11.6.4. Examples

#### Example 1

```
AT 3 LC L1
```

The system responds with the following statement about audit trail 3, leg 1, where 0 is the message number:

```
0-AT 3 L1 FIX F/128/TRK/128
```

To change the maximum number of tracks (*max* parameter) for audit trail 3, leg 1 from 128 to 256, respond with the message number and the required changes: (You must also change the initial memory reserve, *res*, to 256):

```
0 FIX F/256/TRK/256
```

The system then displays the final results to verify your change:

```
AT 3 L1 FIX F/256/TRK/256
```

#### Example 2

```
AT 2 LC L2
```

The system responds with the following statement about audit trail 2, leg 2:

```
0-AT 2 L2 REM F50/128/TRK/128,10P
```

To change the pack-id parameter from 10P to 50P for audit trail 2, leg 2, respond with the message number and the required changes:

```
0-REM F50/128/TRK/128,50P
```

The system displays the final results:

```
AT 2 L2 REM F50/128/TRK/128,50P
```

#### Example 3

```
AT 3 LC L2
```

The system responds with the following statement about audit trail 3, leg 2:

```
0-AT 3 L2 FIX F/200/TRK/200
```

Changes to system defaults for audit trail 3, leg 2, respond with the message number and the required changes:

```
0 FIX
```

*or*

```
0 FIX///
```

The system displays the final results:

```
AT 3 L2 FIX
```

**Example 4**

```
AT 7 LC L1
```

The system responds with the following statement about audit trail 7, leg 1:

```
0-AT 7 L1 REM F50/256/TRK/256,20P
```

To change to system defaults for audit trail 7, leg 1. Respond with the message number and the required changes:

```
0 REM F50/256//256,20P
```

The system displays the final results:

```
AT 7 L1 REM F50///,20P
```

**Example 5**

```
AT 2 LC L2
```

There is an XPC on this system, so the system responds with the following statement about audit trail 2, leg 2:

```
0-AT 2 L2,DOC REM F50/256/TRK/256,20P
```

To change to system defaults for audit trail 2, leg 2, respond with the message number and the required changes:

```
0 REM,DEF F50/256//256,20P
```

The system displays the final results:

```
AT 2 L2 REM,DEF F50/256//256,20P
```

**Example 6**

```
AT LOG MV T34A0
```

Moves the log audit trail from T34A0 to another drive.

The Exec selects a new unit that is most compatible with unit T34A0. The Exec releases device T34A0. If the selected unit cannot be acquired, the original unit remains in use, allowing you to issue the AT *n* MV keyin again.

**Example 7**

```
AT 1 MV T34A1,T34A2
```

Moves audit trail 1 from T34A1 to T34A2.

The Exec switches to tape device T34A2 and releases device T34A1. If the new unit cannot be acquired, the original unit remains in use, allowing you to issue the AT *n* MV keyin again.

### Example 8

```
AT 1 MV T34A2,T34A
```

Moves audit trail 1 from T34A2 to another drive.

The Exec switches to a tape drive (that is configured to control unit T34A) as the new spare tape unit and releases tape unit T34A2. If the new unit cannot be acquired, the original unit remains in use, allowing you to issue the AT *n* MV keyin again.

### Example 9

```
AT 2 MV T34A1,HIC
```

Moves audit trail 2 from T34A1 to another drive.

The Exec selects a tape drive that can satisfy the assign mnemonic HIC, and releases the premounted tape unit T34A1. If the new unit cannot be acquired, the original unit remains in use, allowing you to issue the AT *n* MV keyin again.

### Example 10

```
AT 3 MV T34A0,T34A0
```

Moves file assigned to T34A0. The Exec issues a tape swap for the file assigned to T34A0. A labeled blank reel must be mounted on device T34A0. If the new unit cannot be acquired, a new labeled blank tape might or might not be solicited. Another tape will always be solicited to replace the source unit to preserve the premount minimum value established for audit trail *n* by system generation. If the premount minimum value is still preserved in the event that the destination tape assignment fails, no further replacement is solicited. The actual number of spares for audit trail *n* is reduced by 1.

### Example 11

```
AT 3 OP
```

Audit trail 1 is opened and tape or mass storage file assignments are made as determined by system generation default specifications.

### Example 12

```
AT 3 OP L1
```

Opens mass storage audit trail 3, leg 1.

### Example 13

```
AT 1 OP L2
```

Opens mass storage audit trail 1, leg 2.



**Example 14**

```
AT 1 OP L1/L2
```

Opens mass storage audit trail 1, legs 1 and 2.

**Example 15**

```
AT 1 OP T20A1
```

Opens audit trail 1 as simplex on device T20A1 with no spare.

**Example 16**

```
AT 1 OP T20A,T20A
```

Opens audit trail 1 as simplex on control unit T20A with a spare on the same control unit.

**Example 17**

```
AT 1 OP T20A/T36A
```

Opens audit trail 1 as duplex with one unit on T20A/T36A control unit T20A, one unit on control unit T36A, and two spares, one on control unit T20A and one on control unit T36A.

**Example 18**

```
AT 1 OP T20A/T36A, T20A
```

Opens audit trail 1 as duplex with one unit on control unit T20A, one unit on control unit T36A, and one spare on control unit T20A. The spare is used by the first active unit to swap.

**Example 19**

```
AT n SW unit1
```

Swap from *unit1* without specifying a reel-ID.

If a spare is available, the system selects the swap-to unit. If the audit trail is configured for self-swap and there is no spare assigned, the operator is solicited for a reel-ID. If the trail is not configured for self-swap and there is no spare assigned, the system assigns the swap-to device before releasing the *unit1* tape.

**Example 20**

```
AT n SW unit1 reel-id
```

Swap from *unit1* to the same unit (self-swap) with the *reel-ID* specified. Self-swap must be configured, otherwise reel-ID is ignored and a swap to another unit is issued.

### Example 21

```
AT n SW unit1,unit1
```

Swap from *unit1* to the same unit. The operator is solicited for the reel-ID.

### Example 22

```
AT n SW unit1,unit2
```

Swap from *unit1* to *unit2*.

If a spare is assigned, *unit2* must be a tape assign mnemonic (tape device name, tape assign type, or tape control unit name). If a spare is not assigned, the system assigns an appropriate unit before releasing *unit1*.

### Example 23

```
AT n SW unit1,unit1 reel-id
```

Self-swap on *unit1* with the specified *reel-id*.

### Example 24

```
AT logtrailname FS
```

Displays the status of the audit trail for the system log file, as follows:

```
AT LOG OP
AT LOG L1 FIX(4,28) 23% 00:00:43 CDT
AT LOG L1 BLK 24 TBSN 39
```

where:

|              |                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| OP           | indicates that the trail is open.                                                                                                                         |
| L1           | is the leg-ID.                                                                                                                                            |
| FIX          | indicates fixed mass storage.                                                                                                                             |
| 4,28         | 4 is the number of currently cataloged file F-cycles for leg 1 for the LOG trail. The currently active file F-cycle assigned is absolute file F-cycle 28. |
| 23%          | indicates the percentage of the file that is full.                                                                                                        |
| 00:00:43 CDT | is the time that the file was assigned.                                                                                                                   |
| BLK 24       | indicates the file block number.                                                                                                                          |
| TBSN 39      | indicates the trail block sequence number.                                                                                                                |

### Example 25

```
AT LOG LD
```

Displays the mass storage file assignment parameters for the system log file audit trail, as follows:

```
AT LOG L1 FIX F/5000/TRK/5000
```

where:

|      |                                                                             |
|------|-----------------------------------------------------------------------------|
| L1   | is the leg-ID.                                                              |
| FIX  | indicates fixed mass storage.                                               |
| F    | indicates the addressing mode of the file (in this case, sector formatted). |
| 5000 | is the number of granules (tracks) initially required by the field.         |
| TRK  | is the granule size (TRK = 1 software track).                               |
| 5000 | is the maximum number of granules (tracks) of the file.                     |

### Example 26

```
AT LOG SW L1
```

Closes the current generation of the system log file and opens a new one.

The system displays the following three lines:

```
AT LOG L1 FIX(4,26) *CLOSED* 10:50:30 CST
AT LOG L1 BLK 1791 TBSN 3496
AT LOG L1 FIX(4,26) *SWAP* FIX(5,27) 19:50:21
```

where:

|              |                                                                                                              |
|--------------|--------------------------------------------------------------------------------------------------------------|
| L1           | is the leg-ID.                                                                                               |
| FIX          | indicates fixed mass storage.                                                                                |
| 4,26         | indicates the number of currently cataloged generations for leg L1 and the absolute generation for the file. |
| *CLOSED*     | indicates that the trail is closed.                                                                          |
| BLK 1791     | indicates the file block number.                                                                             |
| TBSN         | indicates the trail block sequence number.                                                                   |
| 10:50:30 CST | is the time that the file was assigned.                                                                      |
| 4,26         | is the file cycle that was closed.                                                                           |
| 5,27         | is the file cycle that you opened.                                                                           |

### Example 27

```
AT logtrailname SW L1
```

Closes the current F-cycle of the system log file and opens a new one.

### Example 28

```
AT 1 FS
```

This keyin displays the following:

```
AT 1 L1 REM(2,1) PL:ATL1P1 16:13:01
AT 1 L1 BLK 17 TBSN 14
```

```
AT 1 S1 REM(2,2) PL:ATL1P2 16:13:03
AT 1 L2 FIX(2,1) PL:*CU20H2 16:13:01
AT 1 L2 BLK 17 TBSN 14
AT 1 S2 FIX(2,2) *SPARE* 16:13:03
```

which shows that L1 is configured as removable and L2 is configured as fixed. Placement (PL:) is defined for fixed and removable files. F-cycle 1 is contained on pack-ID P1 and F-cycle 2 is contained on P2.

```
AT 1 SW L1
```

This keyin swaps L1 removable F-cycle 1 to spare F-cycle 2 and displays the following:

```
AT 1 L1 REM(2,1) *CLOSED* 16:13:02
AT 1 L1 BLK 17 TBSN 14
AT 1 L1 REM(2,1) *SWAP FROM* 16:13:02
AT 1 L1 *SWAP TO* 16:14:40
```

```
AT 1 FS
```

This keyin displays the following:

```
AT 1 L1 REM(3,2) PL:ATL1P2 16:13:06
AT 1 L1 BLK 1 TBSN 14
AT 1 S1 REM(3,3) PL:ATL1P3 16:13:06
AT 1 L2 FIX(2,1) PL:*MS20H2* 16:13:01
AT 1 L2 BLK 17 TBSN 14
AT 1 S2 FIX(2,2) *SPARE* 16:13:03
```

which shows removable F-cycle 2 is contained on P2. The spare message is not changed.

```
AT 1 LC L1
```

This keyin disables removable file placement for AT 1 L1 and displays message 1, as follows:

1. 0-AT 1 L1 REM,DOC F70M/10000/TRK/10000/RFPON
2. REM,DOC F70M/10000/TRK/10000/RFPOFF

The operator responds to read message 1 by changing RFPON to RFPOFF in message 2.

```
AT 1 LD
```

This keyin displays the following:

```
AT 1 L1 REM,DOC F70M/10000/TRK/10000/RFPOFF,ATL1P1/ATL1P2/ATL1P3
AT 1 L2 FIX,DOC F70/10000/TRK/10000/*CU20H2
```

which verifies that the RFP status of L1 has been changed to disabled.

## 11.7. B—Setting the Maximum Number of Batch Runs

The B keyin sets the maximum number of open batch runs allowed in the system (maximum 99,999).

### 11.7.1. Format

B *nnnnn*

where *nnnnn* is 0 through 99999.

## 11.8. BL—Display Runs in Backlog

The BL keyin displays information about runs in the backlog queue.

### 11.8.1. Formats

BL[,NU] [Dnnn] [ID1,ID2,...IDn]

where:

*U*

indicates that the ID arguments are user IDs.

*N*

indicates to negate any mask, that is display all runs except those specified in the ID fields.

(*Dnnn*)

indicates the maximum number of runs displayed.

One or more IDs can be specified, separated by commas and with or without masking. If "U" is not specified, the IDs refer to run IDs; if "U" is specified, the IDs refer to user IDs.

### Examples

|          |                                                                                      |
|----------|--------------------------------------------------------------------------------------|
| BL D     | Detailed output of all runs held in backlog.                                         |
| BL D20   | Detailed information of the first 20 runs held in the backlog.                       |
| BL BA3*  | All runs in the backlog that start with BA3.                                         |
| BL D319? | Details for runs starting with D319 with a maximum of five characters in the run ID. |
| BL,U U1  | All runs that execute under user ID U1.                                              |
| BL,NU U2 | All runs that will not execute under user ID U2.                                     |

BL [D | Dnnn]

When BL is specified alone, the keyin displays all runs in backlog; those in initial facility hold are preceded by an asterisk. When D is specified, the system displays detailed information about each run; runs are ordered by priority (excluding the held runs). When *nnn* is specified with D, up to a maximum of *nnn* runs are displayed.

### 11.8.2. System Response to BL Keyin

The response to a BL keyin is

```
BACKLOG: run-id run-id run-id run-id
```

All backlog runs are displayed, in no particular order. Four run-IDs are displayed per line. Held runs are preceded by asterisks.

### 11.8.3. System Response to BL D Keyin

The response to a BL D keyin is as follows:

```
BACKLOG: CAND./HOLD=num-candidate-runs/num-held-runs yyyy-mm-dd hh:mm:ss zone
run-id/site-id*hold run-id/site-id*hold run-id/site-id*hold
```

All backlog runs except the held runs are displayed in order of priority. Three run-IDs are displayed per line. On BL *Dnnn* keyin, the first *num-candidate-runs* runs of the expected output by BL D are displayed. The possible values for each entry are as follows:

*hold*

is the hold state, which can be any of the following:

|    |                                                                                                                        |
|----|------------------------------------------------------------------------------------------------------------------------|
| AD | The run is in an initial facility hold waiting for access to an arbitrary device.                                      |
| CT | The run is in an initial facility hold waiting for access to a cataloged file.                                         |
| F  | This indicates a FIN statement hold (the @FIN statement has not yet been encountered as the Exec reads the runstream). |
| MS | The run is in an initial facility hold waiting for an available disk or mass storage.                                  |
| O  | A CS H keyin is in effect for this particular run.                                                                     |
| OH | The run is in an initial facility hold waiting for access to specific facilities, other than AD, CT, MS, and TP.       |
| S  | This indicates an S option hold.                                                                                       |
| ST | The run start time is pending.                                                                                         |
| TP | The run is in an initial facility hold waiting for an available tape.                                                  |

*num-candidate-runs*

is the number of candidate runs for opening in backlog at the time of keyin. The value of *num-candidate-runs* can be up to 3 digits; a larger value results in a BL KEYIN ERROR message.

*num-held-runs*

is the number of held runs in backlog at the time of keyin.

*yyyy-mm-dd hh:mm:ss zone*

is the date and time of the BL D keyin. The zone field contains the configured time zone mnemonic.

*run-id*

is the run identification of the run in backlog.

*site-id*

is the symbiont name from which the run was submitted.

\*

is the separator between the site-ID and the hold state.

## 11.8.4. Examples

BL

Displays all run-IDs in backlog, as follows:

```
BACKLOG: JHND2 X0427X WELLT XXPAUF
BACKLOG: SKT4 KILROY DIUUT LBHTLX
BACKLOG: WELLSB JP
```

BL D

Displays detailed information of all runs in backlog, as follows:

```
BACKLOG: CAND./HOLD=64/16 2013-04-26 17:18:26 CDT
SKT4/ WELLL/ WELC/
FREEDQ/*ST BCK022/*ST
```

BL D5

Displays detailed information on the first five runs held in backlog, as follows:

```
BACKLOG: CAND./HOLD=68/16 2013-04-26 16:55:21 CDT
FM513/ QQ38R6/ TAPMPP/
LRBTIP/ RLBQ1/
```

## 11.9. BP—Controlling the Breakpoint Setting

The BP keyin sets, clears, and displays the breakpoint register setting and should be used only at the direction of your site administrator. The breakpoint register is set to trap a suspected problem at a specific main storage location.

### 11.9.1. Formats

#### Display

BP

Displays the current breakpoint register setting.

#### Clear

BP 0

Clears the breakpoint register setting.

#### Set

BP *address operation*

Sets the breakpoint register setting.

where:

*address*

is where the breakpoint is to occur. It has the form

*L, BDI, offset*

where:

*L*

is a symbolic name specifying the address tree level. SYS is the only name supported on all systems and is the default if a level is not supplied.

*BDI*

is either a 1- to 6-digit octal number or the symbolic bank name on an Exec bank. This subfield, along with the L specifies the bank to be changed or displayed.

*offset*

is a 1- to 6-digit octal number specifying the offset into the bank. This offset differs from the relative address used when the segment field is supplied. This is relative to the bank base.



*operation*

is the operation you want to perform with this keyin. The following operations can be entered in any combination, and are interpreted as follows:

- P indicates that a breakpoint interrupt is desired when the specified main storage address is referenced as an instruction.
- R indicates that a breakpoint interrupt is desired when the specified main storage address is referenced as the operand on a read (load) operation.
- W indicates a breakpoint interrupt is desired when the specified main storage address is referenced as the operand on a write (store) operation.

### 11.9.2. System Response

A KEY ERROR message is displayed if the *address* field is missing, contains other than an octal integer, or exceeds 24 bit. A KEY ERROR message is also displayed if the *operation* field contains a letter other than P, R, or W, or it contains any of these letters more than once.

The occurrence of the breakpoint interrupt for the Exec results in an XER 0160.

The system response to a BP 0 keyin is as follows:

BP CLEARED

In response to any other BP keyin, the input is echoed.

## 11.10.C—Display or Modify Main Storage

The C keyin displays main storage locations or modifies them to any value.

Refer to the *Exec System Software Administration Reference Manual* (7831 0323) for information about this keyin.

## 11.11. CJ—Clear Jump Keys

The CJ keyin clears jump keys.

### 11.11.1. Formats

*CJ number1,number2,...numberN*

Clears the jump keys specified by number (1 through 36). See Section 3 for a list and description of the jump keys.

*CJ A[LL]*

Clears all jump keys (1 through 36).

### 11.11.2. System Response

The system displays one of the following messages:

```
JUMP KEYS SET: number1, number2,...numberN
```

```
JUMP KEYS SET: NONE
```

where *number* is the number of a jump key that is set.

A jump key is not cleared until the system displays the above response at the system console indicating the jump key is no longer set.

## 11.12. CK—Checkpoint

The CK keyin saves the state of a batch run. If you need to take the system down for service or if the system fails, the save gives you a point from which to restart the run. The RS keyin is used to restart the run after a CK keyin. See also

- The *System Console Messages Reference Manual* (7833 3275) for the messages that you see on the console during a checkpoint/restart procedure
- The *ClearPath OS 2200 Software Products Installation Guide* (7831 0612) for additional information on the CKRS product.

### 11.12.1. Format

```
CK, [options] [filename,] run-id, [equipment, reel/reel],,,, [CTL-pool]
```

where:

*options*

can be any of the following, and you can specify more than one option:

- |   |                                                                                                                                                                                               |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A | Do not terminate for certain nonfatal errors.                                                                                                                                                 |
| B | Do not save the absolute element being executed. Checkpoint always saves the current state of the program, however. Read access to the program file is required to save the absolute element. |
| C | Checksum the checkpoint data.                                                                                                                                                                 |
| J | Create an unlabeled tape.                                                                                                                                                                     |
| M | Print full diagnostic messages without the diagnostic snapshot while processing the full checkpoint data.                                                                                     |
| N | Do not print error messages or diagnostics in the run's print file. This must be used on production runs that are printing reports.                                                           |
| P | Display diagnostics on the console.                                                                                                                                                           |
| Q | Cause the full checkpoint common bank to put diagnostic snapshots in the run's print file. This must be used only for debugging purposes.                                                     |
| R | Save the symbiont files for the run.                                                                                                                                                          |

- T Terminate the run when the checkpoint is complete. If any errors occur during the checkpoint (for example, the checkpoint is not valid), the run is not terminated.
- Z Cause the partial checkpoint common bank to put diagnostic snapshots in the run's print file. This must be used only for debug purposes.

*filename*

is the file name to which the checkpoint is written. If the file name is omitted, CKRS uses the file name C\KRS\$TAPE. You must type a period at the end of the file name.

*run-id*

is the run-ID of the run to be checkpointed.

*equipment*

is the equipment type on which the checkpoint tape is mounted. The default is T.

*reel/reel*

is the reel number of the tape on which the checkpoint is written.

*CTL-pool*

is the CTL pool name used to allocate a BLANK tape on which the checkpoint is written. Enter NONCTL in this field to use freestanding cartridge tape drives.

## 11.13.CMR—Console Message Redisplay

The CMR keyin redisplays console messages, sets the parameters for how messages are redisplayed, or terminates or interrupts console message redisplay.

The format of the CMR keyin depends on the operation to be performed. There are CMR keyins for requesting message redisplay, controlling CMR use (at console and printer), terminating CMR and interrupting CMR.

### 11.13.1. Format for Requesting Message Redisplay

```
CMR[,option] time1: time2
```

where:

*option*

can be either of the following characters:

- B Breakpoints the requested messages. If this CMR call selects any messages from the CMR database, they are printed on the selected printer.
- R Displays console messages in reverse order. This option is valid only when making a new redisplay request.

*time1*

specifies the desired starting time (or, if you chose the R option, ending time). This time is either the actual time of day (*hhmm* based on a 24-hour clock) or is relative to the current time in minutes (*mm*).

*time2*

specifies the desired ending time (or, with the R option, starting time). This time cannot be relative, and you can use it only when *time1* is the actual time of day.

When specifying the starting and ending times, give the times in chronological order and specify times less than 24 hours old. The current time must be after the starting and ending time or equal to only one of them.

**Note:** Include all four digits when specifying time of day. Relative times must not exceed 60 minutes.

CMR [ +*mm* | -*mm* ]

Skips forward or backward *mm* minutes (*mm* must be between 1 and 60). Skipping past the ending time causes completion of the request. You can adjust the beginning time.

CMR provides special commands to interrupt previous requests from the same console. You cannot, however, cancel messages that are generated but not yet physically displayed on the console. The DELAY parameter slows message redisplay so that you can interrupt a request. To eliminate unwanted remote console output, use the @@X O statement. Refer to the *Executive Control Language (ECL) and FURPUR Reference Manual* (7830 7949) for more information on the @@X statement.

CMR CAN[CEL]

Cancels the previous request. An error message results if there is no active request.

CMR SUS[PEND]

Suspends the current request. An error message results if this console is not active.

CMR RES[UME]

Resumes the previous request. An error message results if this console is not suspended.

**Note:** When the request is breakpointed (using the B option on the CMR keyin), only the CANCEL command is allowed. The CANCEL command stops the redisplay and deletes the alternate print file.

### 11.13.2. Format for Controlling CMR Use

The format of a CMR control keyin is as follows:

*CMR function*

where *function* is one of the following:

TRU[NCATION] ON/OFF

lets you truncate long messages or display them on two lines. Long messages might occur because each redisplayed message has its original time displayed with it. The parameter is initially set to ON for all consoles. You can change the default value at execution time.

DEL[AY] *m/s*

specifies the length of time *m/s* between successive lines of redisplayed output (0 to 2,000 msec). This keeps redisplayed messages from locking out normal console traffic. The default delay time for all consoles is 150 msec. You can change the default at execution time.

PRI[INTER] *devnam*

Specifies the printer name (*devnam*) for breakpointed requests. The default printer for all consoles is PR; you can change the default printer name at the execution time.

TYPE[E] *type*

where *type* can be one or more of the following parameters, separated by commas:

|     |                                               |
|-----|-----------------------------------------------|
| ALL | Redisplays all four types of console messages |
|-----|-----------------------------------------------|

Or, where *type* can be one or more of the following parameters, separated by commas:

|                    |                                            |
|--------------------|--------------------------------------------|
| 1 or MES[SAGE]     | Console read-only messages                 |
| 2 or OUT[STANDING] | Console read-and-reply messages            |
| 3 or REP[LY]       | Replies to console read-and-reply messages |
| 4 or KEY[IN]       | Unsolicited keyins                         |

### 11.13.3. Format for Displaying CMR Configuration

To display the console redisplay configuration, enter the following:

```
CMR STA[TUS] status
```

where *status* can have one of the following values:

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ALL or blank | Displays all the information.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ID           | Displays the site-ID for the remote console from which this request is entered, or the console number (0 to 3) if entered from a system console.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| TRU[NCATION] | Displays ON or OFF based on the parameter setting.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DEL[AY]      | Displays the delay time in milliseconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| TYPE]        | Displays the selected console message type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| PRI[INTER]   | Displays the selected printer for breakpointed requests.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| CON[SOLE]    | Displays the CMR status for the local console or remote console that entered this request. The status is ACTIVE, INACTIVE, or SUSPENDED.<br><br>CMR provides special commands to interrupt previous requests from the same console. You cannot, however, cancel messages that are generated but not yet physically displayed on the console. The DELAY parameter slows message redisplay so that you can interrupt a request. To eliminate unwanted remote console output, use the @@X O statement. Refer to the <i>Executive Control Language (ECL) and FURPUR Reference Manual</i> (7830 7949) for more information on the @@X statement. |

### 11.13.4. Format for Terminating CMR

To terminate CMR, enter

```
CMR TERMINATE
```

The service run is normally active at all times.

### 11.13.5. System Response

If you receive the following message when you enter a CMR keyin, the CMR service run is not open:

```
CMR KEYIN NOT REGISTERED * [,option function]
```

The CMR service run must be open for you to do the CMR keyin.

A CMR keyin to redisplay messages from a console that is currently redisplaying messages immediately cancels the first request and begins processing the new request.

### 11.13.6. Examples

#### Example 1

```
CMR 2330:0115
```

Displays messages beginning at 11:30 p.m. yesterday and ending at 1:15 a.m. today.

#### Example 2

```
CMR,R 2330:0015
```

Displays messages in reverse order for the period just defined.

#### Example 3

```
CMR,B 1300
```

Displays messages from 1:00 p.m. to the time the request was made. Breakpoints the messages to a print file and prints them.

#### Example 4

```
CMR,R 15
```

Redisplays any messages displayed in the past 15 minutes, in reverse order.

#### Example 5

```
CMR 1500:1200
```

Displays 21 hours of messages, from 1500 (3:00 p.m.) yesterday until 1200 (noon) today.

#### Example 6

```
CMR 1000:1015
```

```
CMR -30
```

```
CMR +45
```

This combination of CMR keyins shows how you can change parameters using relative time. The first CMR keyin requests messages starting at 10:00 a.m. and ending at 10:15 a.m. The second entry modifies that request to messages starting at 9:30 a.m. (10:00 - 30 minutes = 9:30). And the final CMR keyin modifies and completes the request by selecting a time that equals the original end time (9:30 + 45 minutes = 10:15).

### Example 7

```
CMR 1000:1015
CMR SUSPEND
CMR -15
```

This combination of CMR keyins show suspension of the request and changing the start time. The first CMR keyin requests messages starting at 10:00 a.m. and ending at 10:15 a.m. The second CMR keyin temporarily halts display. Also the final CMR keyin backs up the start time to 9:45 a.m. (10:00 - 15 minutes = 9:45) and restarts display.

### Example 8

```
CMR TYPE MES,REP
CMR TYPE OUTSTANDING
CMR TYPE REPLY, KEYIN
CMR TYPE ALL
```

These are examples of the CMR TYPE keyin.

## 11.14.CS—Set or Release Hold Conditions

The CS keyin sets or releases various system-specific and run-specific hold conditions that control and direct run execution.

### 11.14.1. Formats

#### CS A

```
CS A
```

Removes the master hold condition set by CS H. Each run in the backlog can be rescheduled except those individually held by CS H *run-ID*.

```
CS[,N] A runid[,runid,runid,...]
```

When the N option is not specified, removes the hold condition established by a CS H or CS H runid keyin on the specified run or runs. If the N option is specified, removes the hold condition established by the CS H or CS H runid keyin on all runs except those specified on the keyin. The list of runids may contain masking and/or wildcard characters.

```
CS,U[N] A userid[,userid,userid,...]
```

When the N option is not specified, removes the hold condition established by a CS H or CS H runid keyin on all runs that have the userid or userids specified. If the N option is specified, removes the hold condition established by the CS H or CS H runid keyin on all runs that do not have one of the userids specified on the keyin. The list of userids may contain masking and/or wildcard characters.



**CS AD**

CS AD

Removes the hold on demand run scheduling set by CS HD.

**CS ALL**

CS ALL

Removes the master hold condition and all individual console hold conditions on backlogged runs. It does not remove the hold set by CS HT or CS HD.

**CS AT**

CS AT

Removes the hold on inactive demand terminals set by CS HT.

**CS H**

CS H

Holds all runs in backlog. The hold set by this keyin is called the *master hold*. This hold does not include installation runs.

CS[,N] H runid[,runid,runid,...]

When the N option is not specified, places a hold on the backlogged run or runs specified on the keyin. If the N option is specified, places a hold on all backlogged runs except those specified on the keyin. The list of runids may contain masking or wildcard characters or both.

CS,U[,N] H userid[,userid,userid,...]

When the N option is not specified, places a hold on the backlogged run or runs that have the userid or userids specified. If the N option is specified, places a hold on all backlogged runs that do not have the userid specified on the keyin. The list of userids may contain masking or wildcard characters or both.

**CS HD**

CS HD

Places a hold on demand run scheduling. Only demand symbiont control statements are accepted. The remote batch input mode can be entered to place runs in the backlog.

### CS HT

CS HT

Places a hold on demand terminals. It prohibits inactive demand terminals from completing the activation process. Active terminals remain active.

### CS *run-id*

CS *run-id*\*[Px*DhhmmShhmmLyF*]

Changes the priority, deadline time, and start time for *run-id*. The P, D, and S options can be used individually or in any combination (such as CS *run-id*\*D1200).

where:

Px

changes the priority scheduling to the letter specified by x (A through Z), except for a critical deadline run.

*Dhhmm*

sets the deadline time to the wall clock specified by the 24-hour clock value hhmm.

*Shhmm*

sets the start time of *hhmm* specified in 24-hour clock time; this is ignored for an open run.

Ly

indicates a change in the processor dispatching priority (A through Z).

F

forces the *run-id* out of backlog. This option works only on runs that are candidates for opening; runs held in backlog that are on facility hold, S option sequencing, or operator hold cannot be forced out. The F option is used alone.

Table 11-1 shows the hold types, the affected run types, and how the hold is imposed and removed.

**Table 11-1. System and Run Hold Types**

| Hold Type                      | Affected Run Type   | Imposed By                                              | Removed By                      |
|--------------------------------|---------------------|---------------------------------------------------------|---------------------------------|
| A – Master hold<br>(all batch) | Batch (in backlog)  | CS H keyin or mass<br>storage limit has<br>been reached | CS A keyin (or CS<br>ALL keyin) |
| D – Demand                     | Demand (signing on) | CS HD keyin                                             | CS AD keyin                     |

Table 11-1. System and Run Hold Types

| Hold Type               | Affected Run Type                                        | Imposed By                                                                                         | Removed By                                                                                                                   |
|-------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| E – EXPOOL              | Batch (in backlog)                                       | EXPOOL space availability is low. (EXPOOL is the scratch area in main memory for the Exec.)        | EXPOOL space freed up by Exec                                                                                                |
|                         | Batch (active)<br>Demand (signing on)<br>Demand (active) | EXPOOL space availability is low. (EXPOOL is the scratch area in main memory for the Exec.)        | Operator rebooting system, if hung<br><b>Note:</b> E holds might indicate a hung system if they remain for a period of time. |
| I – Run-ID suspended    | All batch and demand runs                                | SP <i>run-id</i> keyin                                                                             | RE <i>run-id</i> keyin                                                                                                       |
| L – Demand lines        | All runs                                                 | CS HT keyin                                                                                        | CS AT keyin                                                                                                                  |
| M – Mass storage        | Batch (in backlog)                                       | File in exclusive use by another run                                                               | Other run frees exclusive use of file.                                                                                       |
|                         | Batch (active)                                           | File is rolled out.                                                                                | ROLBAK reloads file from tape to mass storage.                                                                               |
|                         | Demand (active)                                          | Space unavailable on mass storage to create or expand file                                         | ROLOUT unloads mass storage files to tape, making blank space mass available on storage.                                     |
| O – Outstanding message | All batch, demand, and real-time runs                    | System or user program, or hardware problem                                                        | Answer the outstanding message.                                                                                              |
| R – Individual run-ID   | Batch (in backlog)                                       | CS H <i>run-id</i> keyin                                                                           | CS A <i>run-id</i> keyin (or CS ALL keyin)                                                                                   |
| S – Sequence option     | Batch (in backlog)                                       | S option on @RUN statements of more than one runstream that are read into the system consecutively | Previous run finishing execution                                                                                             |
| T – Tape                | Batch (in backlog)                                       | Cataloged tape file in exclusive use by another run                                                | Cataloged tape file's use freed by other run.                                                                                |

**Table 11-1. System and Run Hold Types**

| Hold Type                                             | Affected Run Type                 | Imposed By                                                                                                     | Removed By                                                                                                     |
|-------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                                       | Batch (active)<br>Demand (active) | Tape drives<br>unavailable                                                                                     | Tape drives become<br>available.                                                                               |
| U – Low availability<br>of internal resource<br>areas | All batch and<br>demand runs      | Low availability of<br>one or more<br>resource areas used<br>internally by Exec to<br>support run<br>execution | System clears<br>situation by itself.<br>Contact site<br>administrator if<br>chronic U-holds are<br>occurring. |

### 11.14.2. Examples

CS A MCB

Removes the hold condition established by a CS H or CS H *run-id* keyin on run MCB. The system displays the message

MCB START

CS AD

Removes the hold condition on demand run scheduling that was set with a CS HD keyin. The system displays the message

ABC123/RE4401 START

CS XYZ\*LJ

Changes the processor dispatching priority for run XYZ to J.

## 11.15. CTL—Control CTL System

The CTL keyin changes the state of a cartridge tape library (CTL), displays information about a CTL system, and clears the CTL-controlled status from all tape drives registered with a particular CTL-ID.

### 11.15.1. Formats

#### General Format

The general format of CTL keyin is:

CTL [ctl-id] [FS | HOLD | REJECT | CLEAR ]

If a ctl-id is not specified all registered CTLs are affected, but there are some restrictions. Refer to section 11.15.3 for more information on restrictions.

**Display State of All CTLs**

CTL

Displays the state of all registered CTLs.

**Display State of a Specific CTL**

CTL *ctl-id* FS

Displays the state of the CTL identified by *ctl-ID*.

**Place a CTL in Hold State**

CTL *ctl-id* HOLD

Places the CTL in a hold state. All cartridge tape requests are held until the CTL is reinitialized. Tape assignments can be held by the hold state.

**Place a CTL in Reject State**

CTL *ctl-id* REJECT

Places the CTL in a reject state. The CTL rejects tape requests that require the CTL.

**Clear CTL Status**

CTL *ctl-id* CLEAR

Clears the CTL-controlled status from all tape drives registered with this *ctl-id* that is not registered with another CTL. Subsequent CTL FS keyins for this CTL display NOT REGISTERED.

**11.15.2. System Response**

The system responds to the CTL FS keyin with one of the following messages:

CTL: *ctl-id* NOT REGISTERED

CTL: *ctl-id* ACTIVE

CTL: *ctl-id* HOLD (AND DEREGISTERED)

CTL: *ctl-id* REJECT (AND DEREGISTERED)

### 11.15.3. Restrictions

The following are the restrictions:

- The CTL HOLD keyin can be performed only when the CTL is in an active state.
- The CTL REJECT keyin can be performed only when the CTL is in an active or hold state.
- The CTL HOLD and CTL REJECT keyins must specify a ctl-id if more than one CTL is registered.
- If a ctl-id is specified on the CTL CLEAR keyin, the CTL must be in a reject state.

### 11.15.4. Examples

CTL

In a system with two registered CTLs, the system responds with this information:

```
CTL: CTL-1A ACTIVECTL:
CTL-1B REJECT (AND DEREGISTERED)
```

CTL CTL-1A HOLD

The system responds with the following:

```
CTL: CTL-1A HOLD (AND DEREGISTERED)
```

CTL CTL-1B CLEAR

The system responds with the following:

```
CTL: CTL-1B AND CTL-1B TAPE DRIVES CLEARED
```

CTL CLEAR

The system responds with the following:

```
CTL: ALL TAPE DRIVES CLEARED
```

CTL BAD

The system responds with the following:

```
CTL: INVALID KEYIN. FORMAT IS CTL [ID] [FS HOLD REJECT CLEAR]
```

## 11.16.D—Display or Set System Date and Time

The D keyin is used to display or set the system date and time.

For additional information on D keyins, their operation, and their responses, see the *Exec System Software Administration Reference Manual* (7831 0323).

OS 2200 systems support the internal use of Coordinated Universal Time (UTC). Some D keyins support the use of a UTC time value.

### 11.16.1. Formats

#### D

D[,UTC]

Displays the system date and time. If the UTC option is specified, it displays the current local time, its corresponding UTC value, and all of the time offsets between the two.

#### D ENTER

D[,UTC] ENTER

Bypasses the clock calendar as the time source and lets the operator set the date and time for a specific partition. This keyin is valid only at MODIFY CONFIG when this read-and-reply message is waiting for its response. (The MODIFY CONFIG message occurs on all tape boots and jump key 1 disk boots.)

After a D ENTER keyin, system clock synchronization is disabled for the system session, because the clock calendar was not used to set the dayclock.

If the UTC option is specified, a UTC time value will be solicited from the operator.

The time entered in response to the D ENTER keyin solicitation is only valid for the current boot session. It cannot be recovered across reboots and must be re-entered.

Not supported on systems without a clock calendar.

#### D RESYNC

D RESYNC

Forces Exec to reset the system dayclock from the clock calendar. This keyin is valid only after the system has reached SYS FIN.

D RESYNC is useful for allowing a site to change to or from daylight saving time without rebooting the system. It can also be used for dayclock and clock calendar resynchronization if clock synchronization was previously disabled.

Not supported on systems without a clock calendar.

### D SHIFT

D SHIFT

Controls a seasonal time shift (for example, Daylight Saving Time).

## 11.16.2. System Response

In response to a D keyin, the system displays the date and time in this format:

The current date and time is *day dd mmm yyyy hh:mm:ss [zone]*

In response to a D ENTER keyin, the following message acknowledges that the keyin has been accepted:

The date and time will be solicited during Exec initialization.

In response to a D RESYNC keyin, the following message indicates that the dayclock and clock calendar are within the Exec's tolerances:

The current date and time is *day dd mmm yyyy hh:mm:ss [zone]*  
Exec Dayclock/Clock-Calendar synchronization is now enabled.

In response to a D SHIFT keyin, the system displays the current indicator setting.

Seasonal time shift is ON.

or

Seasonal time shift is OFF.

## 11.16.3. Examples

### Example 1

D

Displays the date and time as follows:

The current date and time is TUE 02 MAY 2000 15:36:08 CDT

### Example 2

D RESYNC

The following message is displayed when the D RESYNC request has been honored:

The current date and time is THU 24 AUG 2000 23:45:01 CDT  
Exec Dayclock/Clock-Calendar synchronization is now enabled.



**Example 3**

D,UTC

The following message is displayed when the D,UTC request has been honored:

```
The current date and time is WED 11 SEP 2002 03:06:40 UTC
Time zone displacement from UTC is: -06:00:00
Additional seasonal displacement is: +01:00:00
The current date and time is TUE 10 SEP 2002 22:06:40 CDT
```

If a D,UTC keyin is entered on a system that does not support Exec's use of UTC, the keyin is rejected with

```
The current system configuration does not allow D,UTC keyins.
```

**Example 4**

D SHIFT ON

Turns seasonal time shift on.

**Example 5**

D SHIFT OFF

Turns seasonal time shift off.

## 11.17.DC—Copy a Disk Pack

The DC keyin copies the contents of one disk pack to another of the same type. It can be used to copy local and shared disks.

### 11.17.1. Format

```
DC devnam1,devnam2
```

where:

*devnam1*

identifies the source disk.

*devnam2*

identifies the destination disk.

**Note:** Processing of the DC keyin can be aborted with the ABORT keyin.

### 11.17.2. System Response

After a DC keyin, disk capacity is compared. If the destination disk capacity is smaller than the source disk capacity, the copy aborts. If the destination disk capacity is larger than the source disk capacity, the following read-and-reply message is displayed:

```
DC WILL SHRINK devnam2 BY xxx SW TRKS CONTINUE COPY? Y,<N>
```

If you respond with a Y, the extra capacity of the destination disk will be lost. (Small differences in size are usually a result of marking the bad tracks on the source disk.) After the size differences are resolved, the following read-and-reply message is displayed to remind you that any data on the destination disk will be overwritten by the new data:

```
DISK COPY WILL DESTROY FILES ON devnam2 CONTINUE COPY? Y,<N>
```

Respond with an N to abort the copy process. If you continue (by entering a Y), the system displays the following message:

```
DC devnam1,devnam2 - USE DEFAULT PARAMETERS? <Y>/N
```

Normally, DC parameters are supplied by configuration tags DC\_MAX\_READ\_ERRORS (DCMAXRERR) and DC\_SPLIT\_DADS\_ON\_ERROR (DCSPLTDAD).

DC\_MAX\_READ\_ERRORS specifies the maximum number of unrecoverable read errors allowed before the copy is aborted. DC\_SPLIT\_DADS\_ON\_ERROR specifies whether device area descriptors (DAD) can be split during read error recovery. To override these default parameters, enter an N and respond to the following two messages. The first message is as follows:

```
DC devnam1,devnam2 - MAXIMUM NUMBER OF UNRECOVERED READ ERRORS? 0-20<xx>
```

where *xx* is the default number. Enter a number between 0 and 20.

The system then displays the second message:

```
DC devnam1,devnam2 - DAD SPLITTING ON UNRECOVERED READ ERRORS? Y/<N>
```

Respond Y if DADs can be split during read error recovery, or respond N if they can not be split.

Once you have responded to these two messages, the copy begins and the following message is periodically displayed as the copy proceeds:

```
DISK COPY IN PROGRESS FROM devnam1 TO devnam2
```

When the copy is complete, the system displays the following messages:

```
devnam1 COPIED TO devnam2
devnam1 DN, devnam2 [UP|SU|RV] [R|F] pack-id
```

where R indicates a removable disk and F indicates fixed disk.

As part of DC keyin processing, the status of the source unit is changed to DN, and the status of the destination unit is changed to the original status of the source unit. The source unit requires reprogramming before the system can use it again.

### **11.17.3. Restrictions**

These restrictions apply to the DC keyin:

- A shared disk can be copied only to another shared disk, and a local disk can only be copied to another local disk.
- If you are copying a shared disk, file sharing must be up on the copying host.
- Your Exec must have either the Network-MHFS or the XPC version of Multi-Host File Sharing configured and installed to copy shared disks. All hosts active in the file-sharing environment must have the shared copy functionality.
- All active shared copies are aborted if a host enters or is verified leaving the file sharing environment. This can happen with an MH,DN keyin or by answering the "Problems Detected" messages to indicate that the host is actually down.
- If a disk is configured as shared and then moved into the local directory (with an MH,AD keyin) it cannot be copied to another local or shared disk, nor can another local or shared disk be copied to it.
- Physically removable disk types cannot be copied.
- The destination disk must be reserved (RV), unassigned, and have an equal or greater capacity than the source disk.
- The source unit must be up (UP), suspended (SU), or reserved (RV). It must not be caching, and it cannot have any outstanding messages.
- Both disks must have the same equipment types, must have equal prep factors, and must be controlled by equal or equivalent handlers.
- Neither disk can be actively unit duplexing. The unit duplexing association must be broken by using the DN keyin before the DC keyin can be successfully used.
- Neither disk can be under the control of the arbitrary device handler (in use by DPREP).

### **11.17.4. User Access During Copy**

While the copy is underway, users can read and write to the source disk. All user writes are duplexed to both the source and destination disks, and all reads access the source disk.

If an error occurs while writing to the destination disk (either copy I/O or duplexed user I/O), the DC keyin is aborted.

### 11.17.5. Read Error Recovery

If an unrecoverable read error occurs on the source disk and configuration tag DC\_MAX\_READ\_ERRORS is zero, the DC keyin is aborted.

Optionally, the copy handler can attempt to complete the copy in spite of unrecoverable read errors. If configuration tag DC\_MAX\_READ\_ERRORS is greater than zero, or you modified the default parameter to allow unrecoverable read errors, the errors in cataloged files are recorded in the system log file. Each entry in the log contains the error's device relative address, file relative address and the name of the file affected, its qualifier, absolute cycle number, and directory ID. A warning message is also sent to the system console.

**Note:** *The information entered into the system log for a cataloged file cannot be determined for temporary files. If the error occurs on a temporary file, only the device relative address and the I/O status are recorded. The DC keyin then aborts.*

The device area descriptor (DAD) that describes the uncopiable area is marked as corrupted. Subsequent attempts to read this area receive an I/O status of 011. If the DAD describes a TIP duplex file, DNR and DNW are also set.

Since the area described by an existing DAD is often larger than the uncopiable area, the damage can be limited by creating a new DAD that describes only the unreadable spot. For sites with recovery procedures that depend on a particular DAD layout, DAD splitting can be defeated by setting CONFIG\$ tag DC\_SPLIT\_DADS\_ON\_ERROR to zero.

If an error occurs while reading the disk label, the MFDF, or the DRS area, the copy is aborted. If the maximum number of unrecoverable read errors is exceeded, the copy is also aborted.

**Notes:**

- *It is not unusual for read errors to occur during a successful DC keyin. For efficiency, the DC keyin handler tries to read the source disk in large pieces without regard for unallocated tracks. If these large I/Os fail, the area is reread in smaller track and record-sized pieces, skipping unallocated areas. An error must occur while reading at the record level before an unrecoverable read error has occurred. If the operator answers G to an I/O error message and the copy completes with no error message, it means all data was successfully copied by avoiding bad spots or areas that were unallocated.*
- *The DC keyin handler uses special device-relative I/O functions RDRA and WDRA to read and write the target disks. I/O error type-and-read messages that result from these functions display a filename of SYSS\$\*MFDF\$\$, despite the actual file-relative location of the error.*

## 11.18.DISPLVL—Change Switching Priority

The DISPLVL keyin changes the switching priority of demand or batch jobs. The keyin has an activation format, a termination format, and two display (report) formats.

### 11.18.1. Formats

#### Activation Format

The activation format changes the switching priority of a specific job or a category of jobs:

*DISPLVL,option operand,level*

where:

*option operand*

is the job category affected by the keyin:

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| <i>A acct#</i>  | All jobs with the specified account number <i>acct#</i>                              |
| <i>P projid</i> | All jobs with the specified project-ID number <i>projid</i>                          |
| <i>R runid</i>  | The job with the run-ID <i>runid</i>                                                 |
| <i>T type</i>   | All jobs of the specified <i>type</i> ( <i>type</i> can be BATCH, DEDLIN, or DEMAND) |
| <i>U userid</i> | All jobs with the specified user-ID <i>userid</i>                                    |

*level*

is a symbolic name. It can be a default or a site-specific name that indicates one of the named switching levels. The default switching-level names assigned by the Exec are TXN, BATCH, DEMAND, and USER. BATCH and DEMAND exist only if the user switching levels were split into batch and demand groups during system configuration. In this case USER does not exist. USER is a default name only if the user levels are not split into separate batch and demand. Refer to the description of the SYSTEMLEVEL SGS in the *Exec System Software Installation and Configuration Guide* (7830 7915).

The operand may include wildcard or masking characters unless the *T type* is specified. Then the operand must include one of the following: BATCH, DEDLIN, or DEMAND.

### Termination Format

The termination format resets activated jobs to their site's normal priority:

```
DISPLVL,option operand
```

where:

*option operand*

is the job category affected by the keyin:

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| A <i>acct#</i>  | All jobs with the specified account number <i>acct#</i>                              |
| P <i>projid</i> | All jobs with the specified project-ID number <i>projid</i>                          |
| R <i>runid</i>  | The job with the run-ID <i>runid</i>                                                 |
| T <i>type</i>   | All jobs of the specified <i>type</i> ( <i>type</i> can be BATCH, DEDLIN, or DEMAND) |
| U <i>userid</i> | All jobs with the specified user-ID <i>userid</i>                                    |

The operand may include wildcard or masking characters unless the T *type* is specified. Then the operand must include one of the following: BATCH, DEDLIN, or DEMAND.

### Partial Display Format

The partial display (report) format displays a list of active DISPLVL keyins:

```
DISPLVL,option
```

where:

*option*

is the job category to be displayed:

|   |                |
|---|----------------|
| A | Account number |
| P | Project-ID     |
| R | Run-ID         |
| T | Run type       |
| U | User-ID        |

### Full Display (Report) Format

The full display (report) format displays a list of all active DISPLVL keyins:

```
DISPLVL
```

### 11.18.2. System Response

After a DISPLVL keyin, the system displays the following message:

```
DISPLVL: Keyin complete
```

## 11.19.DJ—Display Jump Key Settings

The DJ keyin displays the current jump key settings.

### 11.19.1. Format

```
DJ
```

### 11.19.2. System Response

The system displays the following message:

```
JUMP KEYS SET: number1, number2,...numberN
```

or

```
JUMP KEYS SET: NONE
```

where *number* is the number of a jump key that is set.

## 11.20. DN—Bring Down Components and Interfaces

The DN keyin brings down components and interfaces that are not operating properly or that are required for use in another partition. You can use the DN keyin to bring down the following items:

- An instruction processor (IP)
- An input/output processor (IOP)
- An Extended I/O Interface Processor (XIIP)
- An Extended Processor Complex (XPC)
- A memory module
- A channel module
- A control unit
- A device such as a disk drive, system console, printer, or tape drive
- An interface between a specified channel module and control unit

### 11.20.1. Formats

#### Bring Down a Component

To bring down a component, use the following keyin format:

```
DN component[,com2,com3,...]
```

#### Bring Down an Interface

To bring down an interface, use the following keyin format:

```
DN cmod/[intf#]/cu[,cmod//cu2,...]
```

This keyin makes the interface between the specified channel module (*cmod*) and the control unit (*cu*) unavailable. If you do not specify an interface number (*intf#*), the Exec supplies a value. You can also use this keyin for bringing down an interface to the record lock processor (RLP).

#### Bring Down Devices

To bring down all devices attached to a control unit, use the following keyin format:

```
DN,ALL cu[,cu2,cu3,...]
```

#### Disable XPC Caching

To disable XPC caching, use one of the following keyin formats:

```
DN,XCACHE xpcname
```

```
DN,XCACHE xpcname/LOCAL
```

```
DN,XCACHE xpcname/SHARED
```

where *xpcname* is the logical name of the XPC.

Use the DN,XCACHE *xpcname* keyin to disable local and shared caching for each host using the XPC.

Use the DN,XCACHE *xpcname*/LOCAL keyin to disable a host's local caching using the XPC. If local caching is disabled using this keyin, local caching can only be enabled using an UP,XCACHE *xpcname*/LOCAL keyin.

Use the DN,XCACHE *xpcname*/SHARED keyin to disable shared caching using the XPC.



### Bring Down Disk Packs

To bring down one or more removable disk packs, use the following keyin format:

```
DN,PACK pack-id[/[dir-id]/[eqp-mnemonic],[pack-id]...]
```

where:

*dir-id*

is the directory identifier (SHARED or STD).

*eqp-mnemonic*

is the mnemonic name of the unit the pack is on.

This keyin places the specified packs in the DN state. Refer to Section 5. This keyin should only be used by those with a very specific environment.

**Note:** For use of the DN,PACK keyin in a local environment, the NEWCON configuration parameter REQNBPRST0 must not be set to 0. In a shared environment, the parameter REQNBPRST1 must not be set to 0. See the Exec System Software Installation and Configuration Guide (7830 7915) for more information on these configuration parameters.

### Bring Down a Time Source

To make a time source associated with *iopname* unavailable to the Exec, use the following format:

```
DN,TIME iopname
```

This keyin is useful for forcing the Exec to choose another external time source by removing the current one from the partition.

This keyin is applicable only on systems that support an external time source.

## 11.20.2. System Response

Before the system brings down a control unit, it tries to disable all interfaces to the control unit. If the system cannot do this, it brings down the control unit and displays the following message:

```
DISABLE OF cmod/intf#/cu INTERFACE FAILED
```

If the system stops because it cannot disable an interface, you must recover the system from tape.

### 11.20.3. Restrictions

When you bring down a fixed disk with a status of UP or SU, or when you bring down a disk that is both reserved and assigned, you lose access to all files on that disk. A message is displayed that requires you to respond Y or N. Reply Y to continue processing the keyin; this results in an XER 055 system stop. Reply N to tell the system to ignore the keyin.

When you bring down a removable disk that is assigned, a XER 055 system stop occurs and the status change occurs after the next disk boot. When you bring down a removable disk that is not assigned, the status changes immediately and no XER 055 system stop occurs.

When you bring down an active printer, the device stops immediately. If you want to allow the current activity to finish, lock out the device (see the description of the SM *device* L keyin) before bringing it down.

You cannot bring down the last processor.

Bringing down a tape drive that is assigned to a run causes the run to abort.

You cannot use the DN,PACK keyin on unit-duplexed devices, but you can use the DN keyin on one of the devices to break the Unit Duplexing association.

When you bring down the last XIIP on an XPC, the XPC also becomes unavailable (NA).

If a DN IOP results in no path available to the XPC FO, a new type and reply message is displayed as follows:

```
IOP name is connected to <Vchannelname>.
DN OF <IOpname> will disable this host's last path to XPC name.
Continue to DN, enter DISABLE, else CANCEL.
```

If you answer DISABLE to the message, EXEC continues the DN of IOP. Otherwise, the EXEC aborts processing of the DN IOP.

### 11.20.4. Examples

#### Example 1

```
DN DSKT0
```

Brings down disk DSKT0 and displays this message:

```
DSKT0 DN
```

#### Example 2

```
DN PRA,PRB
```

Brings down printers PRA and PRB and displays this message:

```
PRA DN,PRB DN
```

**Example 3**

```
DN,ALL CUTAPD
```

Brings down all of the devices (TAPED0 and TAPED1) attached to the control unit CUTAPD and displays these messages:

```
CUTAPD UPTAPED0 DN,TAPED1 DN
```

**Example 4**

```
DN XPCA
```

Brings down the XPC on host A and displays the XPC status:

```
XPCA DN
```

**Example 5**

```
DN,TIME IOP18
```

Makes all time sources associated with *iopname* unavailable to the Exec

```

IOP18 (TIME) PRIMARY DN
 SECODARY DN

```

When no time sources are installed on the indicated IOP, the following message is displayed:

```

DN,TIME IOP18
IOP18 has no external time sources installed.

```

If the request makes the last time source unavailable to Exec, the following message is displayed:

```

DN,TIME IOP18
Exec was unable to locate a viable external time source.
The SCF Clock-Calendar is being used instead. Answer GO
IOP18 (TIME) PRIMARY UP
 SECONDARY DN NP

```

If entering console keyins related to CSB maintenance on a system that does not support a clock calendar or an external time source, the following message is displayed:

```
DN,TIME keyins are not supported on this system
```

## 11.21. DU MP—Generate an Online Dump

The DU MP keyin generates an online dump of system memory. An online dump lets you gather data without stopping the system, but because parts of the system are still running during the dump process, the data produced can lack integrity.

If the system is hung, an online dump is not a substitute for a normal system dump. (See Section 6 for more information on system dumps.)

Online dumps can be saved to tape only. They cannot be saved to mass storage, because the dump operation breaks linkages between the dump and any working set of files that exist when the dump is taken.

Online dumps produce only selective dumps of main storage. This minimizes the possibility of an online dump causing either a system stop or a system hang by consuming too much memory. If main storage is too tight when the online dump is started, the system responds with the following message:

```
Main storage is tight, no online dump taken
```

In this case, further DU MP processing is terminated.

### 11.21.1. Format

```
DU MP [run-id]
```

where *run-id* is the unique run-ID of a run or TIP transaction.

The DU MP keyin is normally entered without a run-ID. A run-ID is useful only if a particular run or TIP transaction is of interest.

If a run-ID is included in the keyin, the following apply:

- The resident pages associated with the domain of the run or transaction are included in the dump, along with the resident pages associated with the domains of any application-level banks that the run or transaction currently has based.
- The dump process hangs if the named run is waiting for console or terminal input when you enter the DU MP keyin. In this case a second DU MP keyin is rejected.
- The keyin is also rejected in either of these cases:
  - The specified run-ID or transaction does not exist on the system when the keyin is entered.
  - The specified run-ID or transaction is suspended (SP keyin).

In either case, an error message is routed to the console from which the keyin originated, and no dump is produced. See the *System Console Messages Reference Manual* (7833 3275) for the messages.

### **11.21.2. System Response**

While the dump is executing, all processing activity stops. No runs or files are lost, but real-time programs can lose communications data, and response to users can be delayed.

### **11.21.3. Processing an Online Dump**

When you attempt to process the online dump output, keep in mind that the dump output is a snapshot of a moving (dynamic) system. Structures that are linked or chained together (such as EXPOOL buffers or program control tables), or structures that are in the process of initialization (such as the Exec bank descriptor table), are being dumped at a moment when they are either in the process of expanding/contracting, linking/unlinking, or initializing. Because these structures are being captured at a moment when they are not completely updated or initialized, they most likely are not valid when the online dump is taken.

When you use DMPLIB functions on dynamic or improperly initialized structures, the results cannot be useful.

## **11.22.E—Terminate a Run in Error**

The E keyin terminates an active run in error. If the keyin specifies a batch run, the run terminates with an error and allows a postmortem dump (PMD) or the invocation of the Programmers Advanced Debugging System (PADS). If the keyin specifies a demand run, the current task is terminated but the run continues with the next control statement.

Multiple E keyins can be required to terminate a run if the dynamic configuration parameter `run_termination_task` is not blank. When the `run_termination_task` parameter is set, a termination task can start. If a termination task starts, another E keyin is required to terminate that task.

11.22.1. Format

E[,options] run-id

where:

options

are one or both of the following letter options:

- N

Prevents diagnostics from being written and a PMD from being invoked or prevents PADS from being invoked (unless diagnostics were generated by a previous activity of the object module). Absolute (or legacy) programs use PMD to format diagnostic dumps while object modules invoke PADS. If a previous activity of the object module terminated in error and generated diagnostics, then the N option prevents additional diagnostics from being generated—PADS can still be invoked against the diagnostics already generated. The results of using this option vary slightly depending on the type of program terminated.

  - In a batch run, this option always prevents a PMD/PADS for the run.
  - In a demand run, this option prevents a PMD/PADS for the currently executing program.
  - For a transaction, this option prevents diagnostics from being written (unless the transaction is extended mode and diagnostics were generated by a previous activity of the object module). No entry is written to the error history file.
- S

Terminates a suspended run in error. You can only specify this option for a run that has been suspended. If the S option is specified for a run that has not been suspended, an error message is displayed. The effect of the S option, compared to the run state, is as follows:

| Run State     | S Option Specified  | S Option Not Specified |
|---------------|---------------------|------------------------|
| Suspended     | Error terminate run | Keyin rejected         |
| Not suspended | Keyin rejected      | Error terminate run    |

run-id

is the name of the active run to terminate.

### 11.22.2. Restrictions

If a run is suspended and you do not include the S option, the system responds as follows:

```
run-id IS SUSPENDED - RESUME RUN OR SPECIFY S OPTION ON KEYIN
```

If a run is not suspended and you include the S option, the error message is as follows:

```
run-id IS NOT SUSPENDED - S OPTION NOT ALLOWED, KEYIN IGNORED
```

You can terminate a user run at the user's request; however, do not terminate system runs without your site administrator's approval.

### 11.22.3. Example

```
E BOB
```

Terminates the run named BOB. The system response is as follows:

```
BOB ABORT
BOB ABORT FIN
```

## 11.23. ERUNS keyin

The ERUNS keyin lets you terminate multiple runs with a single keyin. It works like the E keyin and some of the keyin options are the same.

### 11.23.1. Format

```
ERUNS[, {B|D}NSUX] ID1[, ID2, ..., IDn]
```

where:

*B*

specifies to terminate batch runs only.

*D*

specifies to terminate the current task for demand runs only. Both B and D can not be specified.

*N*

indicates to negate any mask, that is terminate all runs except those specified.

*S*

has the same meaning as the S option on the E keyin, it specifies that suspended runs are to be terminated.

*U*

indicates that the ID arguments are user IDs.

*X*

has the same meaning as the N option on the E keyin, which prevents diagnostics from being written for the terminated runs.

One or more IDs can be specified, separated by commas and with or without masking. If the U option is not specified, the IDs refer to run IDs; if the U option is specified, the IDs refer to user IDs.

### Examples

|                  |                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ERUNS A1,A2,A3   | error terminate runs A1, A2 and A3.                                                                                                                                                                                    |
| ERUNS MYRUN      | identical to 'E MYRUN'                                                                                                                                                                                                 |
| ERUNS,SXBU USR1* | terminate all batch runs executing under user IDs that start with USR1. The S option indicates to terminate suspended runs. The X option indicates that no diagnostics will be generated for runs that are terminated. |

## 11.24. EU—Display EXPOOL Usage

The EU keyin displays the EXPOOL usage of your system (in decimal words).

### 11.24.1. Format

EU

### 11.24.2. System Response

The system response to the EU keyin is

```
SIZE(I/C/M) - initsz/currsz/maxisz USE(C/M) - curuse/maxuse E11(C)- cure11
```

where:

*initsz*

is the initial size of EXPOOL.

*currsz*

is the current size of EXPOOL.



*maxisz*

is the maximum total size of EXPOOL.

**Note:** Because the maximum basic EXPOOL size is always taken, the *initsz*, *currsz*, and *maxisz* are all equal.

*curuse*

is the current amount of EXPOOL in use.

*maxuse*

is the maximum amount of EXPOOL used since the last boot or midnight.

*cure11*

is the current amount of EXPOOL index 011 buffers available.

### 11.24.3. Example

EU

Displays EXPOOL usage as follows:

```
SIZE(I/C/M) - 176848/176848/176848 USE(C/M) - 25776/27832 E11(C) ũ 283
```

## 11.25.EXPAND—Expand to Larger Disk Capacity

The EXPAND keyin expands the internal OS data structures of a pack to make use of the larger disk capacities. It can be used to expand local and shared disks.

### 11.25.1. Format

```
EXPAND pack-id
```

where:

*pack-id*

identifies the pack to be expanded.

### 11.25.2. System Response

Upon completion of the EXPAND pack keyin, the system displays one of the following informational messages:

```
Disk capacity for directory-id pack pack-id has been increased by
nnnnnn tracks.
```

Or

```
Disk capacity is restricted by the pack's scaling factor.
```

```
Directory-id pack pack-id has been increased by nnnnnn tracks.
```

Or

```
Disk capacity is restricted by the capacity of duplexed device device-
id.
```

```
Directory-id pack pack-id has been increased by nnnnnn tracks.
```

Or

```
Disk capacity is restricted by the maximum supported pack size.
Directory-id pack pack-id has been increased by nnnnnn tracks.
```

Or

```
EXPAND pack for directory-id pack pack-id completed during recovery.
Pack pack-id new capacity is nnnnnn tracks.
```

The pack is identified by *pack-id* and has been increased by nnnnnn tracks. The *directory-id* is either local or shared.

### 11.25.3. Restrictions

The maximum track capacity value that will be supported is the smaller value of either

- the defined capacity for the device (For example, configured in the "bin" file), or
- the largest size allowed for the device's (MBT) Scaling Factor.

If the pack resides on unit-duplexed devices, the track capacity can only be expanded to the maximum capacity of the smaller device.

For shared devices, all hosts must be at a level that supports the expand pack feature.

## 11.26. FA—Simulate an Attention Interrupt

The FA keyin simulates an attention (device ready) interrupt and allows disk pack label inspections.

### 11.26.1. Format

FA *devnam*

where *devnam* is the device name of a disk drive.

## 11.27.FB—Display or Change Flagbox Settings

The FB keyin displays or changes the flagbox settings. The flagbox is a set of software switches that control TIP system operation by turning TIP features on and off.

Refer to the *Transaction Processing Administration and Operations Reference Manual* (7830 7881) for a description of the flagbox settings.

### 11.27.1. Formats

#### FB

FB

Displays the current flagbox bit settings.

#### FB *bit*

FB *bit*

Displays the status of the specified flagbox *bit* (on or off).

#### FB LST

FB LST

Lists all flagbox bits.

#### FB OFF

FB OFF *bit*

Turns off the flagbox bit specified by *bit*.

#### FB ON

FB ON *bit*

Turns on the flagbox bit specified by *bit*.

### FB E

FB,E OFF SCHEDULE

Turns TIP scheduling off and purges the transaction system. All transaction programs are aborted, RTPS and stuck programs are unloaded, and the VINDEX and VALTAB are reloaded from the VALTAB file into memory.

If TIP scheduling is already turned off, the E (error) option still causes the transaction system to be purged. The VINDEX and VALTAB are always reloaded.

The E option is meaningful only with the SCHEDULE flagbox bit.

FB,E ON SCHEDULE

Terminates a transaction-system purge or drain and allows TIP scheduling to resume. This keyin, the counterpart to the FB,E OFF SCHEDULE keyin, is provided for emergencies when scheduling must be resumed before a purge or drain can complete. The VINDEX and VALTAB are not reloaded from the VALTAB file, so it is possible for the online VALTAB and VINDEX to be out of date and cause transaction errors.

The E option is meaningful only with the SCHEDULE flagbox bit.

## 11.27.2. Examples

FB

The system lists all flagbox bits that are currently set, as follows:

```
TIPTRACE ON
SCHEDULE ON
MESSAGES ON
DEBUGXX ON
OP$MAIN ON
```

FB LST

The system displays all flagbox bits, as follows:

|           |           |           |           |
|-----------|-----------|-----------|-----------|
| VTBLOG    | TPURLOG   | SUPURLOG  | D\$FAST   |
| D\$1782   | D\$0432   | OP\$TEST  | OP\$ALT   |
| OP\$MAIN  | CKCKONSCH | DEBUGXX   | CSUPPRESS |
| INHIBSTF  | MAXTIMENQ | MAXTIMEDP | SDEADLOCK |
| NOCMPINIT | STICKING  | MESSAGES  | TRAINFB   |
| TIPLOG    | BTLOADING | RECOVERY  | SCHEDULE  |
| TIPTRACE  |           |           |           |

The system displays the status of flagbox TIPTRACE, as follows:

```
TIPTRACE ON
```

## 11.28. FC—Function Control

The FC keyin enables and disables internal Exec functionality.

### 11.28.1. Formats

#### FC ACTON

FC ACTON

Turns on tracing of activity activations and deactivations.

#### FC ACTOFF

FC ACTOFF

Turns off tracing of activity activations and deactivations.

#### FC ACTCLR

FC ACTCLR

Clears and turns off tracing of activity activations and deactivations.

#### FC ARMON

FC ARMON

Turns on FCNARM tracing during mass storage allocations and releases. For each request, the FCNARM work buffer, the request packet, qualifier, file name, run-ID, and file item is traced. Also, device area descriptor (DAD) entries that are added, changed, or released are traced.

#### FC ARMOFF

FC ARMOFF

Turns off FCNARM tracing during mass storage allocations and releases.

#### FC BPTRACE

FC BPTRACE ON

Acquires a trace container and permit activation of tracing using the breakpoint register.

FC BPTRACE OFF

Releases the trace container and tracing. Using the breakpoint register is no longer permitted.

### **FC DADON**

FC DADON

Turns on DAD table verification for the following:

- Previous and next table links.
- Starting and ending file-relative addresses.
- The number of allocated tracks equal to the sum of the file item Quota counts.
- The number of allocated tracks equal to the number of granules assigned to the file.

If the verification fails, an XER 156 is taken. This keyin impacts mass storage allocation, release performance, and I/O performance.

### **FC DADOFF**

FC DADOFF

Turns off DAD table verification checking.

### **FC MBTON**

FC MBTON

Initiates disk master bit table (MBT) summation checking.

### **FC MBTOFF**

FC MBTOFF

Terminates disk master bit table summation checking.

### **FC PCTON**

FC PCTON

Turns on tracing of program control table (PCT) buffer usage.

### **FC PCTOFF**

FC PCTOFF

Turns off tracing of PCT buffer usage.

**FC RECON**

FC RECON

Engages the bad track/position or MFD sector release error recovery code so that no XER 057 or 157 stops occur.

**FC RECOFF**

FC RECOFF

Disengages the bad track/position or MFD sector release error recovery code so that XER 057 or 157 stops occur on bad track/position or MFD sector release.

**FC UPDON**

FC UPDON

Turns on tracing of IP usage by activities.

**FC UPDOFF**

FC UPDOFF

Turns off tracing of IP usage by activities.

**FC EVTON**

FC EVTON

Turns on tracing of SCIOP event reporting.

**FC EVTOFF**

FC EVTOFF

Turns off tracing of SCIOP event reporting.

**FC INTON**

FC INTON

Activates the processor interrupt tracing. When active, the trace retains copies of the ASP from the ICS frame for the last 8 interrupts on per IP basis. These entries are available in any subsequent Exec dump as a potential analysis aid.

**FC INTOFF**

FC INTOFF

Deactivates the processor interrupt tracing.

### FC STATUS

FC STATUS

Checks the status of FC keyin functions and displays whether they are on, off, or in some other state applicable to that keyin.

### FC RSI\$ON

FC RSI\$ON

Activates tracing of ER RSI\$ requests. The trace records the first five words of the RSI\$ packet on both the call and return.

### FC RSI\$OFF

FC RSI\$OFF

Deactivates tracing of ER RSI\$ requests.

## 11.28.2. System Response

A confirming message is displayed on the console in response to any FC keyin.

## 11.29. FF—Display File Information

The FF keyin displays all run-IDs or common name sections that have a specific cataloged file assigned. The FF keyin is used to determine which user has a file assigned when a hold condition occurs, such as when the file is assigned exclusively to another run. Users with the LIMITED console keyin capability can use this keyin.

**Note:** When using the FF keyin to display information for a file that was assigned prior to a name change, be certain you enter the new file name. If you enter the FF keyin using the old file name, the keyin returns a not found status, because the old file name no longer exists in the file directory. Entering the FF keyin using the new file name returns all users who have the file assigned by either the old or new name.

### 11.29.1. Format

```
FF [dir-id#]qualifier*filename[(F-cycle)] [.]
```

where:

*dir-id*

is the directory of the file (STD or SHARED). If you do not specify the directory-ID on the keyin, all directories are assumed. The STD directory is displayed, followed by the SHARED directory. If MHFS is down on the host, then the SHARED directory is ignored.



*qualifier*

is the file qualifier, from 1 to 12 characters in length. This field can include the characters A to Z, 0 to 9, -, and \$.

*filename*

is the file name, from 1 to 12 characters in length. This field can include the characters A to Z, 0 to 9, -, and \$.

*F-cycle*

is the absolute F-cycle number, expressed as an octal integer (indicated by a leading zero) with a range of 0 to 01747, or as a decimal integer with a range of 0 to 999. An F-cycle of 0 is equivalent to no cycle. Relative F-cycle numbers are neither recognized nor accepted.

If you do not specify an F-cycle, all cycles of the requested file are displayed (up to 32). The keyin displays the information ordered by F-cycle. All name sections with the first file of the F-cycle set assigned are displayed first; then all name sections with the second set, and so on. (In most cases, however, only one or two cycles exist.)

You must specify at least the qualifier and file name when using the FF keyin.

## 11.29.2. System Response

If the file is found, the system displays one of these messages, which explains the file assignment:

```
CNS: c-n-s HAS dir-id#qualifier*filename(f-cycle)
```

```
RUN: run-id HAS dir-id#qualifier*filename(f-cycle)
```

For shared files, the system displays this message:

```
HOST: host-id HAS SHARED#qualifier*filename(f-cycle)
```

If a SHARED file is assigned on other hosts, then host-IDs of the other hosts are displayed (the current host is not included). The message displayed is

```
HOST: host-id HAS dir-id#qualifier*filename(f-cycle)
```

where:

*c-n-s*

is the name of the common name section that has this file assigned.

*run-id*

is the 6-character run-ID of the run or TIP run that has this file assigned. If the operating system has the file assigned, EXEC8 appears in place of the run-ID.

*host-id*

is the 1-character host-ID of the system (other than the one on which the FF keyin was initiated) that has the shared file assigned.

### 11.29.3. Restrictions

No security checking is performed, and no read/write access to the assigned file is verified before the keyin response.

User runs can temporarily stall while opening or assigning files, because locking is involved in FF keyin processing. These periods are normally extremely short and are not necessarily visible to you.

The FF keyin is not allowed until the SYS run has started execution (SYS START).

### 11.29.4. Examples

#### Example 1

```
FF STD#Q*F.
```

Displays the directory-ID of file Q\*F in directory STD. Even if MHFS is not configured, the directory-ID, STD, is displayed.

#### Example 2

```
FF SHARED#Q*F.
```

Displays the directory-ID of file F in directory SHARED with qualifier Q, as follows:

```
RUN: BJP HAS SHARED#Q*F(1)
RUN: BJP HAS SHARED#Q*F(2)
```

#### Example 3

```
FF Q*F(2)
```

If both STD#Q\*F(2) and SHARED#Q\*F(2) are assigned on the same host, the system displays these messages:

```
RUN: BJP HAS STD#Q*F(2)
RUN: BJP HAS SHARED#Q*F(2)
CNS: COMMONNAME01 HAS SHARED#Q*F(2)
```

**Example 4**

FF Q\*F

If cycles 1, 2, and 5 of file STD#Q\*F are assigned, and cycles 74 and 80 of SHARED#Q\*F are assigned, the system displays these messages:

```
RUN: RUN1 HAS STD#Q*F(1)
RUN: RUN3 HAS STD#Q*F(1)
RUN: RUN1 HAS STD#Q*F(2)
RUN: RUN2 HAS STD#Q*F(2)
RUN: RUN1 HAS STD#Q*F(5)
RUN: RUN4 HAS STD#Q*F(5)
CNS: COMMONNAME01 HAS SHARED#Q*F(74)
CNS: COMMONNAME02 HAS SHARED#Q*F(80)
RUN: RUN1 HAS SHARED#Q*F(80)
```

**Example 5**

FF SHARED#Q\*F(2)

If the current host is host A, and hosts B and C have cycles of the file SHARED#Q\*F assigned, the system displays these messages:

```
RUN: RUN1 HAS SHARED#Q*F(2)
RUN: RUN2 HAS SHARED#Q*F(2)
CNS: COMMONNAME02 HAS SHARED#Q*F(2)
HOST: B HAS SHARED#Q*F(2)
HOST: C HAS SHARED#Q*F(2)
```

**Example 6**

FF SHARED#Q\*F

The system displays these messages:

```
RUN: RUN1 HAS SHARED#Q*F(1)
RUN: RUN2 HAS SHARED#Q*F(1)
HOST: B HAS SHARED#Q*F(1)
HOST: C HAS SHARED#Q*F(1)
RUN: RUN3 HAS SHARED#Q*F(2)
HOST: C HAS SHARED#Q*F(2)
HOST: C HAS SHARED#Q*F(5)
```

## 11.30. FS—Display Status of Components and Interfaces

The FS keyin displays the status of a component or interface.

### 11.30.1. Formats

There are many formats for the FS keyin. You can specify an option to display the status of all the components of a certain type, specify a particular component, or specify an interface. You can also specify components from that component down to the device or the control unit level by using the ALL option.

#### Display Status for All Components of a Specific Type

To display the status for all the components of a specific type, use the format:

*FS, component*

where *component* is one of the following:

|      |                                                             |
|------|-------------------------------------------------------------|
| ARB  | All arbitrary devices configured in the system.             |
| CM   | All channel modules (CM) configured in the system.          |
| CU   | All control units (CU) configured in the system.            |
| IOP  | All input/output processors (IOP) configured in the system. |
| IP   | All instruction processors (IP) configured in the system.   |
| MEM  | All main storage modules configured in the system.          |
| TIME | All the time sources installed on the system.               |

#### Display Status of the Disk Device Options

**Note:** Options with [S] designated indicates that plural option is allowed.

|                 |                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|
| ADISK[S]        | Display all UP, RV, or SU disks configured in the system.                                                               |
| DISK[S]         | Display all disks configured in the system.                                                                             |
| DDISK[S]        | Display all downed (DN) disk drives.                                                                                    |
| DRS             | Display the active local and active shared DRS disks.                                                                   |
| DRS,ALL         | Display all UP or SU disks with packs prepped DRS.                                                                      |
| FDISK[S]        | Display all FIXED disks configured in the system.                                                                       |
| MS              | Display all mass storage devices configured in the system.                                                              |
| PACK [/ dir-id] | Display all UP, DN, or PRESENT removable packs in the file directory, dir-id, [STD (local) or SHARED] (Default is STD). |

PACK pack-id [/dir-id],[pack-id] /dir-id...]

Display the status, inhibit status, and assign count of the specified removable disk pack[s] in the UP, DN, or PRESENT state.

PACK pack-id/SHARED

The status, inhibit status, assign count, and hosts that have files assigned to the specified SHARED pack-id.

RDISK[S] Display all REMOVABLE disks configured in the system.

SHRD Display all disks active in the file-sharing environment.

STD Display status of all standard (local) directory disks.

UD Display all unit-duplexed disks in the system.

### Display Status of the Tape Device Options

**Note:** Options with [S] designated indicates that plural option is allowed.

ATAPE[S] Display all available tape drives in a reserved (RV) or up (UP) state, and whether they are ACS-controlled, CTL-controlled, or premounted (PM) drives.

AVTH Display all OS2200 VTH tape devices in a UP or RV state.

DTAPE[S] Display all downed (DN) tape drives.

DVTH Display all OS2200 VTH tape devices in a DN or NA state.

MTAPE[S] Display all tape drives with mounted tapes.

P or PM Display premounted tape devices in the system.

TAPE[S] Display all the tape drives configured in the system and whether they are ACS-controlled, CTL-controlled, or premounted (PM) drives.

UTAPE[S] Display only the tape devices that are in use including assigned, mounted, premounted, and service message waiting.

VTH Display only OS2200 Virtual Tape Handler (VTH) devices.

### Display Status of the Communication Component Options

ACOM Display only UP and RV communication control units (CUs) and devices.

COM Display all communication CUs and devices.

DCOM Display only DN and NA communication CUs and devices.

### Display Status of a Specific Component

To display the status for a specific component, use the following format:

```
FS component
```

where *component* is one of the following variables:

|          |                                                        |
|----------|--------------------------------------------------------|
| cmod     | Display the specific channel module.                   |
| cu       | Display the specific control unit.                     |
| devnam   | Display the specific device.                           |
| iop      | Display the specific input/output processor.           |
| ip       | Display the specific instruction processor.            |
| mmod     | Display the specific memory module.                    |
| xiipname | Display the specific Extended I/O Interface Processor. |
| xpcname  | Display the specific Extended Processing Complex.      |

### Display Status of an Interface

To display the status of an interface, use the following formats:

```
FS cmod/[intf#]/cu
```

Displays the status of the specified channel-module-to-control-unit interface. If you do not specify an interface number, the Exec supplies a value. Also, use this keyin for displaying the status of the interface to the record lock processor (RLP).

```
FS cu/[intf#]/dev
```

Displays the status of the specified control-unit-to-device interface. If you do not specify an interface number, the Exec supplies a value. The status of the control-unit-to-device interface cannot be modified by the operator.

### Display Status of Components to Control Unit or Device Level

To display the status for all components from a specified component name to the control unit or device level, use the following format:

```
FS,ALL component
```

where *component* is one of the following variables:

|      |                                                              |
|------|--------------------------------------------------------------|
| Cmod | Display all CUs and devices on the specified channel module. |
| Cu   | Display all devices attached to the specified control unit.  |
| Iop  | Display the IOP components down to the control unit level.   |

|         |                                                                 |
|---------|-----------------------------------------------------------------|
| ip      | Display all IP components down to the control unit level.       |
| Xpcname | Display the specified XPC and the status of all attached XIIPs. |

### Display Availability of Disk Units

To display the availability of fixed and removable disk units, use the following format:

```
FS,AVAIL [cu1,cu2,...cuN] [device1, device2,...deviceN]
```

where:

*cu*

displays availability of all mass storage devices attached to the control units specified.

*device*

displays availability for the disk units specified.

The FS,AVAIL keyin is not configurable.

### Display Status of External Time Sources

To display the current state of the external time sources in the partition, use the following format:

```
FS,TIME [iopname]
```

This keyin is applicable only on systems that support an external time source.

## 11.30.2. System Response

There are many display formats to accommodate the various system components, as described in the following.

### Instruction Processors

For instruction processors, the display format is

```
processor state [NA] [ACTIVE/JP]
```

where:

*processor*

is the name of the processor whose status is being displayed.

*state*

is the state of the processor (UP, DN, or RV).

NA

indicates that the processor is not available.

ACTIVE

indicates that the processor is running.

### Devices Other Than Tape or Disk

For an IOP, channel module, control unit, or any device other than a tape drive or disk drive, the display format is

```
unit state [NA]
```

where:

*unit*

is the name of the IOP, channel module, control unit, or device whose status is being displayed.

*state*

is the state of the component, as follows:

- UP or DN for an IOP, channel module, or control unit
- UP, DN, SU, or RV for a device
- UP, DN, or RV for a partition

NA

indicates that the component is not available.

### XPCs

For an XPC in local mode, the FS,XPC keyin displays the mode of the XPC and the state of caching. The current status of the master file directory (MFD) file caching and shared system device SHDRUM file caching is also displayed. The following information for each host participating in the multihost environment is displayed:

```
HOST host-id active-state [xpc-interface-id/xiipname]...
```

where:

*host-id*

is the name of the host.

*active-state*

is active or inactive.

*xpc-interface-id*

is the interface ID of the XPC for the particular XIIP.



*xiipname*

is the name of the XIIP (site specific). It also refers to VI interfaces.

For example:

```
XPCA UP MEM MHM - XPC LOCAL CACHING ENABLED FOR THIS HOST
XPC SHARED CACHING ENABLED
LOCAL MFD CACHING IS ENABLED
HOSTA ACTIVE 00/XII1 02/XII2 03/XII4 15/XII5
HOSTB ACTIVE 04/XII1 06/XII4 08/XII5 16/XII5 11/XII7
HOSTC INACTIVE
HOSTD INACTIVE 21/XII1 25/XII2 31/XII5
```

For a file-sharing system, the following example is displayed:

```
XPCA UP MEM MHM - XPC LOCAL CACHING ENABLED FOR THIS HOST
XPC SHARED CACHING ENABLED
LOCAL MFD CACHING IS ENABLED
SHARED MFD AND SHDRUM CACHING IS ENABLED
HOSTA ACTIVE 00/XII1 02/XII2 03/XII4 15/XII5
HOSTB ACTIVE 04/XII1 06/XII4 08/XII5 16/XII5 11/XII7
HOSTC INACTIVE
HOSTD INACTIVE 21/XII1 25/XII2 31/XII5
```

For a specified XPC, FS,ALL *xpcname* displays the status in the following format:

```
xpcname status [NA] mode
xiipname(1) status [NA]
. . .
. . .
. . .
xiipname(n) status [NA]
```

where:

*xpcname*

is the name of the XPC (site specified).

*xiipname*

is the name of the XIIP (site specified).

*status*

is the current status of the component, up or down.

NA

indicates that the component is not available.

*mode*

is the current mode of the XPC, either local or MHM.

### Tape Drives

For a tape drive, the display format is

```
tape state[,ACS][,CTL][,PM] [NA] [* RUNID run-id REEL [reel-#] ring]
[POS [*]ffff{+|-}[*]bbbbbb|POS LOST]
```

where:

*tape*

is the name of the tape drive.

*state*

is the tape drive's status (UP, DN, or RV).

,ACS

indicates the drive is controlled by an ACS (Automated Cartridge System).

,CTL

indicates that the drive is registered to the CTL system (Cartridge Tape Library).

,PM

indicates the tape drive is in a premount state. (See FS,PM keyin)

NA

indicates that the drive is not available.

\*

indicates that the drive is assigned by a run.

*run-id*

is the name of the run with the drive assigned.

*reel-#*

is the number of the reel mounted on the drive. If a BLANK reel request is made, U-BLNK indicates an unlabelled BLANK tape and L-BLNK indicates a labelled BLANK tape is to be loaded.

The following applies only to DVD: If \*\*\*\*\* is displayed as reel-#, a medium that is not accessible by OS 2200 is premounted on the drive. If the drive is subsequently assigned by OS 2200, the medium is unloaded.

*ring*

is either RING or NORING, if specified on assignment.

## POS

indicates that a tape is currently mounted on the drive. The tape position is given as follows:

|                |                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Fffff</i>   | is the current number of files that are extended. For an unlabeled tape, this is the actual number of files. For a labeled tape, the formula for the actual number is<br>$(fffff + 1) / 3$                                                       |
| <i>+bbbbbb</i> | is the current number of blocks that are extended forward from the last tape mark.                                                                                                                                                               |
| <i>-bbbbbb</i> | is the current number of blocks that are extended backward from the last tape mark.                                                                                                                                                              |
| <i>*</i>       | in front of either <i>fffff</i> or <i>bbbbbb</i> means that the actual file or block count is different from the number displayed. Either a Locate Block command was issued, or the actual count is greater than 99,999 files or 999,999 blocks. |

## POS LOST

indicates that a tape is currently mounted on the drive and that tape loss of position has occurred.

## Disks

For a disk, the display format is

```
disk [S] state [CQ] [NA] [[*] R/F PACKID pack-id] [EXEC DRS]
```

where:

*disk*

is the name of the disk device.

*S*

indicates the disk is shared with other hosts. (File Sharing only)

*state*

is the state of the disk device (UP, DN, RV, SU).

*CQ*

indicates the disk is in Command Queuing mode.

*NA*

indicates that the drive is not available.

\*

indicates that the drive is assigned. For disk drives that have removable packs, a pack can be assigned even if the drive is not assigned.

R

indicates that the pack on the drive is prepped as removable.

F

indicates that the pack on the drive is prepped as fixed.

*pack-id*

is the identification of the pack on the drive.

EXEC DRS

indicates the disk where the Exec is stored.

For disks in a Unit Duplexing association, the display format is

```
disk1 -UD- disk2 [S] state [CQ] [NA] [[*] R/F PACKID pack-id] [EXEC DRS]
```

where:

*disk1*

is the source disk.

*disk2*

is the destination disk.

All other fields are the same as previously described.

### Channel-Module-to-Control-Unit Interface

For the channel-module-to-control-unit interface, the display format is as follows in response to an "FS chan//cu" keyin:

```
chan/intf#/cu state [NA]
```

where:

*chan*

is the name of the channel module in the interface.

*intf#*

is the number of the interface supplied by the Exec.

*cu*

is the name of the control unit in the interface.

*state*

is the state of the interface (UP or DN).

NA

indicates that the interface is not available.

### Control-Unit-to-Device Interface

For control-unit-to-device interface, the display format is as follows in response to an "FS *cu*//*devname*" keyin:

```
cu/intf#/devname state [NA]
```

where:

*cu*

is the name of the control unit in the interface.

*intf#*

is the number of the interface supplied by the Exec.

*devname*

is the specified device name.

*state*

is the state of the interface (UP or DN).

NA

indicates that the interface is not available.

### Removable Packs

For removable packs, the display format is

```
pack-id state - INHIBITS = inhibits ASSIGN COUNT = count dir-id HOSTS = hosts
```

where:

*pack-id*

is the name of the pack from the FS,PACK devnam keyin or the pack-ID for each device on the FS,PACK keyin.

*state*

is UP, DN, or PRESENT.

*inhibits*

can be R (read), W (write), A (assign) or NONE (no inhibits assigned).

*count*

is the number of files assigned on the pack.

*dir-id*

is the directory-ID, either STD or SHARED.

*hosts*

is a list of hosts that have files assigned on a shared pack in the DN state.

### **AVAIL**

For the FS,AVAIL keyin, the display format is

```
disk [S] state [CQ] [*] [R/F] PACKID packid
```

```
disk [availability] [LOCAL/SHARED] TRACKS AVAILABLE
```

where:

*disk*

is the name of the disk device.

S

indicates the disk is shared with other hosts. (File Sharing only)

*state*

is the state of the disk device (UP, DN, RV, SU).

CQ

indicates the disk is in Command Queuing mode.

\*

indicates that the drive is assigned. For removable disk drives, a pack can be assigned even if the drive is not assigned.

R

indicates that the pack on the drive is prepped as removable.

F

indicates that the pack on the drive is prepped as fixed.

*pack-id*

is the identification of the pack on the drive.

availability

is the number of tracks available on the disk device.

LOCAL/SHARED

is the file directory where the disk device resides. LOCAL is the standard (STD) directory. SHARED is the shared (SHRD) directory.

### 11.30.3. Examples

#### Example 1

```
FS,PACK
```

Displays the status, inhibit status, and assign count of all removable disk packs in the list of packs in the UP, DN, or PRESENT status, as follows:

```
91Y5 UP - INHIBITS = NONE ASSIGN COUNT = 8
91K5 UP - INHIBITS = NONE ASSIGN COUNT = 168
```

#### Example 2

```
FS,PACK 91Y5
```

Displays the status, inhibit status, and assign count of removable disk pack 91Y5 in the list of packs in the UP, DN, or PRESENT status, as follows:

```
91Y5 UP - INHIBITS = NONE ASSIGN COUNT = 8
```

If removable disk pack 91Y5 is not found in the system list of packs that are UP, DN, or PRESENT, its status is reported as NOT PRESENT:

```
91Y5 NOT PRESENT - INHIBITS = NONE ASSIGN COUNT = 0
```

#### Example 3

```
FS,PACK 91Y6/SHARED
```

Displays the status, inhibits, assign count, and host-ID of hosts that have files assigned on shared pack 91Y6, as follows:

```
91Y6 DN - INHIBITS = NONE ASSIGN COUNT = 8 SHARED HOSTS = AC
```

### Example 4

```
FS IPO
```

Displays information about instruction processor 0, as follows:

```
IPO UP ACTIVE
```

### Example 5

```
FS TAPED0, PRC
```

Displays information about tape unit TAPED0 and device PRC, as follows:

```
TAPED0 UP ,PRC UP
```

### Example 6

```
FS T40P10
```

Displays information about tape device T40P10, on which a tape is currently mounted and extended, as follows:

```
T40P10 UP * RUNID MAPPER REEL 6MRP11 RING POS 5+480
```

If the tape is unlabeled, the actual number of files extended is 5. If the tape is labeled, the actual number is 2. (See the formula given with the POS description, above.)

### Example 7

```
FS, ATAPES
```

Displays information about tapes' drive status and indicates whether they reside in an ACS, as follows:

```
TP1 UP
TP2 RV
TP3,ACS UP
TP4,ACS DN
```

### Example 8

```
FS,ALL CUTAPD
```

Displays information about all devices on control unit CUTAPD, as follows:

```
CUTAPD UP
TAPED3 UP
TAPED4 UP
```



**Example 9**

```
FS,SHRD
```

Displays the status of all devices active in the MHFS environment, as follows:

```
34A8 S UP R PACKID 16P
34A9 S DN
60A19 S UP * F PACKID 50A16
80A0 S UP * F PACKID 70A0
80A2 S DN , 70A3 S DN
```

When there are no shared devices configured, the following message is displayed:

```
NO SHARED DEVICES CONFIGURED
```

**Example 10**

```
FS 70E0
```

Displays the status of unit-duplexed disk 70E0, as follows:

```
70E0 -UD- 70G0 UP * F PACKID F10731
```

**Example 11**

```
FS 50F1,70E1
```

Displays the status of unit 50F1 and unit-duplexed disk 70E1, as follows:

```
50F1 UP
70E1 -UD- 70G1 RV
```

**Example 12**

```
FS 70E3
```

Displays the status of unit-duplexed disk 70E3, as follows:

```
70E3 -UD- 70E4 SU * R PACKID R93024
```

**Example 13**

```
FS,ALL 70EA
```

Displays the status of all devices attached to control unit 70EA, where unit-duplexing has been established with devices on a different disk subsystem, as follows:

```
70EA -UD- 70G0 UP * F PACKID 10730E
70E1 -UD- 70G1 UP * F PACKID 10731E
70E2 SU * R PACKID 10732E
70E3 UP * F PACKID 10731E
```

### Example 14

```
FS 91Y5
```

Displays the status of the removable pack 91Y5, as follows:

```
91Y5 UP * R PACKID 91Y5
```

### Example 15

```
FS,ALL 70B
```

Displays the status of all devices attached to control unit 70B, where both units of a unit-duplexed pair are attached to the same control unit, as follows:

```
70B0 UP
70B1 UP
70B2 -UD- 70B3 UP
70B3 -UD- 70B2 UP
70B4 RV
70B5 DN
```

### Example 16

```
FS,ALL 70EA,70GA
```

Displays the status of all devices attached to the two disk control units 70EA and 70GA, as follows:

```
70E0 -UD- 70G0 UP
70E1 -UD- 70G1 UP
70E2 SU
70E3 SU
70G0 -UD- 70E0 UP
70G1 -UD- 70E1 UP
70G2 DN
```

### Example 17

```
FS,MS
```

Displays the status of all mass storage devices in the system, as follows:

```
70A0 -UD- 70B0 UP
70A1 -UD- 70B1 UP
70A2 -UD- 70B2 UP
70A3 -UD- 70B3 UP
70A4 UP
70A5 DN
70B4 DN
70B5 UP
70C2 UP
70D0 -UD- 70C0 RV
70D1 -UD- 70C1 UP
70D2 DN
70R0 UP
```

```
70R1 UP
70R2 UP
```

**Example 18**

```
FS,UD
```

Displays the status of all unit-duplexed disks in the system, as follows:

```
70A0 -UD- 70B0 F * PACKID DUPLX0 EXEC DRS
70A1 -UD- 70B1 F * PACKID DUPLX1
70C0 -UD- 70D1 F * PACKID DUPLXR
```

**Example 19**

```
FS,DRS
```

Displays the status of UP and SU devices containing packs prepped DRS after shared and local mass storage is initialized or recovered, as follows:

```
20S0 S UP F PACKID 20S0 SHARED DRS
20S1 S UP F PACKID 20S1 DRS
94L0 UP * F PACKID 94L0 EXEC DRS
94L1 UP * F PACKID 94L1 DRS
```

Displays the status of UP and SU devices containing packs prepped DRS after shared and local packs are registered, as follows:

```
94L0 UP * R PACKID 94L0 EXEC DRS
94L1 UP * R PACKID 94L1 DRS
94L2 UP * R PACKID 94L2 DRS
```

Displays the status of the active DRS devices (shared and local) when unit duplexed, as follows:

```
20S0 -UD- 20S2 S UP * F PACKID 20S0 SHARED DRS
94L0 -UD- 94L2 UP * F PACKID 94L0 EXEC DRS
```

Displays the status of the inactive DRS devices (shared and local) when unit duplexed, as follows:

```
20S1 -UD- 20S2 S UP * F PACKID 20S0 DRS
94L1 -UD- 94L2 UP * F PACKID 94L0 DRS
```

**Example 20**

```
FS,AVAIL
```

Displays the status of all mass storage units, as follows:

```
55S17 -UD- 55S18 UP * R PACKID 55A0 EXEC DRS
55S17 123456123456 LOCAL TRACKS AVAILABLE
26S1 UP * F PACKID 26S1
55S17 123456123456 LOCAL TRACKS AVAILABLE
60D0 S UP * F PACKID 60D0
55S17 123456123456 SHARED TRACKS AVAILABLE
```

### Example 21

```
FS,TIME
```

Displays the current state of all external time sources in the partition.

```
FS,TIME
IOP08 (TIME) PRIMARY DN
 SECONDARY DN
IOP10 (TIME) PRIMARY UP EXEC
 SECONDARY UP DEGR
IOP38 (TIME) PRIMARY UP
 SECONDARY DN NP
IOP41 (TIME) PRIMARY DN
 SECONDARY DN NP
```

where:

EXEC        Denotes that Exec has currently selected this time source for its use.

DEGR        Indicates that this time source is operating in a degraded mode and might be unsuitable for use in Exec.

NP          Indicates that this time source is not present.

If a specific IOP is indicated, the following message is displayed:

```
FS,TIME IOP18
IOP18 (TIME) PRIMARY UP EXEC
 SECONDARY UP
```

If no time sources are installed on the indicated IOP, the following message is displayed:

```
FS,TIME IOP18
IOP18 has no external time sources installed
```

If no time sources exist in the partition, the following message is displayed:

```
FS,TIME
No external time sources are available.
```

If entering console keyins related to CSB maintenance on a system that does not support a clock calendar or an external time source, the following message is displayed:

```
FS,TIME keyins are not supported on this system
```

**Example 22**

```
FS,CU
```

Displays the status of all control units (CUs) configured, as follows:

```
1310CU UP , 1320CU UP , 3XCA UP
3XCB UP , 3XSA S UP
T98U3A UP , T98U4A UP , T98U5A UP
TS51AA UP , VT2AA UP NA , VT2AB UP
VT2AC UP
```

**Example 23**

```
FS,STD
```

Displays the status of all standard (STD) directory disks, as follows:

```
3XC2 UP CQ * R PACKID 60W2
3XC3 UP CQ * R PACKID 60W3
3XC4 UP CQ * R PACKID 60W4
3XC5 UP CQ * R PACKID 60W5
3XC6 UP CQ * F PACKID 3XC6
3XC7 UP CQ * F PACKID 3XC7
3XC8 UP CQ * F PACKID 3XC8
3XC9 UP CQ * F PACKID 3XC9
3XC1 UP CQ * F PACKID 60W1
3XC0 UP CQ * F PACKID 60W0 EXEC DRS
```

### Example 24

FS,VTH

Displays only OS2200 Virtual Tape Handler (VTH) devices, as follows:

```
VTPA00 UP , VTPA01 UP , VTPA02 UP
VTPA03 UP , VTPA04 UP , VTPA05 UP
VTPA06 UP , VTPA07 UP , VTPA08 UP
VTPA09 UP , VTPB0 UP , VTPB1 UP
VTPB2 UP , VTPB3 UP , VTPB4 UP
```

### Example 25

FS,COM

Displays all communication control units (CUs) and devices.

```
NC1310 UP * RUNID CPCOMM
NC1320 UP * RUNID CPCOMM
```

### Example 26

FS,PM

Displays premounted tape devices, (same as FS,P), as follows:

```
VT2B14 UP,PM , T98R7 UP,PM , VTPB4 UP,PM
T73A3 UP,PM
```

If no tape units are in the premount state, displays the following:

```
FS KEYIN - NO UNIT IN THE PREMOUNT ONLY STATUS.
```

**Example 27**

```
FS,ALL VTPB
```

Displays the status of the control unit and its devices, as follows:

```
VTPB UP

VTPB0 UP * RUNID ZAJC REEL B22007

VTPB1 UP * RUNID ZAJC REEL B22008

VTPB2 UP * RUNID ZAJC REEL U-BLNK

VTPB3 UP * RUNID ZAJC REEL L-BLNK
```

Reel-id U-BLNK indicates an unlabelled BLANK tape request.

Reel-id L-BLNK indicates a labelled BLANK tape request.

## 11.31. HK—Execute Unsolicited Keyins on Other Hosts

The HK keyin executes an unsolicited keyin on other hosts in an MHFS environment from a single host. The keyin responses appear on the console of the sending host.

See the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469) for more information on the HK keyin.

### 11.31.1. Formats

```
HK keyin
```

Executes a *keyin* on all hosts except the sending host.

```
HK,host-id keyin
```

Executes a keyin on the specified host (*host-id*).

```
HK,ALL keyin
```

Executes a *keyin* on all hosts, including the sending host.

### 11.32. HM—Send Messages to Other Hosts

The HM keyin lets you send messages to other hosts in an MHFS environment.

See the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469) for more information on the HM keyin.

#### 11.32.1. Formats

*HM message*

Sends the *message* to all hosts except the sending host.

*HM,host-id message*

Sends the *message* to the specified host (*host-id*).

*HM,ALL message*

Sends the *message* to all hosts, including the sending host.

If you send a message to the host from which you entered the HM keyin, the message is displayed on the system console but not on remote demand user terminals in console mode. On hosts other than the sending host, the message is displayed on the remote terminals in console mode on those hosts.

### 11.33. HU—Display Host Resource Usage

The HU keyin displays a list of up to 10 run-IDs and the percentage of total available IP SUPs used by each run-ID. The information is displayed on the console 1 second after entering the keyin. The operating system captures the first 10 run-IDs that accumulate IP SUPs after the keyin and displays the run-IDs in descending order of IP SUP usage. This display is preceded and followed by informational header and trailer lines to show that the operating system is responding to the keyin and to show that all appropriate information has been displayed.

If the Software Controlled Performance (SCP) percentage is nonzero (that is, performance is less than 100 percent), the SCP percentage is factored into the total available IP SUPs used by each run. This allows consistent comparisons of IP use across systems with different SCP percentages.

The HU keyin also displays the statistics for application level test and set queued activities. This information can be used to determine which subsystem or common bank is causing the system hang. If a subsystem is causing the system hang, a subsystem deactivate can be executed to clear the bottleneck. If a common bank is causing the system hang, a common bank reload can be done to the bank to clear the bottleneck.



### 11.33.1. Format

HU

### 11.33.2. System Response

RUN-ID: *nn%*

where *nn* is the percentage of total available IP SUPs used by the run during the 1 second after entry of the HU keyin.

If no run-IDs accumulate IP SUPs during the 1 second after the HU keyin is entered, the following message is displayed between the header and trailer lines:

The operating system has used all of the available IP SUPs.

TOTAL TEST AND SET QUEUED ACTIVITIES: *nnn*

where *nnn* is the total number of activities queued in application level subsystems and common banks.

*pp%* SUBSYSTEM      *Subsystem definition element name*

where *pp* is the percentage of *nnn* activities queued in this subsystem.

*pp%* COMMON BANK      *Common bank name* FILE *common bank file*

where *pp* is the percentage of *nnn* activities queued in this common bank.

If you enter the HU keyin while one is already in progress, the system displays the following message:

HU keyin ignored because one is already in progress.

### 11.33.3. Example

HU

Displays host resource usage as follows:

```
PERCENTAGE OF AVAILABLE IP SUPS USED BY RUN-ID:
WEBDMD: 1% SILAS : 1% WEBS22: 1% MCB1 : 1%
MCB16 : 1% MCB5 : 1% CSC : 1% DDPFJT: 1%
*SVC7L: 1% *SVC7K: 1%
127 Total Test and Set Queued Activities
38% Subsystem STD#SYSLIB*CPFTP$LIB(22).SSDEF
29% Subsystem STD#SYSLIB*ASIS-OMS(9).ASIS$SSDEF
13% Common Bank MCBBNK FILE STD#MCB1*MCB$(1).
12% Subsystem STD#SYSLIB*COMAPI$B(56).SSDEF$API$B
END HU KEYIN
```

## **11.34. II—Initiate an Interrupt in the Current Run Task**

The II keyin causes an interrupt in the current task of a run that has registered a contingency for the II keyin.

Use this keyin only under the direction of your site administrator.

### **11.34.1. Format**

```
II run-id [instruction]
```

where:

*run-id*

is the id of the run.

*instruction*

is the data you want to send to the user program (maximum six characters). If you do not specify an instruction, the run can prompt you for it.

### **11.34.2. Example**

```
II MAPPER
```

Causes an interrupt in the MAPPER run. The system responds with

```
0-MAPPER*ENTER DIRECTIVE
0 RLON
MAPPER*RUN LOG ON
```

## **11.35. IN—Initialize Fixed Mass Storage**

The IN keyin destroys all files on the specified mass storage device and makes all of its space available for new files.

### **11.35.1. Format**

```
IN devnam
```

where *devnam* is the name of the fixed mass storage device to initialize.

### **11.35.2. Restrictions**

The status of *devnam* must be UP or SUSPENDED. It cannot be shared mass storage. Use this keyin at MODIFY CONFIG time. If the keyin is used after MODIFY CONFIG time, it is fatal and produces an XER 055 error. Initialization then occurs during the subsequent reboot.

You are notified when the keyin is fatal, and you can decide whether to proceed.

## 11.36. IODELAY—Set an Artificial Time Delay for Unit Duplexing

The IODELAY keyin provides the ability to set an artificial I/O hardware service time delay for secondary units of duplexed pairs. The delay values are not saved when the host operating system is rebooted. In multi-host systems, the keyin must be entered on each host operating system that requires the delay I/O values.

### 11.36.1. Format

```
IODELAY [, OPTION] OPERATOR [, OPERAND] [, OPERAND] [, OPERAND]
```

where:

*OPTION*

is one of the following values:

|        |                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATUS | Displays the values of the delay times for read, write, and RBW operations. This is the default, if no <i>OPTION</i> and no <i>OPERANDs</i> are provided. |
| CLEAR  | Zeros out all delay times. <i>OPERANDs</i> are not allowed.                                                                                               |

*OPERATOR*

UD - I/O delay for secondary legs of unit duplexed devices.

*OPERAND*

*func=vvvvvvtu*

where

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <i>func</i>   | Type of function. It can have one of the following values:<br>R - Read<br>W - Write<br>RBW - Read before write                                |
| <i>vvvvvv</i> | A decimal integer between 0 and 999999 which an operator selects to set the desired time delay.                                               |
| <i>tu</i>     | Time unit of <i>vvvvvv</i> . It can have one of the following values:<br>US - Microseconds<br>MS - Milliseconds<br>S - Seconds<br>M - Minutes |

### 11.36.2. Examples

#### Example 1

```
IODELAY UD,R=300US,W=600US,RBW=1200US
```

Setting the read delay to 300 microseconds, the write delay to 600 microseconds, and the RBW delay to 1200 microseconds.

#### Example 2

```
IODELAY UD,RBW=1200MS,R=400MS
```

Setting the read delay value to 400 milliseconds and the RBW delay value to 1200 milliseconds. This does not change the write delay value.

#### Example 3

```
IODELAY,CLEAR UD
```

*or*

```
IODELAY UD, R=0US,W=0US,RBW=0US
```

Turning the delay logic off.

#### Example 4

```
IODELAY,STATUS UD
```

*or*

```
IODELAY UD
```

Displaying the current delay values.

## 11.37.IOPDUMP—Dump IOP

The IOPDUMP keyin saves the storage of specified IOPs to a tape or mass storage file. If the output medium is tape, the IOPDUMPS for each of the selected IOPs entered in a single keyin are saved onto the same tape medium. If the mass storage (MS) option is specified in the keyin, the dump for each selected IOP is saved in separate mass storage files. The information in an IOPDUMP is used to analyze both software and hardware problems.

For Dorado systems, an automatic IOP dump capability causes an IOP dump when an IOP (SIOP, XIOP, or NIOP) is failing, either through a timeout or because an IOP was internally downed because of a problem. By default, the automatic IOP dump capability is enabled. This feature is available when the IODEBUG configuration tag is set to a value of 1; you can also use an IOPDUMP keyin to turn the automatic capability off or on.

You can also use the IOPDUMP keyin to perform a manual dump of an IOP.

### 11.37.1. Manually Dumping an IOP

#### Format

```
IOPDUMP[,MS[X]] iop-name1,iop-name2, ... iop-nameX
```

where:

*iop-name1, iop-name2, iop-nameX*

are the names of the IOPs to be dumped.

MS

specifies a mass storage file.

X

aborts the previously initiated IOPDUMP.

If the MS option is specified, the dump is to a mass storage file. The mass storage file is cataloged as a private file with a cycle number one greater than that of the most recently cataloged cycle using the file name SYSS\*IOP\$DUMP\$. A mass storage file cannot be specified at "Modify Config" time. A mass storage file cannot be specified unless the system is beyond the point where a mass storage file can be cataloged.

If the MS option is not specified, the dump is to a tape file. The file name used for a tape file is SYSS\*IOP\$DUMP\$. Tape dump files are assigned as labeled tapes without the f option. This means that the filename is written into the label blocks. This same filename must be used when reading the tape file at a later time.

If the X option is specified, the previously initiated IOPDUMP is aborted. The tape or mass storage assignment is freed which causes the original activity to receive an error and terminate. If a IOPDUMP is already in progress and the X option is not specified, an error message is displayed and the keyin is ignored.

If the dump file is a tape, the following read and reply message is displayed:

```
IOP dump: Enter tape device name <exit$>
```

The operator should answer with the name of the tape drive. If no characters are entered for this entry, the IOP dump routine is exited. If an invalid character is entered, an error message is displayed and the operator is solicited again for the tape device name.

After the tape device name has been entered, the following read and reply message is displayed:

```
IOP dump: Enter tape reel ID <exit$>
```

Answer with the reel-ID of the tape to be used. This entry can be from one to six alphanumeric characters. If no characters are entered for this entry, the IOP dump routine is exited.

If an invalid character is entered, an error message is displayed and the operator is solicited again for the tape reel-ID. The program uses the entered tape device name and reel-ID to do an absolute @ASG of the tape device.

After dumping each selected IOP, the following completion message is displayed:

```
IOP dump: Completed dumping iop-name into SYSIOPDUMP$(n).
IOP dump: Block count = xxxxxx, Block size = 1792 words
```

where *n* indicates the file cycle number for mass storage files.

### 11.37.2. Enabling Automatic IOP Dumping

An automatic IOP dump is attempted when a standard IOP is failing. The automatic IOP dump is written to the mass storage file SYS\$IOP\$DUMP\$(*n*).

#### Format

```
IOPDUMP,AUTO { ON | OFF }
```

This keyin allows the operator to turn automatic IOP dumping off or on. The automatic dump feature is on by default.

**Note:** The *IODEBUG* configuration tag must be set to a value of 1 in order for this keyin to have an effect. If *IODEBUG* is not set to 1, automatic IOP dumping is unavailable.

One of the following messages is displayed:

```
IOPDUMP: Automatic IOP dump turned ON
IOPDUMP: Automatic IOP dump turned OFF
IOPDUMP: Automatic IOP dump NOT available, need CONFIG tag IODEBUG = 1
```

## 11.38. IT—Control I/O Traces and Timeout Values

The IT keyin has four main functions:

- Control Host I/O traces
- Control SCIOP/XIIP based I/O traces or display SCIOP/XIIP Microcode Level and Hardware Version
- Set the timeout value for mass storage I/O
- Set the timeout value for realtime tape I/O

The IT keyin has an effect only on the Host on which it is entered. The IT keyin action is not restored to the new session when the Host system is rebooted.

### 11.38.1. Control Host I/O Traces

When used to control I/O traces, the IT keyin can be used to turn I/O initiation and completion traces on and off for debugging and check the status of the traces for all subsystem types configured on the system (except for console subsystems). The keyin can also be used to copy I/O traces to an external user file for subsequent analysis. If you do not utilize the IT-keyin, the trace routines will default to trace I/O initiations and completions for all configured subsystem types. The IT-keyin must be re-entered if the Host system is rebooted.

Format for control of I/O traces:

```
IT[,operation] [operand] [,t]
```

where:

*operation*

is one of the following functions:

|        |                                                        |
|--------|--------------------------------------------------------|
| ON     | Turns trace on.                                        |
| OFF    | Turns trace off.                                       |
| STATUS | Displays the status of the trace. This is the default. |

*operand*

is the trace (or traces) to be affected by the operation. The operand can be any site-defined mnemonic name for a control unit or any device configured as a control unit, in which case the trace for that handler is affected. It can also be one of the following:

|         |                                                       |
|---------|-------------------------------------------------------|
| cuname  | Specifies control unit name and its handler.          |
| devname | Specifies device name and its handler                 |
| MS      | Specifies all mass storage traces.                    |
| DISKS   | Specifies all disk traces.                            |
| TAPES   | Specifies all tape traces.                            |
| ALL     | Specifies all configured traces. This is the default. |

*t*

Indicates whether the initiation or completion traces are affected by the operation field, and can be one of the following values:

|   |                                                                                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C | Affects only the completion traces.                                                                                                                                     |
| I | Affects only the initiation traces.                                                                                                                                     |
| A | Affects both the initiation and completion traces. This is the default value. It is meaningless on a status request because it returns a status of both I and C traces. |

- M** Controls the copying of the trace data to mass storage. If the operation field is ON, then a new F-cycle of file SYSS\$\*IOTRACE\$ is created on fixed mass storage. As trace data identified by the operand is generated, it is copied to this file. The operand field must specify a site-defined mnemonic name for a control unit or any device configured on a control unit. It is not allowed to specify MS, DISKS, TAPES, or ALL. If the operation field is OFF, the copying of the trace data stops and file SYSS\$\*IOTRACE\$ is closed and released.
- T** Controls the copying of the trace data to tape. If the operation field is ON, then a tape drive is assigned using the system default tape mnemonic with a reel-ID of blank. As trace data identified by the operand is generated, it is copied to this tape. The operand field must specify a site-defined mnemonic name for a control unit or any device configured on a control unit. It is not allowed to specify MS, DISKS, TAPES, or ALL. If the operation field is OFF, the copying of the trace data stops and the tape is closed and released.

### Example 1

Displays the current type of I/O tracing for each subsystem.

```
IT

 STANDARD CHANNEL TAPE TRACING INITIATIONS/COMPLETIONS
 STANDARD CHANNEL DISK TRACING INITIATIONS/COMPLETIONS
 ARBITRARY DEVICE HANDLER TRACING INITIATIONS/COMPLETIONS
```

### Example 2

Turns on only completion traces for mass storage devices.

```
IT,ON MS,C

 STANDARD CHANNEL DISK TRACING COMPLETIONS
```

### Example 3

Turns on only initiation traces for all handlers.

```
IT,ON ALL,I

 STANDARD CHANNEL TAPETRACING INITIATIONS
 STANDARD CHANNEL DISKTRACING INITIATIONS
 ARBITRARY DEVICE HANDLER TRACING INITIATIONS
```

### Example 4

Turns off only completion traces for all handlers.

```
IT,OFF ,C

 STANDARD CHANNEL TAPETRACING INITIATIONS
 STANDARD CHANNEL DISKTRACING INITIATIONS
 ARBITRARY DEVICE HANDLER TRACING INITIATIONS
```



**Example 5**

Displays the status of tracing for all disk handlers.

```
IT DISKS

STANDARD CHANNEL DISK TRACING INITIATIONS
```

**Example 6**

Displays the status of tracing for all tape handlers.

```
IT TAPES

STANDARD CHANNEL TAPE TRACING INITIATIONS
```

**Example 7**

Creates the file SYSS\*IOTRACE\$ and begins to copy trace data for DISKA0 and all devices using the same handler to this file.

```
IT,ON DISKA0,M
USE IOT. ,SYSS*IOTRACE$(+1) .
CAT IOT. ,F/10000//10000,,,,ACRDIA
ASG,A IOT.
STANDARD CHANNEL DISK *TRACING INITIATIONS/COMPLETIONS
```

**Example 8**

Turn off DISKA0 trace. The operand field must specify the same unit mnemonic that was used on a previous IT,ON UNIT,M keyin.

```
IT,OFF DISKA0,M
STANDARD CHANNEL DISK TRACING INITIATIONS/COMPLETIONS
FREE IOT.
EXTENDED DEVICE TRACE FILE HAS BEEN CLOSED
```

**Example 9**

Assigns a blank tape and begins to copy trace data for TAPEY1 and all devices using the same handler to this tape.

```
IT,ON TAPEY1,T
ASG,TZJ IOT,T,BLANK
LOAD UNLABELED BLANK T98N0 IOT - 1 EXEC 8
```

After blank tape is loaded

```
STANDARD CHANNEL TAPE *TRACING INITIATIONS/COMPLETIONS
```

### Example 10

Turn TAPEY1 trace off. The operand field must specify the same unit mnemonic that was used on a previous IT,ON UNIT,T keyin.

```
IT,OFF TAPEY1,T

STANDARD CHANNEL TAPE TRACING OFF
FREE IOT.
EXTENDED DEVICE TRACE FILE HAS BEEN CLOSED
```

### Example 11

Assigns a blank tape and begins to copy trace data for DISKA0 and all devices using the same handler to this tape.

```
IT,ON DISKA0,T
ASG,TZJ IOT,T,BLANK
LOAD UNLABELED BLANK T98X3 IOT - 1 EXEC 8
```

After the tape is mounted

```
STANDARD CHANNEL DISK *TRACING ON
```

### Example 12

Turns off the copying of the trace data to tape and releases the tape drive. The operand field must specify the same unit mnemonic that was used on a previous IT,ON UNIT,T keyin.

```
IT,OFF DISKA0,T

STANDARD CHANNEL DISK TRACING OFF
FREE IOT.
EXTENDED DEVICE TRACE FILE HAS BEEN CLOSED
```

## 11.38.2. Control SCIOP/XIIP-Based I/O Traces or Display SCIOP/XIIP Microcode Level and Hardware Version

The IT keyin also provides an interface to control SCIOP/XIIP-based I/O traces. After these traces are enabled and the trace info has been gathered, an IOP dump is taken to retrieve the trace information. Since the trace is enabled in the SCIOP, the trace remains set when the Exec operating system is rebooted. During the reboot, the SCIOP trace settings are retrieved from the SCIOP and saved in Exec memory, so the IT keyin will display the current SCIOP trace settings after the Exec is rebooted. This IT keyin will also display the microcode level and hardware version of the SCIOP.

### Format

```
IT,operation [operand][,iopname]
```

Where:

Operation must be specified and is one of the following functions::

ON Turns trace on.

OFF Turns trace off.

STATUS Displays the status of the IOP traces.

operand is one of the following options. To enable multiple options, repeated IT keyins must be done.

IOPTR IOP Traces

IOPPER Performance Trace

IOPTIM Timer Activity Trace

IOPFMT Formatter Trace

IOPDEB General Debug Trace

ALL Sets, clears, or displays all traces

IOPINQ Displays SCIOP/XIIP microcode level and hardware version

IOPNAME Specifies the SCIOP/XIIP name which is the target of this IT keyin

### Example 1

```
IT,ON IOPTR,IOP01
IOP01 TRACES ENABLED: IOPTR
```

### Example 2

```
IT,ON IOPPER,IOP01
IOP01 TRACES ENABLED: IOPTR,IOPPER
```

### Example 3

```
IT,OFF ALL,IOP01
IOP01 TRACES ENABLED: NONE
```

### Example 4

```
IT,ON ALL,IOP01
IOP01 TRACES ENABLED: IOPTR,IOPPER,IOPTIM,IOPFMT,IOPDEB
```

### Example 5

```
IT,STATUS ALL,IOP01
IOP01 TRACES ENABLED: IOPTR,IOPPER,IOPTIM,IOPFMT,IOPDEB
```

### Example 6

```
IT,STATUS ,IOP01
IOP01 TRACES ENABLED: IOPTR,IOPPER,IOPTIM,IOPFMT,IOPDEB
```

### Example 7

If the operand is not one of the options listed above, the following message is displayed:

```
IT,ON IOPPPP,IOP01
 IT KEYIN INVALID - TRACE DOES NOT EXIST FOR DEVICE TYPE ENTERED
```

### Example 8

If the IOP does not support trace or inquiry functionality, the following error message is displayed:

```
IT IOPINQ,IOP13
 THE IOP DOES NOT SUPPORT THE REQUESTED FUNCTIONALITY
IT IOPINQ,IOP01
 COMPONENT IOP01 MICROCODE: 09.30.30 HARDWARE: 79SCIOP
IT IOPINQ,XIP10
 COMPONENT XIP10 MICROCODE: 15.8.0 HARDWARE: 80-RLP
```

### Example 9

If the IOP name is not valid, the following error message is displayed:

```
IT IOPINQ,XPP10
 THE REQUESTED IOP IS NOT CONFIGURED
```

## 11.38.3. Set the Timeout Value For Mass Storage I/O

The IT,TIMER keyin provides the ability to alter the default disk I/O timer values. The time value specified is used for timing ALL disk I/O. The keyin is provided for maintenance activity when the site needs to override the default disk I/O timing. The I/O timers consist of a hardware timer used by the SCIOP to time disk I/O and a software timer used by the Host to time the SCIOP I/O request. The SCIOP timer default is currently 4 seconds. It can be modified to any number of whole seconds between 4 and 12282 seconds. The Exec software timing uses the periodic timer which is approximately a 6.5 second interval to time events. The time interval calculated by the keyin always covers the hardware timer value requested by the operator. The timer values are not saved if the Host operating system is rebooted; the IT,TIMER keyin must be entered again after the system is rebooted. In multi-host systems, the keyin must be entered on each host operating system requiring the change to the I/O disk timer values.

**Format of the disk timer keyin:**

IT,TIMER operand

where,

operand is the value used to alter the default disk timer values. Valid operand values are:

|            |                                                                                                                                                                                                                                                                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLEAR      | When "CLEAR" is entered as a value, the disk timer values are set back to the default timing for all disk I/O requests.                                                                                                                                                                                                                                                                    |
| 1 – 12282  | These values are the range of seconds which the operator may select to alter the default disk I/O timer. The values are integer representations of the number of seconds to wait for the I/O request to complete before beginning error recovery. Non-zero values which are less than the default disk I/O timer value will cause the disk I/O timer to be set to the default (4 seconds). |
| (No value) | When the operand is not specified, the request is to display the current I/O disk timer values.                                                                                                                                                                                                                                                                                            |

**Example 1**

Set the IOP hardware timer to 7 seconds.

```
IT,TIMER 7
DISK IOP TIMEOUT VALUE IS 7 SECONDS.
DISK HOST TIMEOUT VALUE IS 13 TO 19 SECONDS
```

**Example 2**

Display current values used for disk I/O timers.

```
IT,TIMER
DISK IOP TIMEOUT VALUE IS 7 SECONDS.
DISK HOST TIMEOUT VALUE IS 13 TO 19 SECONDS
```

**Example 3**

Specify time less than 4 seconds.

```
IT,TIMER 2
DISK IOP TIMEOUT VALUE IS 4 SECONDS.
DISK HOST TIMEOUT VALUE IS 6 TO 13 SECONDS
```

### Example 4

If an operand value of zero is entered.

```
IT,TIMER 0
IT ERROR: "0" NOT VALID, USE "CLEAR" TO RESET TIMERS
```

### Example 5

Clear the current disk I/O timers to use system default timer values.

```
IT,TIMER CLEAR
DISK TIMER VALUES RESET TO DEFAULTS
DISK IOP TIMEOUT VALUE IS 4 SECONDS.
DISK HOST TIMEOUT VALUE IS 6 TO 13 SECONDS.
```

## 11.38.4. Set the Timeout Value For Realtime Tape I/O

The IT,TAPTIM keyin provides the ability to change the tape timer value for tape requests made by runs executing at realtime priority. The timer value is not saved when the Host operating system is rebooted. In multi-host systems, the keyin must be entered on each Host operating system requiring the change to tape timer values.

### Format of the keyin:

```
IT,TAPTIM operand
```

Where,

*operand* is the number of integer seconds to wait for realtime tape I/O requests to complete. The value must be greater than 0 and less than 301. If a non-zero value less than 12 is entered, the default of 12 is used. If the value is greater than 300, then 300 is used. If an operand of "CLEAR" is specified, then the realtime timer value is cleared and the non-realtime tape value for the internal tape handler is used. If the operand is not specified, then the current realtime timeout value is displayed.

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLEAR      | Reset the realtime tape timer value to the tape handler's non-realtime timer value.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1 - 999999 | These values are the range of seconds which the operator may select to alter the realtime tape I/O timer value. The values are integer representations of the number of seconds to wait for the I/O request to complete before beginning error recovery. Non-zero values which are less than the default tape I/O timer will cause the tape I/O timer to be set to the default of 12. Non-zero values which are greater than 300 seconds will be set to the maximum timer value of 300 seconds. |

(No value)

When the tape I/O timer value is not specified, the request is to display the current tape I/O realtime timer value.

### Example 1

Set the realtime tape timer value to a Host software timing interval which covers 26 seconds.

```
IT,TAPTIM 26
TAPE REALTIME TIMEOUT VALUE IS 26 SECONDS
```

### Example 2

Displays the current tape realtime timer setting.

```
IT,TAPTIM
TAPE REALTIME TIMEOUT VALUE IS 26 SECONDS
```

### Example 3

Clear realtime timer value setting.

```
IT,TAPTIM CLEAR
CURRENTLY USING NON-REALTIME VALUES FOR TAPE I/O REALTIME TIMER
```

### Example 4

Using 0 as an operand is not allowed and the maximum timer value is 300 seconds.

```
IT,TAPTIM 0
IT ERROR: "0" NOT VALID, USE "CLEAR" TO RESET TIMERS
IT,TAPTIM 999999
TAPE REALTIME TIMEOUT VALUE IS 300 SECONDS.
```

## 11.39. JH—Control Jump History

The JH keyin enables or disables jump history (P trace) and displays whether the jump history is enabled or disabled. The jump history is maintained in a circular buffer in which a limited number of jumps are kept. The jump history is used in dump analysis.

Using the EXTENDED option allows the operating system to collect more jump history entries in the buffer and therefore provides a greater debug capability. Note that turning on JH will have a negative effect on system performance.

### 11.39.1. Format

JH [ON|OFF] [EXTENDED]

where ON enables the jump history, OFF disables the jump history, and EXTENDED increases the number of jump history entries collected.

When you enter JH without an option, a message is displayed indicating whether jump history is enabled or disabled and if the extended jump stack is in use.

## 11.40. KEYINS – Display Registered User Keyins

Programs can be registered to receive unsolicited console keyins where each keyin name must be unique. A program can register for up to four keyins.

The KEYINS keyin displays all keyins that have been registered by such programs.

### 11.40.1. Format

KEYINS *[run-id]*

where:

*run-id*

is the name of a run.

### 11.40.2. System Response

If no run-ID is specified, all registered keyins for all runs are displayed. For example:

```
RUN: IK002 REGISTERED KEYIN(S): ABCDEF7,ABCDEFG*
```

```
RUN: IK003 REGISTERED KEYIN(S): SMA,SMB,SMC,SMD
```

If a run-ID is specified on the keyin, only the keyins registered by the specified run are displayed. The display has the same format as shown above.

## 11.41.LB—Control or Display Logbox Settings

The LB keyin turns TIP Performance Monitor (TPM) selection bits on or off. Two system masks, the mandatory mask and the optional mask, specify the TPM records that are to be collected. Each selection bit can be set in either of the two masks, but not in both.

See the *Transaction Processing Administration and Operations Reference Manual* (7830 7881) for more information on the LB keyin and the TIP Performance Monitor.



These are the selection bits for the TPM system mask:

|         |                                                                                                                                                                                                                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L\$BD   | Allows TPM to collect records for batch and demand programs. Batch and demand information is not collected unless this bit is set.                                                                                                                                                                                |
| L\$BDH  | Allows TPM to record batch and demand header areas for certain TPM records collected on behalf of batch and demand programs. This area is available in all the TPM type 015 records and in all type 012 and 013 records except the P\$SPER record. For this area to be available, the L\$BD bit must also be set. |
| L\$CALL | Permits logging of the types and counts of ERs and CALLs performed by a transaction. This option generates an L\$CALL area within a TPM record type 015, subtype 03.                                                                                                                                              |
| L\$DLCK | Permits logging of FCSS deadlock conditions, which occur on the system. For FCSS deadlocks involving shared (XTC) TIP files to be logged, the TIP FLAGBOX bit SDEADLOCK must also be enabled on the system.                                                                                                       |
| L\$FCSS | Permits logging of each TIP File Control function performed by a program. A record is generated each time the program performs an I/O; therefore, this option should be used with caution.                                                                                                                        |
| L\$FSRL | Allows TPM to collect information about Freespace deferred release processing. This option causes a TPM record type 015, subtype 010 to be generated when an application's Freespace deferred release activity terminates.                                                                                        |
| L\$HVSU | Permits logging of user-passed data. The user passes an ACW on EXIT\$, and the indicated information is written to the TPM audit trail. This option generates an L\$HVSU area within a TPM record type 015, subtype 03.                                                                                           |
| L\$MISC | Permits logging of miscellaneous information regarding a transaction's execution. This option generates an L\$MISC area within a TPM record type 015, subtype 03.                                                                                                                                                 |
| L\$QI   | Allows TPM to record queue-item data areas for TPM records. For message and state-change records, this option is ignored, and QI data are always logged. This area is available for various TPM 012 and 013 audit record types.                                                                                   |
| L\$SUP  | Allows TPM to record double-word SUP data areas for TPM records. This area is available for various TPM 012 and 013 audit record types.                                                                                                                                                                           |
| L\$TDAT | Permits logging of information regarding ER TDATE\$ requests made on the system.                                                                                                                                                                                                                                  |
| L\$TERM | Permits logging of termination information for all transactions. This option generates a termination record type 015, subtype 03.                                                                                                                                                                                 |
| L\$TPIO | Permits logging of the types and counts of TIP I/O performed by a transaction. This option generates less overhead than the L\$FCSS option. This option generates an L\$TPIO area within a TPM record type 015, subtype 03.                                                                                       |
| L\$UDIO | Allows TPM to collect counts of the reads and writes done by IOW\$ and DM\$IOW ERs, as well as the extended-mode call interface UDSS\$IOW, to both Exec and UDS/TIP files. This option generates an L\$UDIO area within a TPM record type 015, subtype 03.                                                        |
| L\$USER | Permits the user to direct information to the audit trail.                                                                                                                                                                                                                                                        |

### 11.41.1. Format

LB[,*option*] [*keyword*] [*recordname*] [,*trailname*]

where:

*option*

is M, valid only with the ON and TRL keywords. With the ON keyword, this option causes the specified *recordname* bit to be set in the mandatory mask. With the TRL keyword, it causes the system trail specification to be set.

*keyword*

is one of the following:

|     |                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ON  | Turns on collection of the specified <i>recordname</i> .                                                                                                                                                                                                                                                                                                                              |
| OFF | Turns off collection of the specified <i>recordname</i> . This option clears the <i>recordname</i> bit in either the mandatory or the optional mask, wherever it is set.                                                                                                                                                                                                              |
| LST | Displays the possible record types in the system mask.                                                                                                                                                                                                                                                                                                                                |
| STA | Displays the status of individual record subtypes.                                                                                                                                                                                                                                                                                                                                    |
| TRL | Specifies the system trail. When used with the STFON parameter, this keyword sets the system trail flag on the specified record and audit trail. Without the STFON parameter, no change is made to the system trail flag.<br><br>The console display of the LB TRL keyin displays the current mask setting for the specified record selection bit: OPT[itional], MAN[datory], or OFF. |

*recordname*

is the name of the selection bit, such as L\$TERM, to be added to a mask or otherwise manipulated, or the group record name L\$ALL, P\$ALL, or U\$ALL.

*trailname*

is the name of the TPM audit trail to which the keyin applies.

### 11.41.2. System Response

These are the allowed format combinations and their effects:

|                           |                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------|
| LB                        | Displays the status of all bits set in the system masks. The bits that are OFF are not displayed. |
| LB <i>recordname</i>      | Displays the status of the specified record selection bit.                                        |
| LB LST                    | Displays all possible record selection bits in the system masks.                                  |
| LB ON <i>recordname</i>   | Sets the specified record selection bit in the optional mask.                                     |
| LB,M ON <i>recordname</i> | Sets the specified record selection bit in the mandatory mask.                                    |

|                                                          |                                                                                                           |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>LB OFF <i>recordname</i></code>                    | Clears the specified record selection bit in the appropriate mask.                                        |
| <code>LB,M TRL <i>recordname</i>,<i>trailname</i></code> | Specifies <i>trailname</i> as the system trail for the specified record and sets the system trail flag.   |
| <code>LB TRL <i>recordname</i>,<i>trailname</i></code>   | Specifies <i>trailname</i> as the system trail for the specified record and clears the system trail flag. |
| <code>LB,M TRL <i>recordname</i></code>                  | Specifies the default TPM trail "TPM" for <i>recordname</i> and sets the system trail flag.               |
| <code>LB TRL <i>recordname</i></code>                    | Specifies the default TPM trail "TPM" for <i>recordname</i> and clears the system trail flag.             |

An LB keyin attempted with any other format displays the following console message:

```
TIP LB keyin syntax error - LB keyin rejected
```

If a record collection bit is already in the state requested by an LB keyin, the keyin displays the current setting, and no change occurs.

Setting a record collection bit in one of the system masks causes the bit to be cleared in the other mask.

### 11.41.3. Restrictions

In an XTC system, TPM record selection bits are not communicated between hosts.

## 11.42. LC—Control TIP Performance Monitor (TPM)

The LC (Logging Control) keyin lets you control the TIP Performance Monitor (TPM). When TPM is on, transactions that have the appropriate VALTAB options set have TPM records written to the specified TPM audit trail.

See the *Transaction Processing Administration and Operations Reference Manual* (7830 7881) for more information on the LC keyin and the TIP Performance Monitor.

### 11.42.1. Format

```
LC [option] [trailname]
```

where:

*option*

is ON or OFF.

*trailname*

is the name of a TPM audit trail.

These are the allowed format combinations and their effects:

|                         |                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LC ON <i>trailname</i>  | Starts collection on the TPM audit trail named <i>trailname</i> , displays the collection state, and shows the current date and time. If <i>trailname</i> is not open, an attempt is made to auto-open it. |
| LC OFF <i>trailname</i> | Turns off collection on the TPM audit trail named <i>trailname</i> , displays the collection state, and shows the current date and time. If <i>trailname</i> is already turned off, no action is taken.    |
| LC ON                   | Starts collection on the default TPM audit trail, displays the collection state, and shows the current date and time. If the default audit trail is not open, an attempt is made to auto-open it.          |
| LC OFF                  | Turns off collection on the default TPM audit trail, displays the collection state, and shows the current date and time. If the default trail is already turned off, no action is taken.                   |
| LC TPM                  | Displays the collection state of the default TPM audit trail.                                                                                                                                              |
| LC                      | Displays the collection state of each configured TPM audit trail.                                                                                                                                          |

### 11.42.2. System Response

The system displays the following output of the LC keyin for TPM ON or OFF:

```
tpm-trail-id ON|OFF [date/time] [error-text] [runid]
```

where:

*tpm-trail-id*

is the ID of the TPM trail (that is TPM).

*date/time*

are the date and times that the last TPM ON or OFF occurred, if any, in the format *yyyy-mm-dd hh:mm:ss zone*. The *zone* field contains the configured time zone mnemonic. Both the local and the UTC times are displayed.

*error-text*

is either "Trail in Recovery" or "State Change in Progress".

*runid*

is the runid that requested the change (not used for console keyin requests).

### 11.42.3. Examples

#### Example 1

```
LC TPM
```

```
TPM ON 2014-01-07 12:20:25 CST 2014-01-07 18:20:25 UTC
```

#### Example 2

```
LC ON TPMVOD
```

```
TPMVOD ON 2014-01-07 14:35:45 CST 2014-01-07 20:35:45 UTC
```

#### Example 3

```
LC OFF
```

```
TPM OFF 2014-01-07 14:40:07 CST 2014-01-07 20:40:07 UTC
```

## 11.43. LG—Make an Entry in the System Log File

The LG keyin lets you make entries in the system log file, which contains information about run starting and completion times, console messages, checkpoints, device assignments and releases, and hardware fault information.

If your site has the optional TeamQuest Log Analyzer (TeamQuest LA) software, you can subsequently request the message report, which extracts the console printer listing from the system log file. The message report includes the text that you entered by using the LG keyin.

### 11.43.1. Formats

```
LG text
```

Enters the specified text into the system log file. The text usually concerns the general operation of the system. (The system records the message in the console printer listing and in the system log file.)

```
LG *run-id text
```

Enters the specified text about a particular run into the system log file. You can use this keyin format only when the run is active. If the *run-id* cannot be found among the active runs, the keyin is rejected. The text also appears with the run's summary information when the run is finished.

### 11.43.2. Additional Information

For more information about the system log file, refer to the *System Log Operations and Support Reference Manual Volume 1* (7831 0315).

### 11.43.3. Examples

```
LG SHIFT CHANGE - NEW OPERATOR
```

```
LG *BOB TAPE 123TST NOT IN LIBRARY
```

## 11.44. MAINT—Change XIIP State

The MAINT keyin moves an XIIP from the DN state to an UP maintenance partition state so that the XIIP microcode can be dumped or loaded.

### 11.44.1. Format

```
MAINT xiip_name
```

where *xiip\_name* is the PCIOP that requires microcode loading or dumping.

The *xiip\_name* is removed from the MAINT partitioning state by entering

```
DN xiip_name
```

When the MAINT *xiip\_name* keyin completes, the status reported on the system console is:

```
UP MT
```

An FS keyin shows the XIIP in the UP MT state.

### 11.44.2. Examples

#### Example 1

```
FS XIP44
```

```
XIP44 UP MT
```

#### Example 2

```
FS,XIIP
```

```
XIP44 UP MT, XIP45 UP
```

### 11.44.3. System Response

If the MAINT *xiip\_name* keyin is entered when the XIIP is not in the DN partition state, the following message is displayed:

```
xiip_name must be DN when the MAINT keyin is entered.
```

If a MAINT request is entered for a non-XIIP component, the following message is displayed:

```
MAINT keyin invalid for logical name component_name, input ignored.
```

If a MAINT keyin is entered for a PCIOP, which is already in the MAINT partition state, the following message is displayed:

```
MAINT keyin already performed for xiip_name.
```

### 11.44.4. Additional Considerations

An FS keyin of the XIIP performed after IOP load has started provides an indicator on the FS display that a microcode load is currently in progress.

UP or DN keyins affecting an XPC-L component while the IOP load is in progress are not allowed and a message indicates this.

When the microcode load or dump operation is completed, the XIIP must be downed to remove the XIIP from the MAINT partition state. An UP of the XIIP moves the XIIP back into the partition for system use.

If the system stops when the XIIP is in MAINT mode, the XIIP remains in MAINT mode when the partition is rebooted. If the XIIP state is moved from UP to DN by the operator at the System Control Facility while the system is stopped, the MAINT mode is cleared when the partition is rebooted.

## 11.45. MD—Modify a Disk Pack

The MD keyin lets you establish, modify, or remove the inhibits on a removable disk pack.

If the pack is unassigned or not currently mounted on any system device, it is given a status of PRESENT. The main benefit of the PRESENT state is that it allows you to place inhibits on a pack not currently on the system, so that when the pack is put on the system, its access is restricted as needed.

### 11.45.1. Format

```
MD pack-id[/[dir-id]/[inhibits],[pack-id]...]
```

where:

*pack-id*

is the name of the pack.

*dir-id*

is the directory-ID supplied on the keyin, either STD or SHARED. STD is for a local pack, and SHARED is for a shared pack. The default is STD.

*inhibits*

is any valid combination of the assign (A), read (R), or write (W) inhibits. Whenever you use the write inhibit, you must also use the read inhibit. If you add inhibits to a pack in the UP or PRESENT state when the assign count is zero, you must include the assign inhibit. To remove existing inhibits from a pack, leave this field blank.

### 11.45.2. System Response

The response to the MD keyin is

```
pack-id state - INHIBITS = inhibits ASSIGN COUNT = count dir-id
```

where:

*pack-id*

is the name of the pack from the keyin.

*state*

is UP, DN, PRESENT, or NOT PRESENT. When you add inhibits to a pack that is not currently on the system, its state is PRESENT. When you remove inhibits from a pack with a status of PRESENT, its state reverts to NOT PRESENT.

*inhibits*

can be R (read), W (write), or A (assign).

*count*

is the number of files assigned on the pack.

*dir-id*

is the directory-ID, either STD or SHARED.

### 11.45.3. Restrictions

You cannot use the MD keyin on a unit-duplexed device.

To use the MD keyin in a local environment, the NEWCON configuration parameter REQNBPRST0 must be greater than 0. To use the MD keyin in a shared environment, the NEWCON configuration parameter REQNBPRST1 must be greater than 0. See the *Exec System Software Installation and Configuration Guide (7830 7915)* for more information on these configuration parameters.



## 11.45.4. Examples

### Example 1

MD DSKF1/STD/RWA

Places the read (R), write (W), and assign (A) inhibits on DSKF1 and places it in the PRESENT state. The response is

DSKF1 PRESENT - INHIBITS = RWA ASSIGN COUNT = 0 STD

### Example 2

MD DSKF1

Removes inhibits on DSKF1.

### Example 3

MD PACKA/STD/RW

Places the read (R) and (W) inhibits on PACKA (assuming PACKA has been brought up and assigned). The system response is

PACKA UP - INHIBITS = RW ASSIGN COUNT = 1 STD

### Example 4

MD PACK1/STD

Removes the inhibits on PACK1, where PACK1 is up and has an assign count equal to 1. The system response is

PACK1 UP - INHIBITS = NONE ASSIGN COUNT = 1 STD

If the pack status is PRESENT and the inhibits are removed, the pack is removed from the system list of UP, DN, or PRESENT packs. The system response is

PACK1 NOT PRESENT - INHIBITS = NONE ASSIGN COUNT = 0

### Example 5

MD PACK2/STD/AWR

Places assign (A), write (W), and read (R) inhibits on PACK2, assuming that the DN,PACK keyin was used to unassign PACK2 and place it in a down state as follows:

DN,PACK PACK2///D8433

The system response is

PACK2 DN - INHIBITS = AWR ASSIGN COUNT = 0 STD

### Example 6

```
MD PACK3/STD/WA
```

Place the write (W) and assign (A) inhibits on PACK3, assuming that PACK3 was assigned and placed in a DN state.

The system response is

```
PACK3 DN - INHIBITS = WA ASSIGN COUNT = 1
```

## 11.46. MH—Control MHFS Activities

The MH keyin controls Multi-Host File Sharing (MHFS) activities and displays information about the facility.

For more information on the Multi-Host File Sharing environment, see the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469).

### 11.46.1. Formats

#### MH FILES

```
MH FILES
```

Displays how many runs on a single host have shared files assigned, and the number of runs assigned to the Exec once the system is up and running.

#### MH RUNS

```
MH RUNS
```

Displays all runs with shared files assigned.

#### MH AD

```
MH,AD
```

Used in a MHFS system at Modify Config time to allow data to be recovered completely from devices that were configured as shared devices but were changed to local devices using the MH,AD STD,*devnam* keyin. The MH,AD keyin causes the boot process to be preempted following successful entrance into the file-sharing environment so that MH,AD STD,*devnam* keyins can be successfully entered. Refer to the *Multi-Host File Sharing (MHFS) Planning, Installation, and Operations Guide* (7830 7469) for complete information.

```
MH,AD dir-id,devnam1,[devnam2,...devnamN]
```

Changes the directory-ID of the specified device to the specified directory. The specified device must be marked down (DN) to perform this keyin.

`MH,AD SHARED|STD,devnam1`

Changes the directory-ID of the disk drive from STD (standard) to SHARED or from SHARED to STD. The disk drive must be marked down (DN) to perform this keyin.

## **MH DN**

`MH,DN`

Removes the host (of the console from which you entered the keyin) from the file-sharing environment. Future shared file assignments are rejected as long as the removed host stays down.

`MH,DN HOST,host-id`

Removes the specified host from the file-sharing environment. Future shared file assignments are rejected as long as the host is down. Removing the host results in the termination of all runs with shared files assigned and prevents future file assignments. Use with extreme caution.

## **MH FS**

`MH,FS HOST`

Displays the host-ID and site-ID for your host and all other hosts active in the file-sharing environment.

`MH,FS HOST,ALL`

Displays the site-ID and status of all hosts in the file-sharing environment.

`MH,FS HOST,host-id`

Displays the status of the specified host-ID in the file-sharing environment.

## **MH IN**

`MH,IN DIR,SHARED`

Marks the shared directory for initialization before the system runs. Enter this keyin after the MODIFY CONFIG message is displayed.

### MH RC

MH,RC DIR,SHARED

Builds the shared master configuration table (MCT). Unlike the MH,IN keyin, the MH,RC keyin causes the shared devices to be recovered, not initialized. Enter this keyin after the MODIFY CONFIG message is displayed.

### MH RV

MH,RV

Removes your host from the file-sharing environment. Runs that have shared files assigned can finish as long as no new shared files are cataloged. This keyin is not allowed until after SYS FIN.

### MH UP

MH,UP

Enters your host into the file-sharing environment while the system is running and notifies the other active hosts of the action.

## 11.47. MIPS—Set Desired MIPS for a Partition

This keyin is used to set the desired MIPS for the partition within the limits set by the maximum MIPS and minimum MIPS specified in the PDB.

### 11.47.1. Format

MIPS [xxxxxx]

where *xxxx* is optional and specifies the new MIPS setting for this partition.

If the system is not running with a MIPS-based or MIPS-metered Exec Image Enabler key, the following message is displayed:

MIPS keyin has no effect since the Image Enabler key registered is not MIPS-based or MIPS-metered.

See the *Exec System Software Administration Reference Manual* (7831 0323) for more information about setting MIPS.

The unsecured and unsolicited MODELNUM keyin is provided to verify and display the SCP Exec key attributes that were SOLAR-registered by way of the SCP Exec key.

## MODELNUM

The MODELNUM keyin output varies depending on the Image Enabler key registered.

```

Key Registered: Model-Based
SCP Key MCN: mmmmmmmmmmmmmmmmmmmmmmmm PDB MCN: pppppppppppppppppppppp
NORMAL KEY Key ID: kkkkkkkkkkkkkk Partitionable: [YES|NO]
 Model: nnnnnnnnnnnnnnnn Number of IPs: ii
 Expiration Date: yyyy-mm-dd Performance Level: jjj%
 Hard Expire Date: yyyy-mm-dd SCN: ssssssssssssss
OPTIONAL KEY (Active)
 Model: nnnnnnnnnnnnnnnn Number of IPs: ii
 Expiration Date: yyyy-mm-dd Performance Level: jjj%
 Hard Expiration Date: yyyy-mm-dd
Other partitions are using ll of the ii IPs allowed.

```

**Note:** The optional key information only appears if an optional key is active. The statement "Other partitions are using xx of the ii IPs allowed" only appears when a partitionable SCP key is used in more than one partition.

```
Key Registered: MIPS-Based
SCP Exec Key MCN: mmmmmmmmmmmmmmmmmmm PDB MCN: pppppppppppppppppp
NORMAL KEY Model: nnnnnnnnnnnnn Performance Level: [jjj%|N/A]
 Key ID: kkkkkkkkkkkk Partitionable: [YES|NO]
 Key MIPS: rrrrrr MIPS SCN: ssssssssssss
 Expiration Date: yyyy-mm-dd Hard Expiration Date: yyyy-mm-dd
OPTIONAL KEY (Active)
 OPTIONAL KEY Model: nnnnnnnnnnnnn Performance: jjj%
 Key MIPS: oooooo MIPS
 Expiration Date: yyyy-mm-dd Hard Expiration Date: yyyy-mm-dd
Desired_mips = xxxxx, Minimum_mips= yyyyy, Maximum_mips= zzzzz
This partition (ddddddddddddd) is using qqqqq MIPS.
No other partitions are using this key.
(or)
Partition eeeeeeeeeeee is using ttttt MIPS
Partition ffffffffffffff is using uuuuu MIPS
```

Partition *gggggggggggg* is using *vvvvv* MIPS  
 Partition *hhhhhhhhhhhh* is using *wwwww* MIPS  
*(Output varies with number of partitions using the key)*

**Note:** The optional key information only appears if an optional key is active. The "Partition xxxx is using" line is displayed for any number of additional partitions that are using the key. If no other partitions are using the key, then "No other partitions are using this key" line is displayed.

## MIPS-Metered Keyin Response

```
Key Registered: MIPS-Metered
```

```
SCP Exec Key MCN: mmmmmmmmmmmmmmmmmmmmmmmmm PDB MCN: ppppppppppppppppppppp
```

```
NORMAL KEY Model: nnnnnnnnnnnnnnn Performance Level: jjj%
```

```
Key ID: kkkkkkkkkkkk Partitionable: [YES|NO]
```

```
Key MIPS: bbbbbb Baseline MIPS aaaaaa Ceiling MIPS
```

```
Expiration Date: yyyy-mm-dd Hard Expiration Date: yyyy-mm-yy
```

```
Contract ID: cccccccccc SCN: ssssssssss
```

```
Desired_mips = xxxxx, Minimum_mips= yyyyyy, Maximum_mips= zzzzz
```

```
This partition (ddddddddddddd) is using qqqqq MIPS.
```

```
No other partitions are using this key.
```

```
(or)
```

```
Partition eeeeeeeeeeee is using ttttt MIPS
```

```
Partition ffffffffffff is using uuuuu MIPS
```

```
Partition gggggggggggg is using vvvvv MIPS
```

```
Partition hhhhhhhhhhhh is using wwwww MIPS
```

```
(Output varies with number of partitions using the key)
```

**Note:** The statements “Partition xxxx is using ....” only appears when a partitionable SCP key is used in more than one partition.

## IP-Based COD Keyin Response

```
Key Registered: IP-Based COD
SCP Key MCN: mmmmmmmmmmmmmmmmmmmmm PDB MCN: pppppppppppppppppppp

NORMAL KEY
Model: nnnnnnnnnnnnnn SCN: ssssssssssss
Key ID: kkkkkkkkkkkk Partitionable: [YES|NO]
Expiration Date: yyyy-mm-dd IP Capacity: jjj%
Hard Expiration Date: yyyy-mm-dd Number of IPs: ii

OPTIONAL KEY (Active)
Model: nnnnnnnnnnnnnn
Key ID: kkkkkkkkkkkk Partitionable: [YES|NO]
Expiration Date: yyyy-mm-dd IP Capacity: jjj%
Hard Expiration Date: yyyy-mm-dd Number of IPs: ii
```

**Note:** The optional key information only appears if an optional key is active.

```
Key Registered: Consumption
SCP Key MCN: mmmmmmmmmmmmmmmmmmmm PDB MCN: pppppppppppppppppppp
NORMAL KEY
Model: nnnnnnnnnnnnnn SCN: ssssssssssssss
Key ID: kkkkkkkkkkkkk Partitionable: [YES|NO]
Expiration Date: yyyy-mm-dd IP Capacity: jjj%
Hard Expiration Date: yyyy-mm-dd Number of IPs: ii
```

```
Key Registered: Unknown
SCP Key MCN: UNKNOWN PDB MCN:
NORMAL KEY
 Model: UNKNOWN SCN: UNKNOWN
 Key ID: UNKNOWN Partitionable: NO
 Expiration Date: NONE IP Capacity: 100%
 Hard Expiration Date: NONE Number of IPs: 32
```

### Explanation of Output

*pppppppppppppppppp*  
is the PDB Manufacturing Control Number.

is the unique key identifier, consisting of the MCN and partition number.

*rrrrrr* is the normal MIPS specified on the MIPS-based Image Enabler key in effect.

000000

is the optional MIPS specified on the MIPS-based Image Enabler key in effect.

*bbbbbb*  
is the prepaid baseline MIPS for MIPS-metering Image Enabler keys.

*aaaaaa*  
is the maximum (ceiling) MIPS for MIPS-metering Image Enabler keys.

*jjj%*

is the normal or optional contracted SCP performance level percentage when the SCP key is registered; otherwise, the default is used. The default performance varies by model family. The registered percentage reflects the contracted performance for this partition. When an optional key is registered, the percentage is shown only in the optional key portion of the output.

If this value is less than 100 percent, the difference represents the performance reserve available for system growth with a new contracted SCP performance level.

*ii*

is the normal or optional contracted IPs when the SCP is registered; otherwise, the default is used.

*//*

is the number of IPs in use by other partitions.

*yyyy-mm-dd*

specifies the normal or optional expiration year, month, and day.

*cccccccccccc*

is the metering contract identifier.

*ssssssssssss*

is the System Control Number (SCN). The SCN is used for entitlement software use.

*xxxxxx*

is the desired MIPS at which you want this partition to be running. If the desired MIPS is 0 when the system is booted, the partition uses all the available MIPS left on the key registered (minimum of MIPS on the key and hardware MIPS available in the partition). Changing the `desired_mips` parameter changes this value.

*yyyyyy*

is the minimum MIPS needed by this partition. If not 0, the MIPS keyin can be used to modify the desired MIPS within the range provided by minimum MIPS and maximum MIPS.

*zzzzzz*

is the maximum MIPS possible for this partition. If not 0 and not equal to the minimum MIPS, the MIPS keyin can change the desired MIPS within the range provided by minimum MIPS and maximum MIPS.

*dddddddddddd*

is the name of the partition displaying the message.



*qqqqq*

is the MIPS being used by the partition displaying the message.

*eeeeeeeeeeee, ffffffffffff, gggggggggggg, hhhhhhhhhhhh*

are the names of other partitions using the MIPS from the same Image Enabler key identifier.

## 11.49. MR—Recall Outstanding Messages

The MR keyin lets you recall outstanding read-and-reply messages that have scrolled off the screen at remote terminal sites. Only local consoles keep read-and-reply messages from scrolling off the screen. When using the remote console capability of the Exec, which permits a demand terminal to be used as a console, messages can scroll off the screen before they can be answered.

Use this keyin in conjunction with the CMR keyins to recall previously displayed console messages.

### 11.49.1. General Format

The general format of the MR keyin is

```
MR function [,subfunction] [identifier]
```

where *function* is TR or RC, and *subfunction* and *identifier* are optional parameters.

### 11.49.2. MR TR Keyin

The MR TR keyin displays information about messages queued for remote consoles. Formats for the MR TR keyin are as follows:

```
MR TR
```

Displays information on all site-IDs in RESPONSE mode if the keyin is entered from an onsite console or a remote console in RESPONSE mode.

```
MR TR site-id
```

Displays outstanding read-and-reply messages for the console specified by *site-id*.

```
MR TR,N
```

Displays the site-IDs of consoles that are in RESPONSE mode, and the number of outstanding read-and-reply messages for each of them.

```
MR TR,N site-id
```

Displays the number of outstanding read-and-reply messages for the console specified by *site-id*.

MR TR, *n*

Displays the site-IDs of consoles that have at least *n* outstanding read-and-reply messages, where *n* is 1–9 or + (+ means 10).

### 11.49.3. MR RC Keyin

The MR RC keyin displays information about active remote consoles that have one or more message groups under their control. Formats for the MR RC keyin are as follows:

MR RC

Displays the site-IDs of all active remote consoles that have one or more message groups under control, and indicates which message groups are under control.

MR RC, *S site-id*

Displays information on the message groups that are under control of the console specified by *site-id*.

MR RC, *M message-group*

Displays the site-IDs of the active remote consoles that have the specified message group under control.

MR RC, *M ALL*

Displays the site-IDs of the active remote consoles that have each message group under control.

### 11.49.4. MR OS Keyin

The MR OS keyin displays information about messages queued for onsite consoles. Following are the formats for the MR OS keyin:

MR OS

Displays information on all read-and-reply messages queued to onsite consoles. The display includes the reply number, the name of the console to which the message is queued, and the text of the message.

MR OS *console-name*

Displays information on all read-and-reply messages queued to the onsite console specified by *console-name*.

## 11.50. MS—Display Mass Storage Information

The MS keyin displays the following information on the availability of fixed and removable mass storage devices:

- A summary of the total accessible (up and suspended) fixed mass storage tracks
- The total available (up and unallocated) fixed mass storage tracks
- The fixed mass storage threshold level at which the system will begin unloading fixed mass storage
- The fixed mass storage availability goal when the unload process completes

### 11.50.1. Formats

#### All Mass Storage

MS

Displays the availability of both local and shared mass storage devices.

#### Local Mass Storage

MS,STD

Displays information on local mass storage devices only.

#### Shared Mass Storage

MS,SHRD

Displays information on shared mass storage devices only.

### 11.50.2. System Response

#### Local Mass Storage

The information displayed for local mass storage is

```
SUMMARY: STD FIXED TRACKS ACCESSIBLE = std-ft-access
STD FIXED TRACKS AVAILABLE = std-ft-avail
STD ROLOUT START THRESHOLD = std-rolout-start
STD ROLOUT AVAILABILITY GOAL = std-rolout-goal
STD FIXED DIRECTORY TRACKS ACCESSIBLE = std-dir-access
STD FIXED DIRECTORY TRACKS AVAILABLE = std-dir-avail
```

where:

*std-ft-access*

is the decimal number of total accessible fixed mass storage tracks (up and suspended) in the directory for the system.

### *std-ft-avail*

is the decimal number of total available fixed mass storage tracks (up and unallocated) in the directory for the system.

### *std-rolout-start;*

is the decimal number of remaining fixed mass storage tracks, derived from the dynamic configuration parameter `STD_ROLOUT_START_THRESHOLD`, at which the system run `ROLOUT` gets started to unload fixed mass storage. It specifies the minimum amount of tracks to unload when *stdstarttrks* is greater than the difference between *stdgoaltrks* and *stdstarttrks*.

### *std-rolout-goal*

is the decimal number of available fixed mass storage tracks, derived from the dynamic configuration parameter `STD_ROLOUT_START_THRESHOLD`, which the system run `ROLOUT` targets as tracks available when `ROLOUT` completes.

### *std-dir-access*

is the decimal number of accessible fixed directory tracks. This is computed as 4096 times the number of fixed disks that are up.

### *std-dir-avail*

is the decimal number of fixed directory tracks currently available for expansion.

## Shared Mass Storage

The information displayed for shared mass storage is

```
SUMMARY: SHRD FIXED TRACKS ACCESSIBLE = shared-ft-access
SHRD FIXED TRACKS AVAILABLE = shared-ft-avail
SHRD SROLUT START THRESHOLD = shared-srolut-start
SHRD SROLUT AVAILABILITY GOAL = shared-srolut-goal
SHRD FIXED DIRECTORY TRACKS ACCESSIBLE = shared-dir-access
SHRD FIXED DIRECTORY TRACKS AVAILABLE = shared-dir-avail
```

where:

### *shared-ft-access*

is the decimal number of total accessible fixed mass storage tracks (up and suspended) in the shared directory for the system.

### *shared-ft-avail*

is the decimal number of total available fixed mass storage tracks (up and unallocated) in the shared directory for the system.

*shared-srolut-start*

is the decimal number of remaining fixed mass storage tracks, derived from the dynamic configuration parameter SHRD\_SROLUT\_START\_THRESHOLD, at which the system run SROLUT gets started to unload fixed mass storage. It specifies the minimum amount of tracks to unload when *shrdstarttrks* is greater than the difference between *shrdgoaltrks* and *shrdstarttrks*.

*shared-srolut-goal*

is the decimal number of available fixed mass storage tracks, derived from the dynamic configuration parameter SHRD\_SROLUT\_START\_THRESHOLD, which the system run SROLUT targets as tracks available when SROLUT completes.

*shared-dir-access*

is the decimal number of accessible fixed directory tracks. This is computed as 4096 times the number of fixed disks that are up.

*shared-dir-avail*

is the decimal number of fixed directory tracks currently available for expansion.

### 11.50.3. Examples

MS,STD

Displays the availability of local mass storage devices, as follows:

```
SUMMARY: STD FIXED TRACKS ACCESSIBLE = 70893754
STD FIXED TRACKS AVAILABLE = 6611732
STD ROLOUT START THRESHOLD = 1430241
STD ROLOUT AVAILABILITY GOAL = 4443225
STD FIXED DIRECTORY TRACKS ACCESSIBLE = 262144
STD FIXED DIRECTORY TRACKS AVAILABLE = 251596
```

MS,SHRD

Displays the availability of shared mass storage devices, as follows:

```
SUMMARY:SHRD FIXED TRACKS ACCESSIBLE = 15463406
SHRD FIXED TRACKS AVAILABLE = 600776
SHRD SROLUT START THRESHOLD = 234294
SHRD SROLUT AVAILABILITY GOAL = 618536
SHRD FIXED DIRECTORY TRACKS ACCESSIBLE = 57344
SHRD FIXED DIRECTORY TRACKS AVAILABLE = 55526
```

## **11.51.MU—Display Central Complex Memory Use**

The MU keyin displays central complex memory use (in decimal words) on a global basis, for an individual run, or for all runs in the system.

### **11.51.1. Formats**

#### **Global Memory Use**

```
MU
```

Displays global memory use.

#### **Memory Use by a Specific Run**

```
MU run-id
```

Displays memory use by a specific run.

#### **Memory Use by All Runs**

```
MU,ALL
```

Displays memory use by all runs.

### **11.51.2. System Response**

The information displayed summarizes the total system memory use. The information is displayed by category. Each category corresponds to storage priorities used by the Exec for memory management. If a particular category has no memory in use (such as no down memory), the category is not displayed.

On systems that do not support paging, insignificant fields are not displayed.

#### **MU**

The response to an MU keyin is as follows:

```
ACTIVE: 32,972,800
NAILED 4,435,968
LOCKED 188,416
FREE: 100,720,640
LOGICALLY AVL: 1,992,134
RESERVED: 586,661
DOWN: 134,217,728

MEMORY UTILIZATION BY CLASS
ABSOLUTE REAL
EXEC: 28,356,608 9,977,856
```

```

USER: 859,070,464 38,326,272
FILE: 11,952,128 7,589,888
QBANK: 769,961,984 14,589,952

```

where:

|                             |                                                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTIVE                      | is the number of words currently used by active runs.                                                                                                                               |
| DOWN                        | is the number of words in the down state.                                                                                                                                           |
| FREE                        | is the number of words ready for immediate use.                                                                                                                                     |
| LOCKED                      | is the number of words locked in memory that can not be unloaded or moved (for example, they are the source or destination of an I/O transfer).                                     |
| LOGICALLY AVL               | is the number of words that are part of a working set that can be unloaded to free additional words or be returned to the active state without use of I/O.                          |
| MEMORY UTILIZATION BY CLASS | is the number of words of memory currently in use by the Exec, user programs, memory files, and queue banks (QBANK).                                                                |
| NAILED                      | is the number of words that cannot be unloaded from real memory, but can be moved around if the physical memory page is needed for some other function (for example, partitioning). |
| RESERVED                    | is the number of words in the reserved state.                                                                                                                                       |

### **MU *run-id***

The information displayed in response to an MU *run-id* keyin has one of four formats. *size* is the total size of all banks contained in the program and activity level bank descriptor tables (BDT).

```
run-id size *prog-name/prog-version
```

The run is using a program with this name and version (if any). The asterisk (\*) indicates that the program is loaded from a Transaction Processing (TIP) program library file.

```
run-id size ** NONE **
```

No program is currently in use in this run.

```
run-id size NAME$
```

The run is using an unnamed program.

```
run-id NOT FOUND IN MEMORY
```

There is no current run with this run-ID.

### MU,ALL

In response to an MU,ALL keyin, the system displays a line of information for each run-ID in one of the formats just described.

If there are insufficient system resources to handle the MU,ALL keyin, the following message is displayed:

```
MU KEYIN ABORTED - INSUFFICIENT RESOURCES
```

If there are no runs in memory at the time of the MU,ALL keyin, the following message is displayed:

```
NO RUNS IN MEMORY
```

## 11.51.3. Examples

### Example 1

#### (Systems that support paging)

```
MU
```

The system responds with

```
ACTIVE: 32,972,800
NAILED 4,435,968
LOCKED 188,416
FREE: 100,720,640
DOWN: 134,217,728

MEMORY UTILIZATION BY CLASS
ABSOLUTE REAL
EXEC: 28,356,608 9,977,856
USER: 859,070,464 38,326,272
FILE: 11,952,128 7,589,888
QBANK: 769,961,984 14,589,952
```

#### (Systems that do not support paging)

```
MEMORY UTILIZATION BY CLASS
ABSOLUTE
EXEC: 69,105,860,608
USER: 859,070,464
FILE: 11,952,128
QBANK: 769,961,984
```

### Example 2

```
MU *TIPMO
```

The system responds with

```
*TIPMO 65536 NAME$
```



**Example 3**

MU BJP

The system responds with

BJP 14784 \*PMD

**Example 4**

MU BJPA

The system responds with

BJPA NOT FOUND IN MEMORY

**Example 5**

MU BJPB

The system responds with

BJPB 2560 \*\* NONE \*\*

**Example 6**

MU CORWIN

The system responds with

CORWIN 262144 TPAS/MCB

**Example 7**

MU,ALL

The system responds with

FPSDMN 54784 FURPUR  
BJPB 2560 \*\* NONE \*\*  
BJP 14784 \*PMD  
TOMBO 25088 IACULL  
OSAMON 57856 OSAM  
\*TIPMO 65536 NAME\$  
CORWIN 262144 TPAS/MCB

**Example 8**

MU,ALL

The system responds with

MU KEYIN ABORTED - INSUFFICIENT RESOURCES

### Example 9

```
MU,ALL
```

The system responds with

```
NO RUNS IN MEMORY
```

## 11.52.MV—Move a Disk Pack

The MV keyin applies to physically removable disk subsystems only, and moves a disk pack from a failing drive to an unassigned working drive that is either suspended or up. The keyin must be entered before the pack is moved. If the disk packs are moved first, I/O errors can occur on the unit from which the pack was moved. The pack can be moved to any unit of the same equipment type and logical channel type.

As part of MV keyin processing, the status of the unit the disk pack is moved from is changed to down (DN). You can change the status to reserved (RV) to run diagnostic tests on the unit.

### 11.52.1. Format

```
MV devnam1,devnam2
```

where:

*devnam1*

is the device from which the pack is to be moved.

*devnam2*

is the device to which the pack is to be moved.

### 11.52.2. System Response

The system response to the keyin is

```
devnam1 MOVED TO devnam2
devnam1 DN, devnam2 [UP/SU] [R/F] pack-id
```

where:

*pack-id*

is the pack-ID.

UP/SU

specifies whether *devnam2* is up (UP) or suspended (SU).

R/F

specifies whether *devnam2* is removable (R) or fixed (F).

A LOAD message and possibly SERVICE messages (if the pack change has not occurred at the expiration of 2-minute intervals) are displayed on the system console.

### 11.52.3. Restrictions

The MV keyin is not allowed on units in a Multi-Host File Sharing environment or on unit-duplexed devices.

## 11.53. OM—Display or Change the Operating Mode

The OM keyin displays or changes the operating mode of a system running SMART, or lets you send a message to the console from SMART. SMART is optional software that automates many functions normally performed by an operator. The operating mode can be attended, casual, or unattended.

For more information about SMART, see the *SMART Administration and Operations Guide* (7830 7337).

### 11.53.1. Formats

```
OM[, {A|C|U|X}] [{ON|OFF}/option[/option]...] [{ON|OFF}/option[/option]...] ...
```

or

```
OM,M message-text
```

where:

A

specifies attended operation mode.

C

specifies casual operation mode.

U

specifies unattended operation mode.

X

terminates the OM keyin handler.

M

sends the specified message text to the console.

ON or OFF

turns the following options on or off.

### *option*

represents the following options, in any order:

|       |                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AAMS  | Turns the autoaction message system (AMS) on or off. This is meaningful only in attended mode, since AMS is always on in casual and unattended mode.                                                                                                                                                                                                    |
| DEBUG | Turns the debug mode version of the AMS on or off. This is a mode of AMS where the actual message response does not take place, but the preferred response is indicated. This mode is used for AMS database debugging or for operator training. Note that turning DEBUG on turns autoresponse off. To return to autoresponse, DEBUG must be turned off. |
| TRME  | Terminates a nonprivileged (no SCONSOLE privilege) user's run with error type termination. This occurs when it issues a message requiring a response and that response is not in the AMS database.                                                                                                                                                      |
| UALM  | Rings an alarm when any user run terminates abnormally.                                                                                                                                                                                                                                                                                                 |

### *message-text*

is the read-only message sent to the console. The message can be 1 to 60 alphanumeric characters in length. You do not need to enclose the message in quotation marks.

## 11.53.2. System Response

The following information is displayed after an OM keyin:

```
OPERATION MODE: {ATTENDED|CASUAL|UNATTENDED}

OM OPTIONS (NEXT UNATTENDED): option [, option]...

OM OPTIONS (CURRENT): option [, option]...
```

In casual mode and unattended mode, all read-and-reply and read-alert messages that do not have a response in the AMS database ring an alarm.

If you enter an incorrect OM keyin, the following message is displayed:

```
OM KEYIN FORMAT ERROR AFTER xx
```

where *xx* are the 2 characters immediately preceding the error.

If you enter an option on the keyin that also has a mode change, and that option is illegal for that mode, the following message is displayed:

```
OM OPTION option NOT CONSISTENT WITH SPECIFIED MODE
```

### 11.53.3. Examples

OM

Displays the current operation mode, as follows:

```
OPERATION MODE: ATTENDED
OM OPTIONS (NEXT UNATTENDED):
OM OPTIONS (CURRENT):
```

OM,A ON/AAMS/DEBUG

Sets attended mode with AAMS and DEBUG options on. The system displays

```
OPERATION MODE: ATTENDED
OM OPTIONS (NEXT UNATTENDED):
OM OPTIONS (CURRENT): AAMS, DEBUG
```

OM ON/UALM

Turns on UALM option for the next change to unattended mode. The system displays

```
OM OPTIONS (NEXT UNATTENDED): UALM
OM OPTIONS (CURRENT):
```

## 11.54. PATHS—Display Paths to a Device

### 11.54.1. Format

PATHS [device1,device2,...deviceN]

where:

device

displays all paths and their states for the device name specified. If the device is not specified, all paths in the system are displayed using one device for each subsystem.

### 11.54.2. System Response

The PATHS display is in the following format:

```
channel-type path-number TO device FOR iop IS cmod/intf#/cu state
```

where:

channel-type

is the type of channel (for example, SBCON, FICON, FCSCSI, SCSI, NTWCA).

path-number

is the sequential number starting at 1 for each path-displayed device.

cmod

is the module name of the channel.

intf#

is the interface number displayed in hexadecimal. Each byte (two digits) is separated by a colon.

The interface number is defined by SCMS II. This number is an aggregation of the following fields:

FCSCSI

- SAN address (Port ID or PID), which comprises
  - 8-bit domain (0-FF values allowed)
  - 8-bit area (also known as port id, 0-FF values allowed)
  - 8-bit ALPA (also known as address (SCMS II), 0-FF values allowed)

FICON

- Logical CU – Logical control unit (also known as CU image number (SCMS II), or subsystem, 0-0xF values allowed)
- SAN address (port ID or PID), which comprises
  - 8-bit domain (0-FF values allowed)
  - 8-bit area (also known as port id, 0-FF values allowed)
  - 8-bit ALPA (also known as address (SCMS II), 0-FF values allowed)

Example:

FICON 1 TO DLEE15 FOR IOP10 IS I10C70/0E:13:19:00/DLEEA UP

The interface number 0E:13:19:00 breaks down into

- LCU = 0x0E
- PID = 0x131900

The SCMS II-defined interface number in a report or *FS cmod/[intf#]/cu keyin* would be 236,132,608.

## SBCON

- Logical CU – Logical control unit (also known as virtual control unit number (SCMS II), or subsystem, 0-0xF values allowed)
- Control unit address (also known as address (SCMS II), 0-FF values allowed)

Example:

SBCON 1 TO MA3A8 FOR IOP11 IS I11C50/ 00:03:C2/MA3B UP

The interface number 00:03:C2 breaks down into

- Unused for SBCON = 0x00
- LCU = 0x03
- Control unit address = 0xC2

The SCMS II -defined interface number in a report or *FS cmod/[intf#]/cu keyin* would be 962.

cu

is the name of control unit.

state

is the state of the path. It may be UP, DN, or SU (suspended). A suspended path is an UP path that is currently not available for general use.

### 11.54.3. Examples

#### Example 1

##### PATHS

Displays all paths, along with their state, in the system using one device for each subsystem.

```

FICON 1 TO DLM300 FOR IOP35 IS I35C50/03:11:1D:00/DLMB3E UP
FICON 2 TO DLM300 FOR IOP21 IS I21C50/03:11:1D:00/DLMB3B DN
FICON 3 TO DLM300 FOR IOP31 IS I31C50/03:0F:18:00/DLMB3B UP
FICON 4 TO DLM300 FOR IOP32 IS I32C00/03:0F:18:00/DLMB3C UP
FICON 5 TO DLM300 FOR IOP21 IS I21C00/03:0F:18:00/DLMB3A DN
FICON 6 TO DLM300 FOR IOP30 IS I30C50/03:0F:18:00/DLMB3A DN
FICON 7 TO DLM300 FOR IOP22 IS I22C10/03:11:1D:00/DLMB3D DN
FICON 8 TO DLM300 FOR IOP32 IS I32C10/03:11:1D:00/DLMB3D SU

FCSCSI 1 TO D4A000 FOR IOP01 IS I01C10/ 03:D2:00/D4B UP
FCSCSI 2 TO D4A000 FOR IOP02 IS I02C10/ 0A:1D:00/D4C UP
FCSCSI 3 TO D4A000 FOR IOP03 IS I03C10/ 05:18:00/D4D UP
FCSCSI 4 TO D4A000 FOR IOP00 IS I00C00/ 0B:F1:00/D4A UP

```

### Example 2

```
PATHS D4A001
```

Displays all paths (along with their state) to D4A001.

```
FCSCSI 1 TO D4A001 FOR IOP01 IS I01C10/ 03:D2:00/D4B UP
FCSCSI 2 TO D4A001 FOR IOP02 IS I02C10/ 0A:1D:00/D4C UP
FCSCSI 3 TO D4A001 FOR IOP03 IS I03C10/ 05:18:00/D4D UP
FCSCSI 4 TO D4A001 FOR IOP00 IS I00C00/ 0B:F1:00/D4A UP
```

## 11.55. PAUSEIO—Interrupt I/O to Mass Storage

The PAUSEIO keyin temporarily halts all I/O to mass storage. The interruption provides a brief quiet period in which to split TimeFinder Business Continuance Volumes (BCV) from their associated standard volumes. Once the split process is complete, I/O is allowed to resume. The interruption normally lasts less than a minute.

More information about TimeFinder, BCVs, and PAUSEIO is available in the *Exec System Software Administration Reference Manual (7831 0323)*.

### 11.55.1. Format

```
PAUSEIO
```

Halts all I/O to mass storage until a DONE or ABORT response is entered.

### 11.55.2. System Response

The response to the PAUSEIO keyin is

```
All disk I/O is paused at yyyy-mm-dd hh:mm:ss zone
0-To resume all disk I/O when done, enter DONE, else ABORT
```

where:

*yyyy-mm-dd hh:mm:ss zone*

is the time at which the PAUSEIO keyin was entered. The zone field contains the configured time zone mnemonic.

**Note:** If Data Integrity Verification (DIV) is active when a PAUSEIO keyin is entered, the system can automatically adjust the *data\_verification\_interval* to a large value after the operator answers DONE to resume all disk I/O. The amount of the adjustment depends on the length of time the PAUSEIO keyin was in effect. You can need to reset the DIV interval to its original value, using dynamic Exec COMUS configuration of that parameter. See Section 16 of the *Exec System Software Administration Reference Manual (7831 0323)* for more information.



This keyin is performed under the control of either an Operations Sentinel script or the system administrator, normally a script. Once it has issued the PAUSEIO, the Operations Sentinel script is designed to wait for messages from EMC Symmetrix TimeFinder and respond to them by issuing a DONE or ABORT keyin. The system then responds with the following console message:

```
All disk I/O was resumed at yyyy-mm-dd hh:mm:ss zone, response = <DONE
| ABORT>
```

where:

*yyyy-mm-dd hh:mm:ss zone*

is the time at which the script issued the DONE or ABORT response. The zone field contains the configured time zone mnemonic.

<DONE | ABORT>

specifies which of the two keyins was issued. DONE indicates that the split completed successfully; ABORT indicates that the configured time allowed for the I/O interruption elapsed before the split completed.

## 11.56. PM—Designate a Tape Unit as Premount Only

The PM keyin designates a tape unit as a premount-only unit, on systems with the Tape Automatic Volume Recognition (TAVR) feature installed.

### 11.56.1. Formats

#### Specified Device

```
PM devnam
```

Places the specified device in premount-only mode.

#### All Devices on a Control Unit

```
PM,ALL cu
```

Places all devices attached to the specified control unit in premount-only mode.

### 11.56.2. Restrictions

The TAVR feature must be installed before you can use the PM keyin. If you try to use it when the feature is not installed, you receive this message:

```
PM KEYIN (TAVR) FEATURE NOT INSTALLED
```

Only tape units can be designated as premount-only. A unit designated for premount-only can be used only when the unit is premounted or aftermounted with the requested reel, or when it is a pool unit of that user. One exception to this is an absolute device or control unit assignment, which overrides the premount-only designation.

## 11.57.PR—Remove Premount-Only Designation

The PR keyin removes the premount-only designation of tape units on systems with the Tape Automatic Volume Recognition (TAVR) feature installed. The premount-only condition is maintained across disk recovery boots. The keyin causes runs held for this type of unit to be removed from the hold queue.

### 11.57.1. Formats

#### Specified Device

```
PR devnam
```

Removes the premount-only designation on the specified device .

#### All Devices on a Control Unit

```
PR,ALL cu
```

Removes the premount-only designation on all devices attached to the specified control unit.

### 11.57.2. System Response

The response to the PR keyin is

```
tape state [NA] [* RUNID run-id REEL [reel-#] ring [POS]]
```

where:

*tape*

is the name of the tape drive.

*state*

is the tape drive's status (UP, DN, or RV).

NA

indicates that the drive is not available.

\*

indicates that the drive is assigned by a run.

*run-id*

is the name of the run with the drive assigned.

*reel-#*

is the number of the reel mounted on the drive.

*ring*

is either RING or NORING, if specified on assignment.

POS

indicates that a tape is currently mounted on the drive.

## 11.58. PURGE—Remove Residue Data from the XPC

The PURGE keyin removes data that has been left in the XPC (typically after testing or after a host is removed from the XTC environment). The keyin makes it unnecessary to reboot the host and destage the data.

### 11.58.1. Format

```
PURGE xpcname/hostid
```

where:

*xpcname*

is the logical name of the XPC.

*hostid*

is the host-ID: A, B, C, and so on.

#### Restrictions

This keyin cannot be used on a host with a status of UP in the XTC environment. The host must be DN before the keyin can be used. An XPC-L device does not support this keyin.

## 11.59. RC—Display Information on a Specific Run

The RC keyin displays specific information on backlog runs, active runs, and runs terminated with output files remaining.

### 11.59.1. Format

```
RC[,option] run-id
```

where:

option

L                If *run-id* is a batch run, displays the run origin information.  
There is no effect for other types of runs.

### 11.59.2. System Response

#### Backlogged or Active

This response indicates that the specified run is backlogged or active:

```
run-id,RP account [site-id] state
mode TIME=hhhh:mm:ss/hhhh:mm:ss PAGES=nnn
CARDS=nnn SIZE=xxx T(DL)=hh:mm
PROJECT=project-id BL TIME=hh:mm:ss or PROGRAM=name
USERID = [user-id] [AG=appl-num]
ORIGINAL RUN-ID = run-id OPENED yyyy-mm-dd hh:mm:ss zone COND = status
ORIG: dir-id#qualifier*element(F-cycle).[element(S-cycle)]
```

where:

*run-id*

is the run-ID from the keyin.

*R*

is the run priority of the run (A to Z or 1 to 5). For demand runs, priority equals 0.  
For batch runs, if the run was removed by the RM *run-id* keyin, priority equals 0.

*P*

is the processor dispatching priority (A through Z).

*account*

is the account number charged for the run.

*site-id*

is the user's site-ID for demand runs; blank for batch runs.

### state

is the state of the run, as follows:

|                        |                                                                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IBL/OPEN]/FACIL/MASS   | The run is waiting for a mass storage I/O facility.                                                                                                                                                                                                                                                                        |
| IBL/OPEN]/FACIL/OTHER  | The run is waiting for an I/O facility other than tape or mass storage.                                                                                                                                                                                                                                                    |
| IBL/OPEN]/FACIL/TAPE   | The run is waiting for a tape I/O facility.                                                                                                                                                                                                                                                                                |
| BL/FINCRD              | The @FIN statement has not yet been read by the input device.                                                                                                                                                                                                                                                              |
| BL/OPER                | The run is held because of the CS H run-ID keyin.                                                                                                                                                                                                                                                                          |
| BL/SOPT/ <i>name</i>   | The @RUN control statement has an S option. This run is held until the previous run (whose run-ID is <i>name</i> ) from the same input device has finished.                                                                                                                                                                |
| BL/T(ST)/ <i>hh:mm</i> | The run is held until the start time specified by <i>hh:mm</i> .                                                                                                                                                                                                                                                           |
| BL/UNOPND              | The run is backlogged and is not being held.                                                                                                                                                                                                                                                                               |
| OPEN/FACIL/CTL         | The run has an outstanding CTL request.                                                                                                                                                                                                                                                                                    |
| OPEN/FACIL/TSWAP       | The run is held because a CTL is in use and a tape swap has occurred. If you need to resolve this TSWAP hold condition quickly instead of letting the run wait for a compatible drive to become available, either bring up another compatible drive or move the tape to the drive that is currently allocated for the run. |
| OPEN/INCOR             | The run is in main storage.                                                                                                                                                                                                                                                                                                |
| OPEN/SUSPENDED         | The run is suspended.                                                                                                                                                                                                                                                                                                      |

### mode

is the mode used: R/T (real time), DEMAND, DEDLIN (deadline batch), or BATCH.

### TIME

is the estimated run time for a backlogged run only. For an open run, it includes both the estimated and used time. For a privileged run (a run that will not abort due to a maximum time condition), the estimated time field is left empty.

### PAGES

indicates both the estimated and used print output page count for all OPEN states; indicates only the estimated count for all other states.

### CARDS

indicates both the estimated and used punch output card count for all OPEN states; indicates only the estimated count for all other states.

### SIZE

is the current size of the program being executed in thousands of words decimal; this field is present for all OPEN states. The size is the total size of all banks contained in the program and activity level BDTs.

### T(DL)

indicates the time the run must start in order to be complete by its deadline time.

### PROJECT

is the project-ID that was specified on the @RUN control statement.

### BL TIME

is the time that the run went into backlog.

### PROGRAM

is the last program loaded if the run is in a facilities hold or if the control card of a run is still being processed; otherwise, signifies the currently executing program. If the program name cannot be found, displays:

NO PROGRAM YET LOADED

The following message is displayed if no element was specified on the last statement to start a program load:

FILE NAME ONLY AVAILABLE

### USER-ID

is the user-ID the run is running under. (If the run was started by an @START,U, the user-ID is not necessarily the same as that of the user who opened the run.)

### AG=appl-num

The application number displayed by this keyin identifies the Integrated Recovery application group in which the program is executing.

### ORIGINAL RUN-ID

is the original run-ID of the run specified on the RC keyin. If the run is still in backlog, this is the only field on the sixth line of RC keyin display.

### OPENED

gives the date and time at which the run specified on the RC keyin opened. The zone field contains the configured time zone mnemonic. This field is not displayed if the run is still in backlog.

**COND**

is the status or condition of the run at this point, that is, whether the run has encountered an error so far. The possible conditions are NORMAL, ERROR, ABORT, and FAC ERROR. These have the same meanings as the identical terms on the *run-id* FIN message when the run ends.

This field is not displayed if the run is still in backlog.

**ORIG**

is the file or element from which batch run originated

**Finished or Never in the System**

This response indicates the specified run was never in the system or has finished printing:

```
run-id NOT FOUND
```

**Finished**

This response indicates that the specified run has finished, and the queued print and punch files are displayed as described for the SQ run-ID \* R keyin:

```
run-id HAS FINNED
run-id: file xx EPGS priority rank/sname
run-id: file xx CARDS priority rank/sname
run-id: file TAPE priority rank/sname
```

**Held**

This response indicates that the specified run has been selected for processing, but is being held because the run's initial facility requests cannot be satisfied at this time:

```
run-id - INITIAL FACILITY HOLD
```

This response indicates that the specified run is being held briefly for an automated cartridge system (ACS) query:

```
RUN HELD FOR:
TAPE ASSIGN, AUTOMATED CARTRIDGE SYSTEM (ASC) QUERY FOR REEL-ID
```

This response indicates why the run is being held:

```
Network device assign, unit not available device = devnam
```

### 11.59.3. Examples

#### Example 1

RC JJPA

Displays the following information for the run JJPA:

```
JJPA,0 5CH091 [RA1402] OPEN/INCOR
DEMAND TIME = 16:39:00/00:00:00 PAGES = 50000/3
CARDS = 10000/0 SIZE = 51
PROJECT = YUR$KP PROGRAM = FURPUR
USERID = [JPARKS]
ORIGINAL RUN-ID = JJP OPENED 2000-07-14 16:06:53 CDT COND = NORMAL
```

#### Example 2

RC MAPPER

Displays the following information for the MAPPER run, which is a real-time run:

```
MAPPER,A CSFMAP OPEN/INCOR
R/T TIME = 72:47:04/26:47:18 PAGES = 50000/979
CARDS = 262000/0 SIZE = 776
PROJECT = CSFMDB PROGRAM = MAPPER
USERID = [CSF1]
ORIGINAL RUN-ID = MAPPER OPENED 2000-01-12 03:06:53 CDT COND = NORMAL
```

#### Example 3

RC DCR

Displays the following information about the DCR run, which is a batch run:

```
DCR ,A 005452 OPEN/INCOR
BATCH TIME = 36:24:31/13:35:11 PAGES = 50000/16
CARDS = 263000/0 SIZE = 91
PROJECT = CED PROGRAM = DCR
USERID = [SMITH]
ORIGINAL RUN-ID = DCR OPENED 2000-08-01 08:01:40 CDT COND = NORMAL
```

#### Example 4

RC GNN

Displays the following message when GNN is not found:

```
GNN NOT FOUND
```



**Example 5**

```
RC GNAB
```

Displays the following message if GNAB has finished:

```
GNAB HAS FINNED
GNAB: PR 2EPGS J 1ST/PR
```

**Example 6**

```
RC,L YE004
```

Displays the following about the YE004 run, which is a batch run:

```
YE004 ,MM 164140 OPEN/INCOR
BATCH TIME = 16:40:00/ 00:00:08 PAGES = 50000/16
CARDS = 9999/0 SIZE = 36
PROJECT = ETP PROGRAM = PMD
USERID = [RGW]
ORIGINAL RUN-ID = YE004 OPENED 2007-12-31 13:10:28 CDT COND = NORMAL
ORIG: STD#ETP*SOURCE$(1).YE004(0)
```

**11.60. RD—Remove Disk Packs from Pack List**

The RD keyin removes unassigned packs in the DN or PRESENT state from the list of packs. A typical use for this keyin is to remove packs to make room for other packs when the pack status table is full. The maximum number allowed in the table is defined by configuration parameters REQNBPRST0 and REQNBPRST1.

**11.60.1. Format**

```
RD pack-id[/[dir-id],[pack-id]...]
```

where:

*pack-id*

is the name of the pack.

*dir-id*

is the directory-ID supplied on the keyin, either STD or SHARED. STD is for a local pack, and SHARED is for a shared pack. The default is STD.

### 11.60.2. System Response

When the RD keyin is successful, the pack is removed from the pack status table, and the following message is displayed:

```
pack-id REMOVED
```

where *pack-id* is the name of the pack.

An RD keyin is not allowed on a pack in the UP state. The RD keyin is rejected, and the following message displayed:

```
RD OF PACK pack-id FAILED - pack-id IS UP
```

An RD keyin is not allowed on a pack that does not exist in the system list of UP, DN, or PRESENT packs (that is, its status is NOT PRESENT). The RD keyin is rejected, and the following message displayed:

```
RD OF PACK pack-id FAILED - pack-id IS NOT PRESENT
```

The following error message is displayed if you attempt any of the preceding keyins for packs mounted on unit-duplexed devices:

```
keyin KEYIN NOT ALLOWED - PACK pack-id MOUNTED ON UNIT DUPLEXED DEVICE.
```

where:

*keyin*

is the keyin entered.

*pack-id*

is the pack-ID specified on the keyin.

### 11.60.3. Restrictions

You cannot use the RD keyin on a unit-duplexed device.

To use the RD keyin in a local environment, the NEWCON configuration parameter REQNBPRST0 must be greater than 0. To use the RD keyin in a shared environment, the NEWCON configuration parameter REQNBPRST1 must be greater than 0. See the *Exec System Software Installation and Configuration Guide* (7830 7915) for more information on these configuration parameters.

### 11.60.4. Example

```
RD DKSF1
```

Removes pack DKSF1 from the list of packs in the DN or PRESENT state. After the pack is removed, the following message is displayed:

```
DKSF1 REMOVED
```

## 11.61.RE—Resume a Run

The RE keyin resumes a run that is suspended with the SP keyin.

### 11.61.1. Format

```
RE[{B|D}U] ID1[,ID2,...,IDn]
```

where:

*B*

specifies resume batch runs only.

*D*

specifies resume demand runs only. Both B and D cannot be specified.

*U*

indicates that the ID arguments are user IDs.

One or more IDs can be specified, separated by commas and with or without masking. If the U option is not specified, the IDs refer to run IDs; if the U option is specified, the IDs refer to user IDs.

For Examples, See SP keyin.

### 11.61.2. System Response

If the keyin is accepted, the system responds as follows:

```
run-id RESUMED yyyy-mm-dd hh:mm:ss zone
```

If the keyin is not accepted, the system displays a message indicating why it was not accepted.

### 11.61.3. Restrictions

Remote demand console users must have the FULL, DISPLAY, or RESPONSE console capability to use the RE keyin.

### 11.62. RL—Common Bank Reload

The RL keyin reloads a common bank from a library file. Use it only at the direction of your site administrator.

For more information on common banks, refer to the *Exec System Software Common Banks Programming Guide* (7830 7386).

#### 11.62.1. Format

```
RL bankname
```

where *bankname* is the name of the bank. The name is defined when the common bank is created.

#### 11.62.2. System Response

When you enter an RL keyin, the system responds with an RLOAD message as follows (if the message is repeated after several minutes, the system has exceeded the time limit for reloading the common bank):

```
0-RLOAD bankname IEST
```

or

```
0-RLOAD bankname FROM FILE filename IEST
```

Respond by entering an I, E, S, or T to designate the type of reload, as follows:

- |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I | Immediate request. Current users are not notified.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| E | Error request. Users are given contingency.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| S | Stall request. A stall request waits for all users of all common banks currently in use to finish using the common banks. During the waiting period, all new attempts to access the common bank to be reloaded are queued. In a busy system, the stall (wait) period is likely to expire before reaching the point where the common bank can be reloaded. If the wait period expires, you are notified. You can then decide whether to repeat or terminate the load request. |
| T | Terminate the reload request.                                                                                                                                                                                                                                                                                                                                                                                                                                                |

During processing of the keyin, the system displays an information message that gives the status of the reload.

If the reload is successful, the system displays this confirming message:

```
RLOAD bankname COMPLETED
```

The *System Console Messages Reference Manual* (7833 3275) lists the common bank error messages and the corrective action to take. All common bank error messages contain the symbol -#, where # is an integer between 0 and 9 that helps you to determine which lines of a message belong together when other console messages appear between the lines of common bank messages.

For example

```
COMMON BANK bank-name
-2- RELOAD ERROR.
-2- ALTERNATE FILE COMMON BANK STALL RELOAD NOT ALLOWED.
```

If there is more than one common bank on the system with the name specified on the RL keyin, the system responds with a message that solicits the name of the file from which the common bank is to be loaded. When you respond, provide both the qualifier and file name. If more than one file with the same qualifier and file name exists, you must specify the file cycle (F-cycle). If the file name is unknown, enter B. This response allows the system to continue processing reload requests.

### 11.62.3. Restrictions

You cannot use the RL keyin to reload an alternate file common bank (AFCB) unless the bank has already been reloaded or based by a program. If you attempt to reload an AFCB with this keyin, using the I response for an immediate reload, and the old and new copies of the AFCB are of different execution modes, the keyin is rejected, the AFCB is not reloaded, and this message is displayed on the console:

```
RLOAD bankname MODE ERROR
```

For an uninstalled NCCB, the response would be

```
bankname NOT FOUND
```

## 11.63. RM—Remove a Run from the Backlog Queue

The RM keyin is used to remove runs from the backlog queue.

### 11.63.1. Format

```
RM[,ALL|NU] ID1[,ID2,...IDn]
```

where:

*ALL*

specifies to remove all runs held in the backlog. Other options and IDs cannot be specified with this option.

*U*

indicates that the ID arguments are user IDs.

*N*

indicates to negate any mask that is remove all runs except those specified in the ID fields.

One or more IDs can be specified, separated by commas and with or without masking. If the U option is not specified, the IDs refer to run IDs; if the U option is specified, the IDs refer to user IDs.

### Examples:

|                |                                                                                                             |
|----------------|-------------------------------------------------------------------------------------------------------------|
| RM,ALL         | removes all runs held in the backlog                                                                        |
| RM,ALL R3      | invalid keyin                                                                                               |
| RM R1,RX?A,RV* | removes run R1, all four-character runs that start with RX and end with A, and all runs that begin with RV. |

## 11.63.2. System Response

When the run is removed, it is opened and then terminates in error. The following message is printed in the run's output listing:

```
OPERATOR REMOVED RUN
```

If the run is in a sequencing hold (S option on the @RUN statement), the run is not removed from backlog until the preceding run from the same device has ended.

If the run is removed, the system displays messages in this format:

```
run-id START
run-id ERROR FIN
```

### Already Executing

If the run is already executing, the system displays a message in this format:

```
run-id ACTIVE - NOT REMOVED
```

In this case, use an E or X keyin to terminate the run.

### Finished

If the run has finished and output is queued, the system displays a message in this format:

```
run-id FINNED - NOT REMOVED
```

### Finished or Never in the System

If a run has finished or was never in the system, the system displays a message in this format:

```
run-id NOT FOUND
```

### Scheduling Error

If a scheduling error has occurred and the run cannot be removed, the system displays a message in this format:

```
run-id INTERNAL ERROR - KEYIN IGNORED
```

### 11.63.3. Example

```
RM CJM321
```

Removes the CJM321 run. The system displays the messages

```
CJM321 START
CJM321 ERROR FIN
```

## 11.64. RP—Remove Pack

The RP keyin removes an unassigned removable pack's file information from the master file directory. After using this keyin, the pack must be reregistered (see the FA keyin) before files on the pack can be accessed.

### 11.64.1. Format

```
RP pack-id[/dir-id]
```

or

```
RP pack-id
```

where:

*pack-id*

is the name of the pack.

*dir-id*

is the directory-ID supplied on the keyin, either STD or SHARED. STD is for a local pack, and SHARED is for a shared pack. The default is STD.

### 11.64.2. System Response

After the keyin, the system displays one of the following messages:

```
PACK pack-id number FILES REMOVED
```

```
dir-id PACK pack-id number FILES REMOVED
```

Shows the number of removed files. The variable number indicates the number of files.

```
PACK pack-id NO FILES CATALOGED
```

```
dir-id PACK pack-id NO FILES CATALOGED
```

Informs you that the pack is registered but contains no cataloged files or contains only multipack files.

PACK *pack-id* NOT CURRENTLY REGISTERED

*dir-id* PACK *pack-id* NOT CURRENTLY REGISTERED

Informs you that the pack has not been previously registered.

## 11.65. RS—Restart a Run

The RS keyin restarts a run after a CK keyin. See also

- The *System Console Messages Reference Manual* (7833 3275) for information about the messages displayed on the console during a checkpoint/restart procedure
- The *ClearPath OS 2200 Software Products Installation Guide* (7831 0612) for additional information on the CKRS product

### 11.65.1. Format

```
RS [,priority/options/pdp] run-id, [account-nbr][/ user-id], [project-id],
filename, [checkpoint number, [equipment,] reel/[reel]]
```

where:

*priority*

is the scheduling priority of the run.

*options*

can include any of the following. You can use more than one option.

|   |                                                                                                                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| J | Specifies the checkpoint tape is unlabeled.                                                                                                                                                                                                                       |
| N | Specifies do not print diagnostics in the run's print file.                                                                                                                                                                                                       |
| P | Prints diagnostics on the console. This should always be used with the RS keyin.                                                                                                                                                                                  |
| Q | Causes the full restart common bank to put diagnostic snapshots in the run's print file. This is for debugging purposes.                                                                                                                                          |
| R | Causes the saved symbiont files to be restored. This is useful when a system crash causes a loss of these files.                                                                                                                                                  |
| U | Specifies that CKRS files should not be verified after a restart. Use this option to bypass file verification if the CKRS_FILE_VERIFY parameter is set to TRUE. If both the U and V options are used on the restart request, the U option overrides the V option. |
| V | Specifies that CKRS files should be verified after a restart. Use this option to force file verification if the CKRS_FILE_VERIFY parameter is set to FALSE. If both the U and V options are used on the restart request, the U option overrides the V option.     |
| Y | Causes the restart initiation processor to put diagnostic snapshots in the run's print file. This is for debugging purposes.                                                                                                                                      |



**Z** Causes the partial restart common bank to put diagnostic snapshots in the run's print file. This is for debugging purposes.

*pdp*

is the processor dispatching priority

*run-id*

specifies the run-ID to be used for the restart run. This run- ID must be the same as the generated run- ID of the checkpointed run.

*account-nbr*

specifies the account number of the user- ID for the restarted run.

*user-id*

is required when user- ID s are configured or the terminal security system is configured.

*project-id*

is the project- ID for the restarted run. The default is Q\$Q\$Q\$.

*filename*

is the file name that contains the checkpoint.

*checkpoint number*

is the checkpoint number to be restarted. This must be specified if the checkpoint is on tape.

*equipment*

is the equipment type on which the checkpoint tape is mounted. The default is T.

*reel/reel*

is the reel number of the tape on which the checkpoint is written.

## **11.65.2. System Response**

The user-ID and clearance level are required for runs restarted with the RS keyin. After the system has accepted the keyin, the following message is displayed at the system console:

```
ENTER CLEARANCE LEVEL FOR : run-id
```

where *run-id* is the one specified on the RS keyin.

The clearance level is optional. If you do not enter a clearance level, the minimum clearance level for your user-ID is assumed. If the supplied user-ID or clearance level is not valid, the system displays the following message:

```
run-id:INVALID CLEARANCE LEVEL
```

## 11.66. RV—Reserve a Component

The RV keyin reserves a component. Reserved components are available for use only by users that request them by name, such as online maintenance.

### 11.66.1. Formats

The following keyins specify a particular component:

```
RV ip
```

Reserves the specified instruction processor for dedicated activities.

```
RV mmod
```

Reserves the specified memory module for dedicated activities.

```
RV devnam
```

Reserves the specified device for a particular assignment.

The following keyin specifies all devices for a control unit:

```
RV,ALL cu
```

Reserves all devices attached to the specified control unit for specific assignments.

### 11.66.2. Restrictions

XIIPs and XPCs cannot be reserved.

### 11.66.3. Example

```
RV,ALL CUTAPD
```

Reserves all devices attached to CUTAPD. These lines are displayed:

```
CUTAPD UP
TAPED3 RV,TAPED4 RV
```

## 11.67. SB—Control SIP Performance Data Collection

The SB keyin controls the collection of performance data with the Software Instrumentation Package (SIP) and I/O trace, including

- Initiating and terminating SIP and I/O trace data collection, either together or separately
- Controlling the specific functional areas of SIP that are to be collected
- Controlling the composition of the SIP output file
- Inhibiting the freeing of SIP and I/O trace tape files when data collection terminates
- Displaying the current status of SIP and I/O trace on the system console
- Specifying device names for SIP and I/O trace tape files

Use this keyin only under the direction of your site administrator.

### Caution

The duration of a SIP data collection session should not exceed 24 hours. The validity of the data cannot be guaranteed after 24 hours because certain counters can overflow. The counters are cleared when you turn SIP off (SB OFF); you can then immediately turn SIP on again (SB ON).

### 11.67.1. Format 1: Initiating SIP ON Mode

SB ON

This format initiates SIP ON mode, which starts SIP data collection unconditionally. To prevent data overflow, enter the SB OFF keyin within 24 hours after entering SB ON.

A SIP data collection can be one of two modes—ON or AUTO. These modes are initiated by the SB ON and SB AUTON keyins. For these keyins, if I/O trace is configured (SIPIO=TRUE), I/O trace data collection in the ON mode unconditionally initiates unless the SB NOIOT keyin is in effect. The OFF keyin terminates the SIP data collection.

The following rules apply to ON processing:

- If the SIP output file is configured to mass storage (SBFILE=0), the mass storage file is created except in the following instances:
  - Timed output of SIP data is not configured (CONTIM=0 and SBTIMEON=0).
  - Timed output is configured (SBTIMEON>0), but the output time interval is configured as zero (CONTIM=0) and has not been altered by the SB Tx keyin.

- Timed output is configured, and the output time interval has been set to zero by the SB T0 keyin.
- If the SIP data file is not created during SIP initialization, it is created prior to the first write of a data block to the file. This data block can be one of the following types:
  - Time interval block

This type of data block is written to the file if the output time interval is changed to a nonzero value.
  - Run termination block

This type of data block is written to the file if it is configured (SBRUN>0) and not inhibited by the SB RUNOFF keyin.
  - Summary block

This type of data block is written to the file at SIP termination. The following statements create the SIP data file on mass storage:

```
@CAT,PZ SYSBALLOG(+1),F2/1/POS/n
@ASG,AZ SYSBALLOG
```

where n is the value of the configuration parameter SIP\_FILE\_MAX\_SIZE (default value is 10).

**Note:** When *SENTRY\_CONTROL=TRUE*, the *SYSBAL\$LOG\$* file is not cataloged as public; instead, the ACR ACRDIA is attached to the file.

- If the SIP output file is configured to tape (SBFILE=1), the tape file is assigned during initialization except in the following instances:
  - Timed output of SIP data is not configured.
  - Timed output is configured, but the output time interval was configured as zero and has not been altered.
  - Timed output is configured, but the output time interval has been set to zero.
  - The SIP output file is currently assigned.

If the file is not assigned at the time of the first data block write, the assign takes place at that time. The following statement creates the SIP data file on tape:

```
@ASG,TZ SYSBALLOG,xxx
```

where xxx is the value of the system standard equipment type for tape assignments.

- Data collection is initiated for the SIP functional areas not inhibited by the SB FLGOF keyin (except IO+). It continues until terminated by the SB OFF keyin.
- If the SIP file was assigned, it remains assigned throughout the collection unless both run termination and time interval output are disabled. When this occurs, a summary block is written and the SIP file is freed. Whenever the file is not assigned and a data block write is initiated, the assignment is made.

- If the I/O trace feature is configured, I/O trace data collection in the ON mode is initiated, unless the SB NOIOT keyin is currently in effect or an I/O trace data collection is currently in progress.

### 11.67.2. Format 2: Initiating SIP AUTO Mode

SB AUTON [x] or SB AUTON T[y]

This format initiates SIP AUTO mode, where *x* can contain up to six decimal digits, and *y* can contain up to five decimal digits.

In this mode, SIP data collection takes place only when the number of open runs in the system is equal to or greater than a specified threshold. When the first form (*x*) is used, the threshold value is set to the total number of runs open at the time of the SB AUTON keyin, plus *x*. If *x* is blank, it is assigned a value of 0. When the second form (T[*y*]) is used, the threshold value is set to *y*. If *y* is blank, it is assigned a value of 1.

The first data collection begins when a run opens and the number of open runs in the system is equal to or greater than the threshold value. The following message is displayed:

SYSTEM BALANCE: ON *yyyy-mm-dd hh:mm:ss zone*

When an AUTO mode data collection is in progress and the number of runs drops below the threshold, SIP data collection terminates and the following message is displayed:

SYSTEM BALANCE: OFF *yyyy-mm-dd hh:mm:ss zone*

When the number of open runs again meets the threshold value, a data collection is initiated and the SYSTEM BALANCE: ON *yyyy-mm-dd hh:mm:ss zone* message is displayed.

The SIP data space is never reinitialized while AUTO mode remains in effect. That is, the SIP data always contains the cumulative values from all collection intervals during the AUTO session.

Each AUTO mode data collection creates an individual SIP data file F-cycle when SIP is configured to mass storage. When configured to tape, a summary block is written at the end of each AUTO session, and the same reel remains mounted until filled or SIP is turned off.

I/O trace data collection in AUTO mode is continuous and is initiated in the same manner as in ON mode.

AUTO mode remains in effect until terminated by the SB OFF keyin.

### 11.67.3. Format 3: Terminating ON and AUTO Modes

SB OFF

This format terminates ON and AUTO modes.

#### **Caution**

The duration of a SIP data collection session should not exceed 24 hours. The validity of the data cannot be guaranteed after 24 hours because certain counters can overflow. The counters are cleared when you turn SIP off (SB OFF); you can then immediately turn SIP on again (SB ON).

When an SB OFF keyin is entered with SIP in the ON or AUTO mode, that mode is terminated and the following events occur:

- If SIP data collection is in progress, it terminates and a summary block is written to the SIP data file.
- If the SIP file is configured to mass storage, it is freed and the following message displayed:

SIPFILE FREE

- If the SIP file is configured to tape, the tape is freed unless an SB NOFREE keyin is in effect. The following message is displayed:

SIPFILE FREE

- If I/O trace is in the ON mode, I/O trace data collection is terminated (see SB IOFF keyin). The following message is displayed:

IOTRACE FILE SYS\$\*IODATFILE FREE

### 11.67.4. Format 4: Controlling I/O Trace Data Collection

SB { IOMAN | IOFF }

This format initiates and terminates I/O trace data collection independent of SIP data collection. These keyins are invalid if I/O trace is not configured.

If I/O trace is configured to mass storage (IOTRACE\_FILE\_MAX\_SIZE > 0) and the file is not assigned at the time the keyin is entered, the following statements are performed:

```
@CAT,PSZ IODATFILE(+1),F2/n/POS/n
@ASG,AZ IODATFILE.
```

where *n* is the value of the configuration parameter IOTRACE\_FILE\_MAX\_SIZE.

I/O trace data collection is always initiated unless an I/O trace is in progress. MANUAL mode continues until terminated by the SB IOFF keyin.

If I/O trace is in ON mode when the IOMAN SB keyin is entered, the SB IOMAN keyin places I/O trace in MANUAL mode. When I/O trace is in MANUAL mode, the SB IOFF keyin is honored, but the SB OFF keyin has no effect upon I/O trace.

SB IOFF terminates the MANUAL mode of I/O trace data collection. When this keyin is entered and I/O trace is in MANUAL mode, the data collection terminates. If writing to tape, the tape file is freed unless the SB NOFREE keyin is in effect. If writing to mass storage, the file is freed regardless of the SB NOFREE keyin. If I/O trace is not in MANUAL mode, no action is taken.

The following keyins are used to exercise control over a current or future data collection or to display information about the current status of SIP and I/O trace.

### **11.67.5. Format 5: Controlling I/O Trace Data Collection in ON or AUTO Mode**

```
SB { NOIOT | IOT }
```

This format inhibits and enables I/O trace data collection in ON or AUTO mode. Normally, both SIP and I/O trace data collection are initiated when an SB ON or SB AUTO keyin is entered.

SB NOIOT inhibits I/O trace data collection in ON or AUTO mode; SB IOT reverses the effect of the NOIOT keyin and enables I/O trace data collection in ON or AUTO mode.

These keyins can be entered whether or not SIP data collection is in progress, but they affect only subsequent data collections.

### **11.67.6. Format 6: Controlling the Freeing of SIP and I/O Trace Tape Files**

```
SB { NOFREE | FREE }
```

This format inhibits and enables the freeing of the SIP and I/O trace tape files. These keyins have no effect upon the SIP data file or the I/O trace data file when they are not configured as tape files. These keyins are honored whenever the I/O trace or SIP output files are configured to tape (IOTFILMAXSIZ>0 or SBFILE=1).

Normally, the SIP tape file is freed when ON or AUTO mode is terminated, and the I/O trace tape is freed when I/O trace data collection is terminated. When the SB NOFREE keyin is entered, however, these files are not freed as long as this keyin remains in effect, unless an I/O error or end-of-tape condition occurs. The use of the SB NOFREE keyin enables stacking of multiple files on one tape file.

The SB FREE keyin negates the effect of the SB NOFREE keyin. In addition, if a SIP tape is assigned and SIP is not in ON or AUTO mode, the SIP file is freed. Likewise, if an I/O trace tape is assigned and an I/O trace data collection is not in progress, the I/O trace file is freed.

### 11.67.7. Format 7: Controlling SIP Data Collection for Functional Areas

SB { FLGOF | FLGON } *flag*

This format inhibits and enables the collection of SIP data for a specific functional area or for all functional areas. These keyins are allowed only if a SIP data collection is not in progress when they are entered.

Most of the SIP statistics are assigned to one of the 10 functional areas of SIP. Data collection for each of these areas is controlled by a specific SIP flag, which can be turned on and off by the SB FLGOF and SB FLGON keyins. To turn all flags on or off, enter ALL for *flag*. At system initialization, all flags default to on, except IO+ which is off. The flags are as follows:

| Flag | Area                                                                                                                                                                            |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CAU  | Processor activity                                                                                                                                                              |
| EXP  | EXPOOL                                                                                                                                                                          |
| FNC  | Not used                                                                                                                                                                        |
| HM   | Not used                                                                                                                                                                        |
| IO   | I/O. Turning on IO only turns on IO. Turning off IO also turns off IO+.                                                                                                         |
| IO+  | Controls the same statistics as the IO flag plus statistics for block multiplexer devices. Turning on IO+ also turns on IO. Turning off IO+ only turns off IO+ (IO remains on). |
| MEM  | Main storage use, program and bank loading activity                                                                                                                             |
| MIS  | System activity, interrupts, Executive requests and calls                                                                                                                       |
| QTM  | Switching activity                                                                                                                                                              |
| RSP  | Demand response times                                                                                                                                                           |

The *Exec System Software Administration Reference Manual* (7831 0323) shows the statistics summarized by SIP level and lists each SIP data item and the flag associated with it. Some low-overhead SIP statistics, such as the idle statistics, are not associated with any flag and are therefore collected whenever SIP is on, regardless of the state of the SIP flags.

### 11.67.8. Format 8: Altering the Interval of Writing Data Blocks

SB T[*x*]

When SIP time interval output is configured (SBTIMEON>0), this format alters the interval at which time interval data blocks are written to the SIP output file, where *x* is a one-digit to five-digit integer specifying the new interval size in seconds. If *x* is missing, an invalid keyin message is displayed. If *x* is 0, time interval output is turned off. This keyin is honored whether or not SIP is turned on; it remains in effect until altered by a subsequent keyin. The new interval size must be greater than or equal to 0, and must be less than or equal to 34359.



### 11.67.9. Format 9: Controlling the Writing of Run Termination Data Blocks

```
SB { RUNOFF | RUNON }
```

SB RUNOFF and SB RUNON inhibit and enable the writing of run termination data blocks to the SIP file. These keyins are not valid when run termination blocks are not configured (SBRUN=0).

When run termination blocks are configured, the writing of a SIP data block occurs and is initially enabled each time a run terminates. The writing of run termination data blocks can be inhibited by the SB RUNOFF keyin.

The SB RUNON keyin reverses the effect of the SB RUNOFF keyin and enables the writing of run termination data blocks.

These keyins are honored whether or not SIP data is being collected when they are entered.

### 11.67.10. Format 10: Generating a SIP and I/O Trace Status Display

```
SB STATUS
```

This format generates a display of the current status of SIP and I/O trace on the system console.

The STATUS display has the following format (all flags shown in their default status):

```
FLAG: MIS* IO * MEM* CAU* RSP* IO+
FLAG: FNC* EXP* QTM* HM *
SIP-status : [XPMON|XPMOFF]
TIME : x
IO TRACE IS trace-status
```

where:

FLAG:

displays the status of each of the SIP flags listed in the description of SB FLGOF and SB FLGON. An asterisk (\*) following a flag name indicates that the flag is enabled.

**Note:** All flags except IO+ are enabled by default.

SIP-status

indicates the current status of SIP. SIP-status can be one of the following:

|           |                     |
|-----------|---------------------|
| AUTO T[y] | SIP is in AUTO mode |
| ON        | SIP is in ON mode   |
| OFF       | SIP is off          |

### XPMON | XPMOFF

indicates whether this host is allowed to collect XPC Performance Monitor data (XPMON) or is prevented from collecting it (XPMOFF). If neither XPMON nor XPMOFF is displayed, the SB XPMONIXPMOFF keyin was not issued on this host, and the host can collect.

### X

displays the current time interval in seconds for the output of SIP time interval data blocks. This line is displayed when time interval data blocks are configured.

### *trace-status*

displays the current status of I/O trace. This line is present on the STATUS display only when I/O trace is configured. One of the following values appears for trace-status:

|        |                                                                                |
|--------|--------------------------------------------------------------------------------|
| ON     | I/O trace is on and was initiated by the SB ON or SB AUTON keyin.              |
| MANUAL | I/O trace is on and was initiated or set to MANUAL mode by the SB IOMAN keyin. |
| OFF    | I/O trace is not on.                                                           |

## 11.67.11. Format 11: Specifying a Device Name for a SIP Tape File

```
SB SBDEV [device]
```

This format specifies a logical device name for the SIP tape file, SYSBAL\$LOG\$. If *device* is not specified, then the system default type is used. The file (SYSBAL\$LOG\$) must be configured as a tape file.

If an SB NOFREE keyin is in effect, then the tape device cannot be changed until an SB FREE keyin is entered. The device specified is saved until the SB FREE is entered, and the next time SYSBAL\$LOG\$ is assigned the specified device is used on the assignment statement.

## 11.67.12. Format 12: Specifying a Device Name for the I/O Trace Tape File

```
SB IODEV [device]
```

This format specifies a logical device name for the I/O trace tape file, IODATFILE. If *device* is not specified, then the system default type is used. The file (IODATFILE) must be configured as a tape file.

If an SB NOFREE keyin is in effect, then the tape device cannot be changed until an SB FREE keyin is entered. The device specified is saved until the SB FREE is entered, and the next time IODATFILE is assigned the specified device is used on the assignment statement.

### 11.67.13. Format 13: Specifying Hosts on Which XPM Data May Be Collected

```
SB { XPMON | XPMOFF }
```

Controls collection of XPC Performance Monitor (XPM) data. The SB XPMON keyin enables XPM data to be collected on the host issuing the keyin. SB XPMOFF prevents the issuing host from collecting XPM data.

In a multihost environment, you should issue SB XPMOFF on all hosts except the one designated to collect XPM data. On that host, issue SB XPMON. Both keyins should be completed before SIP data collection is turned on.

### 11.67.14. System Response

When a valid SB keyin is entered, the Exec echoes the keyin by displaying the message

```
SYSTEM BALANCE: text yyyy-mm-dd hh:mm:ss zone
```

where *text* contains the text of the keyin following the SB prefix, up to a maximum of 13 characters.

When you enter an invalid SB keyin, the system responds

```
SYSTEM BALANCE: text IS INVALID yyyy-mm-dd hh:mm:ss zone
```

The keyin can be invalid because you did not enter it correctly or because the particular keyin pertains to a SIP feature that is not configured in the operating system.

When SIP is not configured in the operating system, the response to any SB keyin is

```
SB KEY ERROR
```

SB keyins are processed sequentially. When you enter an SB keyin and the system does not echo the input, that keyin has been queued. When queuing occurs, it is generally because SIP is waiting for an operator action such as a tape mount or a reply to a console message.

### 11.67.15. Examples

Following are examples of sequences of SB keyins that generate SIP or I/O trace data collections. These examples illustrate techniques for generating a variety of different types of data collections. Unless otherwise noted, the following assumptions are made about the configuration of SIP:

- Run termination and time interval blocks are configured.
- I/O trace is configured.

Some of the effects of other configurations are noted in the examples.

### Example 1: SIP and I/O Trace Data Collection

```
SB ON [1]
.
.
.
SB OFF [2]
```

1. Initiate SIP and I/O trace data collection. Both files are assigned. A SIP data block is written to the SIP file every  $x$  seconds, where  $x$  is the value of the configuration parameter CONTIM.
2. Terminate SIP and I/O trace data collection. A summary block is written to the SIP file. The SIP and I/O trace files are freed.

### Example 2: SIP Data Collection Based on the Number of Runs Open

```
SB NOIOT [1]
SB FLGOF MEM [2]
SB AUTON T15 [3]
.
.
.
SB OFF [2]
```

1. Inhibit I/O trace data collection.
2. Disable the collection of main storage statistics.
3. Assign the SIP file. SIP data collection is initiated when the number of open runs becomes equal to or greater than 15; data collection terminates when the number of open runs becomes less than 15. SIP continues to go off and on, depending upon the number of runs open. If the SIP file is configured to mass storage, the file is freed when a data collection terminates, and a new cycle is assigned each time a data collection is initiated. If SIP is configured to tape, the tape file remains assigned throughout the AUTO session.
4. If a SIP data collection is currently in progress, terminate it and write a summary block to the SIP file. The SIP file is freed.

**Example 3: Two SIP Data Collections in One SIP File**

The following example shows two SIP data collections in one SIP file. The SIP file is configured to tape.

```
SB NOIOT [1]
SB RUNOFF [2]
SB T60 [3]
SB ON [4]
SB NOFREE [5]
SB OFF [6]
.
.
.
SB ON [7]

.
.
.
SB FREE [8]_
SB OFF [9]
```

1. Inhibit I/O trace data collection.
2. Inhibit the writing of SIP run termination blocks.
3. Change the output time interval for the writing of SIP time interval blocks to 60 seconds.
4. Assign the SIP tape and initiate SIP data collection.
5. Inhibit the freeing of the SIP tape.
6. Terminate SIP data collection and write a summary block to the SIP file.
7. Initiate SIP data collection.
8. Enable the freeing of the SIP file at SIP termination.
9. Terminate the SIP data collection, write a summary block to the SIP file, and free the file.

**Example 4: I/O Trace Data Collection**

```
SB IOMAN [1]
.
.
.
SB IOFF [2]
```

1. Assign the I/O trace file and initiate I/O trace data collection.
2. Terminate I/O trace data collection and free the file.

### **Example 5: SIP and I/O Trace Data Collection with Independent Terminations**

```
SB ON [1]
.
.
.
SB IOMAN [2]
SB IOFF [3]
.
.
.
SB OFF [4]
```

1. Assign the SIP and I/O trace files and initiate SIP and I/O trace data collection.
2. Enable MANUAL mode for I/O trace.
3. Terminate I/O trace data collection and free the I/O trace file.
4. Terminate the SIP data collection, write a summary block to the SIP file, and free the file.

### **Example 6: SIP Data Collection with a File Assigned for Summary Block Write Only**

The file created in this example is configured to tape and has a summary block as its only data block. This type of collection can be useful when the availability of tape units is a concern.

```
SB NOIOT [1]
SB RUNOFF [2]
SB T0 [3]
SB ON [4]
.
.
.
SB OFF [5]
```

1. Inhibit I/O trace data collection.
2. Inhibit the writing of run termination blocks.
3. Inhibit the writing of time interval blocks.
4. Initiate SIP data collection. The SIP file is not assigned until a data block write is initiated.
5. Terminate SIP data collection. The SIP file is assigned and the summary block is written. The file is then freed.

**Example 7: SIP Data Collection with an Altered Collection Control Flag**

```
SB NOIOT [1]
SB FLGOF ALL [2]
SB FLGON CAU [3]
SB FLGON IO+ [4]
SB ON [5]
SB STATUS [6]
.
.
.
SB OFF [7]
```

```
SB FLGON ALL [8]
```

1. Inhibit I/O trace data collection.
2. Disable all collection control flags.
3. Enable processor activity statistics collection.
4. Enable I/O and IO+ statistics collection.
5. Assign the SIP file and initiate SIP data collection.
6. Display SIP status.
7. Terminate the SIP data collection, write a summary block to the SIP file, and free the file.
8. Enable all collection control flags.

## 11.68. SEC—Save or Restore Security Files on Tape

The SEC keyin provides a single keyin procedure for saving and restoring all the security files. The files are saved and restored automatically in the following order:

1. SEC@USERID\$
2. SMDTF\$
3. SEC@ACR\$
4. ACCOUNT\$R1
5. SEC@ACCTINFO

**Note:** The SC and SF keyins are no longer supported, and the SV and SR keyins can no longer be used with the security files.

The following considerations apply:

- SEC,RESTORE is allowed after SYS FIN.
- SEC,SAVE writes a version 2 tape.
- SEC,RESTORE accepts only a version 2 tape when it is entered after SYS FIN.

### 11.68.1. Formats

#### SEC SAVE

`SEC,SAVE [device, reel/reel, exp-days/mmspec, CTL-pool name]`

Saves all security files to tape.

where:

*device*

can be an assign mnemonic, a tape device name preceded by an asterisk (\*), or the name of an automated cartridge system (ACS).

*reel/reel*

specifies up to four reels. The field is optional.

*exp-days/mmspec*

specifies the default tape expiration and Media Manager specification.

*CTL-pool name*

specifies the cartridge tape library (CTL) pool to be used.



**SEC RESTORE**

```
SEC,RESTORE [device,reel/reel]
```

where:

*device*

can be an assign mnemonic, a tape device name preceded by an asterisk (\*), or the name of an ACS.

*reel/reel*

specifies the SEC,SAVE tape or tapes. The field is optional on systems configured to allow reading of tapes assigned as BLANK.

**Note:** All files saved using the SEC,SAVE keyin can be restored at reboot. See Section 8.

**11.68.2. System Response**

If the Authentication and Session Initiation Subsystem (ASIS) is not active, the system responds to a SEC,RESTORE keyin by prompting you with the following message:

```
SEC,RESTORE Do you really want to restore security file <file>? Y/N
```

If N is the response, that file is skipped and the system prompts you about the next file on the tape.

If the ASIS is active, the system responds to a SEC,RESTORE keyin by displaying the following message:

```
The keyin to restore the SEC@USERID$ may be delayed while awaiting
coordination with the Authentication Subsystem (ASIS)
```

Then the system waits approximately 2 minutes until the ASIS responds. If the ASIS responds appropriately, the system performs the restore and the ASIS performs any necessary cleanup.

If the ASIS does not respond appropriately, the system prompts you with the following message:

```
The Authentication Subsystem (ASIS) has not yet authorized the SEC@USERID$
update. Do you want to wait again, or terminate the ASIS? W or T
```

If W is the response, the timeout period starts over.

If T is the response, the system aborts the ASIS as follows:

- All pending authentications are returned to VALA as rejected.
- The system erases its knowledge of the ASIS' domain, causing subsequent SERVER\$GET and SERVER\$PUT requests to be rejected, and resulting in an ASIS shutdown.

- The system reactivates any waiting SERVER\$GET activity and returns "ASIS not registered" status.
- The system changes the ASIS state to "Temporarily Not Available".

Contact the site security officer if you see the following message after using SEC,RESTORE to load all security files from a tape created with SEC,SAVE:

```
SEC,RESTORE *ERROR* Tape block count error. reel-id.
Restore terminated. Answer - Go.
```

Contact the site security officer if you see the following message after using SEC,RESTORE to load all security files from a tape created with SF:

```
BLOCK COUNT ERROR IN SECURITY SAVE TAPE - GO
```

**Note:** If you restore ACCOUNT\$R1 while runs are open, cataloged mass-storage quota values can be incorrect afterward.

### 11.68.3. Examples

```
SEC,SAVE
```

Saves all security files to a BLANK cartridge tape.

```
SEC,SAVE HICL,S2998
```

Saves all security files to tape S2998.

```
SEC,SAVE HICL,,,POOL-A
```

Saves all security files to a BLANK cartridge tape from CTL POOL-A.

```
SEC,SAVE ,,7
```

Saves all security files to a BLANK, labeled cartridge tape in which the label expires in 7 days.

```
SEC,RESTORE HICL,S2998
```

Restores all five security files from tape S2998.

## 11.69. SJ—Set Jump Key

The SJ keyin lets you set jump keys.

### 11.69.1. Formats

*SJ number1,number2,...numberN*

Sets the specified jump keys (1 to 36). See Section 3 for a list and description of the jump keys.

*SJ A[LL]*

Sets all jump keys (1 to 36).

*SJ O octal-number*

Sets the jump keys by their octal representation. (The bit pattern is right-justified and zero-filled if you use fewer than 12 digits.)

### 11.69.2. System Response

After the system sets the jump keys requested, it displays the following message:

*JUMP KEYS SET: number1, number2,...numbern*

where *number* is the number of a jump key that is set.

A jump key is not set until the system displays the above response at the system console.

## 11.70. SM—Control Symbiont Devices

The SM keyin displays the status of symbiont devices and control symbiont operation. Symbiont devices include card readers, card punches, printers, and remote terminals.

### 11.70.1. Formats

*SM sname*

Displays the status of symbiont device *sname*.

*SM sname operation*

Controls the symbiont devices.

where:

*sname*

is the symbiont device name.

### *operation*

is the operation to be performed. If you do not specify an operation, the system displays the device status: suspended, locked, active, or inactive. You can use one of the following values:

|       |                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| B     | Displays the current print band cartridge-ID or allows dynamic changes of the default cartridge-ID for the specified device. |
| C     | Displays the current page format for a printer or changes the current page format for a printer.                             |
| E     | Creates an end-of-file (EOF), terminating the active symbiont file.                                                          |
| I     | Initiates an inactive onsite device; resumes onsite device operation; simulates an attention interrupt.                      |
| L     | Locks out an onsite device.                                                                                                  |
| Q     | Requeues the current file and locks out the device.                                                                          |
| R     | Creates an EOF, removing the runstream that the input device is currently reading.                                           |
| Raaa  | Reprints or repunches <i>aaa</i> pages or cards, where <i>aaa</i> is a decimal number.                                       |
| R+aaa | Advances <i>aaa</i> pages or cards and then begins printing or punching, where <i>aaa</i> is a decimal number.               |
| RALL  | Reprints or repunches the entire file.                                                                                       |
| S     | Suspends device operation.                                                                                                   |
| T     | Terminates the device with an EOF, losing the remainder of the file. The device is locked out.                               |

## 11.70.2. Operations

The operations are described in detail in the following.

### **B[AND]—Display Cartridge-ID**

*SM sname B[AND]*

This operation displays the cartridge-ID of the print band or font currently on a printer. The printers in your system can use a variety of print bands, which you can change at a user's request. Each print band has a cartridge-ID that identifies it to the system.

The system displays this message:

*sname CARTRIDGE-ID, id*

where *id* is the current 2-character cartridge-ID.

**B[AND]—Enter New Cartridge-ID**

Your system maintains a default print band for each printer. Whenever it processes a file, the system reads the current cartridge-ID and checks it against the default. If the two do not match, the system assumes that you installed the wrong cartridge and issues an error message.

If you did install the wrong cartridge, replace it with the correct one. If you want to use the cartridge, however, use this SM keyin to enter the current cartridge-ID as the new system default:

```
SM sname B[AND],id
```

where *id* is a 2-character cartridge-ID (hexadecimal).

The new cartridge-ID takes effect the next time the printer is inactive, and it is assumed for the printer specified until one of the following events occurs:

- You enter another SM B[AND] keyin for the device, specifying a new cartridge.
- You perform a system boot from tape.

When you issue a valid SM B[AND] keyin, the new values overwrite the values issued by a previous SM B[AND] keyin. You cannot restore those previous values (other than the original configuration values) except by specifying them with a new keyin.

**C[HANGE]—Display Page Format**

```
SM sname C[HANGE]
```

This operation displays the current page format of a printer.

The response displayed depends on the type of printer represented by *sname*. For the 0770 and 0776 impact printers, the system displays this message:

```
sname PSIZ=psiz, TOP=top, BOT=bot, LPI=lpi
```

where:

*sname*

is the printer name.

*psiz*

is the number of lines per page.

*top*

is the number of lines for the top margin.

*bot*

is the number of lines for the bottom margin.

*lpi*

is the number of lines per inch.

### **C[HANGE]—Change Page Format**

The C operation also changes the current page format for your printer. To change a parameter, include its new value in the appropriate position.

```
SM sname C[HANGE],[siz,top,bot,lpi]
```

where:

*sname*

is the name of the printer.

*siz*

is the number of lines per page. The maximum is 999.

*top*

is the number of lines for the top margin. The maximum is 999.

*bot*

is the number of lines for the bottom margin. The maximum is 999.

*lpi*

is the number of lines per inch (6, 8, or 12). The standard number of lines per inch is 8.

Use an asterisk (\*) to specify the system default value for any of the parameters. Your site coordinator must know the defaults. To skip a parameter (leave it as it is), leave the parameter field blank.

### **C[HANGE]—Restore Defaults**

```
SM sname C[HANGE],DEF[AULT]
```

or

```
SM sname C[HANGE],*,*,*,* (9246, 0770, and 0776 impact printer)
```

This operation restores all printer values to system standards.

After the SM C[HANGE],*siz,top,bot,lpi* keyin is accepted, the new values specified become the controlling values for the device. The new values remain in effect until one of the following events occurs:

- An SM *sname* C[HANGE],*siz,top,bot,lpi* keyin specifying new values is issued for the device.
- An SM *sname* C[HANGE], DEF[AULT] keyin is issued for the device.

- You perform a system boot from tape.

**E—Terminate an Active File**

SM *sname* E

This operation terminates an active file. Use the E operation for on-site devices only. Use the E operation when the remaining part of the input or output file is not desired. A card reader bypasses the remainder of the card deck and processes the next card deck. If no more cards remain, the card reader becomes inactive. A printer or card punch bypasses the rest of the current file and then processes the next available file.

**I—Initiate or Resume Operation**

SM *sname* I

This operation initiates or resumes operation on a symbiont device. The system initiates an inactive onsite device; resumes operation of an onsite device after a previous S, L, T, R (card reader), or Q operation; or simulates an attention interrupt (that is, the device status changes from stop to run).

**L—Lock Out**

SM *sname* L

This operation locks out a symbiont device. If the device is inactive, it is locked out immediately. Otherwise, the device is locked out after the input file (card reader) or the output file (card punch or printer) finishes processing. Use the L operation only for onsite devices. You can resume device operation by using the SM keyin with the I operation specified.

**Q—Requeue**

SM *sname* Q

This operation requeues files and lock out of a symbiont device. This operation has no meaning for card readers. Use this operation when abnormal printing or punching occurs due to a device malfunction. The complete file is reentered into the symbiont queue and rescheduled for output. When the system is printing a tape, the Q operation releases the print tape and reassigns it when a printer is available. The Q operation is used only for onsite devices. You can resume device operation by using the SM keyin and specifying the I operation.

**R—Reprint or Repunch on Input Device**

SM *sname* R

Specifies reprinting or repunching on an input symbiont device. The R operation without a number or ALL can be used only for input devices. The R response removes the runstream that the input device is currently reading. The remainder of the batch input (card deck) is bypassed until a new @RUN card or @RUN statement is encountered. For more information on the @RUN control statement, see the *Executive Control Language (ECL) and FURPUR Reference Manual* (7830 7949).

### **Raaa—Reprint or Repunch Specified Pages**

SM *sname* Raaa

Specifies a number of pages to reprint or cards to repunch on a symbiont device, where aaa is a decimal number equal to or less than 504 for symbiont handlers. The output can be incorrect if the file contains multiple changes of forms, margins, print bands, and so on, since the whole file is not scanned from the beginning. The keyin Raaa cannot be used for full-duplex remote (FDR) devices.

### **R+aaa—Advance for Reprinting or Repunching**

SM *sname* R+aaa

Advances a specific number of pages for printing or cards for punching, where aaa is a decimal number equal to or less than 504 for symbiont handlers. The output can be incorrect if the file contains multiple changes of forms, margins, print bands, and so on, since the whole file is not scanned from the beginning. Do not use the keyin R+aaa for FDR devices.

### **RALL—Reprint or Repunch an Entire File**

SM *sname* RALL

Reprints or repunches an entire file. The symbiont device reprints or repunches the file as follows:

- Printers immediately start printing from the beginning of the file without aligning the paper, unless you have changed the paper to a different physical length.
- The system displays the following message, allowing you to empty the punch hopper:

*sname* - READY PUNCH - A

Tape files that the symbiont device is printing or punching are rewound to the previous EOF mark, or to the start-of-reel; printing or punching begins from that point. If the tape file is a multireel file and a previous EOF mark is on the previous reel, the operator begins at the beginning of the current reel.

### **S—Suspend Operation**

SM *sname* S

This operation suspends the operation of a symbiont device. When you specify the suspend device (S) operation, the device is locked out immediately. Use this option for quick-stop situations only, such as aligning print or reloading cards into a punch unit. The S operation is used only with onsite devices. You can resume device operation by executing an SM keyin and specifying the I operation.



## T—Terminate a Device

SM *sname* T

This operation terminates the device, losing the remainder of the file; the device is locked out. If the device is a card reader, the run that the card reader is reading is removed from the system. When the device is either a remote device or a demand terminal, both the site and the run are terminated. If a run is suspended and you enter an SM *site-id* T from another terminal, the run will complete (FIN) first. Demand terminal users must sign on again. If the device is a printer, only the file that is printing is terminated, unless you also indicate a run-ID (*sname*). You can restore the onsite symbiont device by executing an SM keyin with the I operation.

### 11.70.3. System Response

When you enter an SM *sname* keyin, the system responds with one of the following status messages:

*sname state1, state2, PAGES PRINTED = numpag*

where:

*sname*

is the symbiont device name.

*state1*

is one of the following states:

|           |                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------|
| ACTIVE    | Specifies in use.                                                                                          |
| INACTIVE  | Specifies available, but not in use.                                                                       |
| LOCKED    | Specifies unavailable, because the device is locked. Use the SM <i>sname</i> I keyin to unlock the device. |
| SUSPENDED | Specifies available for I/O operations only.                                                               |
| WAITING   | Specifies waiting for input from the print queue.                                                          |

*state2*

is one of the following states:

|            |                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------|
| IO ERROR   | Specifies recovering from an I/O error.                                                                                         |
| PRINTING   | Specifies printing a file.                                                                                                      |
| REPRINTING | Specifies repositioning a file because an SM <i>sname</i> R <i>n</i> or SM <i>sname</i> RALL (reprint file) keyin is in effect. |
| SKIPPING   | Specifies repositioning a file because an SM <i>sname</i> R+ <i>n</i> (advance file) keyin is in effect.                        |
| STOPPED    | Specifies inactive or locked.                                                                                                   |

*numpag*

is the number of pages in the current file that the symbiont device printed.

### 11.70.4. Examples

#### Example 1

```
SM PRC B
```

Displays the cartridge-ID of the print band on printer PRC. The system displays this message:

```
PRC CARTRIDGE-ID, 01
```

#### Example 2

```
SM PRC
```

Displays the status of printer PRC. The system displays this message:

```
PRC ACTIVE, PRINTING, PAGES PRINTED=3
```

#### Example 3

```
SM PRA B,02
```

Changes the default cartridge-ID for printer PRA. The system displays this message:

```
PRA CARTRIDGE-ID, 02
```

#### Example 4

```
SM PRA C
```

Displays the current page format of printer PRA, an 0776 imprint printer. The system displays this message:

```
PRA PSIZ=66, TOP=6, BOT=3, LPI=6
```

#### Example 5

```
SM PRC C,66,2,1,8
```

Changes the current page format values for printer PRC, an 0776 impact printer. The system displays this message:

```
PRC PSIZ=66, TOP=2, BOT=1, LPI=8
```

#### Example 6

```
SM PRC C,DEF
```

Restores all values to system standard for the printer (0776 impact printers). The system displays this message:

```
PRC PSIZ=88, TOP=3, BOT=3, LPI=8
```

**Example 7**

```
SM PRC C,*,2,*,8
```

Restores the system-standard values only for lines per inch and for the bottom margin (0776 impact printers). The system displays this message:

```
PRC PSIZ=88, TOP=2, BOT=3, LPI=6
```

**Example 8**

```
SM PRC E
```

Terminates an active print file on printer PRC.

**Example 9**

```
SM PRA I
```

Reactivates printer PRA. The system displays this message:

```
PRA UNLOCKED
```

**Example 10**

```
SM PRA J ON
```

Initiates jogging of output on printer PRA.

**Example 11**

```
SM PRC L
```

Locks out printer PRC. The system displays this message:

```
PRC LOCKED
PRC LOCKED , STOPPED
```

**Example 12**

```
SM PRA Q
```

Requeues print files and lock out printer PRA. The system displays this message:

```
PRA REQUEUED AND LOCKED
```

**Example 13**

```
SM PRA R12
```

Reprints 12 pages on printer PRA. The system displays this message:

```
PRA NOW REPRINTING 12 PAGES
```

### Example 14

```
SM PRC R+2
```

Skips two pages on printer PRC. The system displays this message:

```
PRC COMPLETED SKIPPING 2 PAGES
```

If the keyin advances the file to the end-of-file mark, the system displays this message:

```
sname SKIPPED n PAGES - EOF! REPRINT? PAGES OR N
```

### Example 15

```
SM PRC R+17
```

Skips 17 pages on printer PRC. There are only 14 pages left in the file, so the system displays this message:

```
0-PRC SKIPPED 14 PAGES - EOF! REPRINT? PAGES OR N
```

To have three pages reprinted, respond with

```
0 3
```

The system displays this message:

```
PRC NOW REPRINTING 3 PAGES
```

### Example 16

```
SM PRC RALL
```

Reprints the entire file on printer PRC. Only 12 pages of the file have been printed, so the system displays this message:

```
0-PRC ONLY 12 PAGES PRINTED - REPRINT ALL? - YN
```

To have the 12 pages reprinted, respond with

```
0 Y
```

The system displays this message:

```
PRC NOW REPRINTING 12 PAGES
```

### Example 17

```
SM PRC S
```

Completely suspends the operation of printer PRC. The system displays this message:

```
PRC SUSPENDED
```

**Example 18**

```
SM PRC C,88,,,
```

Changes only the size field on a Model 0770 printer.

The system-standard values are taken from the system-standard configuration, not from the standards for the particular device involved. The page size, top margin, and bottom margin system-standard values are controlled by system software configuration parameters. However, no software configuration parameter defines the system-standard value for lines per inch; the system value is always 8.

**11.71.SP—Suspend a Run**

The SP keyin lets you suspend active runs.

**11.71.1. Format**

```
SP[, {B|D}U] ID1[, ID2, ..., IDn]
```

where:

*B*

specifies to suspend batch runs only.

*D*

specifies to suspend demand runs only. Both B and D cannot be specified.

*U*

indicates that the ID arguments are user IDs.

One or more IDs can be specified, separated by commas and with or without masking. If the U option is not specified, the IDs refer to run IDs; if the U option is specified, the IDs refer to user IDs.

**Examples**

|             |                                                                                                      |
|-------------|------------------------------------------------------------------------------------------------------|
| SP,BU JUDGE | Suspends all batch runs executing under user ID JUDGE                                                |
| SP,D R2D2*  | Suspends all demand runs that start with R2D2                                                        |
| SP R1*,R2?  | Suspends all runs that start with R1 and all runs that start with R2 and that are up to 3 characters |

### 11.71.2. System Response

When the keyin is accepted, the system responds as follows:

```
run-id SUSPENDED yyyy-mm-dd hh:mm:ss zone
```

If the activity you are suspending is running with the Exec key set, it is performing an extended mode Exec Service. Extended mode services cannot be suspended until the service has completed. The standard system method of waiting for extended mode Exec services completion usually suspends the job.

The following are instances of extended mode Exec services for which the standard method of waiting for completion does not apply:

- ILES\$MANAGE
- CARTTAPELIB\$
- TIP\$FILE\$FUN
- SERVER\$GET

If an activity of the job you intend to suspend is performing any of these extended mode Exec services, the job is not suspended and the system displays the following message:

```
run-id CANNOT BE SUSPENDED NOW - TRY AGAIN LATER
```

If the keyin is not accepted for any other reason, the system displays a message that indicates why it could not be suspended.

### 11.71.3. Restrictions

The E and X keyins cannot be used to terminate suspended runs. You must resume them before they can be terminated. If a run is suspended, it will complete (FIN) only if a \$\$\$CLOSE or @@TERM is issued from the run's demand terminal or an SM *siteid* T is issued from another terminal.

Only demand console users with a console mode capability of FULL, DISPLAY, or RESPONSE can enter the SP keyin.

## 11.72.SQ—Display Print File Status

The SQ keyin displays the status of the print files in the print queue.

### 11.72.1. Formats

#### All Devices and Symbiont Groups

SQ

Displays one line of information for each device and symbiont group. The system output is as follows:

```
sname: nn FILES nnn [PAGES|CARDS] nn TAPES
```

#### Specified Symbiont Queue

SQ *sname*

Displays one line of information for the specified symbiont queue. The system output is as follows:

```
sname: nn FILES nnn PAGES nn TAPE
```

#### Total Number of Files Queued to User-IDs

SQ USR\*ID

Displays the total number of files queued to all user-IDs. The system output is as follows:

```
USR*ID HAS NO PRINT/PUNCH FILES QUEUED
```

or

```
USR*ID: nnnn FILES nnnn PRINT nnnn PUNCH nnnn TAPE
```

#### Each File Queued by User-ID

SQ USR\*ID \*

Displays file names, number of pages or cards, and priority for all files queued to all user-ids. The system output is as follows:

```
user-id: run-id filename nn EPGS priority
```

where *priority* is one of the following:

- |   |                            |
|---|----------------------------|
| 0 | ROLOUT/ROLBAK runs         |
| 1 | Critical deadline runs     |
| A | Run priorities A, B, and C |

|   |                            |
|---|----------------------------|
| D | Run priorities D, E, and F |
| G | Run priorities G, H, and I |
| J | Run priorities J, K, and L |
| M | Run priorities M, N, and O |
| P | Run priorities P, Q, and R |
| S | Run priorities S, T, and U |
| V | Run priorities V, W, and X |
| Y | Run priorities Y and Z     |

### Files Queued to Symbiont Queue

*SQ sname \**

Displays the file names, number of pages or cards, and priority for a specific device or group. The system output could be one or several of the following lines:

```
sname: run-id file nnn EPGS priority active mmm PPGS
sname: run-id file nnn EPGS priority active
sname: run-id file nnn ECDS priority active mmm PCDS
sname: run-id file TAPE priority active
```

### Each File Queued by Run-ID

*SQ run-id \* R*

Displays file names, number of pages or cards, and rank for the given *run-id(s)*. The system output is as follows:

```
run-id: file nnnnn EPGS priority rank/sname
run-id: file nnnnn CDS priority rank/sname
run-id: file TAPE priority rank/sname
```

For files restored with an SR keyin, the *run-id* is the original run-ID for primary print or punch files, or EXEC8 for user-created print or punch files.

*SQ,R run-id1[,...run-idn] [filename]*

Displays run-id, file name, number of pages or cards, priority and rank for the given set of run-ids.



**User-ID**

```
SQ user-id * U
```

Displays file names, number of pages or cards, and priority for the user-id(s). The system output is as follows:

```
user-id: run-id file nnn EPGS priority
user-id: run-id file nnn CDS priority
user-id: run-id file TAPE priority
```

```
SQ,U[A] user-id1[,...user-idn] [filename]
```

If A option is not specified, displays user-id, run-id, file name, number of pages or cards, priority and device for the given set of user-ids.

If A option is specified, displays user-id, run-id, file name, priority and rank for the given set of user-ids.

**Priority**

```
SQ run-id priority
```

Changes the priority order for all files of the run-id.

```
SQ run-id filename priority
```

Changes the priority order for a specified file.

**Redirection (QTO)**

```
SQ sname QTO sname
```

Redirects all files queued to *sname* to queue *sname*.

```
SQ sname1 QTO sname2
```

Redirects present queues for *sname1* to *sname2*.

```
SQ run-id file QTO sname
```

Redirects a single file to a different queue.

```
SQ run-id file QTO sname
```

Redirects a specific file to *sname*.

```
SQ run-id file QTO user-id/U
```

Redirects a specific file to user-id

### Redirection (TO)

```
SQ sname1 TO sname2
```

Redirects future files for *sname1* to *sname2*, where *sname1* is a specific on-site device that is locked out and idle and *sname2* is a specific device that receives the future files for *sname1*. The system displays a confirming message:

```
SQ - FUTURE FILES FOR sname1 WILL BE DIRECTED TO sname2
```

Do not use this keyin for remote batch runs.

### Printerless Systems

```
SQ PR AQ
```

Allows files to be queued to the default print queue (printerless systems only).

```
SQ PR PQ
```

Prohibits queuing on the default print queue (printerless systems).

## 11.72.2. Examples

```
SQ
```

Displays information for each device and symbiont group, as follows:

```
RG2621: 21 FILES 1022 PAGES
B112X: 7 FILES 1793 PAGES
PR: 1 FILE 1153 PAGES
NOT777: 2 FILES 10 PAGES
```

```
SQ PR
```

Displays information for printer PR, as follows:

```
PR: 1 FILE 1153 PAGES
```

```
SQ PR *
```

Displays information for each file in queue PR, as follows:

```
PR: JJP JJP*A 1022 EPGS D
PR: DPE PRRRIN*0129240 30 EPGS J
PR: SLATTE IOUE*PRINT 1040 EPGS M
```

### 11.72.3. Operator Interfaces

The following rules apply for the SQ keyins:

The `run-id` and `user-id` can contain wildcard characters ('\*', matching zero or any number of characters) and masking characters ('?', matching any single character). The `run-id` can also be specified as 'EXEC 8' and 'TIP\$'.

1. PR or PR/ or PR/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary print files.
2. PU or PU/ or PU/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary punch files.
3. qualifier\*filename(cycle) having the standard syntax of file names. Qualifier and asterisk may be missing, which is interpreted as any qualifier.
4. qualifier'\*' filename(cycle). In addition to standard characters, qualifier and filename may also contain wildcard (\*) and masking (?) characters. Both qualifier and filename must be provided.

**Note:** The '\*' following the qualifier must be enclosed in quotes as an indicator that:

1. Either wildcard or masking characters or both are present
2. This particular \* is not interpreted as a wildcard character.
5. qualifier\'\*filename is alternate syntax to step 4). The backslash character indicates that either wildcard or masking characters or both are present.

## 11.73.SR—Restore Saved Symbiont Files

The SR keyin restores symbiont files that were previously saved to mass storage so you can queue them for printing (or punching).

If your site has Security Level 2 or 3 or a Security Level 3 trusted environment, you can transport labeled print files from system to system by using the SR and SV keyins, in addition to saving the current print queue for a single host. If you save and restore a nonstandard file (using the SV and SR keyins), the banner from the @SYM control statement is lost. The `run-ID` is used as the banner.

As with the SV keyin, use a labeled tape if your system was configured to require a prelabeled tape.

### 11.73.1. Formats

```
SR ,unit/reeln A/F
```

Restores and queues all symbiont files for the specified reel, reeln.

```
SR ,unit/reeln A/F sname
```

Restores all files for the specified symbiont group or device.

```
SR ,unit/reeln run-id,...,run-id
```

Restores all files for specific runs (up to five run-IDs).

```
SR ,unit/reeln run-id file
```

Restores specified files.

**Note:** To restore Exec files, use the SEC, RESTORE keyin.

### 11.73.2. System Response

If there is more than one reel of tape to restore, the system prompts you with the following message:

```
REEL NUMBER FOR SRKYINfilename nnnnnn or N
```

You can add a new tape number, or you can answer N to terminate the SR keyin.

As each file is restored to mass storage, the system displays this message:

```
SRKYINfilename: run-id file/sname
```

In this case, the run-IDs are internally named EXEC 8 to avoid duplication of run-IDs in the system or in the output queue. The exceptions are primary PRINT\$ or PUNCH\$ files, where the displayed run-ID is the original run-ID.

**Note:** When the Exec restores a primary print file and displays the file name, it substitutes a dollar sign ( \$ ) for the ampersand ( @ ) in the name.

When all files have been restored, the system displays a message giving you the total number of restored files.

Contact the site administrator if you see the following message while restoring files:

```
SRKEYINfilename BLOCK COUNT ERROR - GO
```

The preceding message indicates the data restored into storage is incorrect because the block count calculated during the restore operation does not match the block count stored on the tape. Responding GO allows SR processing to continue in spite of the data being corrupt. The corrupt data remains in storage and should not be used. Files restored to the print queue will be printed even though the block count error indicates they are in error.

### 11.73.3. Restrictions

Files restored by an SR keyin cannot be deleted by the SX run-id or the SX run-id filename keyins. They can be deleted only by the SX A/F or SX A/F sname keyins.

### 11.73.4. Operator Interfaces

The following rules apply for the SR keyins:

The `run-id` and `user-id` can contain wildcard characters ('\*', matching zero or any number of characters) and masking characters ('?', matching any single character). The `run-id` can also be specified as 'EXEC 8' and 'TIP\$'.

1. PR or PR/ or PR/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary print files.
2. PU or PU/ or PU/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary punch files.
3. qualifier\*filename(cycle) having the standard syntax of file names. Qualifier and asterisk may be missing, which is interpreted as any qualifier.
4. qualifier'\*' filename(cycle). In addition to standard characters, qualifier and filename may also contain wildcard (\*) and masking (?) characters. Both qualifier and filename must be provided.

**Note:** The '\*' following the qualifier must be enclosed in quotes as an indicator that:

1. Either wildcard or masking characters or both are present
  2. This particular \* is not interpreted as a wildcard character.
5. qualifier\\*filename is an alternate syntax to Step 4). The backslash character indicates that either wildcard or masking characters or both are present.

## 11.74. SS—Display System Status

The SS keyin displays information describing the system status. The SS keyin response is seven lines long. The first three lines are the fixed portion of the SS keyin response; the last four lines are the same as those shown continuously at the top of the system console display.

The four lines of the continuous console display are actually displayed as two lines in a rotating sequence. Each sequence is updated twice (6 seconds apart) before the other sequence is displayed. The continuous console display lines are sent to the console printer or a remote console only as a result of the SS keyin.

The instruction processor percentages in the keyin response are based on instruction processor usage. Instruction processor usage is determined by the amount of instruction processor time a given program type used compared to the total amount of instruction processor time. These values cannot total 100 percent because the Exec always uses some percentage of instruction processor time.

Asterisks appear in display fields when the field's value exceeds its size.

### 11.74.1. Format

SS

## 11.74.2. System Response

The response to the SS keyin is as follows:

```
STATUS: OMSG nnnn, STRT nnn, DDLN nnn, yyyy-dd-mm hh:mm:ss zone
HELD: OPER nnnn, TAPE nnnn, MASS nnnn, SOPT nnn
ACTV: OPEN nnnn, ITLD nnnn SITE: PART-A HOST: A
IP: BAT xx% DEM xx% R/T xx% TIP xx% IDLE xx%
OPEN: BAT nnnn DEM nnnn UNOPND: nnnn HOLDS holds
PAGE IO/M: nnnn MASS: nnnnnnnn KWEM: nnnnnnnn yyyy-dd-mm hh:mm:ss zone
EXP: xx%-nnnnnn E11:nnn ID: level TIP: fnnnnnn/r Q: nnnn A: nnnn
```

Table 11-2 lists the parameters displayed in response to the SS keyin.

**Table 11-2. Parameters Displayed by the SS Keyin**

| Parameter                       | Description                                                                                                                                                                                |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>yyyy-dd-mm hh:mm:ss</i> zone | date and time of the keyin in standard display format—that is year, month, day, hours, minutes, seconds, and configured time zone mnemonic.                                                |
| OMSG <i>nnnn</i>                | Number of outstanding read-and-reply messages sent to all consoles, regardless of whether they are actually displayed yet.                                                                 |
| STRT <i>nnn</i>                 | Unopened runs waiting for start time.                                                                                                                                                      |
| DDLN <i>nnn</i>                 | Unopened deadline runs.                                                                                                                                                                    |
| OPER <i>nnn</i>                 | Runs held by the CS H run-ID keyin.                                                                                                                                                        |
| TAPE <i>nnnn</i>                | Runs held for tape or other peripheral facilities.                                                                                                                                         |
| MASS <i>nnnn</i>                | Runs held for mass storage facilities or cataloged files.                                                                                                                                  |
| SOPT <i>nnnn</i>                | Runs waiting for other runs to finish due to an @RUN statement S option.                                                                                                                   |
| OPEN <i>nnn</i>                 | Runs in main storage.                                                                                                                                                                      |
| ITLD <i>nnnn</i>                | Runs waiting for initial load of a task.                                                                                                                                                   |
| IP: BAT <i>xx%</i>              | Percentage of IP SUPs used by user programs using batch and deadline batch processing.                                                                                                     |
| DEM <i>xx%</i>                  | Percentage of IP SUPs used by user programs using demand processing.<br>If demand processing is not configured, this entire field is left blank.                                           |
| R/T <i>xx%</i>                  | Percentage of IP SUPs used by user programs using real-time processing.                                                                                                                    |
| TIP <i>xx%</i>                  | Percentage of IP SUPs used by TIP transactions.<br>If TIP is not configured, this entire field is left blank.                                                                              |
| IDLE <i>xx%</i>                 | This parameter is present only when the aggregation of IP SUPs used by user programs is negligible. Note that this does not include IP utilization by the Exec on behalf of user programs. |

Table 11-2. Parameters Displayed by the SS Keyin

| Parameter                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UNOPD <i>nnn</i>                                           | Number of backlogged runs, including runs that are waiting to be opened and runs that are being held.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| HOLDS <i>holds</i>                                         | <p>Indicates the reasons why runs are being held. The value holds can contain any combination of the following letters. If none applies, the entire field is blank.</p> <p>A All batch runs are held. This results from a CS H keyin or scheduling being suspended because the mass storage limit has been reached.</p> <p>D Demand runs are being held due to a CS HD keyin.</p> <p>E Insufficient EXPOOL space is available to open additional runs. This occurs when EXPOOL space availability has reached the critical threshold.</p> <p>I At least one run is suspended on the system.</p> <p>L Demand terminals are being held due to a CS HT keyin.</p> <p>M Mass storage facilities or cataloged files are not available.</p> <p>O Indicates at least one outstanding read-and-reply message for an onsite or remote console.</p> <p>R Operator hold by means of CS H run-ID keyin.</p> <p>S The S option appeared on some @RUN statements, but the previous run from the same peripheral device has not terminated.</p> <p>T Tape or other peripheral facilities not available. Includes tape units, removable disk units, reel-IDs, ACS queries, and facilities other than mass storage space and cataloged mass storage files.</p> <p>U Low availability of one or more resource areas used internally by Exec to support run execution.</p> |
| PAGE IO/M<br><i>nnnn</i><br>or<br>PAGE IO/S<br><i>nnnn</i> | For systems that support paging. Indicates the number of page input/outputs per minute; if the number of paging operations per minute exceeds 999, it is reported as operations per second.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| PAGE PG/M<br><i>nnnn</i><br>or<br>PAGE PG/S<br><i>nnnn</i> | For systems that do not support paging. Displays the rate at which 4096-word areas ("pages") of memory have been written to, or read from, the Swapfile by the infrastructure over the most recent reporting interval.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| KWMEM<br><i>nnnnnnn</i>                                    | For systems that support paging. Amount of memory available, measured in kilowords (Kwords).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| MEM<br><i>Pnnn%</i><br>or<br><i>Snnn%</i>                  | <p>For systems that do not support paging. Indicates two percentages:</p> <p>P is the percentage of Physical space currently in use.</p> <p>S is the percentage of Swapfile space currently in use.</p> <p>If either value is a negative number, it indicates the value is decreasing or remaining constant. If either value is a positive number, it indicates the value is increasing.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**Table 11-2. Parameters Displayed by the SS Keyin**

| Parameter                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MASS<br>nnnnnnnn                 | <p>Number of local mass storage tracks available (does not include shared mass storage).</p> <p>This field is limited to 8 characters in length.</p> <ul style="list-style-type: none"> <li>If the number of available tracks is larger than 99,999,999, the value is displayed in 1,000-track increments, with a "K" appended to the value as the eighth character.</li> <li>If the number of available tracks is larger than 9,999,999,999, the value is displayed in 100,000-track increments, with an "M" appended to the value as the eighth character.</li> </ul> <p>For example</p> <p>MASS: 2345678K</p> <p>indicates that between 2,345,678,000 and 2,345,678,999 tracks are available.</p>                                                                                                                                                                                                                       |
| yyyy-dd-mm<br>hh:mm:ss zone      | is the date and time of the last continuous display update in standard display format—that is year, month, day, hours, minutes, seconds, and configured time zone mnemonic.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| EXP xx%-<br>nnnnnn               | <p>The percentage and amount of EXPOOL available:</p> <p>xx%            Percentage of EXPOOL available</p> <p>nnnnnn        Number of words of EXPOOL available</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| E11 nnn                          | Number of EXPOOL index 011 size buffers available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ID iiiiiiiiii                    | Level identification of the Exec on which the system is running.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| TIP<br>fnnnnnn/r<br>Q:nnn A:nnnn | <p>If TIP is not configured, this entire field is left blank.</p> <p><i>f</i>            Processing rate indicator. When the percentage difference between the Transaction Processing rate in the most recent 6-second period and the previously displayed rate is at least 10%, an indicator (either + or -) is appended to the displayed processing rate. If the difference is less than 10%, the <i>f</i> field is left blank.</p> <p><i>nnnnnn</i>    Indicates the transaction processing rate measured in transactions per minute, if there are fewer than three transactions per second. Otherwise it is measured in transactions per second.</p> <p><i>r</i>            Indicates either seconds (S) or minutes (M).</p> <p><i>Q:nnnn</i>    Number of input transactions queued for processing.</p> <p><i>A:nnnn</i>    Number of active transaction programs. Stuck programs are not included in this count.</p> |



### 11.74.3. Examples

#### Example 1

SS

For systems that support paging. The system displays the following:

```
STATUS: OMSG 0000, STRT 000, DDLN 000, 2013-05-15 15:03:33 CDT
HELD: OPER 0000, TAPE 0000, MASS 0000, SOPT 000
ACTV: OPEN 0042, ITLD 0000, SITE: PART-A HOST: A
IP: BAT 0% DEM 0% R/T 1% TIP 0%
OPEN: BAT 38 DEM 4 UNOPND: 1
PAGE IO/M: 0 MASS: 1252142 KWMEM: 1788796 2013-05-15 15:03:28 CDT
EXP: 90%-138632 E11:255 ID: 49R1.1 TIP: 0/M Q: 0 A: 0
```

The system displays the following message when Multi-Host File Sharing (MHFS) is not installed:

```
STATUS: OMSG 0000, STRT 000, DDLN 000, 2013-05-15 15:03:33 CDT
HELD: OPER 0000, TAPE 0000, MASS 0000, SOPT 000
ACTV: OPEN 0042, ITLD 0000, SITE: PART-C
IP: BAT 0% DEM 0% R/T 1% TIP 0%
OPEN: BAT 38 DEM 4 UNOPND: 1
PAGE IO/M: 0 MASS: 1252142 KWMEM: 1788796 2013-05-15 15:03:28 CDT
EXP: 90%-138632 E11:255 ID: 49R1.1 TIP: 0/M Q: 0 A: 0
```

See 1.4.4 and 9.1.1 for more information on MHFS.

For systems that do not support paging. The system displays the following:

```
STATUS: OMSG 0000, STRT 000, DDLN 000, 2013-05-15 15:03:33 CDT
HELD: OPER 0000, TAPE 0000, MASS 0000, SOPT 000
ACTV: OPEN 0042, ITLD 0000, SITE: PART-C
IP: BAT 0% DEM 0% R/T 1% TIP 0%
OPEN: BAT 38 DEM 4 UNOPND: 1
PAGE PG/S: 984 MASS: 6207500 MEM:P-87% S-06% 2013-05-29 15:57:29 CDT
EXP: 90%-138632 E11:255 ID: 49R1.1 TIP: 0/M Q: 0 A: 0
```

#### Example 2

The following lines are shown continuously at the top of the system console display. One pair of lines is shown for 6 seconds, then the other pair is shown for 6 seconds.

For systems that support paging. The system displays the following:

```
IP: BAT 1% DEM 0% R/T 0% TIP 0%
OPEN: BAT 2 DEM 0 UNOPND: 0 HOLDS 0
```

```
PAGE IO/M: 0 MASS: 1252142 KWMEM: 1788796 2013-05-15 15:03:28 CDT
EXP: 90%-138632 E11:255 ID: 49R1.1 TIP: 0/M Q: 0 A: 0
```

For systems that do not support paging. The system displays the following:

```
IP: BAT 1% DEM 0% R/T 0% TIP 0%
OPEN: BAT 2 DEM 0 UNOPND: 0 HOLDS 0

PAGE PG/S: 984 MASS: 6207500 MEM:P-87% S-06% 2013-05-29 15:57:29 CDT
EXP: 90%-138632 E11:255 ID: 49R1.1 TIP: 0/M Q: 0 A:0
```

## 11.75.ST—Start a Stored Runstream

The ST keyin submits a runstream stored in a file or element for execution. The runstream will be scheduled and executed as an independent run.

Also see the *System Console Messages Reference Manual* (7833 3275) for information about the console messages related to the ST keyin.

### 11.75.1. Format

```
ST,priority/run-options/pdp/name,setc,run-id,acct-id/user-id,
project-id,est-run-time/deadln,pages/cards,start-time,sequence-id
```

where:

*priority*

If specified, replaces the scheduling priority given on the @RUN statement for the stored run.

**Note:** When the Exec is holding a run in backlog because resources are not available (such as a file), other lower priority runs are not allowed to start. To avoid holding up the start of runs, start as many runs as possible at the same priority.

*run-options*

If specified, replace those on the @RUN statement for the stored runstream. If the S option is used on the @RUN statement, it is ignored. The remaining options (Z, U, and V) are described in 0.

*pdp*

If specified, replaces the processor dispatching priority on the @RUN statement for the stored runs.

*name*

Identifies the file or element that contains the stored runstream. The name must include the name of a file. If the name is a file name only, the file must be a data file.

*setc*

Specifies a positive octal number (not more than 4 digits) to be placed in T2 of the condition word of the run being scheduled.

*run-id*

If specified, this run-ID replaces the one on the @RUN statement for the stored runstream.

*acct-id*

If specified, this account number replaces the one on the @RUN statement for the stored runstream.

*user-id*

If specified, this user-ID replaces the one on the @RUN statement for the stored runstream.

**Note:** Remote ST keyins that specify a user-ID other than the issuer's user-ID must have the SSSTRTZOPT privilege to successfully start the run. For example, if @@CONS ST qual\*file.elt,,,acct/B is entered by user-ID A, then user-ID A must have the SSSTRTZOPT privilege to start the run.

*project-id*

If specified, this project-ID replaces the one on the @RUN statement for the stored runstream.

*est-run-time*

If specified, this estimated run time replaces the one on the @RUN statement for the stored runstream.

*deadln*

If specified, this deadline replaces the one on the @RUN statement for the stored runstream.

*pages*

If specified, this estimate of output replaces the one on the @RUN statement for the stored runstream.

*cards*

If specified, this estimate of output replaces the one on the @RUN statement for the stored runstream.

*start-time*

If specified, this start time replaces the one on the @RUN statement for the stored runstream.

*sequence-id*

If specified, this identifier is displayed as [id] at the end of an ST keyin error message if the keyin fails. It is displayed in the format

```
ST KEYIN ERROR - text [id]
```

The *sequence-id* can be 1 or 2 alphanumeric characters. This identifier can be helpful in correlating the error message with the failing keyin.

## **11.75.2. Options**

### **U Option**

The U option on the ST keyin (ST,/U) allows a run to be started and executed under the user-ID of the @RUN statement in the stored runstream, instead of under the user-ID specified on the ST keyin.

Rules for the U option are as follows. Note that the rules concerning execute access and file ownership can be bypassed by means of a security privilege (see 4 and 5 below). The bypass allows runstreams with different user-IDs to be stored in the same file.

1. The U option is valid only when user security is configured (SENTRY\_CONTROL is TRUE).
2. Both the user-ID and account number must be present on the stored @RUN statement.
3. The user-ID on the ST keyin must have execute access to the user-ID on the stored @RUN statement.
4. The user-ID on the ST keyin must have execute access to the start file unless the following two conditions hold:
  - The ST keyin is entered from the physical console.
  - The user-ID on the ST keyin has the SSSTUBYPASSO privilege.
5. The user-ID on the stored @RUN statement must own the file containing the stored runstream, unless the user-ID on the ST keyin has either the SSSTUBYPASSO or the SSBYPASSOWNR privilege.
6. The user-ID and account number of the run to be started must be specified on the ST keyin.

### **V Option**

The V option on the ST keyin allows a run to be started with the minimum clearance level and default compartment set of the user-id on the @RUN statement. Without the V option, the started run assumes the default compartment set of the user-ID on the ST keyin, and the operator is solicited for the minimum clearance level.

When the V option is combined with the U option (ST,/UV), the clearance level and compartment set of the start file must equal the minimum clearance level and the default compartment set of the user-ID on the ST keyin. There are two exceptions to this rule:

- If the user-ID on the ST keyin is sufficiently privileged to change the start copy file to the minimum clearance level and default compartment set of the user-ID on the @RUN image, then the run is started at that minimum level with the default compartment set.
- If the start file is system low (clearance level 0 and NULL compartment set), then the run is started at system low.

The purpose of the V option is to let a trusted user-ID—that on the ST keyin—start a more privileged run. It is useful for starting runstreams that can run productively below their top security level.

### Z Option

The Z option on the ST keyin (ST,/Z) lets you initiate runs that execute at system high, provided that you specify a user-ID that can execute at system high, and that you have full console capability. If you specify the U option, the run that was started must possess the Z-option privilege.

When SENTRY\_CONTROL is TRUE, the following read-and-reply message is suppressed for ST keyins that specify the Z option:

```
ENTER C.L. FOR: filename
```

### Starting Runstreams for Other Software Products

The ST keyin also starts runstreams for other software products. Use the following ST keyin formats:

```
ST DPREP
```

Starts the runstream that executes DPREP software. For information on preparing disks, refer to the *Exec System Software Installation and Configuration Guide* (7830 7915).

```
ST INSTALL
```

Starts a batch run to execute the INSTALL element for COMUS.

## 11.75.3. System Configuration Parameters and the ST Keyin

The format for the ST keyin depends on the values for the system configuration parameters SENTRY\_CONTROL and QUOTA\_ENFORCEMENT\_BY\_USERID.

The following configurations are possible:

|   | SENTRY_CONTROL | QUOTA_ENFORCEMENT_BY_USERID |
|---|----------------|-----------------------------|
| 1 | FALSE          | FALSE                       |
| 2 | FALSE          | TRUE                        |
| 3 | TRUE           | FALSE                       |
| 4 | TRUE           | TRUE                        |

You can use the ST keyin with different configuration parameters, as described in the following:

### 1. Both Parameters FALSE

When SENTRY\_CONTROL and QUOTA\_ENFORCEMENT\_BY\_USERID are both FALSE, use the following keyin formats:

```
ST element
```

The system searches for the element in these files:

```
SYSS$*RUN$.
SYSSLIB*RUN$.
```

If the element is found, the account number and user-ID are taken from the @RUN statement, and the run is started. For more information on the @RUN control statement, refer to the *Executive Control Language (ECL) and FURPUR Programming Reference Manual* (7830 7949).

```
ST element,,,account
```

or

```
ST element,,,account/user-id
```

The system ignores the user-ID on the keyin. The user-ID comes from the @RUN statement, and the account number comes from the keyin. If there is no user-ID on the @RUN statement, then a MISSING USERID error is displayed, and the keyin terminates.

```
ST qualifier*filename.element,,,account
```

The run is started with the user-ID from the @RUN statement, and account is taken from the keyin. *ST qualifier\*filename.element,,,account/user-id*

Both *account* and *user-id* come from the keyin. If the user-ID is the security officer's user-ID, then the system displays the following read-and-reply message:

```
ENTER PASSWD/C.L. FOR: qualifier*filename.element
```

Using a space in the reply (in other words, no password) indicates a blank password. If TSS\_CONTROL is FALSE, the keyin is allowed and the started run executes with the user-ID specified, but using the minimum security attributes. If TSS\_CONTROL is TRUE, then a blank password is accepted, but only if the TSS configuration parameter ALLID is set.

## 2. SENTRY\_CONTROL=FALSE and QUOTA\_ENFORCEMENT\_BY\_USERID=TRUE

When SENTRY\_CONTROL is FALSE and QUOTA\_ENFORCEMENT\_BY\_USERID is TRUE, the ST keyin is the same as when QUOTA\_ENFORCEMENT\_BY\_USERID is FALSE, except that the following keyin is not allowed without the user-ID:

```
ST qualifier*filename.element,,,account
```

## 3. SENTRY\_CONTROL=TRUE and QUOTA\_ENFORCEMENT\_BY\_USERID=FALSE

When SENTRY\_CONTROL is TRUE and QUOTA\_ENFORCEMENT\_BY\_USERID is FALSE, you must specify a valid account-ID and user-ID on the keyin. No password solicitation occurs. The system displays this read-and-reply message:

```
ENTER C. L. FOR: qualifier*filename.element
```

A blank or null response is assigned the minimum clearance level. The only allowed keyins are as follows:

```
ST element,,,account/user-id
```

The account and user-ID come from the keyin.

```
ST qualifier*filename.element,,,account/user-id
```

The account and user-ID come from the keyin.

## 4. Both Parameters TRUE

When both SENTRY\_CONTROL and QUOTA\_ENFORCEMENT\_BY\_USERID are TRUE, keyins have the same effect as when SENTRY\_CONTROL is TRUE and QUOTA\_ENFORCEMENT\_BY\_USERID is FALSE (see the immediately preceding topic).

# 11.76. SU—Suspend a Component

The SU keyin suspends devices from being allocated. Use the keyin if you are getting frequent I/O errors on a component. Suspend permits access to files already on the component, but prevents any more from being placed on it. The SU keyin is a facility keyin.

## 11.76.1. Formats

```
SU devnam
```

Inhibits further allocation on the component while still permitting I/O operations.

```
SU,ALL cu
```

Suspends further allocation on all devices attached to the specified control unit while still permitting I/O to the devices.

## 11.77.SV—Save Queued Output Files to Tape

The SV keyin saves queued print and punch output files to magnetic tape. Files queued to USR\*ID cannot be unloaded with the SV keyin.

**Note:** You must use the SEC,SAVE keyin to save EXEC-8 files.

### 11.77.1. SV Keyin Format

```
SV [, [*[unit]] [/reeln] [, [options] [, CTL-pool] [, expiration] [/MMspec] [, data-
comp] [/buf-write]] file-spec.
```

where:

comma as the first character of the keyin text indicates that details and options of the tape assignment follow the comma. If the comma is not the first character then the keyin text contains *file-spec* only. In this case a blank tape from the standard system equipment type is assigned for the save operation.

\*

is optional and a specific tape unit.

*unit*

is optional and a specific tape unit if preceded by '\*'. Otherwise, unit is tape assign mnemonic. If not specified, the configured standard system equipment type is used.

*reeln*

specifies the tape reel number. If no number is specified, the system requests a blank tape.

*option*

options can be one of the following:

- |   |                                                                                                                                                                             |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| N | Keeps the file in the queue and saves the contents on tape. If you save a file once with this option, you cannot save it again. Subsequent SV keyins bypass the file.       |
| R | Allows removable disk files also to be saved on tape. Note that using the SR keyin to restore these removable disk files will restore them to fixed mass storage.           |
| S | Writes the tape in 224-word blocks instead of the default 1,792-word blocks. These tapes can be directly queued for printing or punching from a run with an @SYM statement. |
| U | <i>file-spec</i> contains user-ids, not run-ids. If this option is specified, A/F and A/F <i>sname</i> are invalid in <i>file-spec</i> field.                               |

*CTL-pool*

is optional and specifies the CTL pool used for blank tape allocation.



*expiration*

is optional and specifies the number of days that the tape is retained.

*MMspec*

is the optional Media Manager specification, when the Media Manager is used.

*data-comp*

is the optional data compression specification; for example, CMPON

*buf-write*

is the optional buffered write specification; for example, BUFTAP, BUFFIL

*file-spec*

specifies the files that are saved to tape. It can be one of the following:

|                                                        |                                                                                                                                                                                    |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A/F                                                    | Saves all files on all queues to tape. Files queued to user-IDs are not saved.                                                                                                     |
| A/F sname                                              | Saves all files queued to symbiont group or device name.                                                                                                                           |
| <i>run-id1[,runid...,run-idn]</i><br><i>[filename]</i> | Saves files from the specified runs. This can be many files if any run-id and/or filename contains masking or wildcard characters.                                                 |
| <i>user-id1[,...,user-idn]</i><br><i>[filename]</i>    | Saves files from the specified user-IDs. This can be many files if any <i>user-id</i> and/or <i>filename</i> contains masking/wildcard characters. The U option must be specified. |

## 11.77.2. System Response

For each SV keyin, the system requests a tape SVKYINnnnnnn to load; each tape has a unique number.

As each file is read to tape, the system displays this message:

```
SVKYINtapename: run-id file/sname
```

If the file to be saved is a removable disk file and the R option was not specified, the file is not saved and a message with the reel number or disk pack is displayed in one of the following formats:

```
SVKYINtapename: run-id file TAPE-reeln
```

or

```
SVKYINtapename: run-id file PACK-pack-id
```

where SVKYINtapename identifies the file to the Exec.

When all files have been read, the system displays the total number of files saved.

The following error cases can occur when saving removable disk files (with the R option specified):

- If the main item address within the smoque entry is not the correct main item address, then the smoque entry is not processed and the following E\$ORMSG read-only message (message number 0350) is displayed.

*SVKYINTapename*: The symbiont queue entry for file *filename* from run *run-id* is not valid for SV processing; therefore, it has been bypassed.

This can occur when the RP keyin has removed the file from the master file directory (MFD). When the file has been removed, the main item in the smoque entry is no longer valid. A subsequent FA keyin can reestablish the file, but the file will receive a new main item address.

This error case can also occur because of a small timing window after the SV keyin has verified that the smoque entry is valid during which the RP keyin can be performed before the file is actually assigned. This can also generate an I/O error 021 with the file in a "not cataloged" state.

- If there is an I/O error 021 while reading the removable disk file during SV processing and the files status is "not cataloged," the following E\$ORMSG read-only message (message number 0150) is displayed.

*SVKYINTapename*: The queued file *filename* from run *run-id* is on a removable pack but could not be assigned for SV processing.

This case can occur if a file spans multiple packs and one of the packs is down.

This case can also occur even when the file is cataloged—when the file is exclusively assigned by a user at the time of the SV keyin.

- When shared removable files are queued and file sharing is down, SV processing skips saving such files and displays an appropriate message at the conclusion of SV processing.

*SVKYINTapename*: The queued file *filename* from run *run-id* is on a removable pack and received I/O error *errornum* during SV processing.

*SVKYINTapename*: Remove queue entry (may delete the file) for *filename* Y/N?

### 11.77.3. Restrictions

You cannot mount an unlabeled tape for an SV keyin if the parameter PRELABELED\_TAPES\_REQUIRED was turned on when your system was configured.

If you save and restore (using the SV and SR keyins) a nonstandard file, the banner from the @SYM statement is lost. The run-ID is used as the banner.

You must use the SEC,SAVE and SEC,RESTORE keyins to save and restore Exec files.

### 11.77.4. Examples

```
SV ,HICL/PRO231 A/F
```

This keyin saves all queued print and punch output files to reel PRO231 on any cartridge tape unit.

```
SV ,HICL/PRO234,R GARTH REPORT*PRINT9
```

This keyin saves the removable disk file REPORT\*PRINT9, which was queued by the run GARTH.

### 11.77.5. Operator Interfaces

The following rules apply for the SV keyins:

The `run-id` and `user-id` can contain wildcard characters ('\*', matching zero or any number of characters) and masking characters ('?', matching any single character). The `run-id` can also be specified as 'EXEC 8' and 'TIP\$'.

1. PR or PR/ or PR/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary print files.
2. PU or PU/ or PU/nnn where nnn is a decimal number in the range 0 through 511, inclusive. This form is used for primary punch files.
3. qualifier\*filename(cycle) having the standard syntax of file names. Qualifier and asterisk may be missing, which is interpreted as any qualifier.
4. qualifier'\*' filename(cycle). In addition to standard characters, qualifier and filename may also contain wildcard (\*) and masking (?) characters. Both qualifier and filename must be provided.

**Note:** The '\*' following the qualifier must be enclosed in quotes as an indicator that:

1. Either wildcard or masking characters or both are present
  2. This particular \* is not interpreted as a wildcard character.
5. qualifier\\*filename is alternate syntax to step 4). The backslash character indicates that either wildcard or masking characters or both are present.

## **11.78. SX—Delete Queued Files**

The SX keyin deletes queued print and punch files from the system.

### **11.78.1. Formats**

#### **All Queued Files**

```
SX A/F
```

Deletes all print/punch files except those queued to user-IDs and generates the following message:

```
REALLY WANT TO DESTROY ALL PRINT/PUNCH FILES? Y OR N-
```

Answer Y to continue the keyin. Answer N to cause the keyin to be ignored.

#### **Files for a Symbiont Group or Specified Device**

```
SX A/F sname
```

Deletes all print/punch files queued to the specific symbiont name.

#### **Files for a User-ID**

```
SX A/F USR*ID
```

Deletes all print/punch files queued to all user-IDs.

```
SX A/F user-id *U
```

Deletes all print/punch files queued to a specific user-ID.

```
SX,U user-id1[,...,user-idn] [filename]
```

Delete print/punch files for the specified user-IDs. If the optional file name is specified only that file is deleted (or matching files if wildcard/masking is used).

#### **Files for a Run-ID**

```
SX run-id,...,run-idn
```

Deletes all files for up to five specified run-IDs.

```
SX run-id[1,...,run-idn] [filename]
```

Deletes print/punch files for the specific run-id. If the optional file name is specified only that the file is deleted (or matching files, if wildcard/masking is used).

## 11.78.2. System Response

When the SX keyin processing is finished, the system displays the following message when one file is deleted:

```
1 FILE DELETED VIA SX KEYIN
```

The system displays the following message when multiple files are deleted:

```
number FILES DELETED VIA SX KEYIN
```

When the symbiont queues are large, it can take several minutes for the Exec to complete an SX keyin. The SX A/F and SX A/F *sname* keyins are especially slow. Any symbiont control keyins (SR, SV, SX, SQ) entered while an SX keyin is in progress cause further delays, and in some cases could cause an EXPOOL hold.

## 11.78.3. Restrictions

You cannot use the SX keyin on a print file that is exclusively assigned to a run.

## 11.78.4. Examples

### Example 1

```
SX A/F
```

Deletes all files queued.

### Example 2

```
SX A/F PR1
```

Deletes all files queued to printer PR1.

### Example 3

```
SX RUN1
```

Deletes all queued files belonging to run-ID RUN.

### Example 4

```
SX RUN1,ABC,DICK
```

Deletes all queued files for run-IDs RUN1, ABC, and DICK.

### Example 5

```
SX JIP YUR$KO*DDRB$PR$NT
```

Deletes the user file YUR\$KO\*DDRB\$PR\$NT for run-ID JIP.

### Example 6

```
SX ABC QUAL*FILE
```

Deletes the user file QUAL\*FILE for run-ID ABC.

### Example 7

```
SX A/F ABC *U
```

Deletes all queued files for user-ID ABC.

If any shared symbiont files are queued for printing (@SYM) and then deleted from the queue while file sharing is down, the following message is displayed:

```
xx FILES BYPASSED, FILE SHARING DOWN
```

where *xx* is the number of files that could not be deleted.

## 11.79. T\_Display Demand and Batch Run Information

The T keyin displays information about active demand runs, active batch runs, held runs, and suspended runs.

Use the B option with the T keyin if you want only the run-IDs of active batch runs; use the D option if you want only the run-IDs and site-IDs of active demand runs. If you do not specify the B or D option, you get information on both batch and demand runs. If you want more detailed information, specify the D parameter in addition to either of those options.

### 11.79.1. Format

```
T[, {B|D}UN] [D|H|S] [ID1, ID2, ..., IDn]
```

where:

B and D

is batch or demand. Both B and D cannot be specified.

U

indicates that the ID arguments are user IDs.

N

indicates to negate any mask that is displays all runs except those specified.

D

specifies run details.

H

specifies held runs.

S

specifies suspended runs.

One or more IDs can be specified, separated by commas and with or without masking. If the U option is not specified, the IDs refer to run IDs; if the U option is specified, the IDs refer to user IDs.

### Examples:

|          |                                                                                              |
|----------|----------------------------------------------------------------------------------------------|
| T D      | Detailed output of all runs in the system                                                    |
| T D,RUN? | Information for run D and all runs that start with RUN and maximum of 4 characters in length |
| T A*     | All run IDs that start with A                                                                |
| T *A     | All run IDs that end with A                                                                  |
| T,U D U1 | Details of all runs executing under user ID U1                                               |
| T,UND U1 | All demand runs not executing under user ID U1.                                              |
| T ?????? | Same as 'T' or 'T *' keyins                                                                  |

**Note:** If the first field (DIHIS) is not specified, but a single character ID is specified, and that character is D, H, or S, then the ID field is ignored and the single ID character is taken to be the DIHIS field. You can use the character and the mask character to achieve the desired result. For example, "T D?" to display the run "D" instead of just "T D" which can be unclear.

## 11.79.2. System Response

### T

The response to a T keyin contains one or more lines with several run-IDs and (for demand runs) site-IDs in the following format:

```
ACTIVE: run-id[+] [/site-id] run-id[+] [/site-id]
```

A plus (+) after run-ID indicates that the run is in main storage; /site-id indicates that it is a demand run initiated from device *site-id*.

### T,B

The response to a T,B keyin contains one or more lines with several run-IDs of batch runs in the following format:

```
BATCH: run-id[+] run-id[+] run-id[+]
```

### T,D

The response to a T,D keyin also has one or more lines with several run-IDs and site-IDs of demand runs in the following format:

```
DEMAND: run-id[+]/site-id run-id[+]/site-id run-id[+]/site-id
```

### D Parameter with T, T,B, or T,D

If you use the D parameter with a T, T,B, or T,D keyin, the run-IDs are displayed in three columns, and detailed storage information is displayed for the runs. The run-IDs and (for demand runs) site-IDs are displayed in this format:

```
run-id/[site-id] run-state storage-size
```

```
run-ID1/[site-id] state size run-ID2/[site-id] state size run-ID3/[site-id] state size
```

where:

*run-state*

is one of the following:

- + in main storage
- swapped out
- \* in facility wait

*storage-size*

is decimal storage size in thousands of words. It is the total size of all banks contained in the program and activity level BDTs.

For a T D keyin, the first line of the display includes information in this format:

```
keyword: BATCH/MAXOPN=active-batch/max-open DEMAND=active-demand
yyyy-mm-dd hh:mm:ss zone
```

where:

*keyword*

is ACTIVE for the T D keyin, BATCH for the T,B D keyin, and DEMAND for the T,D D keyin.

*active-batch*

is the number of active batch runs.

*max-open*

is the maximum number of open batch runs permitted.

*active-demand*

is the number of active demand runs.



*yyyy-mm-dd hh:mm:ss zone*

is the date and time the keyin was submitted in standard display format—that is, year, month, day, hours, minutes, seconds, and configured time zone mnemonic.

## **T H**

The response to a T H keyin contains one or more lines with several run-IDs of held runs in the following format:

HELD: *run-id run-id run-id run-id*

## **T S**

The response to a T S keyin contains one or more lines with several run-IDs of suspended runs in the following format:

SUSPENDED: *run-id run-id run-id run-id*

## **11.79.3. Examples**

### **Example 1**

This example of a T display shows nine active batch runs and four active demand runs:

```
ACTIVE: MSCP+ DCR+ RUNNER+ MAP8TM+ AP8TIM+
ACTIVE: SDTTIM SILAS+ CPCOMM+ WLSCON+/RE4323 AA085B+/RE4201
ACTIVE: JTNDAT+/RY0726 MCB1+ JTND AU+/RY0725
```

### **Example 2**

Here is a partial response to a T D display, showing information on 8 of the 29 active batch runs and 4 of the 89 active demand runs. These are the same run-IDs that are shown in the example of the T display:

```
ACTIVE: BATCH/MAXOPN=29/30 DEMAND=89 2013-05-22 17:24:35 CDT
CPCOMM/+46024 SILAS/+3141 ARC/+76
CSC/+3575 MSCPRT/+35 CIPHER/+470
WEBDMD/G00002+447 RDO/RDO002+33 GOE/RGOE23+33
RDOA/RDO003+33 JHND/RJMH23+33
```

If you enter a T,B D keyin, the format of the first display line is the same as for the T D keyin, except that it begins with BATCH instead of ACTIVE. Similarly, the first display line of the T,D D keyin begins with DEMAND.

### **Example 3**

If you enter a T D keyin when no runs are active, the system displays the message:

ACTIVE: - NONE

### Example 4

If you enter a T,B D keyin when no batch runs are active, the system displays:

```
BATCH: - NONE
```

### Example 5

If you enter a T,D D keyin when no demand runs are active, the system displays the message:

```
DEMAND: - NONE
```

### Example 6

To see the run-IDs of held runs, use the T H keyin. Here is an example of the response:

```
HELD: JAKEY CR JJIP PLPPL
HELD: ALOG BKC03 MARKKE TT
HELD: JOKU WEBQ BEWNE
```

### Example 7

To see the run-IDs of suspended runs, use the T S keyin. Here is an example of the response:

```
SUSPENDED: TQAGGA XB0001 7PCCHY XB0000
SUSPENDED: XE0000 XS0000 TQAGG XM0001
SUSPENDED: XM0000 XS0001 XE0001
```

## 11.80. TB—Send a Message to Demand Terminals

The TB (terminal broadcast) keyin sends a message to all active demand terminals.

### 11.80.1. Formats

```
TB text
```

Displays the specified text at all active demand terminals.

```
TB /I text
```

Used for emergency messages; it overrides any holds at the terminal.

### 11.80.2. Example

```
TB 7-015 TAPE UNIT WILL HAVE MAINTENANCE AT 18:00
```

The system sends this message to all active demand terminals:

```
TB 7-015 TAPE UNIT WILL HAVE MAINTENANCE AT 18:00
```

## 11.81.TF—Initialize TIP File Control

The TF keyin recovers or initializes TIP File Control. Use the keyin after the SYS FIN message.

### 11.81.1. Formats

TF ALL D

Initializes TIP File Control and deletes and recatalogs Exec files that contained TIP system files.

TF ALL I

Initializes TIP File Control and catalogs Exec files used for TIP system files.

TF ALL R

Recovers TIP File Control.

### 11.81.2. System Response

Requests for initialization are rejected if the component is not in the deferred initialization state. TIP can be in a deferred state as a result of two actions: (1) During a boot, a message is displayed (if jump key 7 is set), asking you whether you want to start TIP or defer the start. If you respond D, the start is deferred. (2) An Exec error will put TIP in a deferred state so that you can fix the error and restart TIP without booting the system.

## 11.82. TIP—Display or Change TIP Automatic Recovery Status

The TIP keyin displays or changes the status of TIP automatic recovery, which lets you automatically restore the TIP system environment after a system failure. Control information saved from a previous session lets the system restore TIP to its previous state.

For more information on the TIP keyin, refer to the *Transaction Processing Administration and Operations Reference Manual* (7830 7881).

### 11.82.1. Format

TIP *action*

where *action* is one of the following:

DF

changes state of TIP automatic recovery from DN (down) to DF (deferred).

DN

disables TIP automatic recovery.

FS

displays the status of TIP automatic recovery. The status is one of the following:

DF            TIP automatic recovery is in a deferred state.

DIP          TIP automatic recovery is in the process of being disabled.

DN            ARF has been freed from the Exec, and TIP application group state changes are not being tracked.

EIP          TIP automatic recovery is in the process of being enabled.

MSG          A read-and-reply message relating to restarting TIP has not been answered.

RS            The TIP application group is in the process of being restarted.

UIP          The Exec is in the process of updating the ARF.

UP            Currently tracking TIP application group state changes in the ARF.

RS

restarts the TIP application group; reads the ARF and reestablishes the TIP environment.

UP

enables TIP automatic recovery and saves the control information for a future recovery.

## 11.83. TIPSS—Display TIP Session Control Statistics

The TIPSS keyin displays TIP Session Control statistics.

### 11.83.1. Format

```
TIPSS application-group [function]
TIPSS OPEN
```

Displays the number of open sessions, the number of memory pages in use, the number of input and output messages processed per minute, and the paging success rate.

where:

*application-group*

is either the application group number or ALL. ALL causes the information to be displayed for each application group in sequence.

*function*

indicates the type of TIP Session Control data you want displayed:

|          |                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SESSIONS | Displays one line of statistics for open TIP sessions.                                                                                                                          |
| PAGES    | Displays one line of paging statistics for TIP Session Control.                                                                                                                 |
| MESSAGES | Displays one line describing the number of messages processed in a minute.                                                                                                      |
| STATUS   | Displays all the TIP Session Control statistics (all three of the lines).                                                                                                       |
| OPEN     | displays the number of active local clients, active remote clients, and close-pending clients in OPEN TIP sessions. These statistics are also displayed by the TIPSS ALL keyin. |

### 11.83.2. System Response

The system displays the information described in the following for the different keyin function options.

#### SESSIONS

```
APP n SESSIONS:s TERMINALS:t CAPACITY:x%
```

where:

*s*

is the number of active TIP sessions, including all open terminal sessions, the number of terminal sessions marked closed but retained until outstanding messages are processed, and the number of batch and demand runs connected to TIP under this application group.

*t*

is the number of terminals assigned to active TIP sessions, including sessions where the user is in the process of signing on.

*x%*

is the fraction of the configured maximum number of TIP sessions that are currently in use.

#### PAGES

```
APP n PAGES:p CAPACITY:y% SUCCESS RATE:r%
```

where:

*p*

is the number of memory pages that are currently allocated. The size of the main storage paging area is dynamically adjusted by the system to respond to system load.

$y\%$

is the fraction of the configured maximum number of memory pages that are currently allocated.

$r\%$

is the probability that the system can obtain space from the main storage paging area without displacing an existing page. The amount of mass storage input/output necessary to support the application group is inversely related to the paging success rate. The success rate, in turn, is affected by the number of active sessions and the configured size of the main storage paging area.

### MESSAGES

```
APP n MESSAGES PER MINUTE- INPUT:i OUTPUT:o
```

where:

*i*

is the number of input messages received from the communications network and pass-off messages generated by application group programs.

*o*

is the number of output messages generated by the application group in 1 minute. This statistic refers only to the output message creation rate, not to the message delivery rate, which depends on the state of the communications network hardware and software.

### STATUS

displays each of the preceding one-line messages, in the order shown. If you specify ALL instead of an application group number, you see the complete set of messages for each application group in numerical order; you might see the message APP *n* IS NOT CONFIGURED or APP *n* IS NOT UP instead of the set of messages for a particular application group.

### OPEN

```
Sessions: xxxxxx (Active Remote: yyyyyy; Active Local: zzzzzz)
```

where:

xxxxxx

is the total number of all OPEN TIP sessions: active remote sessions, active local sessions, and close-pending sessions. If the sum of the active remote and active local does not equal xxxxxx, then the difference is the number of close-pending sessions.

yyyyyy

is the number of all remote sessions not in a close-pending state; remote sessions originate from the network.

ZZZZZZ

is the number of all local sessions not in a close-pending state; local sessions are created by OS 2200 program requests to an Open CMS handler, such as OLTP.

### Errors

If the system detects an error in the TIPSS keyin processing, it might display one of the following messages:

TIPSS KEY ERROR: APPLICATION NUMBER IS NOT NUMERIC

TIPSS KEY ERROR: FUNCTION IS INVALID

TIPSS KEYIN NOT REGISTERED\*

Open session control is not initialized.

## 11.84. TM—Send a Message to a Site-ID, Run-ID, or User-ID

The TM keyin sends a message to a site-ID, run-ID, or user-ID.

### 11.84.1. Formats

TM *site-id text*

Sends the message *text* to the specified site-ID.

TM *run-id/R text*

Sends the message *text* to the specified run-ID.

TM *user-id/U text*

Sends the message *text* to the specified user-ID.

### 11.84.2. System Response

The system displays an error message if any of the following conditions occur:

- The site-ID, run-ID, or user-ID is not active.
- The site-ID is not a demand terminal.
- The addressee's message queue is full.

### 11.84.3. Example

TM JIP/R I CAN'T FIND YOUR TAPE.

The following message is sent to run-ID JIP:

\*TM\* I CAN'T FIND YOUR TAPE.

## 11.85. TP—Control Transaction Processing

The TP keyin controls Transaction Processing. The TP keyin has several formats to

- Get the status of the transaction code
- Clear indicators
- Load and unload copies of RTPS programs
- Display the status of RTPS programs
- Control maximum concurrency
- Display the length of queues
- List run condition information
- List the status of transaction programs
- Start and stop the collection of continuous statistics.

Table 11-3 lists the TP format variations.

**Table 11-3. TP Format Variations**

| Format                                               | Description                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TP A [ <i>trans-code</i> ] or [ <i>#app-number</i> ] | Lists the status of the currently active transaction program specified by transaction code. If you do not specify a transaction code or application number, the keyin lists the transaction code and status of all transaction programs that are currently active in the system.<br>More detailed information follows this table. |
| TP CLR                                               | Resets all the counters displayed by the TP TIP keyin to zero, except MOUT. The keyin also resets the total COMPOOL request counters to zero, and resets the lowest available COMPOOL counts to the current available counts.                                                                                                     |
| TP D RTPS#, <i>program#</i> , <i>program-name</i>    | Deletes resident transaction program system (RTPS) programs; they can also be controlled by the RPINIT utility program. The TP S keyin displays the status of RTPS programs.                                                                                                                                                      |



Table 11-3. TP Format Variations

| Format                                                                                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TP I RTPS#, <i>program#</i> , <i>program-name</i> ,<br>[, <i>number_of_copies</i> ][, <i>RTPS_type</i> ]                                             | The TP I keyin loads copies of a transaction program as an RTPS program. This keyin also lets you change the number of resident copies without unloading any online programs (that is, without using the TP D keyin). The field for the number of copies is optional. If it is used and the value exceeds the number of copies listed in the program's VALTAB entry, the number from the VALTAB entry is used. If the field is not used, the value defaults to the number of copies in the VALTAB entry.<br><br>RTPS_type defaults to D for dedicated; its other possible value is N for nondedicated. |
| TP MAX<br>TP MAX <i>number</i><br>TP MAX OFF<br>TP MAX ON                                                                                            | Displays a console message indicating the current state of TP MAX enforcement.<br><br>Sets a new concurrency maximum, which overrides the default limit set by TIPTPS and any previous TP MAX limit.<br><br>Ends enforcement of the current TP MAX limit and reinstates enforcement of the system default limit (the TIPTPS value).<br><br>Reinstates the enforcement of the most recent TP MAX limit after it has been disabled by a TP MAX OFF keyin.<br><br>More detailed information follows this table.                                                                                           |
| TP QUE                                                                                                                                               | Tells the system to display the length of current queues and the status of the system counters.<br><br>In a multihost environment, the TP QUE keyin displays only values that are local to the host.<br><br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                              |
| TP RA [ <i>trans-code</i> ] or [ <i>#app-number</i> ]<br>TP RS [ <i>trans-code</i> ] or [ <i>#app-number</i> ]<br>TP RH [ <i>number-of-seconds</i> ] | Used to help identify bottlenecks and other problems in high-volume transaction environments by displaying detailed information about specific transactions.<br><br>TP RA displays information about active transactions.<br><br>TP RS displays information about stuck or resident transaction program system (RTPS) deactivated transactions.<br><br>TP RH displays information about all active transactions that have remained active for the specified number of seconds.<br><br>More detailed information follows this table.                                                                    |
| TP RC * <i>run-id</i>                                                                                                                                | Lists run condition information about a specific execution of a transaction program, which you identify by the TIP generated run-ID.<br><br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                                                                                              |
| TP RL                                                                                                                                                | Displays all FCSS locks on the system that have runs or transactions queued to them.<br><br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Table 11-3. TP Format Variations**

| <b>Format</b>                                                                                                                                                                                                                    | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TP XL                                                                                                                                                                                                                            | Displays all shared (XPC-L) TIP FCSS locks on the system that have runs or transactions queued to them.<br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TP S [RTPS#]                                                                                                                                                                                                                     | Displays the status of one or all RTPS programs, which you load with the TP I keyin and delete with the TP D keyin. If you do not specify an RTPS program number in the keyin, you see the status of all loaded resident online programs.<br><br>In a multihost environment, RTPS programs are a local resource. The TP S keyin displays only the status of RTPS programs that are local to the host.<br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                                                        |
| TP T [ <i>prog-name</i> ] [ <i>#app-number</i> ]                                                                                                                                                                                 | Lists the status of a currently active transaction program, which you specify by program name. If you do not specify a program name or application number, the keyin lists the status of all names or all programs currently active in the system. The output appears as it does for the TP A keyin, except that the program name is displayed instead of a transaction code. As with the TP A keyin, the system displays the information in the format consistent with the highest application group number configured on your system.<br><br>The application group displayed by this keyin is derived from the AUDIT field in the VALTAB for the transaction. This number might differ from the application group displayed by the TP RC keyin. TP RC displays the application group in which the transaction is scheduled. |
| TP TA [ <i>trans-code</i> ] or [ <i>#app-number</i> ]<br>TP TS [ <i>trans-code</i> ] or [ <i>#app-number</i> ]<br>TP TQ [ <i>trans-code</i> ] or [ <i>#app-number</i> ]<br>TP TR [ <i>trans-code</i> ] or [ <i>#app-number</i> ] | Displays information based on transaction codes.<br><br>TP TA displays information about transaction codes that have at least one currently active transaction.<br><br>TP TS displays information about transaction codes that have at least one currently stuck or RTPS deactivated transaction.<br><br>TP TQ displays information about transaction codes that have at least one currently queued request.<br><br>TP TR displays information about transaction codes that had an available nonzero completion rate over the last complete DAPA time interval. (A DAPA interval is approximately 6.5 seconds.)                                                                                                                                                                                                               |
| TP TIP                                                                                                                                                                                                                           | Tells the system to display the status of transaction programs. In a multihost environment, the TP TIP keyin displays only values that are local to the host.<br>More detailed information follows this table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### 11.85.1. TP A Keyin

TP A [*transaction-code*] or [*#application-number*]

Lists the status of the currently active transaction program specified by *transaction code*. If you do not specify a transaction code or application number, the keyin lists the transaction code and status of all transaction programs that are currently active in the system.

The system displays the information in the following format:

```
ACT=transaction-code AG=appl-num Q=q *runid [, *runid]
```

where :

*appl-num*

is the application group number.

The application group displayed by this keyin is derived from the AUDIT field in the VALTAB for the transaction. This number can differ from the application group displayed by the TP RC keyin. TP RC displays the application group in which the transaction is executing.

*q*

is the number of scheduling requests (messages) queued within the Exec TIP complex for that transaction code. This count does not include scheduling requests for transactions which have not been dequeued from the Step Control input tree.

*runid*

is the TIP-generated run-ID.

*s*

is the status of the run:

|   |                       |
|---|-----------------------|
| = | Copy not in storage   |
| + | Copy in execution     |
| - | Copy stuck            |
| * | Copy terminating      |
| ! | RTPS copy deactivated |

If no TIP transaction programs are active, the system displays the following message:

```
TIP TP keyin found no active transaction programs
```

### 11.85.2. TP DTQ Keyin

TP DTQ "selection criteria" <A>

The TP DTQ keyin will delete TIP scheduling requests (TIP transaction input messages) queued on the system. This includes TIP scheduling requests queued in the TIP scheduling complex and optionally (if the "A" parameter is specified) queued on step control application input/scheduling tree(s).

The "selection criteria" parameter must be specified as part of the TP DTQ keyin in one of the following formats:

TP DTQ txncod

Delete TIP scheduling requests queued for the specified TIP transaction code (in this case, transaction "txncod").

TP DTQ #1

Delete queued TIP scheduling requests associated with the specified application group number (in this case, application number 1).

TP DTQ #ALL

Delete all queued TIP scheduling requests.

If the optional "A" parameter is specified after the required "selection criteria" parameter, step control application input/scheduling tree(s) are first searched. If any TIP input messages (queue items) are found which match the "selection criteria" they are deleted. Examples of specifying the "A" parameter on the TP DTQ keyins previously described:

TP DTQ txncod A

TP DTQ #1 A

TP DTQ #ALL A

In all cases the TIP scheduling complex is then searched. If any TIP input messages (scheduling packets) are found which match the "selection criteria" they are deleted. If any TIP scheduling requests were deleted the following console message is displayed ("xxx" is the total number deleted):

```
TIP TP DTQ keyin deleted xxx queued TIP input messages.
```

One of the following console messages will be displayed if no queued TIP scheduling requests are found which satisfy the selection criteria or the submitted selection criteria is missing or invalid:

```
TIP TP DTQ did not find/delete any TIP input messages.
```

```
TIP TP DTQ selection criteria required but not specified.
```

```
TIP TP DTQ application number specified is invalid.
```

```
TIP TP DTQ keyin selection criteria specified is invalid.
```

\*TIP\* TP DTQ keyin syntax error - TP DTQ keyin rejected.

The total number of TIP scheduling requests currently queued on the system is reported as part of the continuous console display ("Q:" field) or as output from the "SS" system console keyin.

The "TP TQ" system console keyin will display additional information about TIP scheduling requests queued in the TIP scheduling complex as well as the total number of TIP scheduling requests queued to step control application input/scheduling trees.

**Notes:**

- *When a TIP transaction code is specified as the "selection criteria" and the "A" option is also specified, information from the TIP transaction's current VALTAB/VINDEX entry (AUD, QPR, and REC fields) is used to search the appropriate step control application input/scheduling tree.*
- *To be located on the step control input tree by transaction code, MCB transactions must have the audit field (AUD,x) and the "J" status (STA,J) specified on their VALTAB entries.*
- *As part of deleting queued TIP scheduling requests, any associated input messages (MCB or COMPOOL) are released.*

### 11.85.3. TP MAX Keyin

TP MAX

Displays a console message indicating the current state of TP MAX enforcement.

TP MAX *number*

Sets a new concurrency maximum, which overrides the default limit set by TIPTPS and any previous TP MAX limit. The system displays a message confirming the enforcement of this new value.

You can set TP MAX to zero even if there are RTPS programs registered on the system, but you cannot set TP MAX to a value other than zero that is less than the total number of dedicated RTPS transactions registered on the system. Such an attempt results in an appropriate error message.

TP MAX OFF

Ends enforcement of the current TP MAX limit and reinstates enforcement of the system default limit (the TIPTPS value). The system displays a message confirming this action, or a TP MAX KEY ERROR message if TP MAX enforcement is already off.

### TP MAX ON

Reinstates the enforcement of the most recent TP MAX limit after it has been disabled by a TP MAX OFF keyin. This limit overrides the system default limit (the TIPTPS value). The system displays a message confirming the enforcement of the TP MAX limit, or a TP MAX KEY ERROR message if TP MAX enforcement is already on.

Use the TP MAX keyin to change the maximum number of transaction programs that can be concurrently active in the system. You can also use this keyin to find out the current limit and the number of currently active programs. After a disk recovery of the system, TIP scheduling enforces the TP MAX concurrency limit set by the last TP MAX keyin or, if no value is currently in effect (TP MAX OFF), the system default limit (the TIPTPS configuration parameter value).

After a tape recovery of the system, if TIP Automatic Recovery has been enabled from the previous session, TIP scheduling enforces the TP MAX concurrently limit set by the last TP MAX keyin. If TIP Automatic recovery is not enabled, the system default limit (the TIPTPS configuration parameter value) is enforced by TIP scheduling.

In a multihost environment, concurrency is supported on a local basis to each host. The TP MAX keyin controls concurrency only on the local host and displays values only for that host.

The system displays a message confirming enforcement of a TP MAX limit if you enter one of the following keyins:

- A TP MAX *number* keyin
- A TP MAX ON keyin
- A TP MAX keyin without any parameters while a TP MAX limit is in effect

A console message confirming TP MAX enforcement has this format:

```
TPMAX mmmmm, USE uuuuu, ALL aaaaa, RQC cccccccccccc, STK ssss,
RTPS-D dddddd, RTPS-N nnnnn, DACT-D ddact, DACT-N ndact
Last TPMAX change at YYYY-MM-DD HH:MM:SS zone
```

where:

*mmmmm*

is the maximum number of concurrently active programs allowed in the system.

*uuuuu*

is the number of currently active programs.

*aaaaa*

is the total number of transactions programs, active or not, currently existing on the system, plus any programs being loaded or flushed.

*cccccccccccc*

is the number of times the scheduling of transaction programs was deferred because of TP MAX limit enforcement. This requeue count is reset to zero after each TP MAX number keyin and after each system reboot.

*yyyy-mm-dd hh:mm:ss zone*

is the date the current TP MAX limit was established, in year, month, day, hours, minutes and seconds. The zone field contains the configured time zone mnemonic.

*ssss*

is the total number of stuck transactions currently on the system.

*dddd*

is the total number of dedicated RTPS transactions, active or not, currently on the system.

*nnnn*

is the total number of nondedicated RTPS transactions, active or not, currently on the system.

*ddact*

is the total number of dedicated RTPS transactions in the RTPS-deactivated state that are currently on the system.

*ndact*

is the total number of nondedicated RTPS transactions in the RTPS-deactivated state that are currently on the system.

The system displays a message indicating that no TP MAX limit is in effect if you enter one of the following keyins:

- A TP MAX OFF keyin
- A TP MAX keyin without any parameters sometime after a TP MAX OFF keyin.

A message indicating no TP MAX enforcement has the following format:

*TP MAX number \*OFF\* yyyy-mm-dd hh:mm:ss zone*

On the date and time (*yyyy-mm-dd hh:mm:ss zone*) indicated in the message, a TP MAX OFF keyin ended enforcement of that TP MAX limit. TIP is now enforcing the system default limit (the TIPTPS value).

To ensure that a TP MAX limit is not exceeded, TIP calculates the number of currently active programs (the USE value):

total number of transaction programs (the ALL value),

minus the number of programs stuck in main storage (the STK value),

minus the number of RTPS-deactivated nondedicated RTPS transactions (the DACT-N value)

The system displays a warning message the first time when the enforced TP MAX limit is exceeded:

```
TIP TPMAX number *initially exceeded*
```

If you use a TP MAX *number* keyin to lower the concurrency maximum during online operation, the number of active programs can temporarily exceed the new limit. However, TIP will not force any programs to abort in order to satisfy the new limit. If you enter a TP MAX 0 keyin, you effectively suspend TIP scheduling.

If RTPS programs are being used, you must not specify a nonzero value for the concurrency maximum below the total number of dedicated RTPS copies requested on the TP I keyin. If you attempt to do so, the system displays the following error message:

```
TIP MAX keyin error - TPMAX value is greater than zero and less than
the total number of dedicated RTPS copies requested.
```

### 11.85.4. TP QUE Keyin

This format tells the system to display the length of current queues and the status of the system counters. There can be more than one Communications Management System (CMS) program registered, and the CMS queue and counter values reflect total counts.

In a multihost environment, the TP QUE keyin displays only values that are local to the host.

When you enter a TP QUE keyin, the system displays a TIP STATUS message:

```
TIP STATUS: <EXEC level> <site name> hh:mm:ss mm:dd:yy zone
```

The rest of the display consists of labels and a 2-digit value for each. The labels represent the following:

|      |                                            |
|------|--------------------------------------------|
| ALT  | SUB-Qs built from ALTERNATE library cycle. |
| AVL  | Current available count.                   |
| CNEC | Batch-connected programs.                  |
| CORE | COMPOOL storage.                           |



|      |                                                                       |
|------|-----------------------------------------------------------------------|
| D432 | SUB-Qs built from ONLINED0432A file.                                  |
| D782 | SUB-Qs built from ONLINED1782A file.                                  |
| DAQ  | Outstanding dynamic allocation (DA) main storage requests.            |
| DF2  | SUB-Qs built from ONLINEFAST2A file.                                  |
| INQ  | SUB-Qs not yet built.                                                 |
| LODQ | SUB-Qs on initial load chain.                                         |
| LOW  | Lowest available count.                                               |
| MAIN | SUB-Qs built from MAIN library cycle.                                 |
| MSQ  | RTRANO/RTRANU schedule packets waiting for mass storage COMPOOL read. |
| OPEN | Open runs (batch and demand).                                         |
| OUTQ | Messages on CMS output queues.                                        |
| OVQ  | Entries on CMS overflow queues.                                       |
| PRIM | Primary mass storage COMPOOL.                                         |
| REQ  | Total request count.                                                  |
| SEC  | Secondary mass storage COMPOOL.                                       |
| SEXQ | Scheduling packets queued for SEXEM.                                  |
| STK  | Stuck program copies.                                                 |
| SUBS | SUB-Q count.                                                          |
| TACT | Online transaction programs.                                          |
| TEST | SUB-Qs built from TEST library cycle.                                 |
| TOT  | Total count specified at system generation.                           |
| TRWT | Waiting transaction programs.                                         |
| UNOP | Runs in backlog (system wide).                                        |

### **11.85.5. TP RA, TP RS, and TP RH Keyins**

These formats help identify bottlenecks and other problems in high-volume transaction environments by displaying detailed information about specific transactions.

The TP RA keyin displays information about active transactions. When no transaction code or application number is specified, the keyin displays information about all active transactions. When a transaction code is specified, information is displayed only about active copies of the specified transaction code. When an application number is specified, information is displayed only about transactions (active copies) currently associated with the specified application group.

The TP RS keyin displays information about stuck or resident transaction program system (RTPS) deactivated transactions. When no transaction code or application number is specified, the keyin displays information about all stuck or RTPS deactivated transactions. When a transaction code is specified, information is displayed only about stuck or RTPS deactivated transaction program copies of the specified transaction code. When an application number is specified, information is displayed only about stuck or RTPS deactivated transaction copies associated with the specified application group.

The TP RH keyin displays information about all active transactions that have remained active for the specified number of seconds. If no *number-of-seconds* parameter is specified, a default of 5 seconds is used. The maximum value that can be specified is 999999 seconds.

Information for each transaction is displayed in the following format:

```
run-id ag t-code t-type program lv state ex-time sups i/os pid
```

where:

*run-id*

is a unique system-generated transaction run identifier.

*ag*

is the application group number or 0 if the run is not associated with an application group.

*t-code*

is the transaction code (same as ACTION keyword from the associated VALTAB entry).

*t-type*

is the transaction program type, which can be any of the following:

|      |                                                                |
|------|----------------------------------------------------------------|
| BTCH | Online batch program                                           |
| HVTP | HVTIP program                                                  |
| RENT | Reentrant TIP program                                          |
| REN* | Reentrant TIP program requested but TIP loaded program as SINT |
| SDES | Self-destructive TIP program                                   |
| SINT | Self-initializing TIP program                                  |

*program*

is the program name for non-HVTIP transactions. For HVTIP transactions, this is the current library/bank number.

*lv*

If *state* is ILOAD, the initial switching level (same as LEVEL keyword from associated VALTAB entry).

If *state* is not ILOAD, the current switching level. This can be less than LEVEL in the VALTAB entry.

*state*

is the current program execution state, which can be any of the following:

|        |                                                                                                                                                                                                      |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACTIVE | Active TIP program.                                                                                                                                                                                  |
| ERROR  | TIP program in error termination.                                                                                                                                                                    |
| ILOAD  | TIP program with initial load status. Values for <i>ex-time</i> , <i>supr</i> , <i>i/os</i> , and <i>pid</i> are not available for transactions in this state.                                       |
| RTPSD  | Active dedicated RTPS program.                                                                                                                                                                       |
| RTPSN  | Active nondedicated RTPS program.                                                                                                                                                                    |
| DACTD  | RTPS-deactivated dedicated RTPS program.                                                                                                                                                             |
| DACTN  | RTPS-deactivated nondedicated RTPS program.                                                                                                                                                          |
| STUCK  | Stuck TIP program. When a transaction is in this state, the <i>pid</i> , <i>i/os</i> , <i>supr</i> , <i>lv</i> , and <i>ex-time</i> fields contain values for the last execution of the transaction. |
| TERM   | Terminating TIP program.                                                                                                                                                                             |

*ex-time*

is the execution time (current time minus run start time) in *hh:mm:ss* format. If this field overflows, the value is displayed as \*\*\*\*\*. If the *state* of the transaction is ILOAD, this value is unavailable (N/A).

*sups*

is the number of standard units of processing (SUP) used by this program. The value is either total accumulated SUPs or IP SUPs, depending on the VALTAB S indicator setting and the MAX\_TIME\_IP\_ONLY configuration parameter. If the value is the number of IP SUPs, an asterisk is appended to the value.

If this field overflows, the value is displayed as **\*\*\*\*\***. If the *state* of the transaction is ILOAD, L-I-P, or SWAP, this value is unavailable (N/A).

*i/os*

is the total number of I/O requests. If this field overflows, the value is displayed as **\*\*\*\*\***. If the *state* of the transaction is ILOAD, this value is unavailable (N/A).

*pid*

is the position id of the input terminal. If the *state* of the transaction is ILOAD, this value is unavailable (N/A).

### 11.85.6. TP RC Keyin

This format lists run condition information about a specific execution of a transaction program, which you identify by the TIP generated run-ID.

The system then displays a two-line to five-line status summary for that program. Lines 1 and 2 are always reported. Line 3 is reported only for HVTIP programs. Lines 4 and 5 are reported only if the Program Control Table (PCT) for the transaction program is currently in main storage.

Line 1:        *runid trans-code state=state program-type Prog=pppppp [RTPS=rrrr]*

Line 2:        *NAME=name [FILE=name [program-library-cycle]] Level=# [LOCAL]*

Line 3:        *[ICPLib=# ICPBank=# Lib=# Bank=# LLib=# LBank=#]*

Line 4:        *[Pid=PID Size=# SUPS=# [Pages=#] [mode]]*

Line 5:        *[Start=time Activities=# I/O=# AG=appl-num] | [Size=#] [AG=appl-num]*

For a TIP program in a SUPUR program file, the second line has this format:

*Name=name File=SUPUR-file# program-library-cycle*

If the program is in fast-load format, a period appears after the file number:

*FILE=SUPUR-file#.*

The *program-library-cycle* is MAIN, ALTERNATE, or TEST.

These fields appear, or might appear, in a four-line response to a TP RC keyin:

*runid*

is the TIP-generated run-ID.

*trans-code*

is the transaction (action) code by which the program is scheduled.

*state*

is the current program state:

|        |                       |
|--------|-----------------------|
| XQTING | Executing             |
| INTLOD | Initial load          |
| TERM   | In termination        |
| STUCK  | TIP stuck copy        |
| RTPDAC | RTPS deactivated copy |
| ERROR  | In error termination  |

*prog-type*

is the program type:

|        |                   |
|--------|-------------------|
| RE-ENT | Reentrant         |
| S-INIT | Self-initializing |
| S-DEST | Self-destructive  |
| ONBTCH | TIP online batch  |
| HVTIP  | High-Volume TIP   |

PROG=#

is the TIP program number from the VALTAB entry for the transaction code.

RTPS=#

is the RTPS number of the program.

NAME=*name*

is the name of the absolute element in the load file.

MAIN

is the MAIN HVTIP library file cycle if the program is an HVTIP program.

FILE=*name*

is the name of the standard online program file or the number of the SUPUR program file from which the absolute element was loaded. If the SUPUR program is in fast-load format, a period follows the file number. For standard online file loading, the file name is one of the following:

|        |              |
|--------|--------------|
| D0432A | ONLINED0432A |
| D1782A | ONLINED1782A |
| FAST2A | LINEFAST2A   |

*program-library-cycle*

is the cycle of the SUPUR program library file if the program is an absolute. The value is MAIN, ALTERNATE, or TEST.

LEVEL=#

is the execution level of the transaction program as specified in the VALTAB entry for the transaction code.

LOCAL

is an indicator that the transaction's VALTAB entry has been registered as local in an XTC environment by means of the HOST VALTAB parameter. The LOCAL field is reported for all transaction program types.

ICPLIB=#

is the HVTIP library number of the initial control program I-bank.

ICPBANK=#

is the HVTIP bank number of the initial control program I-bank.

LIB=#

is the HVTIP library number of the current I-bank.

BANK=#

is the HVTIP bank number of the current I-bank.

LLIB=#

is the logged library number provided by the TPUR COPY,L option.

LBANK=#

is the logged bank number provided by the TPUR COPY,L option.

PID=#

is the position-ID of the requesting workstation.

SIZE=#

is the program size in decimal 64-word blocks.

SUPS=#

is the number of standard units of processing (SUP) used by this program. The value is either total accumulated SUPs or IP SUPs, depending on the VALTAB S indicator setting and the MAX\_TIME\_IP\_ONLY configuration parameter. If the value is the number of IP SUPs, an asterisk is appended to the value.

PAGES=#

is the total actual pages.

*mode*

is the mode in which the transaction program is executed, if other than normal mode; TESTMODE, TRAINING mode, or TEST/TRAINING mode.

START=*time*

is the start date and time in yyyy-mm-dd hh:mm:ss zone ( that is year, month, day, hours, minutes, seconds, and configured time zone mnemonic).

ACTIVITIES=#

is the number of activities currently in execution for this program.

I/O=#

is the number of I/O operations performed by this program.

AG=*appl-num*

The application number displayed by this keyin identifies the application group in which the transaction is executing. This number can differ from the application group displayed by the TP A or TP T keyins. These keyins display the application group specified by the transaction's VALTAB entry.

### Example

```
TP RC *BBBBB
```

The system displays the following console messages:

```
*BBBBB XICPC State=XQTING HVTIP Prog=608
Name=XICPC MAIN Level=14
ICPLib=13 ICPBank=1 Lib=13 Bank=1 LLib=13 LBank=1
PID=0 Size=359 SUPS=0
Start=2013-04-25 16:33:13 CDT Activities=1 I/O=0 AG=0
```

If the transaction program copy identified by the run-ID is not active, the system displays the following message:

```
TIP run-id not found
```

### 11.85.7. TP RL Keyin

This format displays all FCSS locks that have runs or transactions queued to them on the system. The run-ID of the lock's owner and the run-ids of all queued runs/transactions are displayed. The TIP file number and locked record range are also displayed.

On XTC systems, the run-IDs of local runs or transactions queued to shared (XPC-L) FCSS locks are displayed. If the owner (run or transaction) of a shared lock exists on the host its run-ID is also displayed. If the owner of a shared lock cannot be found on the host its owner is displayed as -SHRD-.

#### Example 1

TP RL

The system displays the message

| OWNER  | FILE# | START-REC | END-REC | QUEUED-TRANSACTION/RUN                                   |
|--------|-------|-----------|---------|----------------------------------------------------------|
| *BBBBG | 103   | 151       | 151     | *BBBBJ *BBBBL *BBBBN *BBBBQ<br>*BBBBS *BBBBV *BBBBX RDOB |
| RDOA   | 106   | 124       | 124     | *BBBBH *BBBBK *BBBBM                                     |
| -SHRD- | 109   | 5211      | 5211    | *BBBBG                                                   |

If there are no runs/transactions queued to FCSS locks on the system, the output of the TP RL keyin is as follows:

\*TIP\* No transactions/runs found for TP RL keyin.

#### Example 2

TP RL *file#*, *START-REC#* <*END-REC*>

This form of the keyin displays the owner of an FCSS lock regardless of whether there are runs or transactions queued to the lock. If the end record number is not specified, it is assumed to be the same as the start record number. The output of this keyin is as follows:

| OWNER  | FILE# | START-REC | END-REC | QUEUED-TRANSACTION/RUN |
|--------|-------|-----------|---------|------------------------|
| *BBBBF | 254   | 300       | 310     | -NONE-                 |

If an invalid TIP file is specified, the following console message are displayed:

\*TIP\* Invalid TIP file number specified for TP RL keyin.



### 11.85.8. TP XL Keyin

This format displays all runs and transactions that are queued to shared (XPC-L) TIP FCSS locks on a host in an XTC environment. With the exception of the "H/Q" field, the information displayed is the same as the "TP RL" keyin (see 11.85.7 for description). The "H/Q" field displays the following information:

- H - The HOSTID where the lock owner ("Owner" field RUNID) resides.
- Q - Indicates whether the lock owner ("Owner" field RUNID) is also queued to a shared (XPC-L) TIP FCSS lock (Y = Yes, N = No).

#### Example

TP XL

The system displays the following console messages:

| Owner  | H/Q | FILE# | START-REC | END-REC | QUEUED-TRANSACTION/RUN-HOST-A |
|--------|-----|-------|-----------|---------|-------------------------------|
| *BBBBG | A/Y | 103   | 151       | 151     | *BBBBJ *BBBBL *BBBBN *BBBBQ   |
| *BBBBL | A/Y | 106   | 124       | 124     | *BBBBH *BBBBK *BBBBM          |
| *BBBBX | B/Y | 109   | 5211      | 5211    | *BBBBK                        |
| *BBBBY | C/N | 109   | 5300      | 5300    | *BBBBN *BBBBL *BBBBG          |
| *BBBBZ | D/Y | 110   | 200       | 200     | *BBBBP                        |

If there are no runs or transactions queued to the shared (XPC-L) TIP FCSS locks on the system, the output of the TP XL keyin is as follows:

\*TIP\* No transactions/runs found for TP XL keyin.

Only one TP XL keyin is allowed to be active on a system at any given time. If a TP XL keyin is already active at the time another one is issued, the output of the TP XL keyin is as follows:

\*TIP\* TP XL keyin already in progress - try again later.

### 11.85.9. TP S Keyin

This format displays the status of one or all RTPS programs, which you load with the TP I keyin and delete with the TP D keyin. If you do not specify an RTPS program number in the keyin, you see the status of all loaded resident online programs.

In a multihost environment, RTPS programs are a local resource. The TP S keyin displays only the status of RTPS programs that are local to the host.

The system displays a response message in the following format:

```
RTPS#-rrrr Prg#-pppppp Name-name Actual=aaa Desired=ddd Type=t
```

where:

*rrrr*

is the RTPS program number.

*pppppp*

is the TIP program number.

*name*

is the TIP program name.

*aaa*

is the number of RTPS copies actually loaded.

*ddd*

is the desired number of RTPS copies.

*t*

is the RTPS program type indicator:

D      for dedicated RTPS programs

N      for nondedicated RTPS programs

## 11.85.10.TP T Keyin

TP T [program-name] or [#application-number]

Lists the status of the currently active transaction program specified by program name. If you do not specify a program name or application number, the keyin lists the program name and status of all programs on the system.

The system displays the information in the following format:

```
ACT=program-name AG=appl-num Q-q *runid [, *runid]
```

where:

*appl-num*

is the application group number.

The application group displayed by this keyin is derived from the AUDIT VALTAB for the transaction. This number can differ from the application displayed by the TP RC keyin. TP RC displays the application group in transaction it is executing.

*q*

is the number of scheduling requests (messages) queued within the complex for that transaction code. This count does not include scheduling for transactions that have not been dequeued from the Step Control, runid is the TIP generated run-ID.

*s*

is the status of run:

|   |                       |
|---|-----------------------|
| = | Copy not in storage   |
| + | Copy in execution     |
| - | Copy stuck            |
| * | Copy terminating      |
| ! | RTPS copy deactivated |

If no TIP transaction programs are active, the system displays the following message:

```
TIP TP keyin found no active transaction programs
```

## 11.85.11. TP TA, TP TS, TP TQ, and TP TR Keyins

TP TA [*transaction-code*] or [#*application-number*]

TP TS [*transaction-code*] or [#*application-number*]

TP TQ [*transaction-code*] or [#*application-number*]

TP TR [*transaction-code*] or [#*application-number*]

These keyins display information based on transaction codes, as follows:

|       |                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TP TA | Displays information about transaction codes that have at least one currently active transaction.                                                                                     |
| TP TS | Displays information about transaction codes that have at least one currently stuck or RTPS deactivated transaction.                                                                  |
| TP TQ | Displays information about transaction codes that have at least one currently queued request.                                                                                         |
| TP TR | Displays information about transaction codes that had an available nonzero completion rate over the last complete DAPA time interval. (A DAPA interval is approximately 6.5 seconds.) |

If a transaction code is specified, only information about the specified transaction is displayed. If an application number is specified, only information about transactions that have the AUDIT field in their VALTAB entries set to the specified application group are displayed. The transactions must also meet the selection criteria of the specific keyin. If no transactions are found that meet the selection criteria, the following console message is displayed for the TP TA, TP TS, and TP TR keyins:

```
No transactions found for the TP Tx keyin.
```

The last line displayed for a TP TQ keyin will be a summary line as follows (if no transactions meet the selection criteria this is the only line displayed):

Total queued: x TIP queue: y Step Control queue(s): z

where x,y,z are defined as follows:

x

number of TIP scheduling requests (schedule packets) queued in the TIP scheduling complex and input messages (queue items) queued to nodes of Step Control input tree(s).

y

number of TIP scheduling requests (schedule packets) queued in the TIP scheduling complex.

## Z

number of input messages (queue items) queued to nodes of Step Control input tree(s).

If a syntax error is detected on a TP TA, TP TS, TP TQ, or TP TR keyin, a console message is displayed:

```
TP keyin syntax error - TP keyin rejected.
```

All these keyins display output in the same format, as in the following example:

| T-CODE  | TIPQ | ACTIVE | MAXACT | STUCK | DEACT | TYPE | LEVEL | STAT | RATE |
|---------|------|--------|--------|-------|-------|------|-------|------|------|
| TIPTX1  | 33   | 102    | 102    | 0     | 0     | HVTP | 2     | 0    | 37/S |
| TIPTX2  | 5    | 15     | 15     | 0     |       | SINT | 7     | 0    | 11/S |
| *TIPTX3 | 0    | 1      | 1      | 0     |       | SDES | 14    | 0    | 9/M  |

Definitions of the output fields are as follows:

|        |                                                                                                                                                                                                                                                                                                                                      |      |              |      |       |      |           |      |                  |      |                   |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------|------|-------|------|-----------|------|------------------|------|-------------------|
| T-CODE | Transaction code.<br>An asterisk ( * ) preceding the transaction code indicates a TEST mode transaction.                                                                                                                                                                                                                             |      |              |      |       |      |           |      |                  |      |                   |
| TIPQ   | Number of scheduling requests for the transaction code currently queued in the TIP scheduling complex. This does not include input messages queued to nodes of Step Control input trees.                                                                                                                                             |      |              |      |       |      |           |      |                  |      |                   |
| ACTIVE | Number of copies of the transaction currently active.                                                                                                                                                                                                                                                                                |      |              |      |       |      |           |      |                  |      |                   |
| MAXACT | Maximum number of copies of the transaction allowed to be concurrently active.                                                                                                                                                                                                                                                       |      |              |      |       |      |           |      |                  |      |                   |
| STUCK  | Number of transaction copies currently stuck.                                                                                                                                                                                                                                                                                        |      |              |      |       |      |           |      |                  |      |                   |
| DEACT  | Number of copies of the transaction currently RTPS deactivated.                                                                                                                                                                                                                                                                      |      |              |      |       |      |           |      |                  |      |                   |
| TYPE   | Transaction type, as follows: <table> <tr> <td>BTCH</td><td>Online batch</td></tr> <tr> <td>HVTP</td><td>HVTIP</td></tr> <tr> <td>RENT</td><td>Reentrant</td></tr> <tr> <td>SDES</td><td>Self-destructive</td></tr> <tr> <td>SINT</td><td>Self-initializing</td></tr> </table>                                                       | BTCH | Online batch | HVTP | HVTIP | RENT | Reentrant | SDES | Self-destructive | SINT | Self-initializing |
| BTCH   | Online batch                                                                                                                                                                                                                                                                                                                         |      |              |      |       |      |           |      |                  |      |                   |
| HVTP   | HVTIP                                                                                                                                                                                                                                                                                                                                |      |              |      |       |      |           |      |                  |      |                   |
| RENT   | Reentrant                                                                                                                                                                                                                                                                                                                            |      |              |      |       |      |           |      |                  |      |                   |
| SDES   | Self-destructive                                                                                                                                                                                                                                                                                                                     |      |              |      |       |      |           |      |                  |      |                   |
| SINT   | Self-initializing                                                                                                                                                                                                                                                                                                                    |      |              |      |       |      |           |      |                  |      |                   |
| LEVEL  | Transaction's initial switching level.                                                                                                                                                                                                                                                                                               |      |              |      |       |      |           |      |                  |      |                   |
| STAT   | Current transaction scheduling status. If this value is nonzero, the system copy of the transaction's VALTAB entry is in the process of being rebuilt (scheduling for this transaction code is currently suspended). This value corresponds to the TIP sticking status contained in the TIP termination (type 103) system log entry. |      |              |      |       |      |           |      |                  |      |                   |

|      |                                                                                                                                                                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RATE | Approximate transaction rate computed or extrapolated from the number of completions over the last complete DAPA time interval. (A DAPA time interval is approximately 6.5 seconds.) This value is displayed in transactions per second (/S) or per minute (/M). If a complete DAPA time interval of data has not yet been collected, N/A is displayed. |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.85.12. TP TIP Keyin

This format tells the system to display the status of transaction programs.

In a multihost environment, the TP TIP keyin displays only values that are local to the host. When you enter a TP TIP keyin, you see a TIP STATUS message.

The rest of the display contains labels and a 11-digit value for each. The labels represent the following:

|        |                                          |
|--------|------------------------------------------|
| BXFR   | Block transfer loads                     |
| CGY    | Contingencies generated/presented        |
| ERR    | TIP scheduling errors                    |
| ILOD   | Initial loads                            |
| INIT   | Multiple INITAL call iterations          |
| MOUT   | Messages placed on all CMS output queues |
| OPFAIL | Transaction scheduling failures          |
| RACT   | Stuck programs reactivated               |
| RSTR   | Transaction program restarts             |
| SUBB   | SUB-Qs built                             |
| SXMA   | Times SEXEM activated for TP error       |
| TRAN   | Transaction programs executed            |

#### Example 1

```
TP D 1,40,TTA1
```

The system displays the message:

```
RTPS- 1,TIP PROG- 40,NAME-TTA1 UNLOADED
```

#### Example 2

```
TP I 1,40,TTA1
```

The system displays the message:

```
RTPS- 1,TIP PROG- 40,NAME-TTA1 LOADED
```

**Example 3**

```
TP MAX OFF
```

The system displays the message:

```
TIP TPCMAX mm *OFF* YYYY:MM:DD HH:MM:SS Zone
```

**Example 4**

```
TP RC *TPC456
```

The system displays the message:

```
TPC456 T12345 State=XQTING Rentrant Prog = 40 RTPS=1
Name=TTA1 File=1000 Level=1
Size=16 SUPS=0 Pages=555
```

**Example 5**

```
TP T
```

The system displays the message:

```
Name= SECSVR AG=9 Q=0 : *BBBD9+
Name= XICPC AG=0 Q=2 : *BBBHD- *BBBHF- *BBBHG-
Name= XICPC : *BBBHH-
Name= XICPC : *BBBHI- *BBBHJ- *BBBHK- *BBBHL- *BBBHM-
```

## 11.86. TS—List Disabled Terminals and Enable Terminals

The TS keyin checks which terminals are disabled and then restores any or all terminals you specify.

### 11.86.1. Format

```
TS[,ALL] command [,id] [,id] . . .
```

where:

ALL

in conjunction with the LIST command, displays the status of every terminal either disabled or waiting for entry of a user-ID/password.

In conjunction with the ENABLE command, ALL indicates that every waiting or disabled terminal should be enabled.

*command*

is either LIST or ENABLE.

LIST displays the status of specified terminals on the system console. You can abbreviate LIST to L, LI, or LIS. In conjunction with ALL, LIST causes the security console to display the terminal-IDs that are either waiting for a sign-on solicitation or are disabled.

ENABLE restores disabled terminals or terminals waiting for a sign-on solicitation to a normal sign-on sequence. You can abbreviate ENABLE to E, EN, ENB, ENAB, or ENABL. In conjunction with ALL, ENABLE enables every waiting or disabled terminal.

*id[,id] ...*

specifies the terminal-IDs to list or enable. The valid characters for this field are A through Z and 0 through 9. If you have specified ALL, this field is ignored.

### 11.86.2. Examples

#### Example 1

```
TS,ALL L[IST]
```

Lists all site-IDs or Position Identifiers (PID) that are disabled or are waiting to solicit the message

Enter your user-id/password and clearance level:

#### Example 2

```
TS L[IST] [id], [id],...[id]
```

Lists the status of the terminals specified by terminal-ID.

#### Example 3

```
TS,ALL E[NABLE]
```

Returns every terminal to the normal sign-on sequence.

```
TS E[NABLE] [id], [id],...[id]
```

Returns the specified terminals to the normal sign-on sequence.

### 11.86.3. Restrictions

The configuration parameter DELAYED\_SIGN\_ON\_SOLICITATION must be TRUE in order to execute the TS keyin.



## 11.87. TU—Release a Tape Unit

The TU keyin unconditionally releases a tape unit from exclusive use. This keyin can force another system to lose exclusive use of the device. The keyin should be used only when the host that has exclusive use of the tape devices is down or has not been able to perform channel initialization for some reason. Do not use this keyin if the host is still up and functioning. If channel initialization has been performed, other systems in the tape subsystem environment can acquire the devices exclusively without the TU keyin.

After the keyin is complete, any system can then acquire the unit by bringing up the tape unit.

Use this keyin with caution, because it could stop another system from booting by removing exclusive use of the tape unit from the system.

### 11.87.1. Format

```
TU tape-unit-name [, tape unit name,...]
```

where *tape-unit-name* is the name of the tape unit.

### 11.87.2. System Response

If an error occurs during the processing of the TU keyin and the unit is not released, the following message is displayed:

```
TU OF tape-unit-name FAILED: UNABLE TO RELEASE THE UNIT
```

The keyin aborts, and no further processing takes place.

## 11.88. UD—Establish Unit Duplexing

The UD keyin establishes a unit-duplexed environment in a running system by joining a fixed or removable disk pack to an identical drive that is unassigned and reserved. It does this by copying the source unit to the destination unit.

You can use Unit Duplexing only with identical equipment types (identical in the Exec configuration). Unit Duplexing of dissimilar equipment types is not permitted.

For more information on Unit Duplexing, see the *Unit Duplexing Planning, Installation, and Operations Overview* (7830 7592).

### 11.88.1. Format

`UD devnam1,devnam2`

where:

*devnam1*

identifies a disk unit that is up (UP) or suspended (SU), in use, and not currently in Unit Duplexing mode.

*devnam2*

identifies an unassigned unit that is reserved (RV).

**Note:** Use the ABORT keyin to stop processing of the UD keyin.

## 11.89. UL—Rewind Tape Units (with Interlock)

The UL keyin rewinds a tape unit or all tape units connected to a control unit, and clears the premount status of the tape units. The UL keyin is used only on systems with TAVR. The keyin is allowed if the unit is up (UP) or reserved (RV).

The UL keyin, which is applicable only to tape units, does not cause a change in the status of a unit. Rather, it makes a unit containing a premounted reel available to you for a subsequent reel mount.

Use this keyin instead of manually rewinding and removing a premounted reel at the tape unit. It ensures that the system is aware that the reel is no longer premounted.

Also use the UL keyin:

- When you need to unload premounted reels that you expect cannot be assigned in the near future
- When a run terminates in error before all anticipated reels have been used
- When the run does not start as you expected

### 11.89.1. Formats

`UL devnam/cu`

Rewinds a specified tape unit connected to a control unit and clears the premount status.

`UL,ALL cu`

Rewinds all tape units connected to a control unit and clears the premount status.

## 11.90. UP—Bring Up Components and Interfaces

The UP keyin brings up a component that is down, suspended, or reserved. The keyin works for any component with a status of down (DN) or reserved (RV), or any disk drive with a status of suspended (SU). You cannot use the component, however, unless a path to the component is available.

### 11.90.1. Formats

There are many formats to the UP keyin. You can specify a particular component, specify a path, or use an option with or without a particular component.

#### Bringing Up Components

The following keyins specify a component:

`UP ip`

Adds the specified instruction processor to the partition.

`UP iop`

Adds the specified IOP to the partition.

`UP xiipname`

Adds the specified XIIP to the partition.

`UP xpcname`

Adds the specified XPC to the partition.

`UP mmod`

Adds the specified memory module to the partition.

`UP cmod`

Adds the specified channel module to the partition.

`UP cu`

Adds the specified control unit to the partition.

`UP devnam`

Adds the specified device to the partition.

`UP,TIME iopname`

Makes the time sources associated with *iopname* available to Exec.

If entering console keyins related to CSB maintenance on a system that does not support a clock calendar or an external time source, the following message is displayed:

```
UP,TIME keyins are not supported on this system
```

### Bringing Up an Interface

The following keyin specifies an interface:

```
UP cmod/[intf#]/cu
```

Makes the interface from the specified channel module to the control unit available. If you do not specify an interface number, the Exec supplies a value. Also use this keyin for bringing up an interface to the record lock processor (RLP).

### Bringing Up Devices

The following keyins specify an option:

```
UP,ALL cu
```

Adds all devices attached to the specified control unit to the partition.

```
UP,PACK pack-id//[dir-id]// [pack-id]...
```

Places the specified packs in the UP state.

**Note:** To use the UP,PACK keyin in a local environment, the NEWCON configuration parameter REQNBPRST0 must be greater than zero. To use the UP,PACK keyin in a shared environment, the NEWCON configuration parameter REQNBPRST1 must be greater than zero. See the Exec System Software Installation and Configuration Guide (7830 7915) for more information on these configuration parameters.

### Enabling XPC Caching

```
UP,XCACHE xpcname
```

```
UP,XCACHE xpcname/LOCAL
```

```
UP,XCACHE xpcname/SHARED
```

Enables caching with the specified XPC (*xpcname*).

Use the UP,XCACHE *xpcname* keyin to enable local and shared caching for each host using the XPC.

Use the UP,XCACHE *xpcname*/LOCAL keyin to enable a host's local caching using the XPC. If local caching was disabled using the DN,XCACHE *xpcname*/LOCAL keyin, local caching can only be enabled using an UP,XCACHE *xpcname*/LOCAL keyin.

Use the UP,XCACHE *xpcname*/SHARED keyin to enable a host's shared caching using the XPC. Once specified, shared caching can be disabled using a DN,XCACHE *xpcname*/SHARED keyin or a DN,XCACHE *xpcname* keyin. The UP,XCACHE *xpcname*/SHARED keyin can be issued when the XPC is in MHM mode.

UP,MHM *xpcname*

Initializes the XPC for use in a multihost environment.

## 11.90.2. System Response

Some messages warn of undesirable consequences of your keyin and ask for a yes/no response.

## 11.90.3. Restrictions

If your site has Security Level 2 or 3 or a Security Level 3 trusted environment, and the printout labeling feature is configured, do not use the UP keyin to bring up a printer. If you do, print files do not have security labels, are treated as system high, and the printout labeling feature is overridden. Use the RV keyin instead.

Remember, when you bring up a fixed disk, you initialize the disk and that initialization erases all disk files.

Before the system can bring up a control unit, it must enable all usable interfaces to the control unit. If the system cannot enable all the interfaces, it displays this message:

```
ENABLE OF channel/module/cu INTERFACE FAILED
```

Bring down the failing interface and bring up the control unit again.

Bringing up a disk that is prepped as removable causes the files on that disk to be registered with the system.

If your system uses the Tape Automatic Volume Recognition (TAVR) feature, do not perform the UP keyin on a tape unit with a mounted reel when the MODIFY CONFIG message appears. Wait until the SYS START message is complete. If you try to premount a tape before this point, the system displays this message:

```
device [ipnam]/iopnam/cmname/cuname NOTRDY REWI run-id A,B,M
```

Answer B to this message.

Bring the reel to load point before you attempt input to or output from the tape unit. If you attempt input or output to the tape first, the system displays this message:

```
tape-unit IS SELECTED
```

The FS keyin shows the identifier of the reel on the mounted tape, but it does not show the status as mounted.

## 11.90.4. Examples

### Example 1

```
UP BMCA
```

Brings up component BMCA and displays this message:

```
BMCA UP
```

### Example 2

```
UP ZOT0.ZOT1
```

Brings up tapes ZOT0 and ZOT1 and displays this message:

```
ZOT0 UP , ZOT1 UP
```

### Example 3

```
UP,ALL CUTAPD
```

Brings up all the devices (TAPED3 and TAPED4) attached to control unit CUTAPD and displays this message:

```
CUTAPD UPTAPED3 UP , TAPED4 UP
```

### Example 4

```
UP XPCA
```

Brings up XPC on host A and displays the mode of the XPC:

```
XPCA UP LOCAL - XPC LOCAL CACHING ENABLED FOR THIS HOST
```

### Example 5

```
UP,TIME IOP18
```

Both time sources are installed in IOP18. However, the primary time source is shown as having been selected by Exec as its time reference.

```
IOP18 (TIME) PRIMARY UP EXEC
 SECONDARY UP
```

When only the primary time source is installed, the following message is displayed:

```
UP,TIME IOP18
IOP18 (TIME) PRIMARY UP
 SECONDARY DN NP
```

where NP indicates that the secondary time source hardware is not present.

If entering console keyins related to CSB maintenance on a system that does not support a clock calendar or an external time source, the following message is displayed:

```
UP,TIME keyins are not supported on this system
```

When the remaining external source has lost access to its time reference, the following message is displayed:

```
UP,TIME IOP18
IOP18 (TIME) PRIMARY DN
 SECONDARY UP DEGR EXEC
```

When the request fails, the following message is displayed:

```
UP,TIME IOP18
x Primary time source on IOP18 failed (NZCC 006). Answer GO
IOP18 (TIME) PRIMARY DN
 SECONDARY UP
```

## 11.91.X—Terminate a Run

The X keyin aborts the specified batch run. It terminates the current task on a demand run, but allows the run to continue with the next control statement.

Use the X keyin only when the E keyin is ineffective. The E keyin allows shared software subsystems and common banks (for example Checkpoint/Restart) to remove the run from internal tables. The X keyin does not notify shared software subsystems nor common banks, and as in the example, a new checkpoints or restarts cannot be processed because they are queued.

You can abort a user run at the user's request; however, do not abort system runs that use UDS unless directed to do so by your site administrator.

Multiple X keyins can be required to terminate a run because the first terminates the run, but can start a termination task. If the operating system's dynamic configuration parameter `run_termination_task` is not blank, then a program will run after the first X keyin.

### 11.91.1. Format

```
X run-id
```

where *run-id* is the name of the active run to terminate.

### 11.91.2. System Response

If you try to terminate a run that is suspended, the system responds as follows:

```
run-id IS SUSPENDED - X KEYIN NOT ALLOWED
```

You must resume the run-ID before you can terminate it.

### 11.91.3. Example

X BOB

The system responds with

BOB ABORT

BOB ABORT FIN

## 11.92. XPCDUMP—Dump XPC Storage

The XPCDUMP keyin dumps selected nonvolatile storage (NVS) segments from the XPC to tape or mass storage.

### 11.92.1. Format

```
XPCDUMP[,MS|X] [xpc-name]
```

where:

MS

indicates the dump is to mass storage. If MS is not specified, the dump is to tape.

X

aborts a previously initiated XPCDUMP.

*xpc-name*

is the XPC name. This field is optional, but if it is entered, the name is checked for validity. If no XPC is identified, the currently configured XPC is selected.

**Note:** Use the ABORT keyin to stop processing of the XPCDUMP keyin.

### 11.92.2. Restrictions

If an XPCDUMP is initiated at MODIFY CONFIG time, the MODIFY CONFIG should not be answered "done" until the XPCDUMP is complete. If it is answered "done" before the XPCDUMP completes, the XPCDUMP can fail and the dump path can be downed.

The XPCDUMP keyin is not supported on XPC FO environment and will be rejected with a new console message. The message is as follows:

```
XPCDUMP keyin is not supported on this model of XPC
```

Other keyins relative to XPC caching terminate without any action and display existing console messages that indicate that XPC caching is not supported.



### 11.92.3. Example

XPCDUMP

The system responds with

XPC dump: Enter tape device name <EXIT\$>

Reply with the name of the tape device to be used for the XPC dump.

The system responds with

XPC dump: Enter tape reel ID <EXIT\$>

Premount the tape on the unit selected and reply with the reel ID of the tape to be used for the XPC dump.

The system responds with

XPC dump: Enter dump script <default>, directory indexes, EXIT, or ?

Reply with one of the following dump scripts or a string of Directory Table indexes or index ranges:

|                                                                          |                                                                                                                                                                                          |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DIRECTORY                                                                | Dumps only the header blocks out to the dump file. The Directory Table is included in the header blocks.                                                                                 |
| DEFAULT                                                                  | Dumps the NVS areas defined by Directory Table indexes 1 through 300. This dump script is automatically selected if you respond to the read and reply message with no output.            |
| ALL                                                                      | Dumps the NVS areas defined by all entries in the Directory Table.                                                                                                                       |
| EXIT                                                                     | Terminates the XPCDUMP process.                                                                                                                                                          |
| ?                                                                        | Displays a list of the valid dump scripts (DIRECTORY, DEFAULT, and ALL).                                                                                                                 |
| <i>string of<br/>Directory<br/>Table indexes<br/>or index<br/>ranges</i> | Dumps the storage areas specified in the Directory Table indexes. For example, enter 2–20, 27,30, to dump the storage areas defined by Directory Table indexes 2 through 20, 27, and 30. |

At the completion of the dump, the system responds with the following:

```
XPC dump: Completed writing block-cnt blocks
XPC dump: File name is SYS$*XPC$DUMP$.
XPC dump: Tape reel ID is reel-id
XPC dump: Do you want to dump more on same tape? Y/<N>
```

where:

*block-cnt*

Indicates the number of blocks written to the file.

*reel-id*

Indicates the reel number of the tape file.

If you reply with an N, the XPCDUMP process is terminated.

# Appendix A

## **Restrictions and Operating Considerations**

This appendix describes restrictions and operating considerations that do not appear elsewhere in this manual.

### **A.1. Restrictions**

If the compartment security feature is installed on your system, and you execute the CK keyin and then the RS keyin, the run is not necessarily restarted with the compartment set in execution at the time of the checkpoint. If you perform the RS keyin, the default compartment set for the user-id entered is given to the restarted run.

If the common bank protection feature is installed on your system, you cannot execute the CK or RS keyins. The CKRS product is not compatible with the common bank protection feature. If you attempt to do the CK or RS keyins, the system displays an error message.

See the *ClearPath OS 2200 Software Products Installation Guide* (7831 0612) for additional restrictions.

### **A.2. Operating Considerations**

This subsection describes permanent limitations on the operation of a system, product, or feature.

#### **A.2.1. Dumps**

While booting the system, you cannot access user files on mass storage and copy the contents of those files into a dump file. You also cannot obtain a dump listing of those files.

#### **A.2.2. Symbiont Control**

You cannot use the SX keyin on a print file that is exclusively assigned to a run.

### **A.2.3. System Logging**

For systems with log configured to tape, if the system log tapes are not mounted quickly, an XER 070 system stop results.

### **A.2.4. TIP Session Control**

TIP Session Control is valid only if the highest application group number configured on the system is nine or less.

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